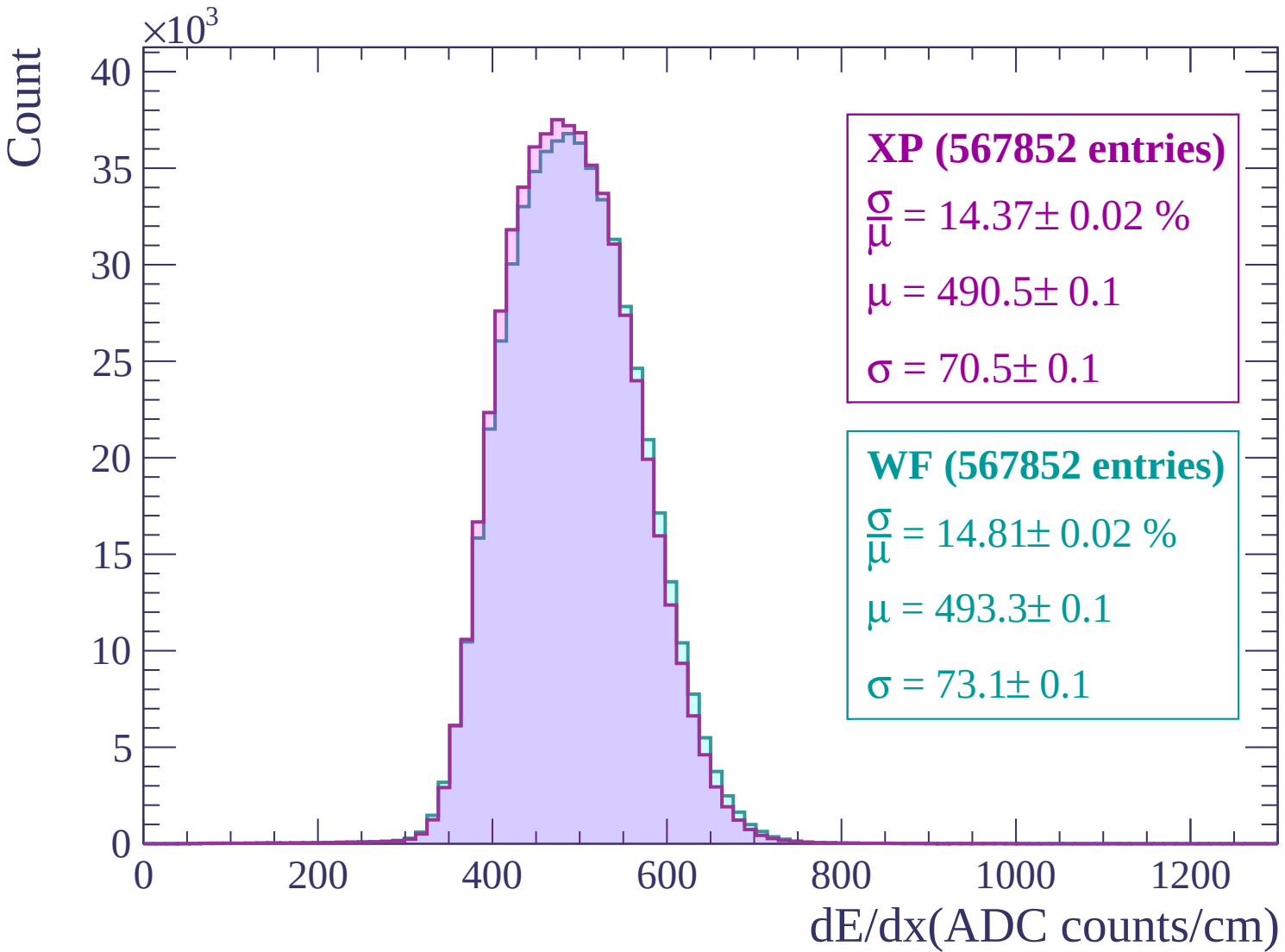
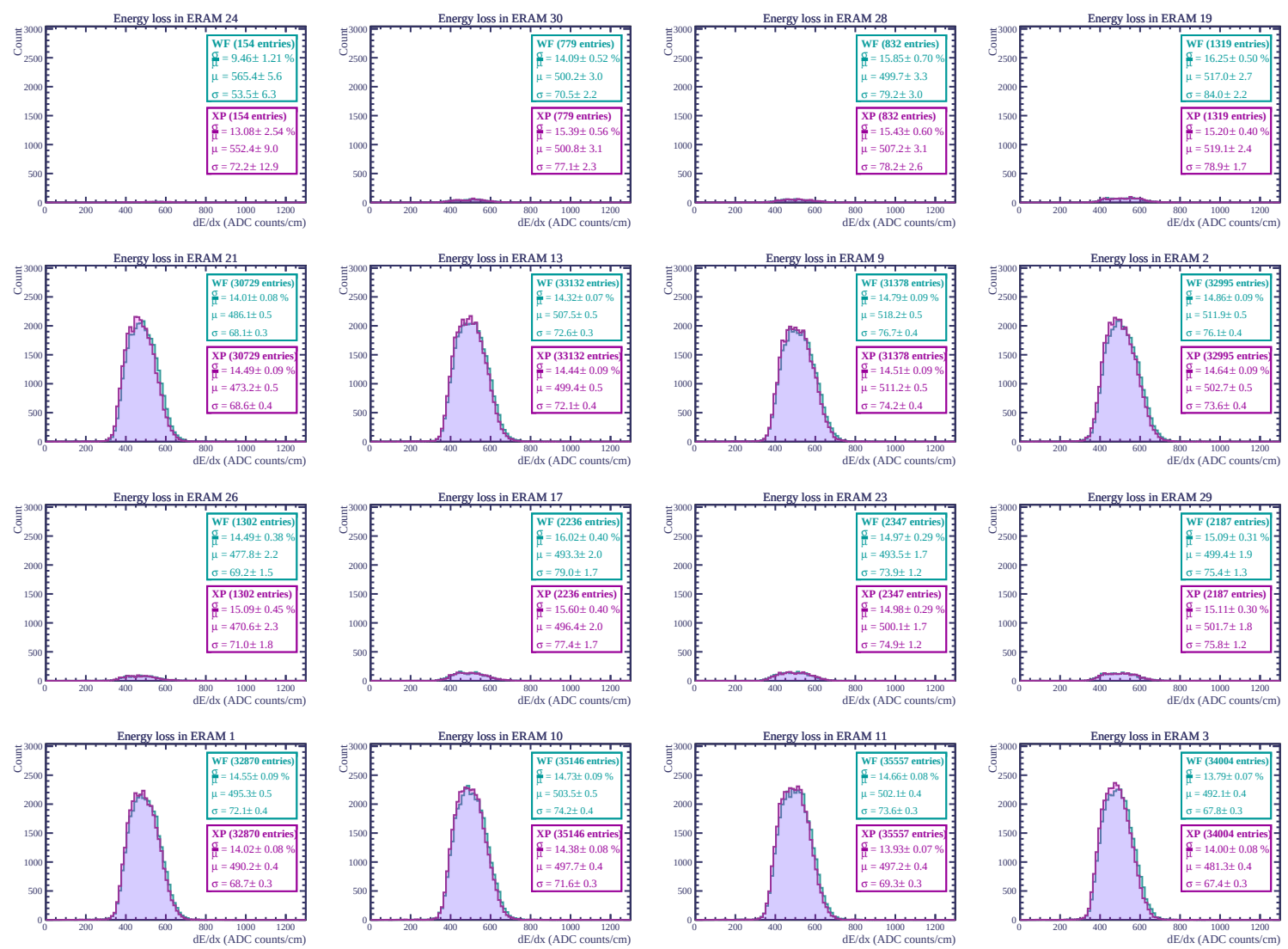
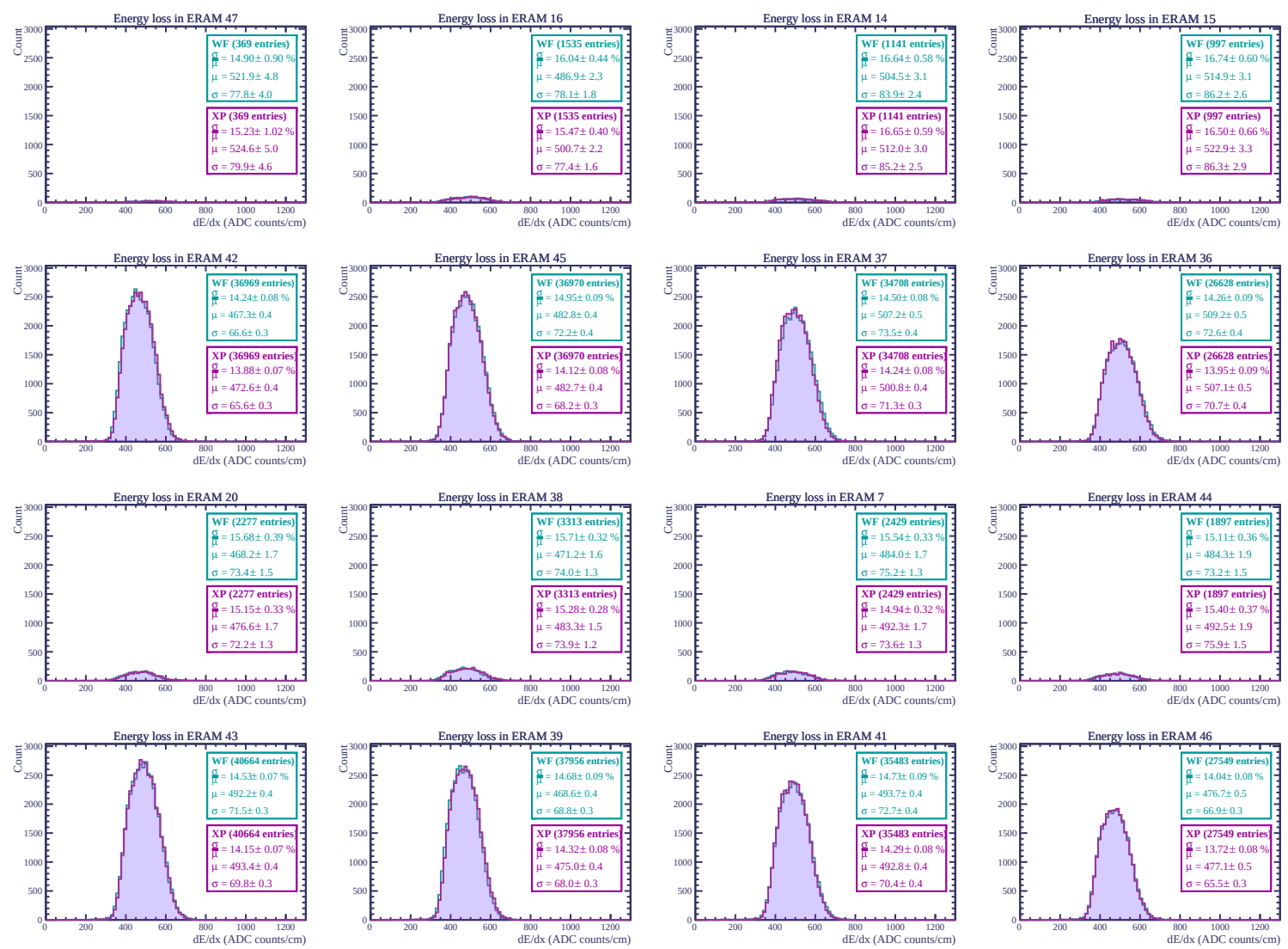
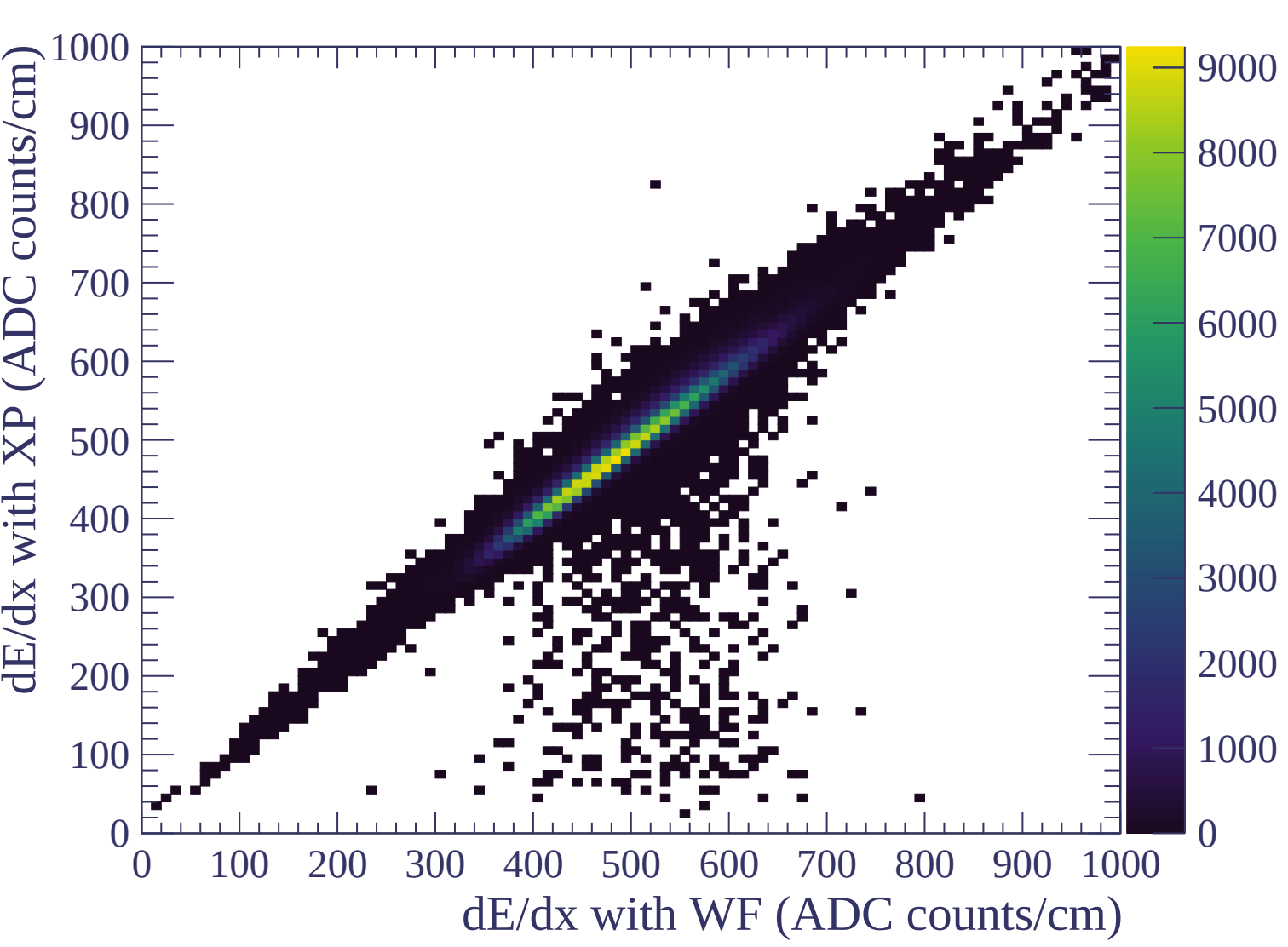
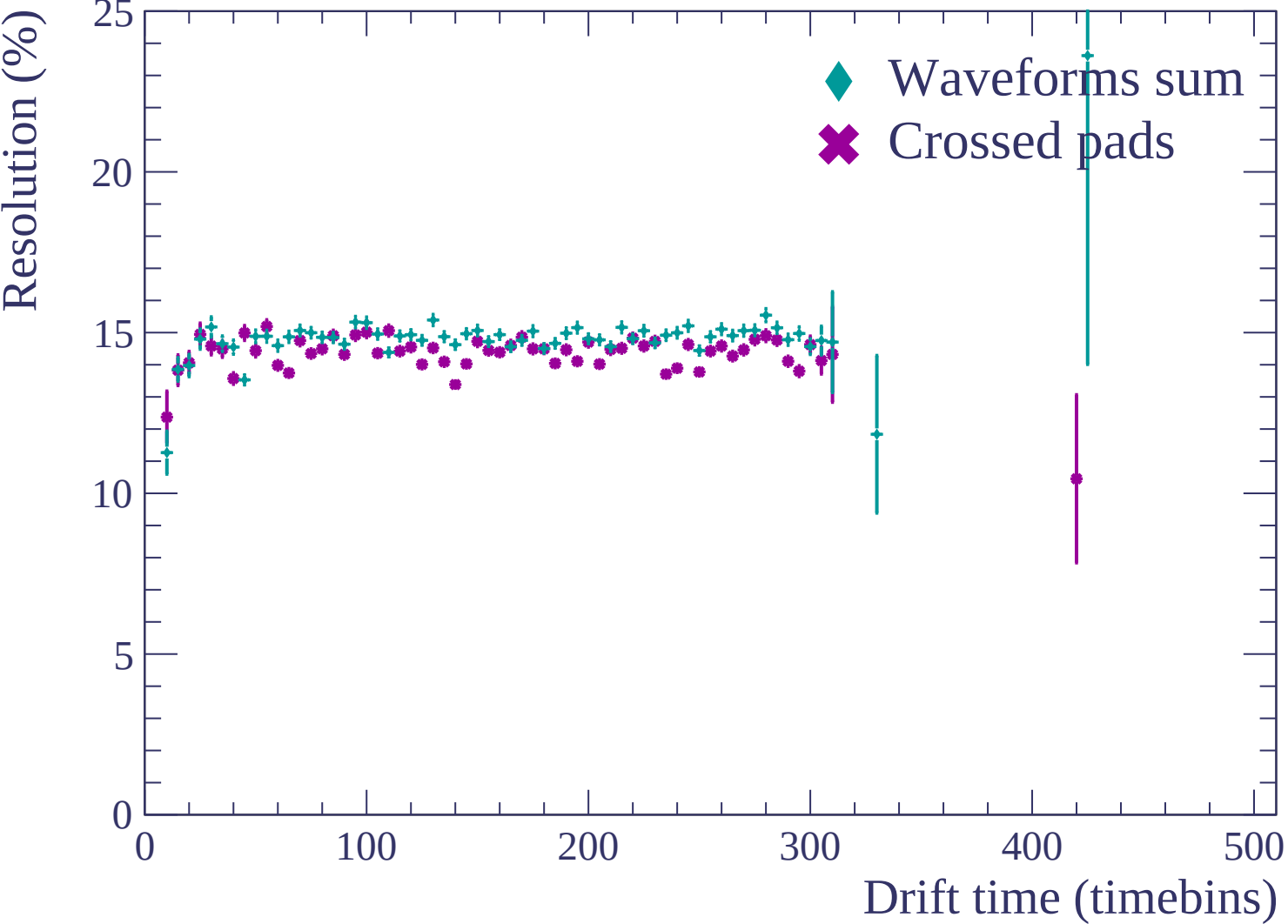


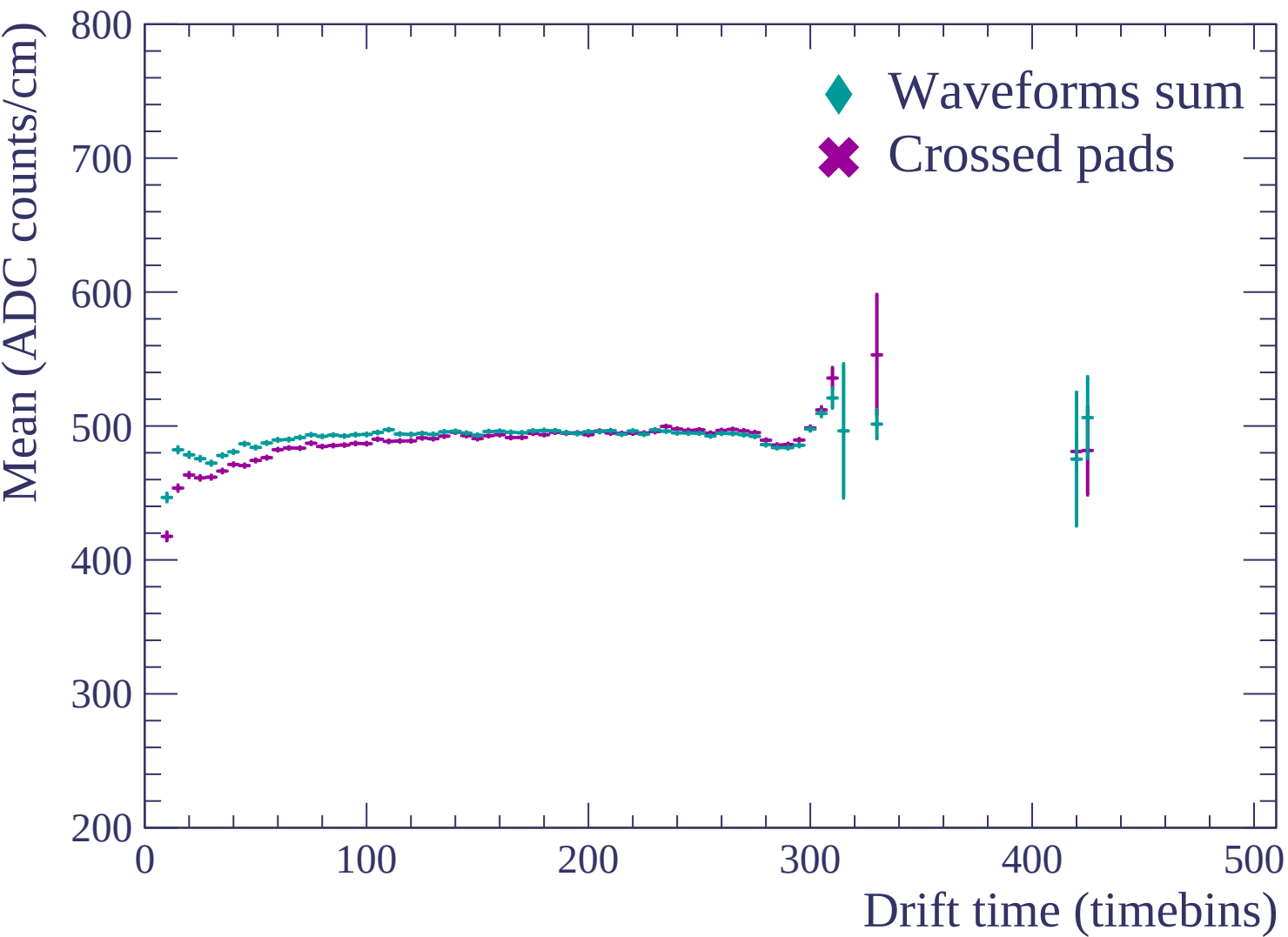


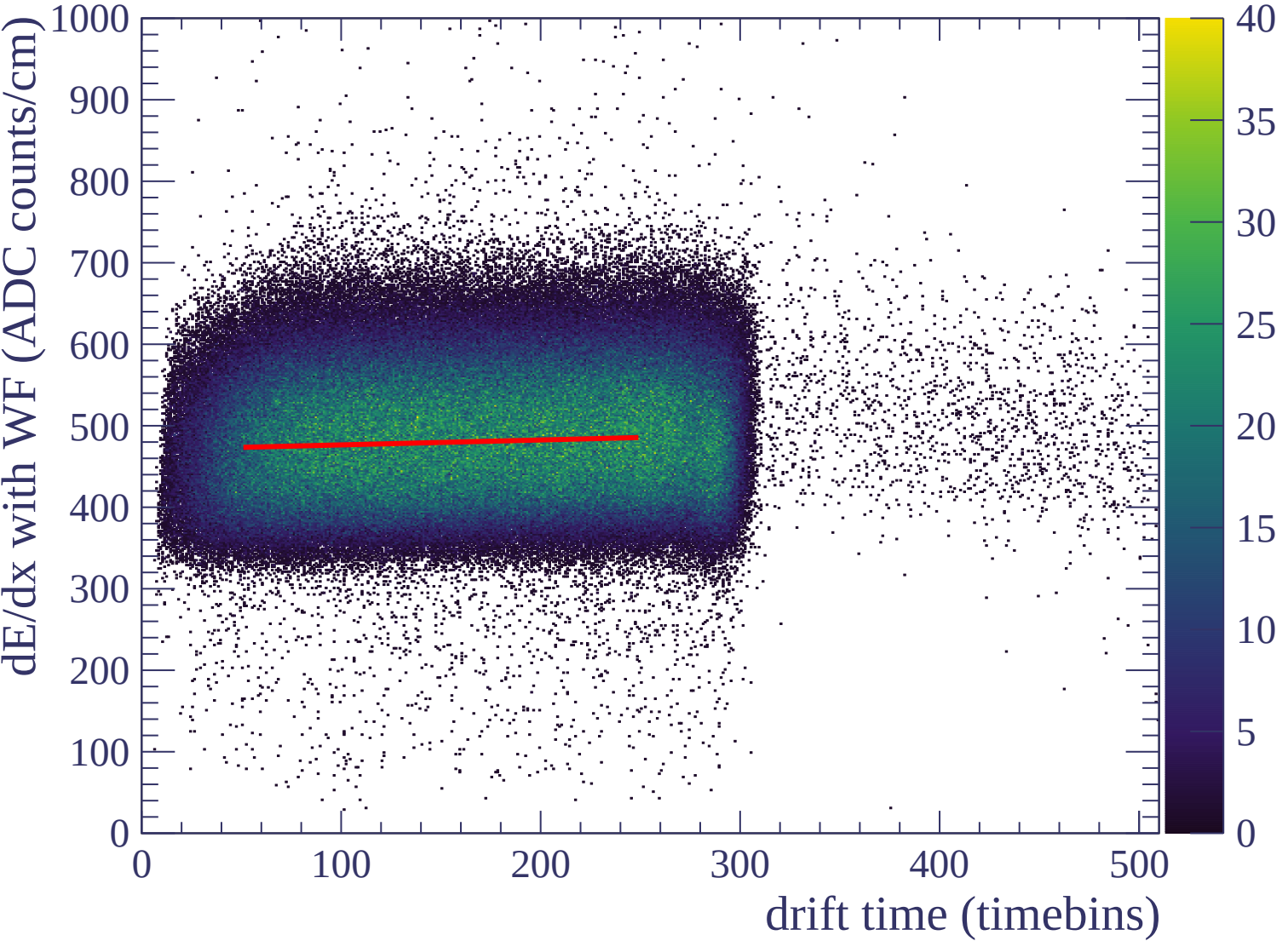


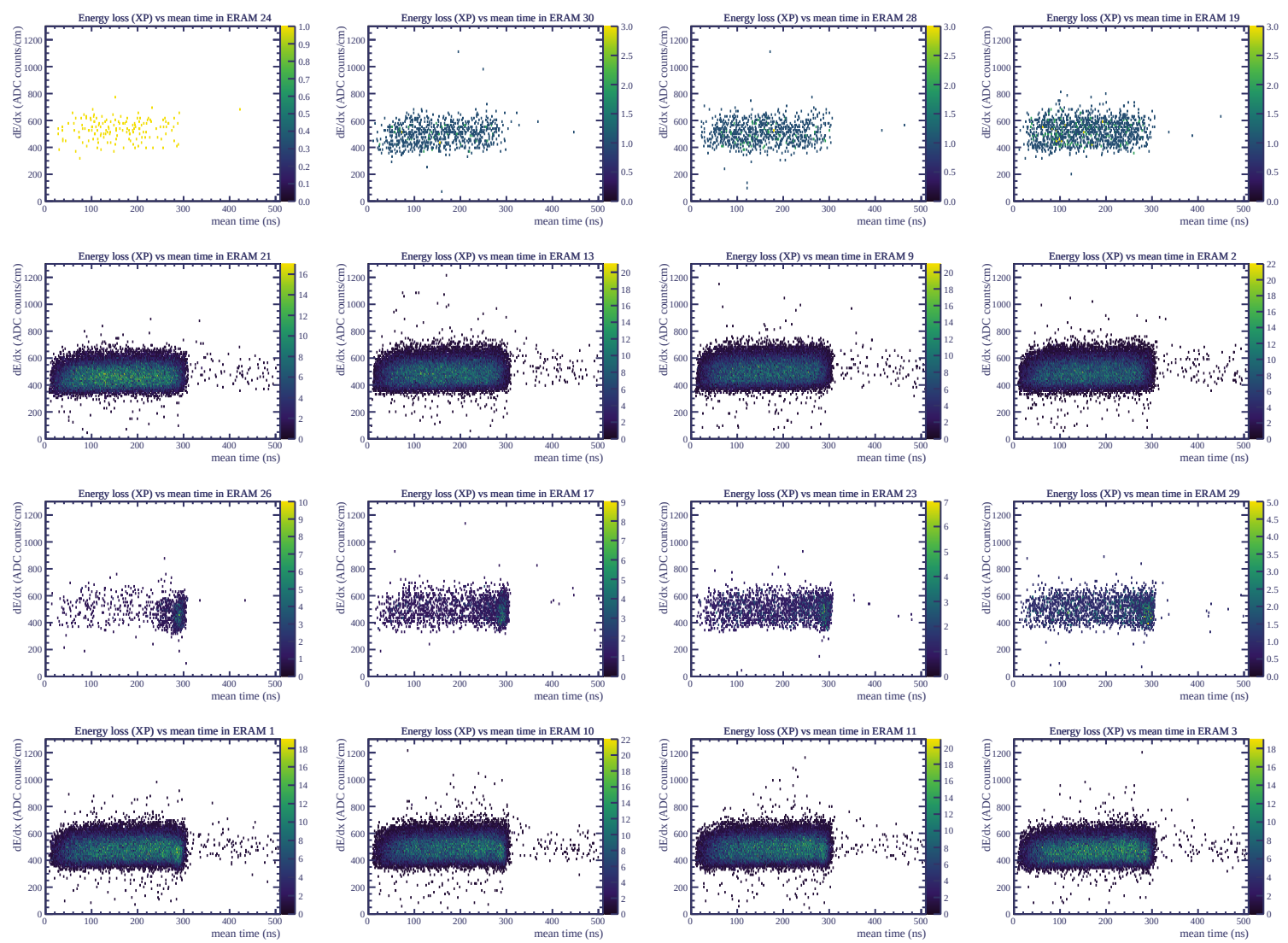


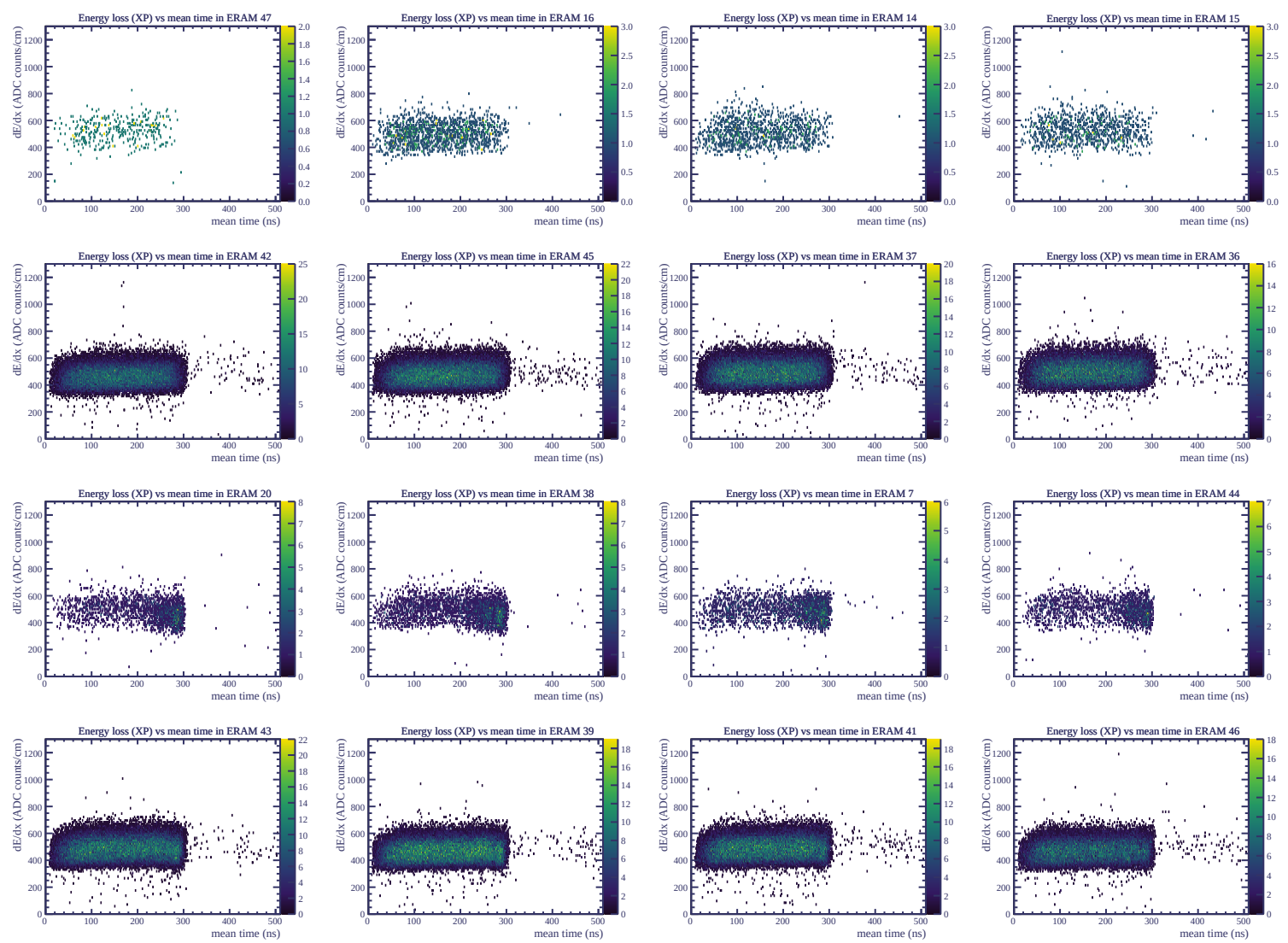


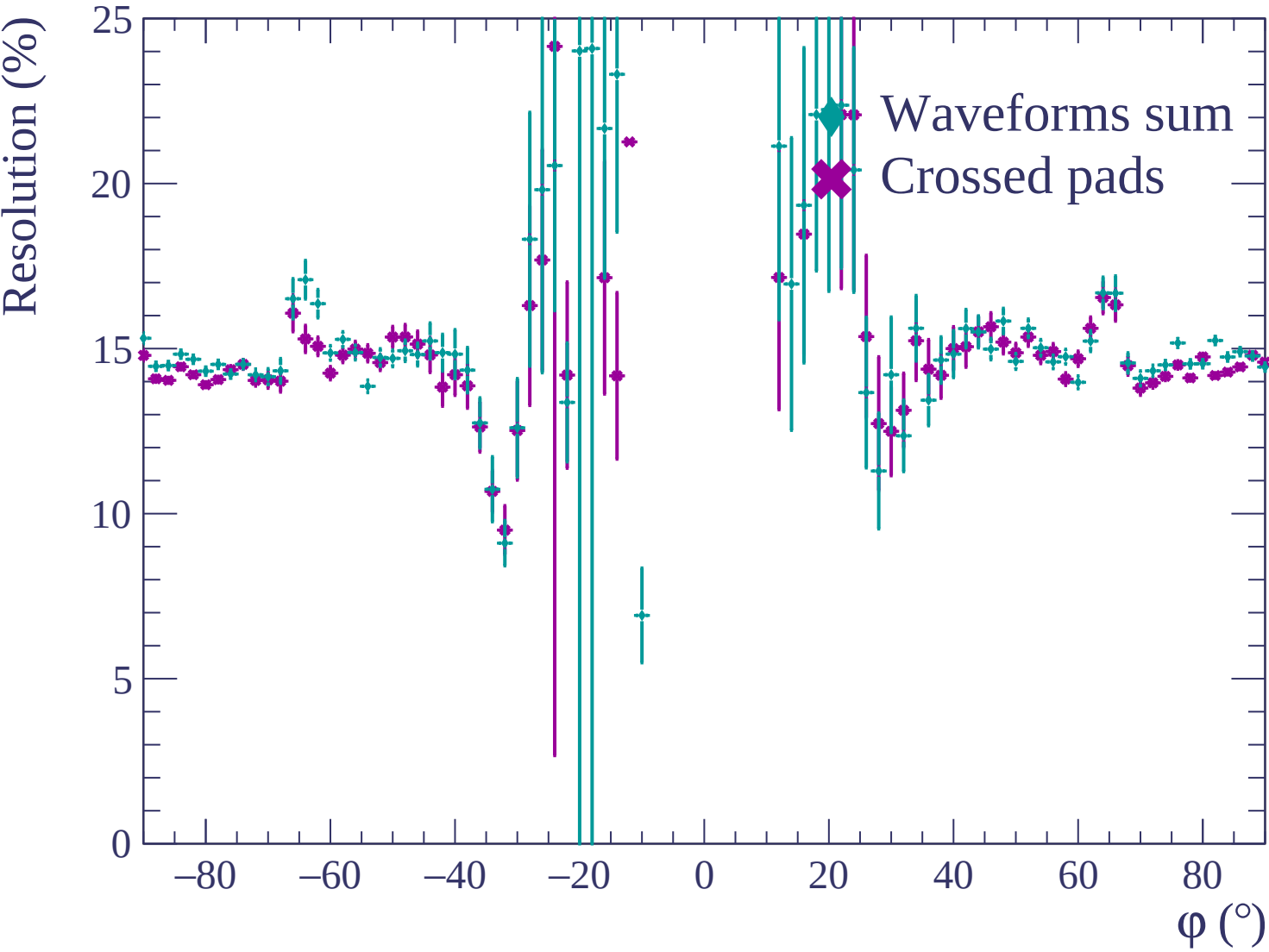


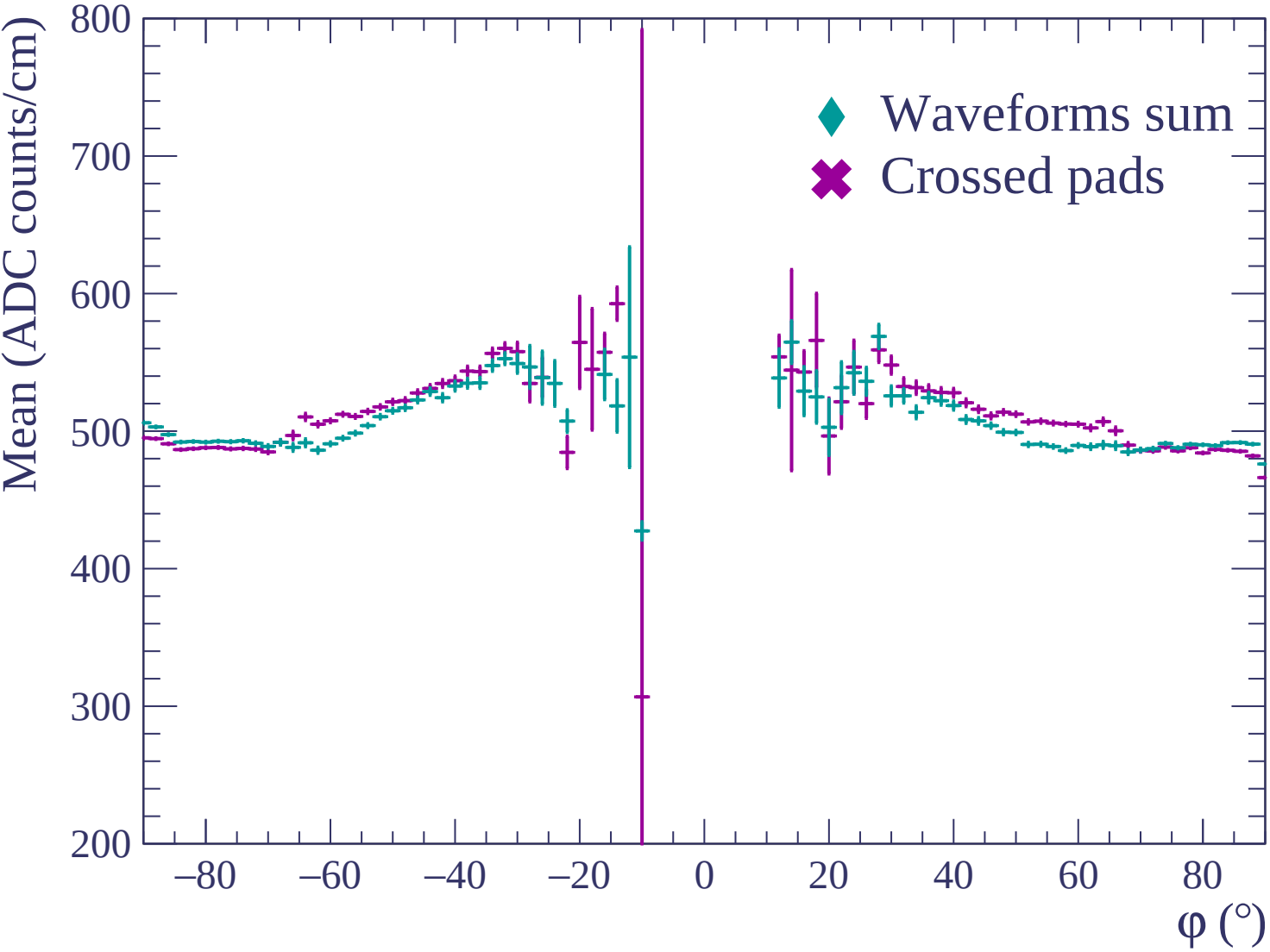


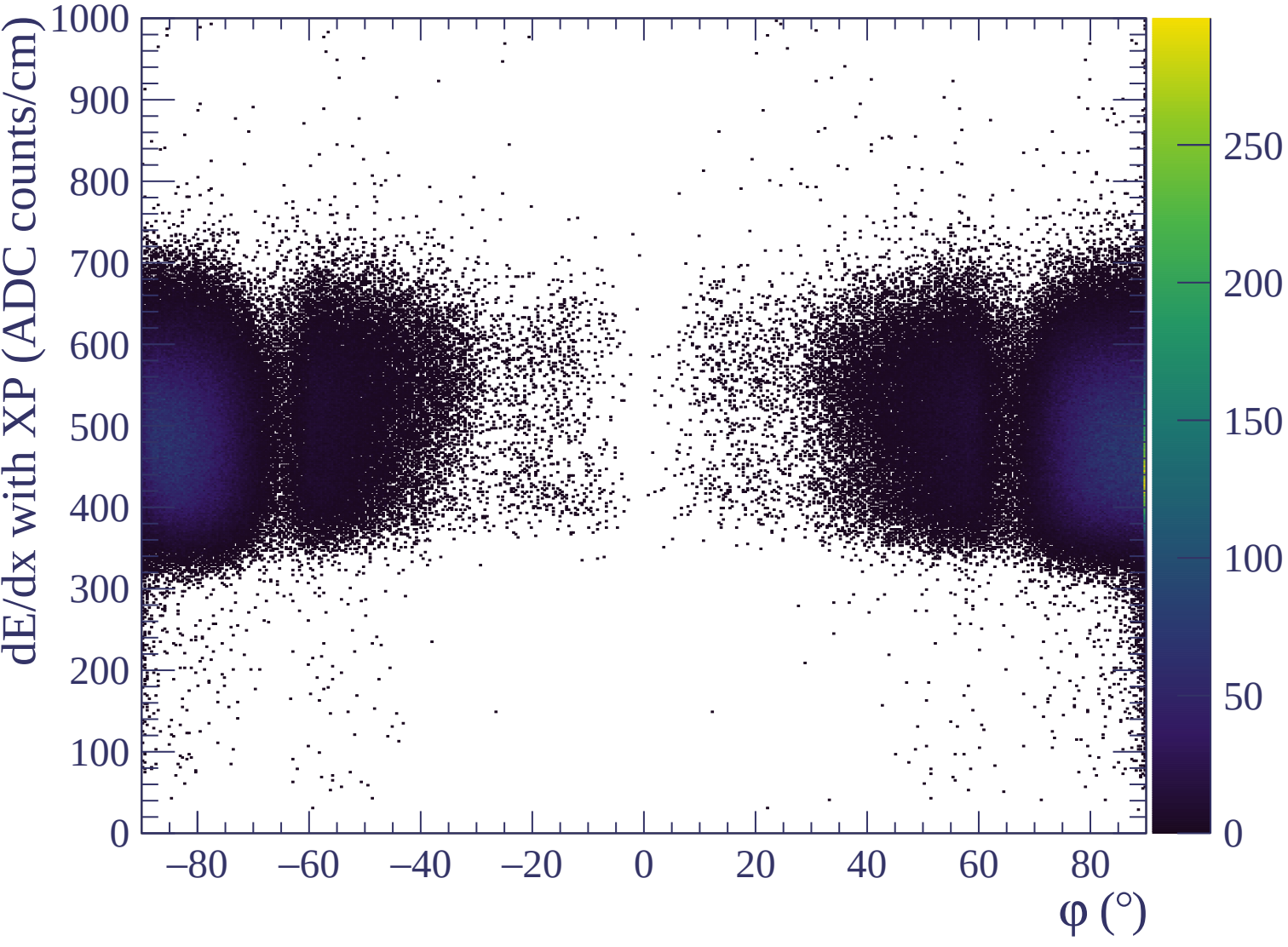


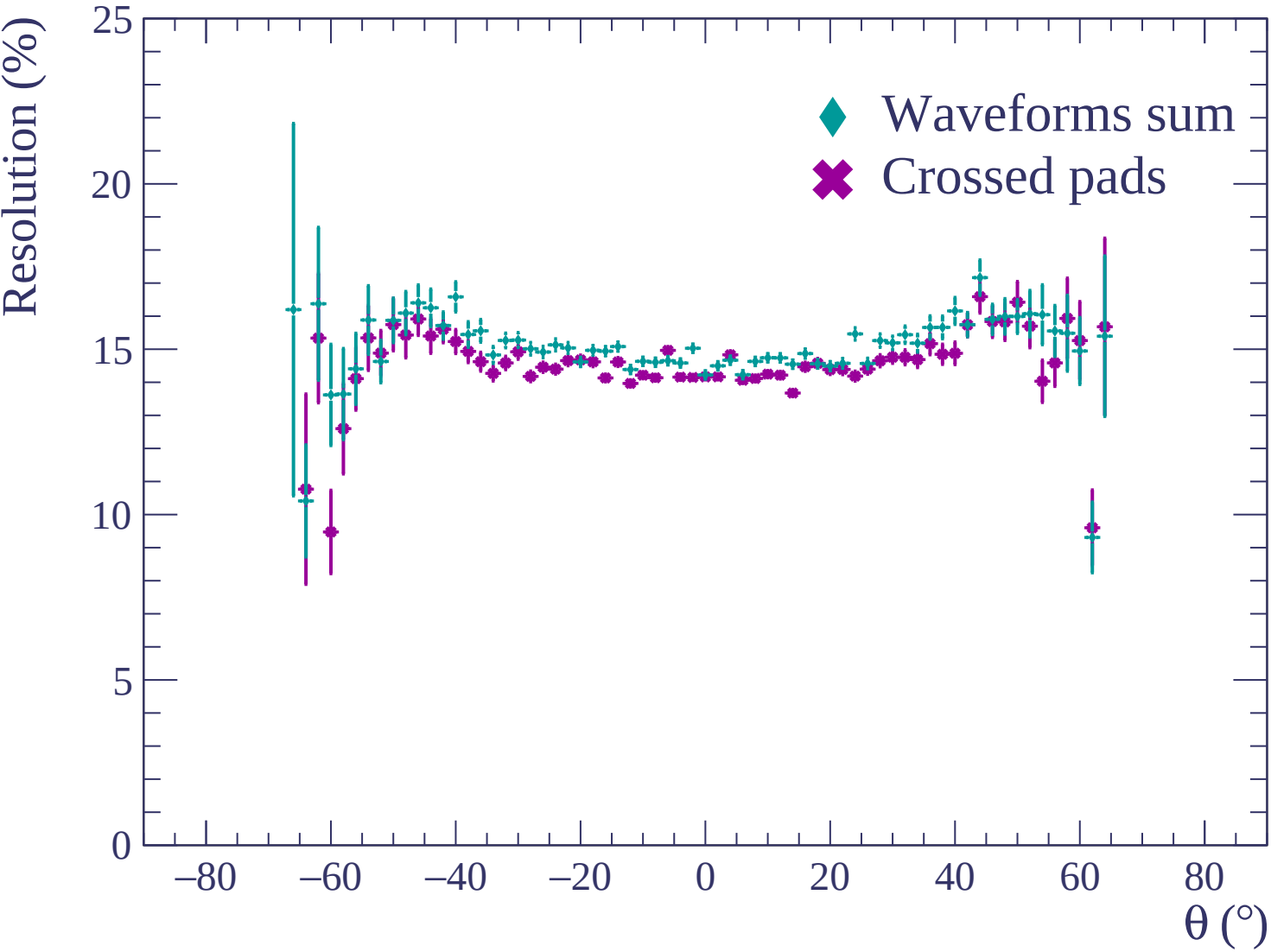


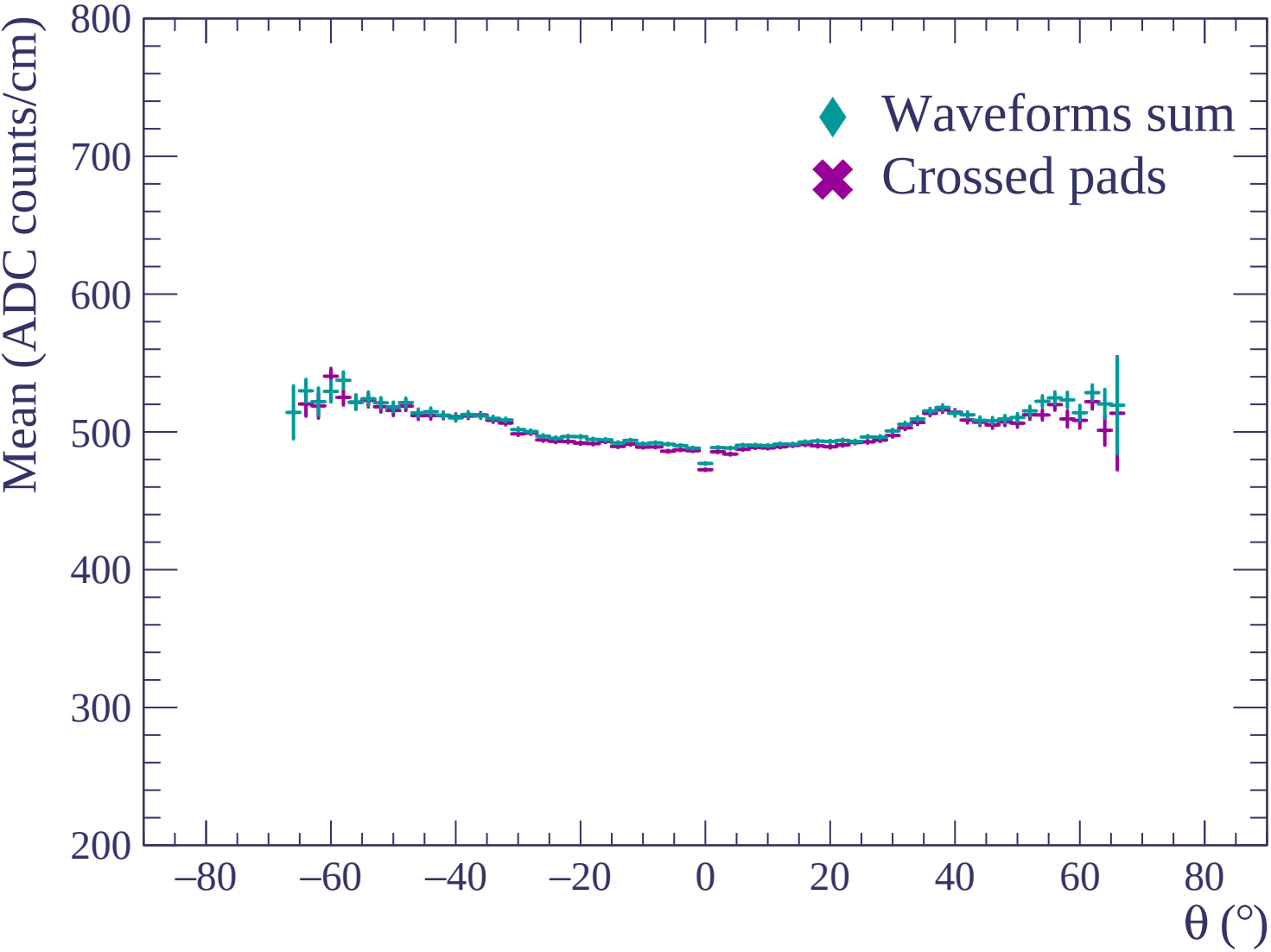


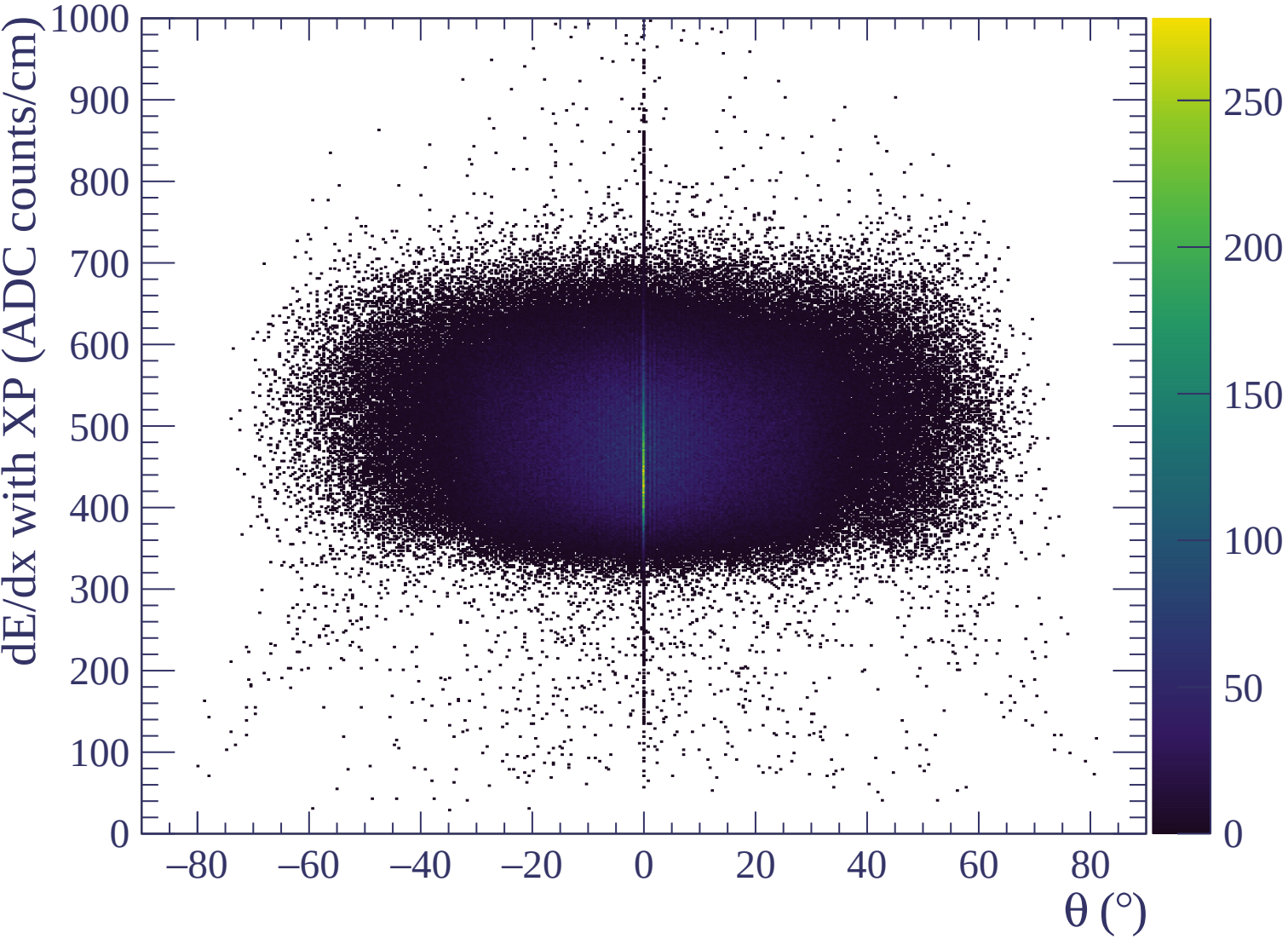


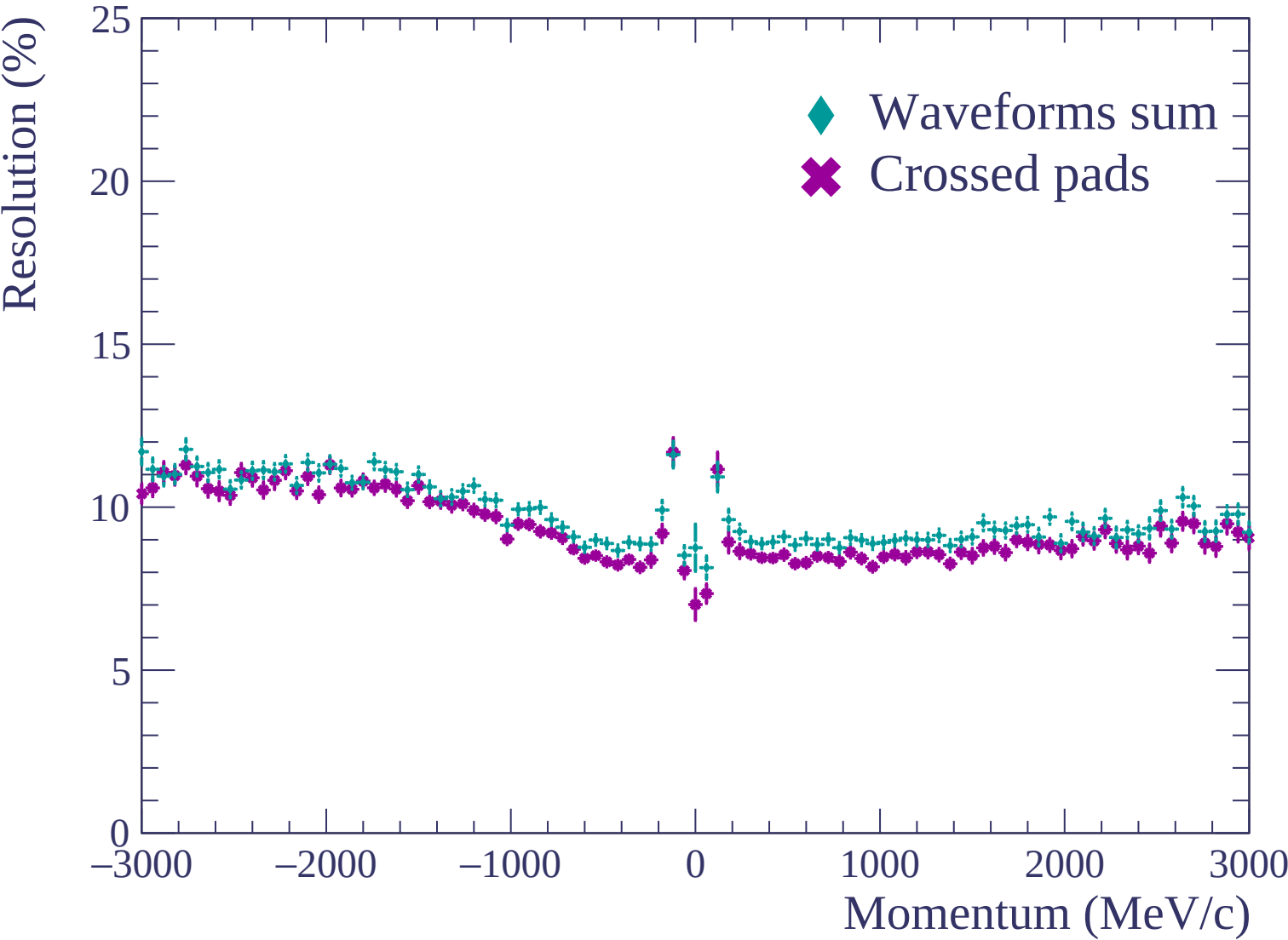


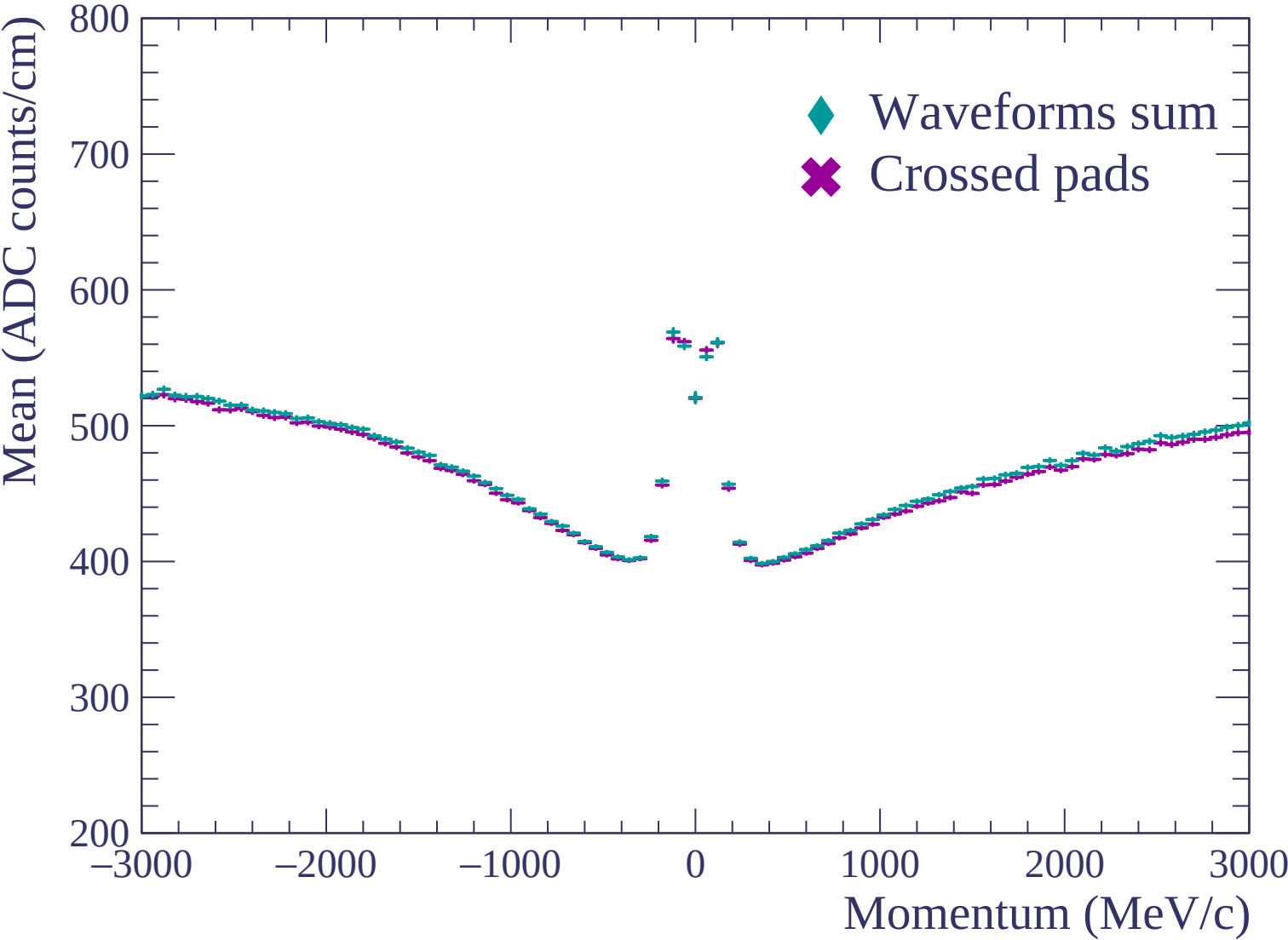


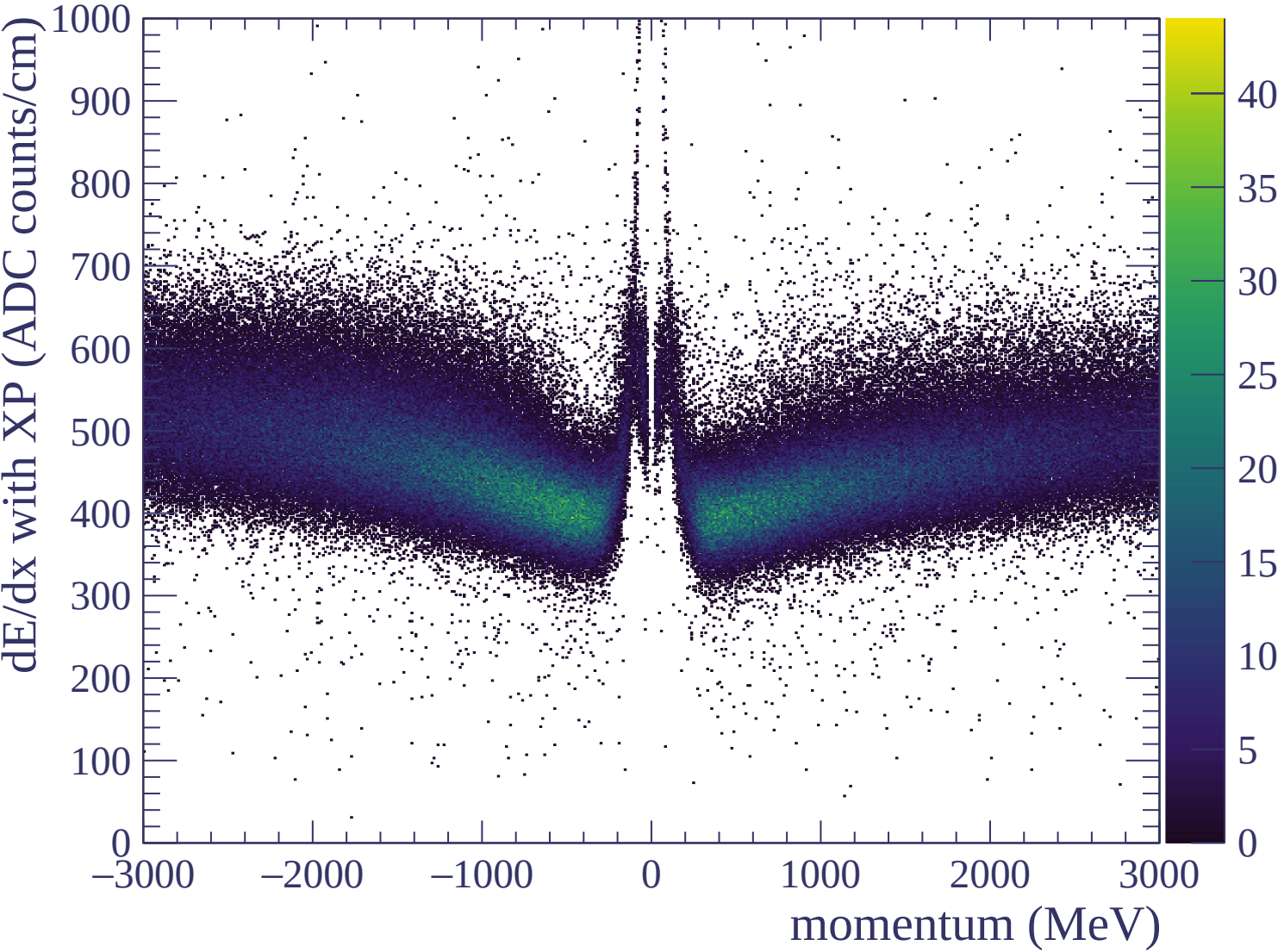


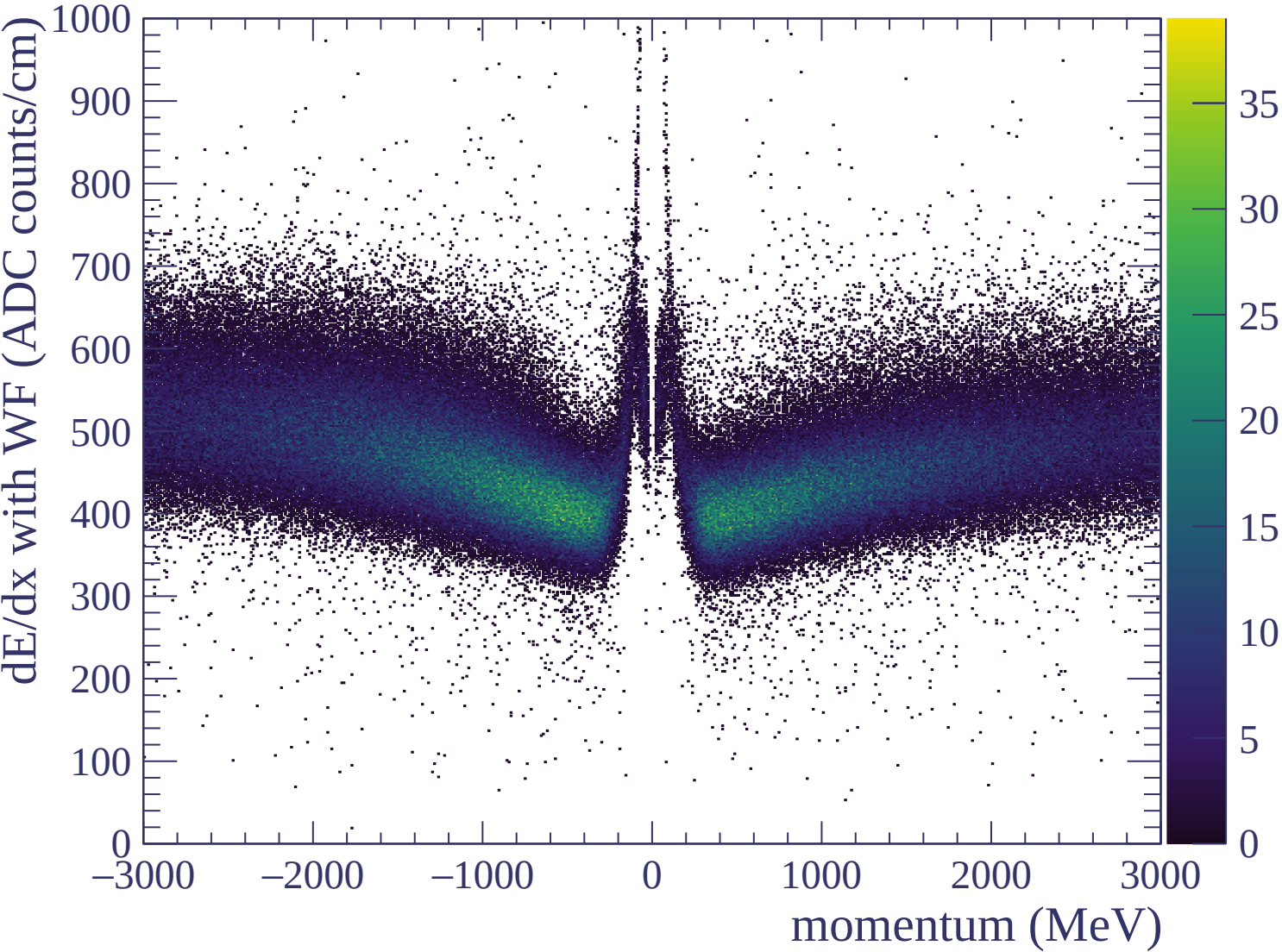


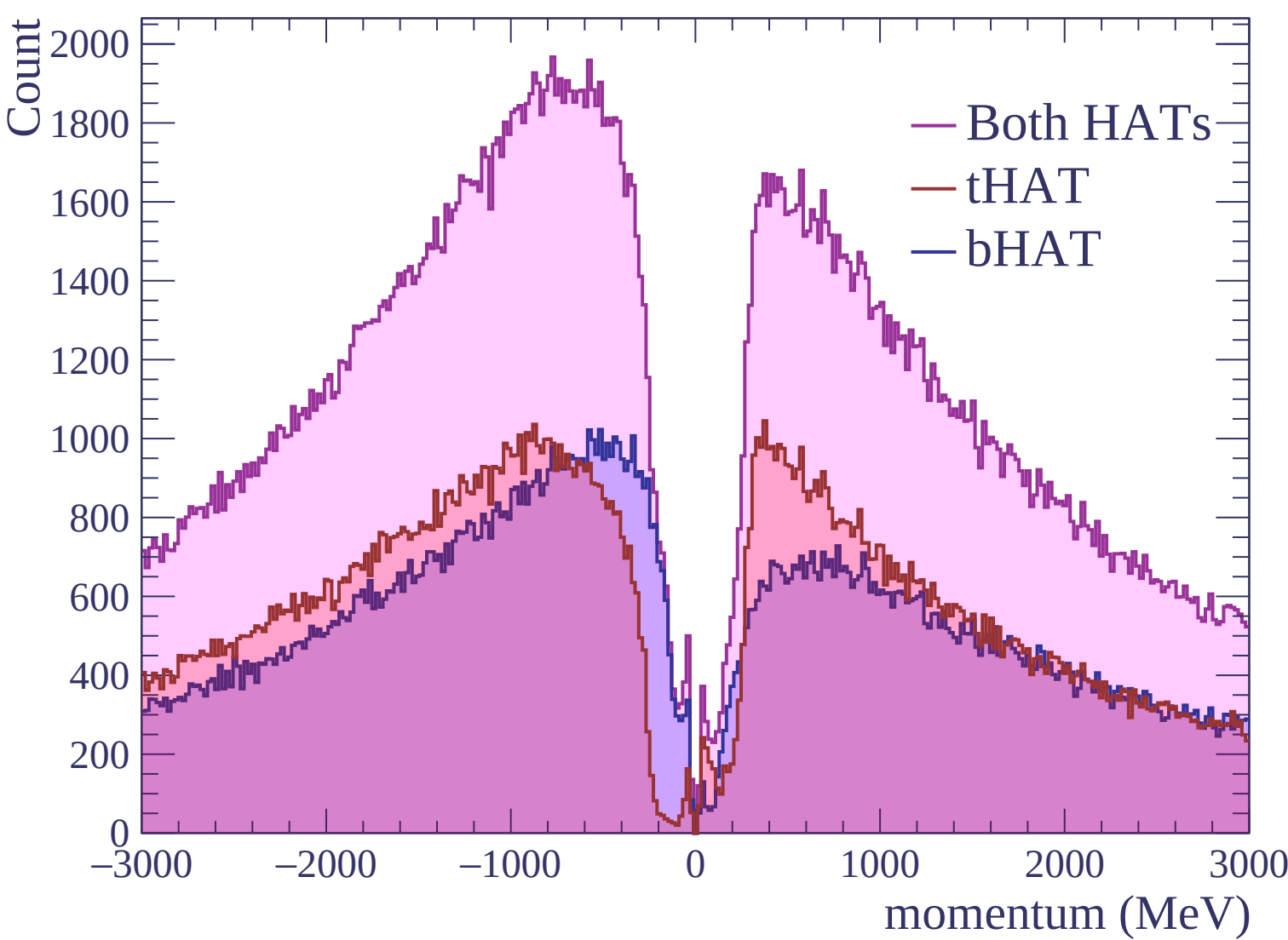


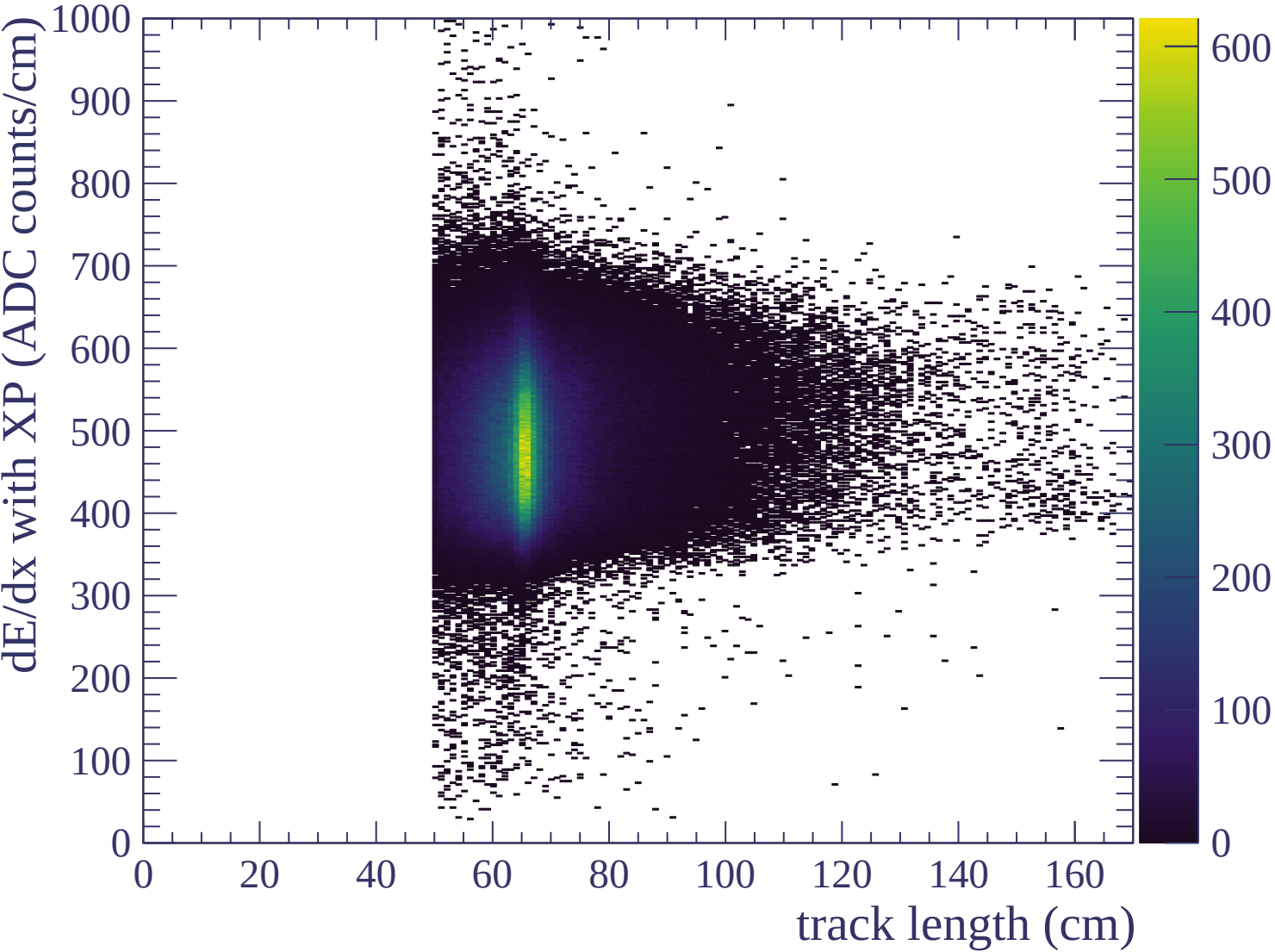


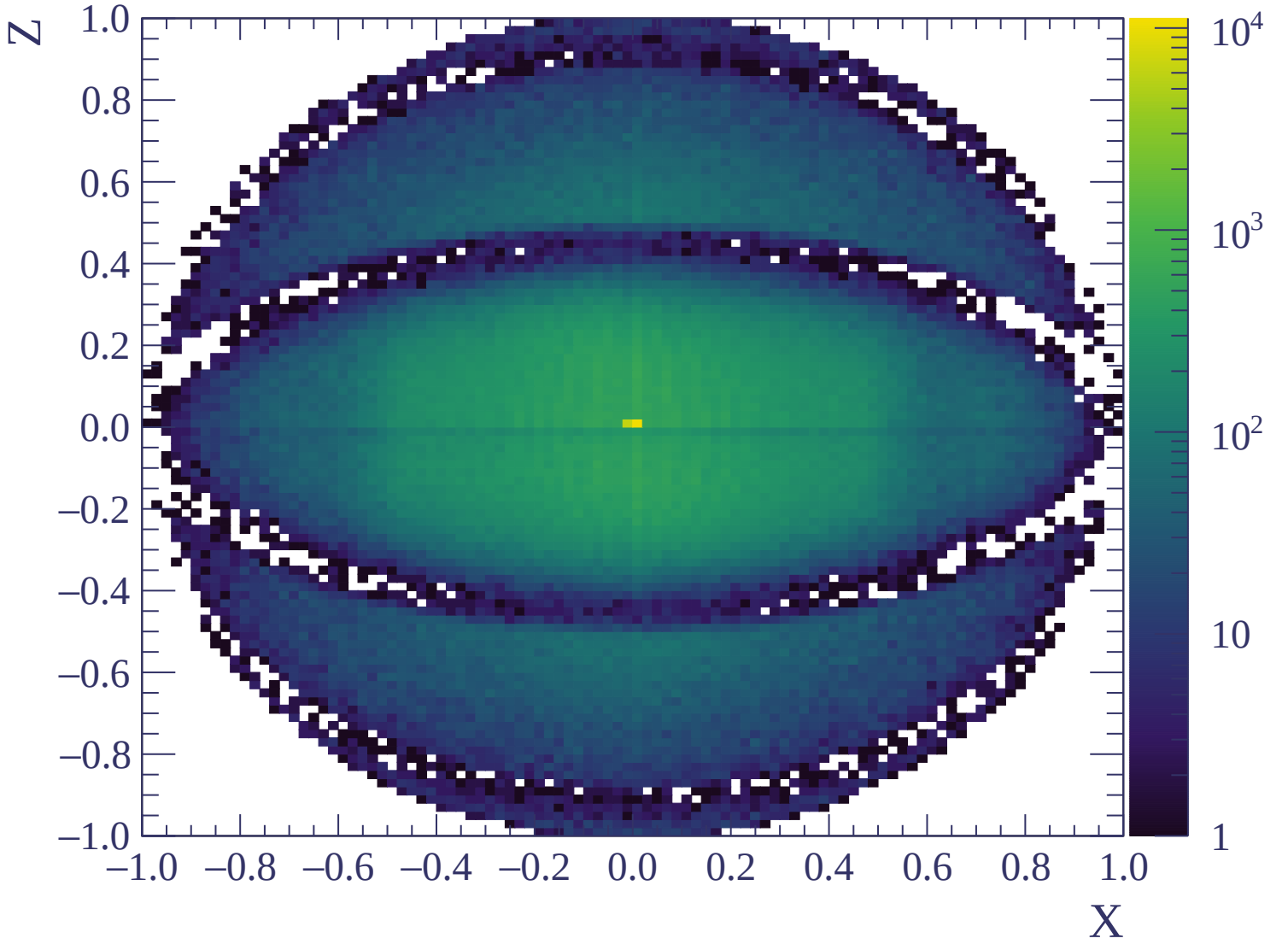


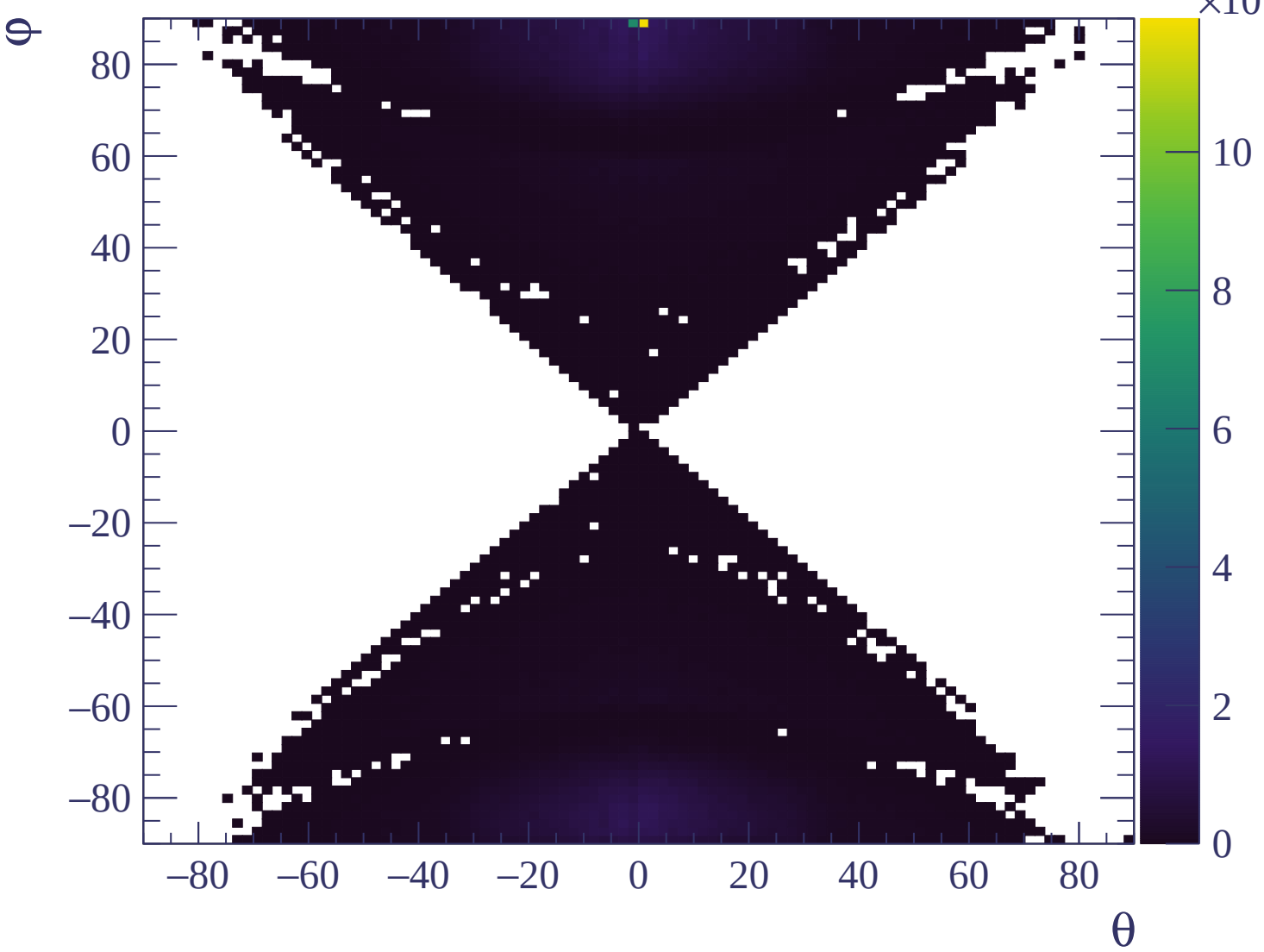


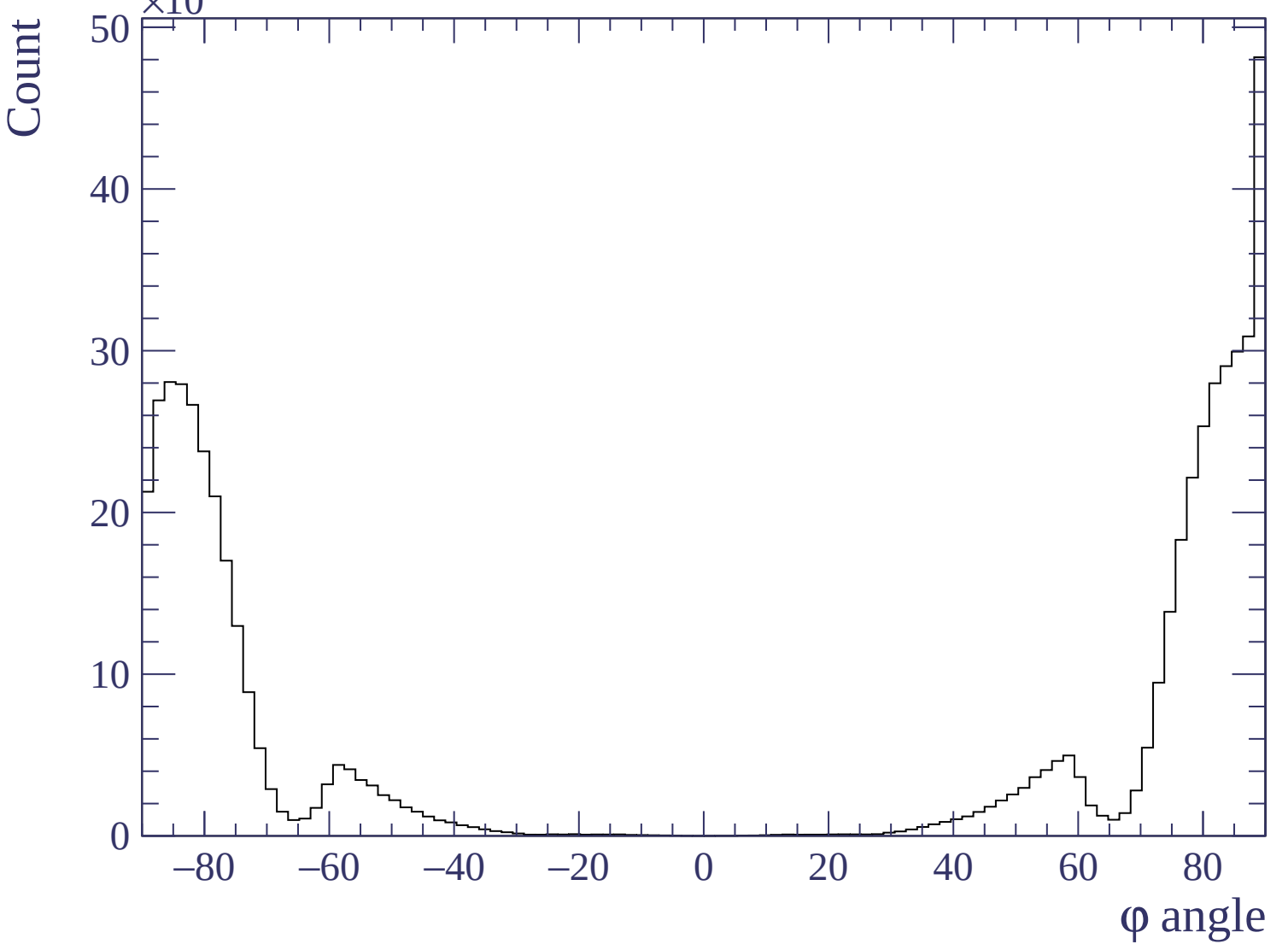


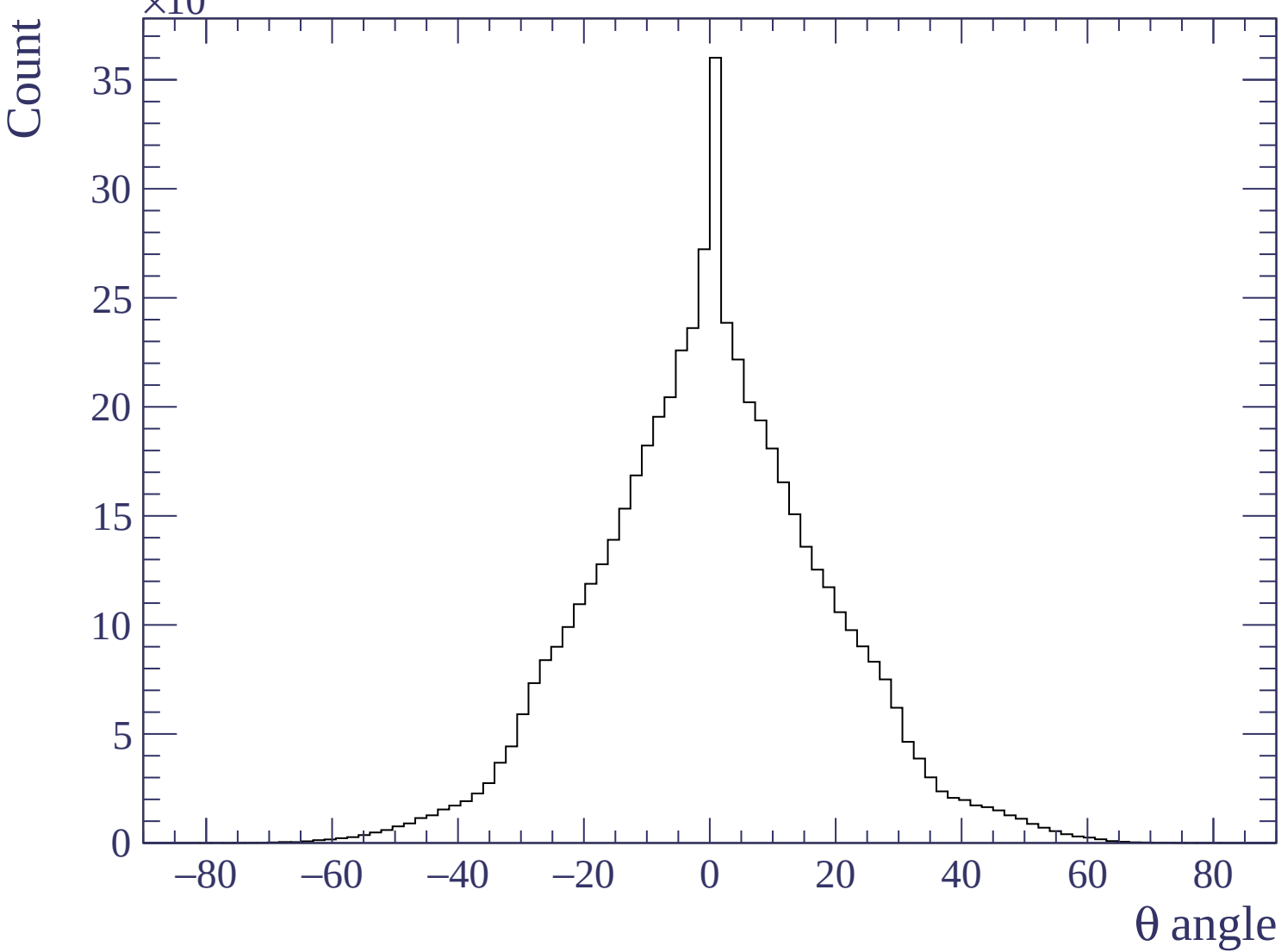


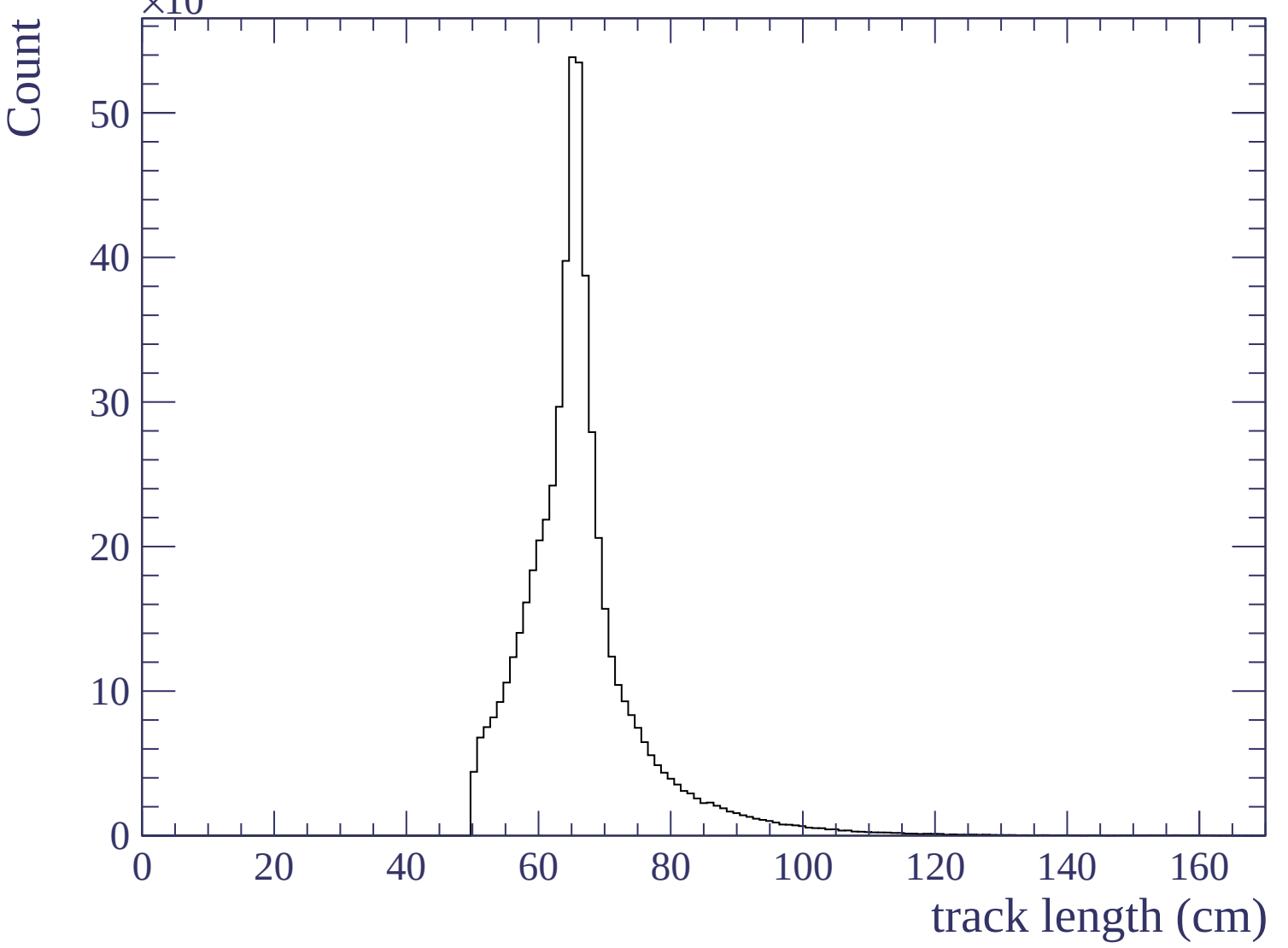


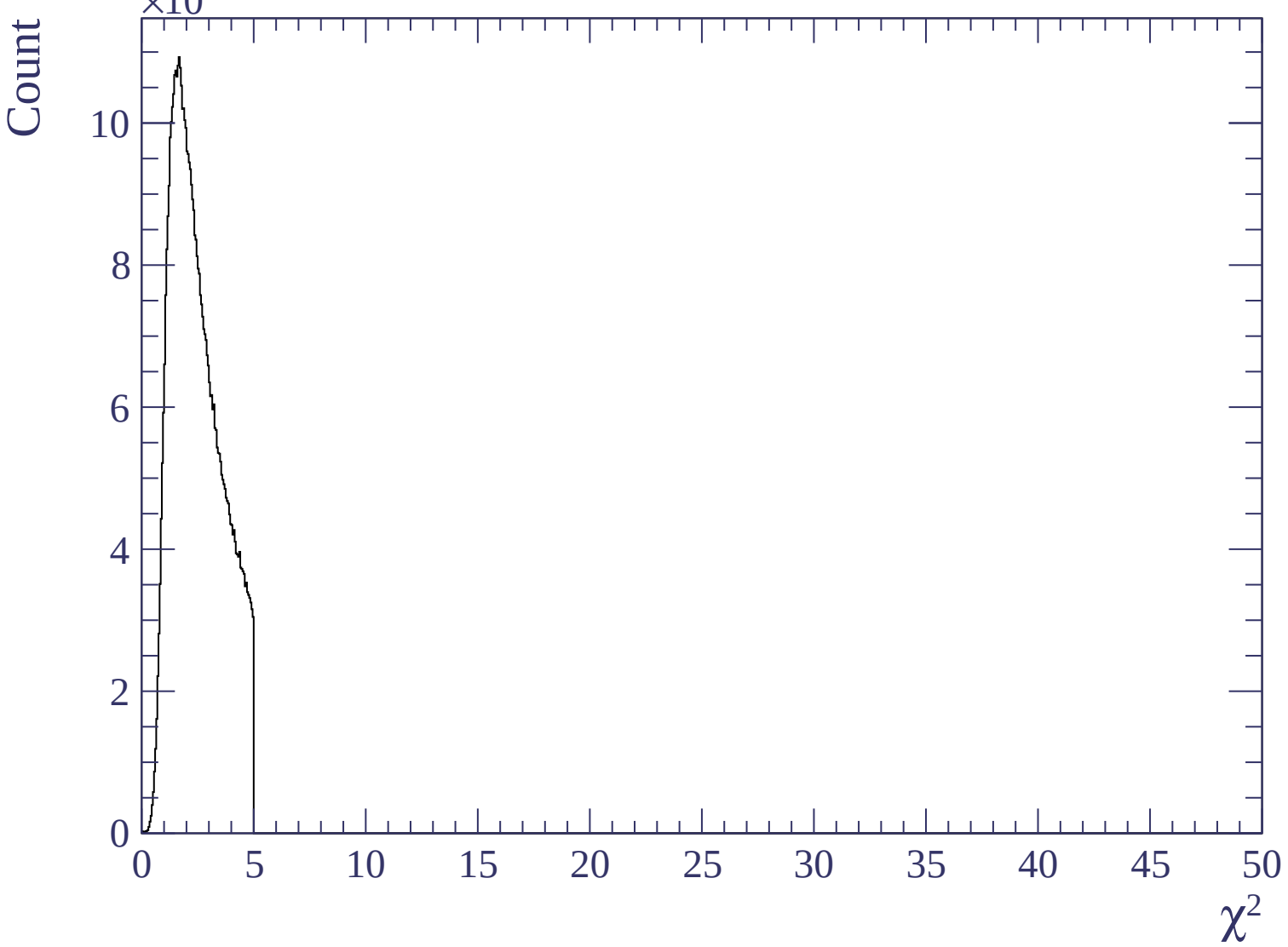


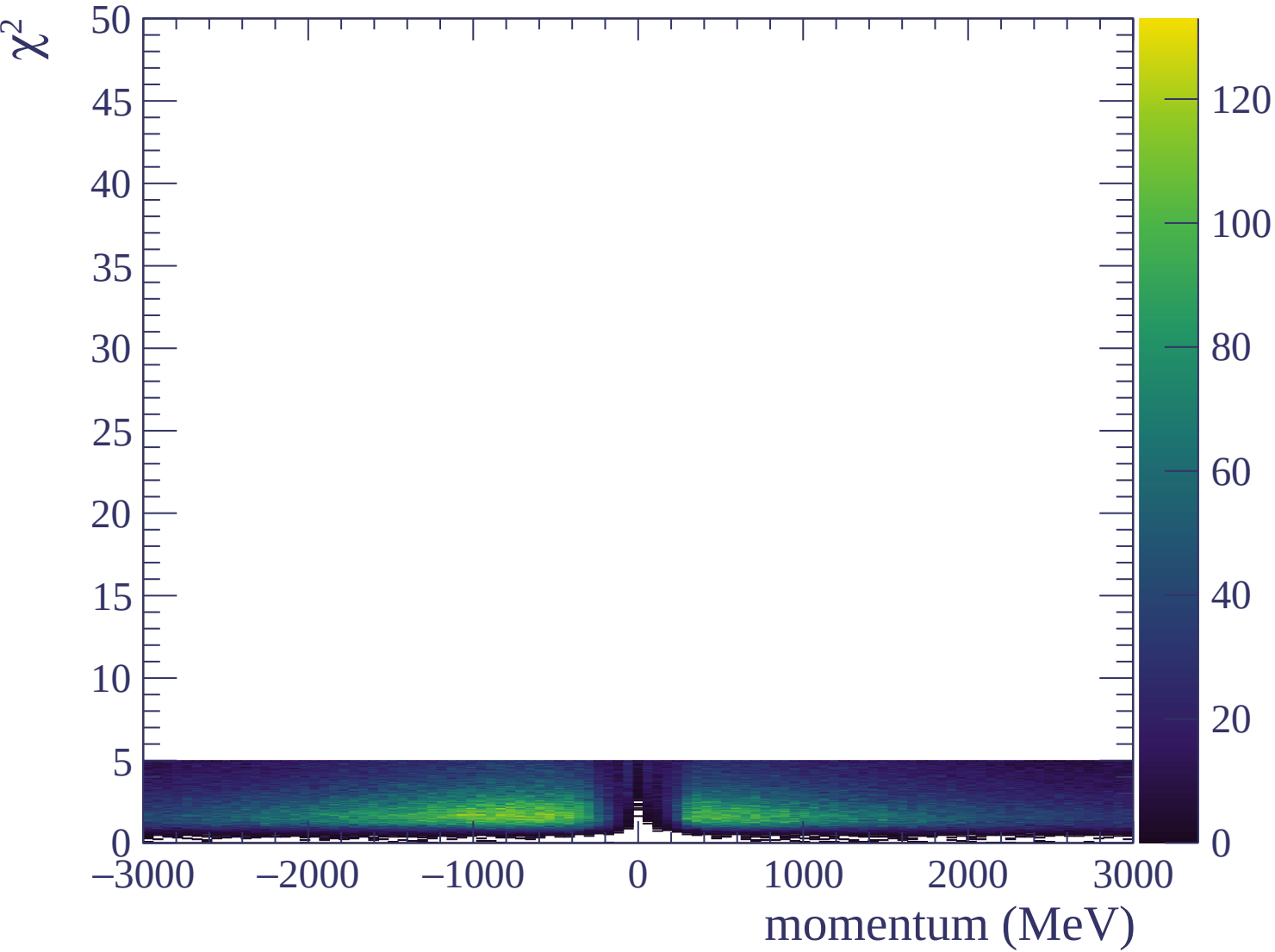


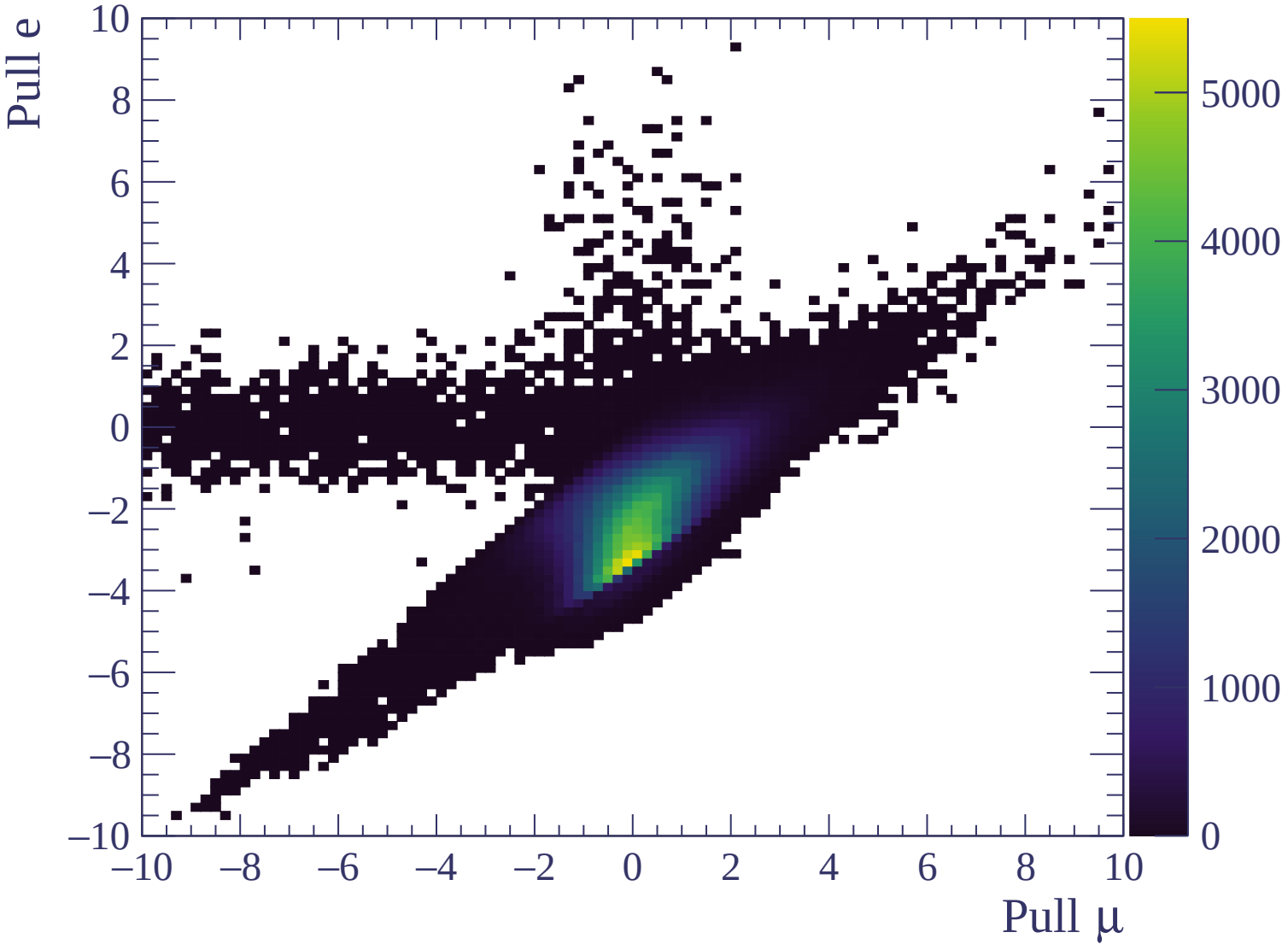


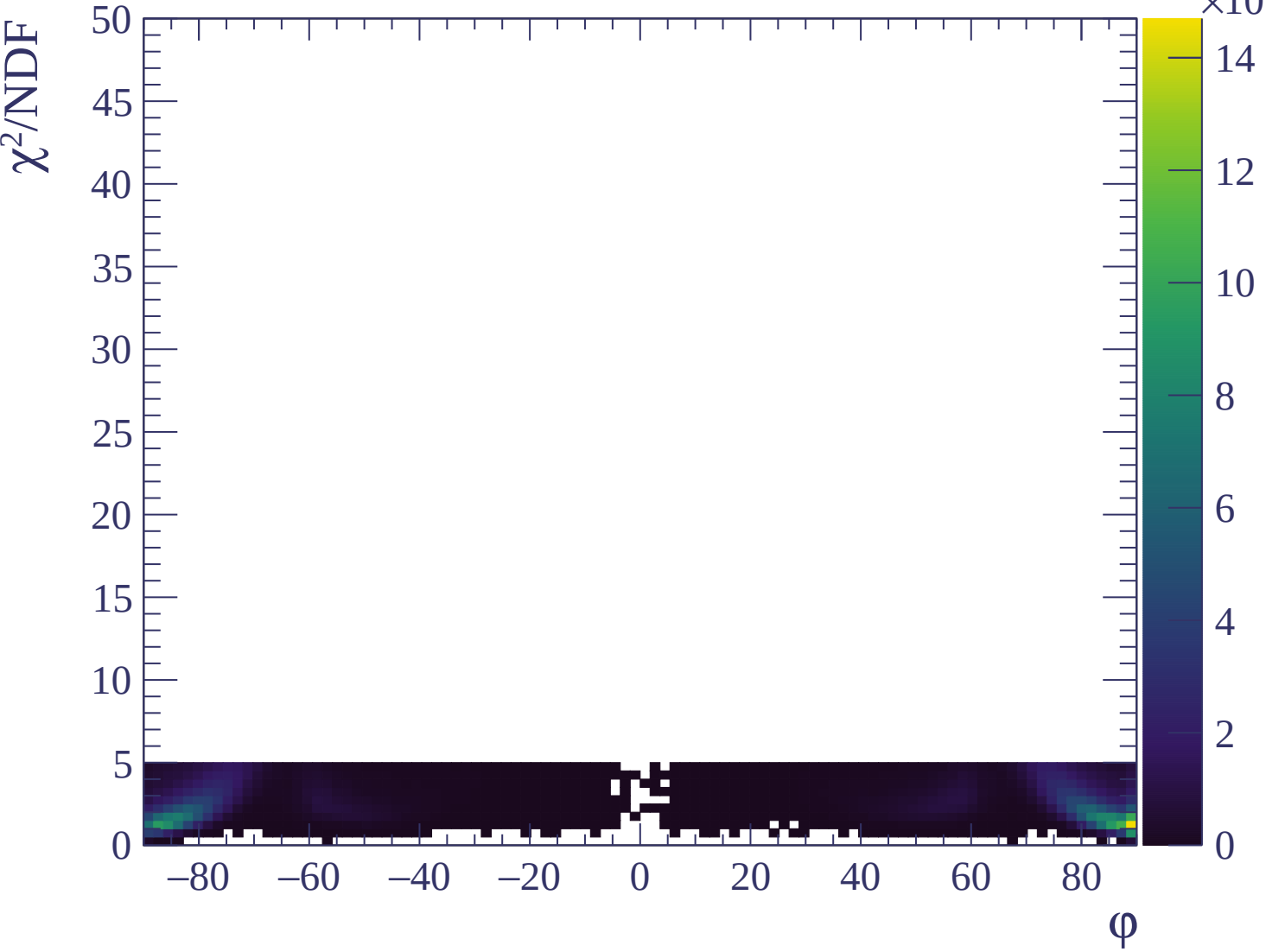


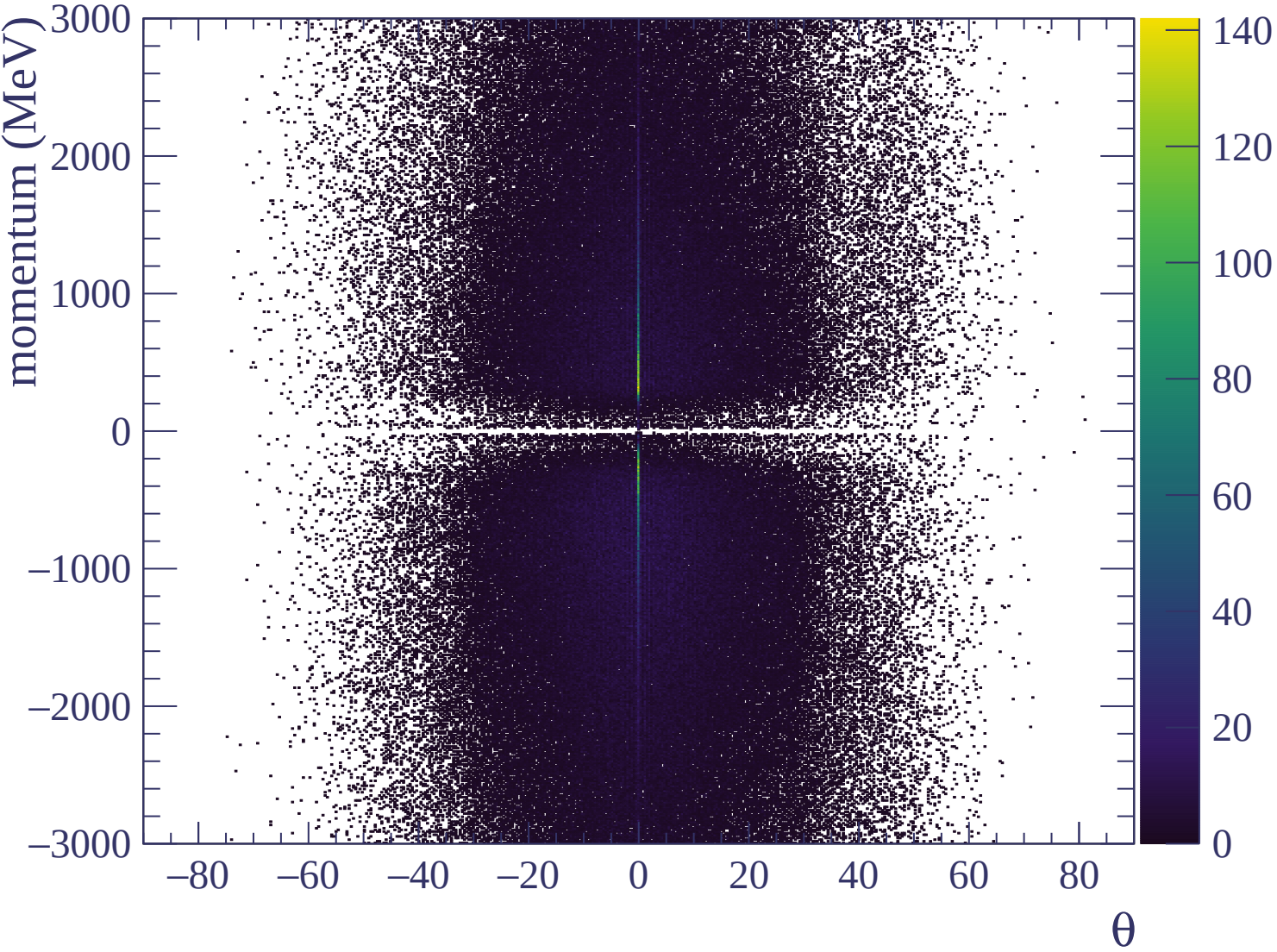


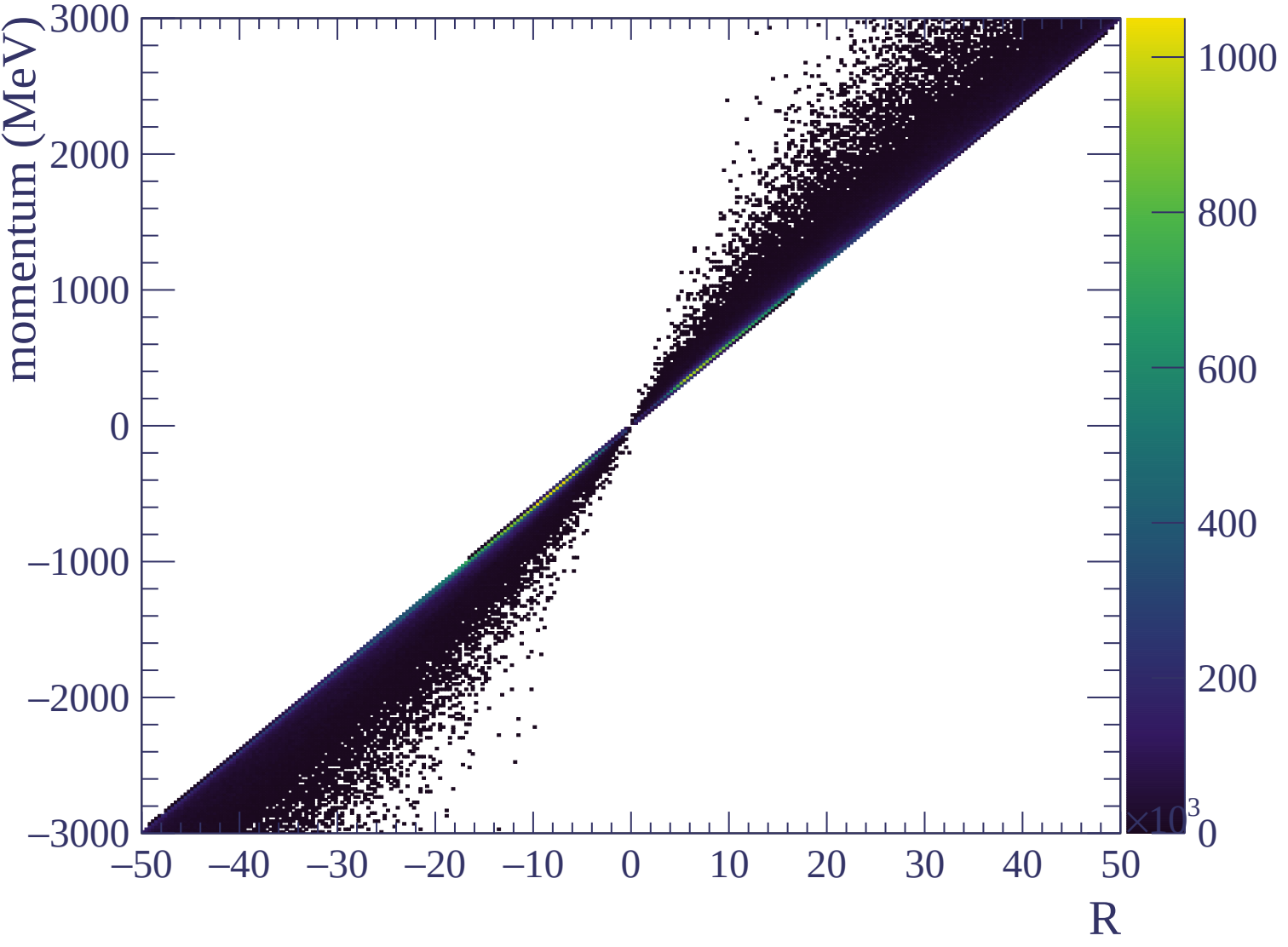


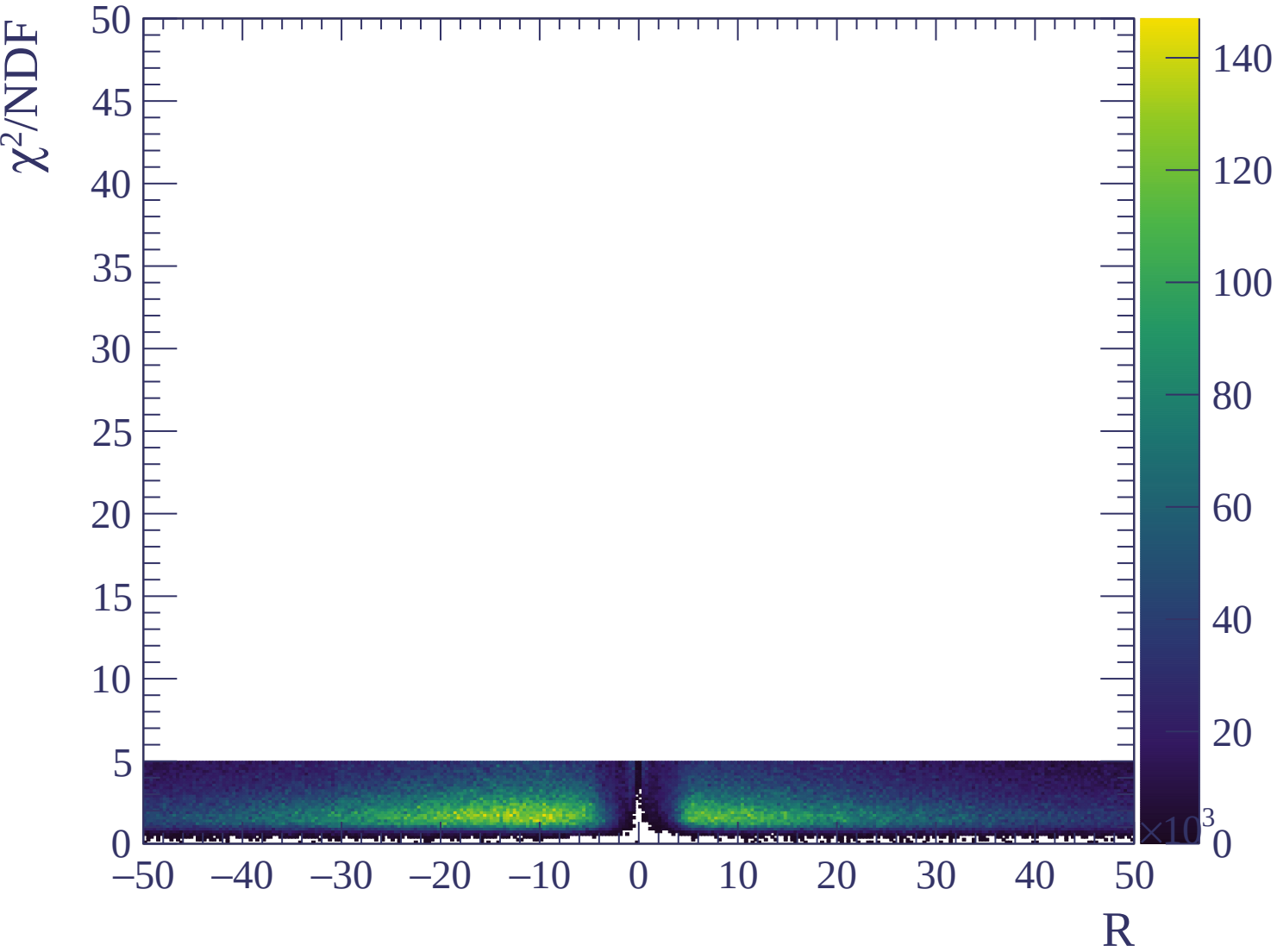


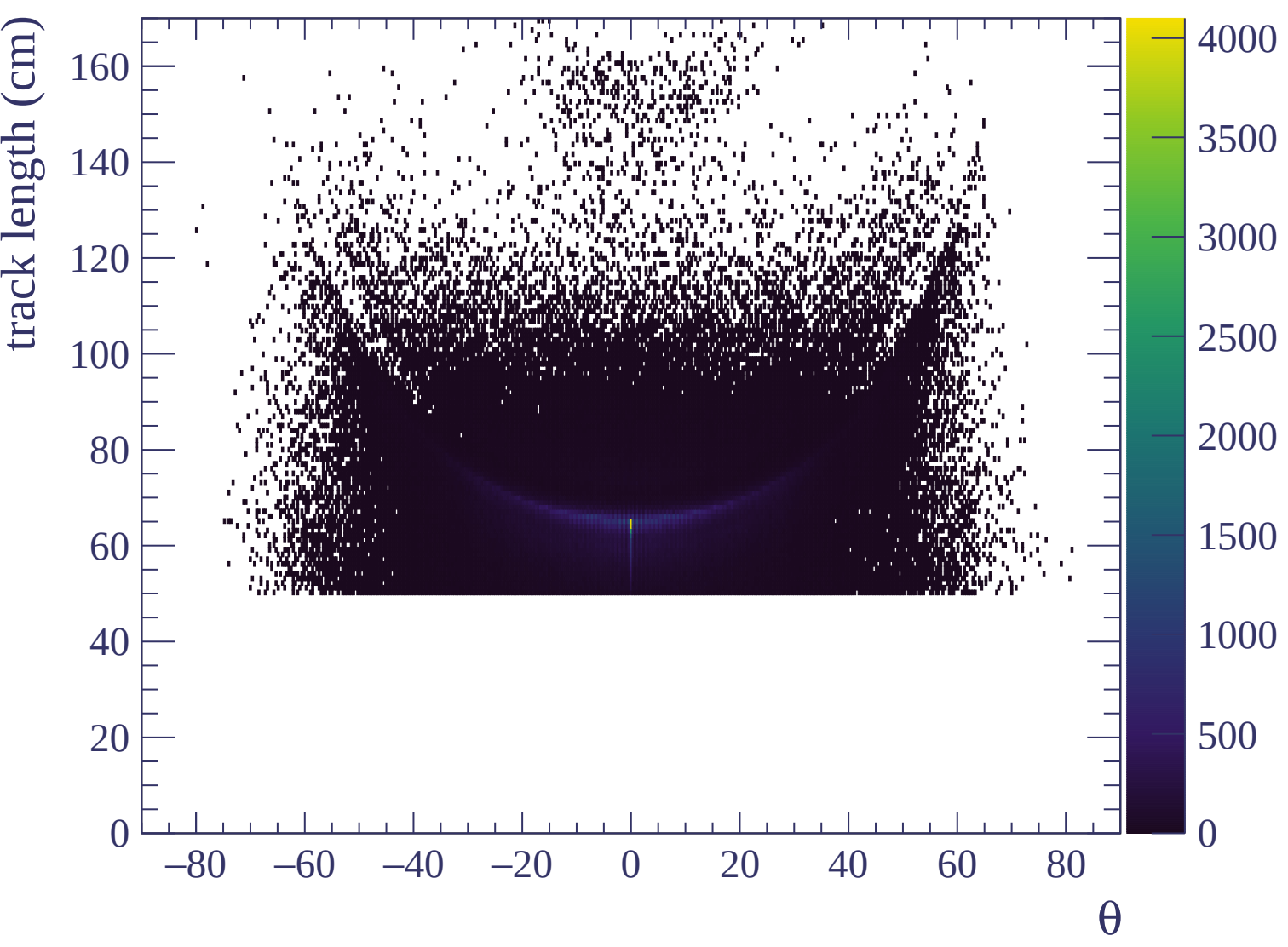


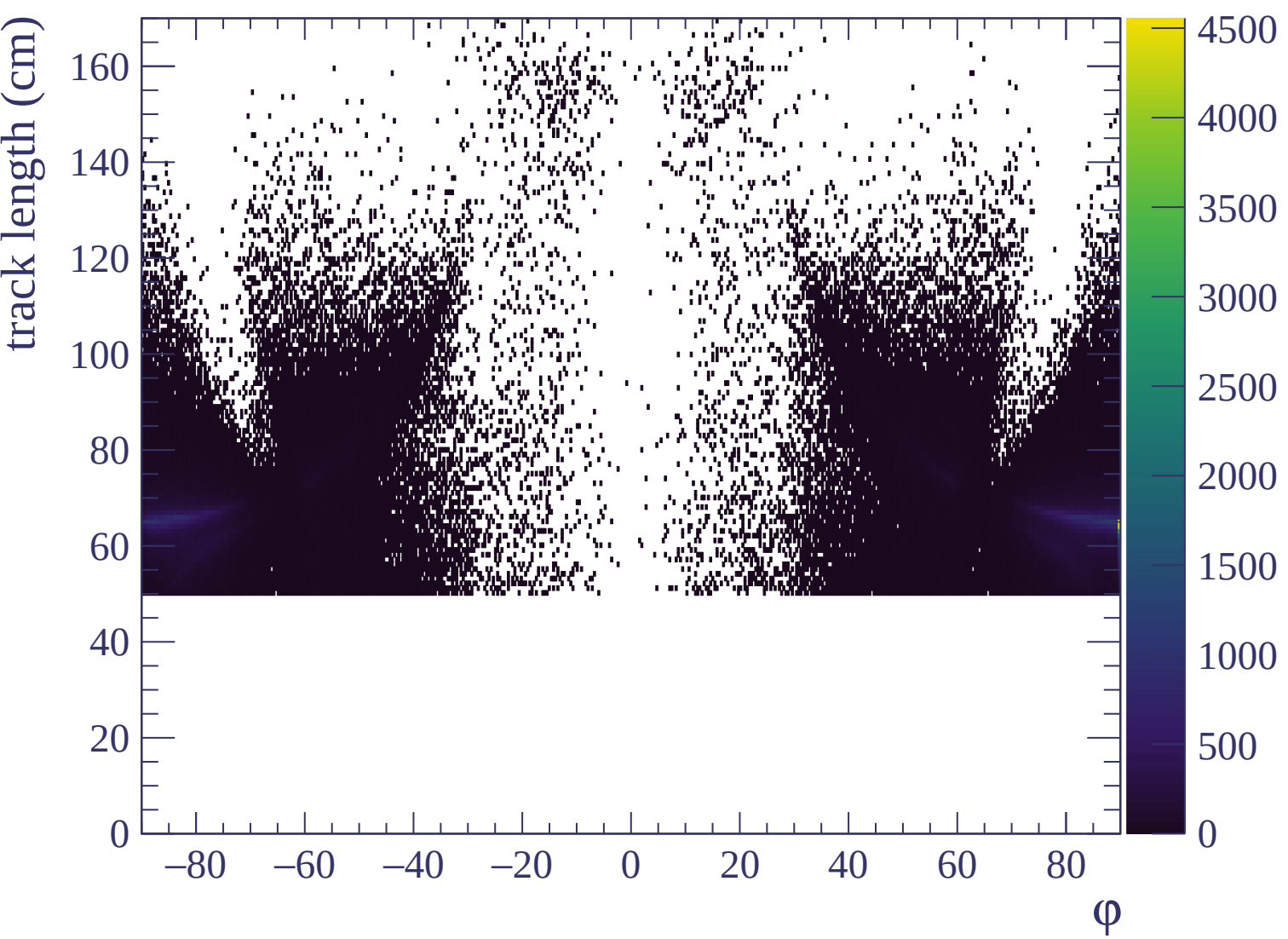






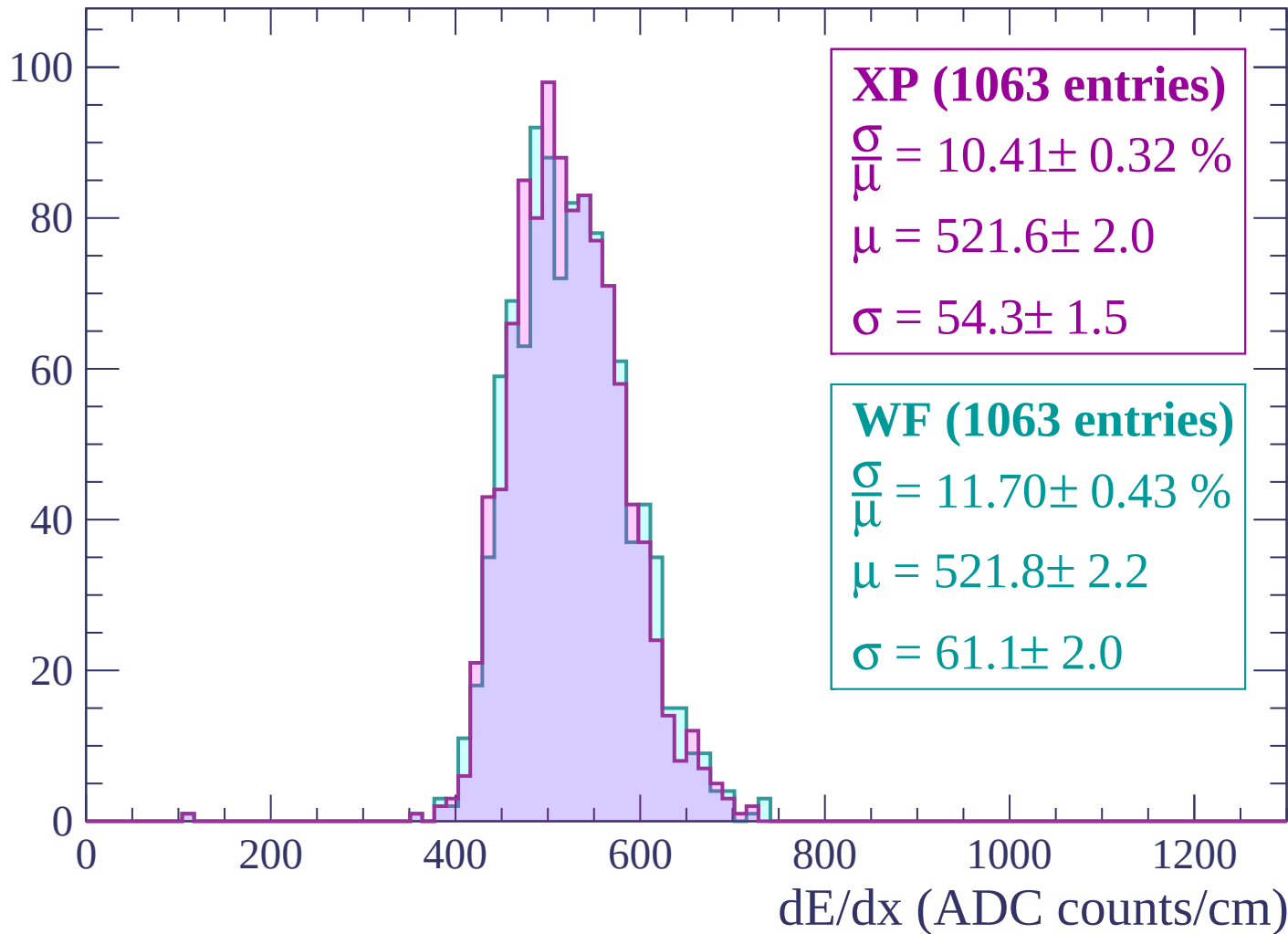






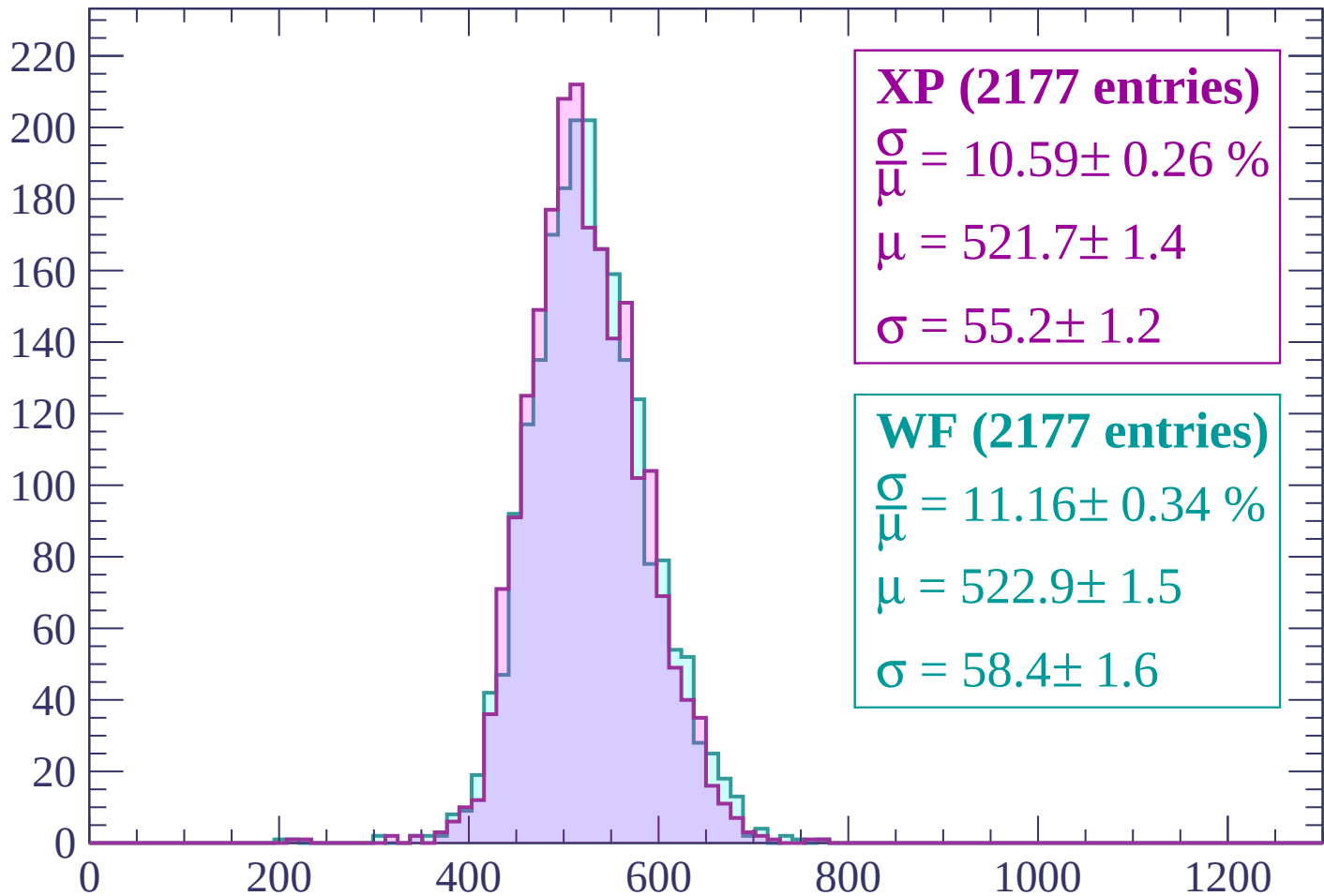
Energy loss | $-3000 < p < -2940$

Count



Energy loss | $-2940 < p < -2880$

Count



XP (2177 entries)

$$\frac{\sigma}{\mu} = 10.59 \pm 0.26 \%$$

$$\mu = 521.7 \pm 1.4$$

$$\sigma = 55.2 \pm 1.2$$

WF (2177 entries)

$$\frac{\sigma}{\mu} = 11.16 \pm 0.34 \%$$

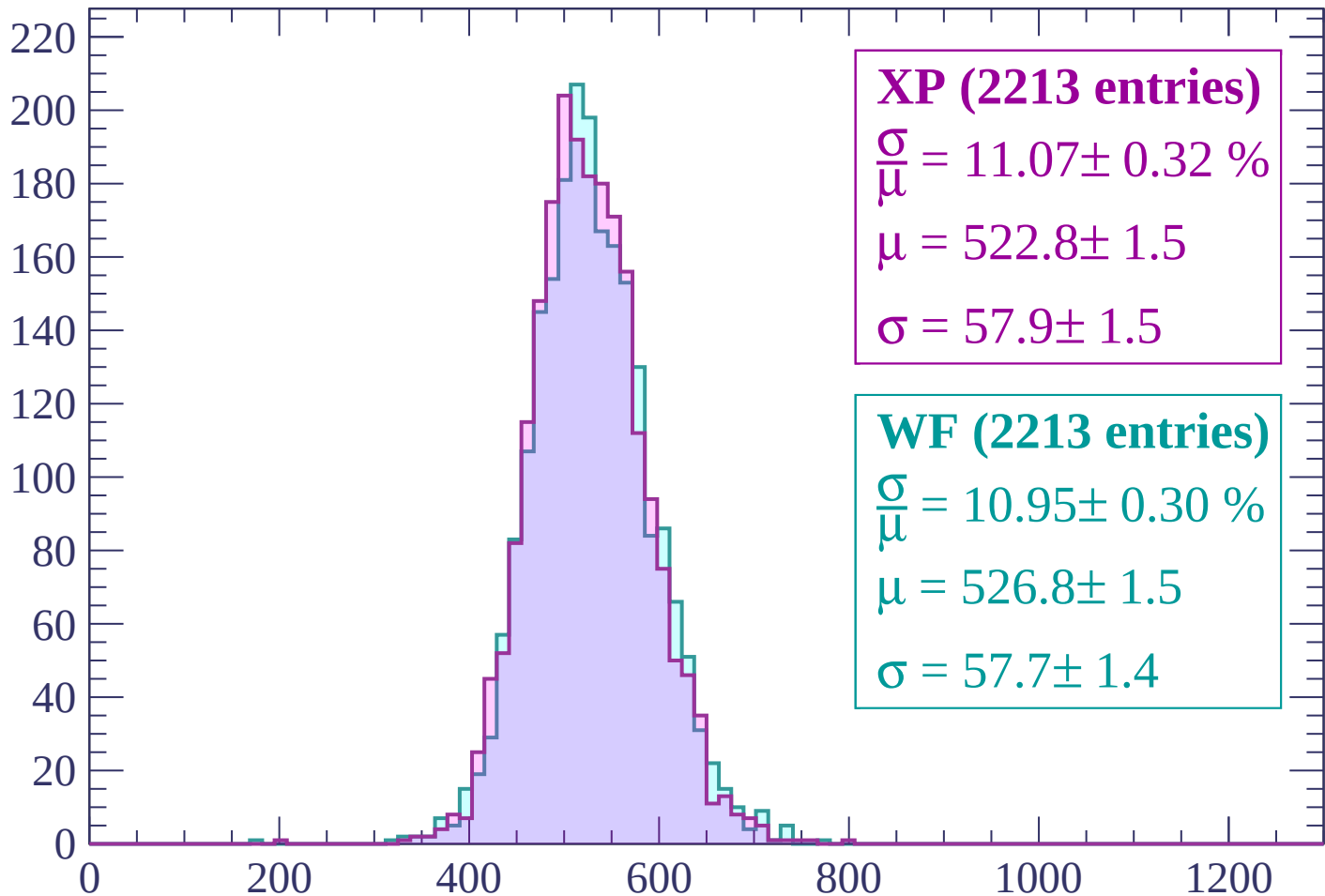
$$\mu = 522.9 \pm 1.5$$

$$\sigma = 58.4 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-2880 < p < -2820$

Count



XP (2213 entries)

$$\frac{\sigma}{\mu} = 11.07 \pm 0.32 \%$$

$$\mu = 522.8 \pm 1.5$$

$$\sigma = 57.9 \pm 1.5$$

WF (2213 entries)

$$\frac{\sigma}{\mu} = 10.95 \pm 0.30 \%$$

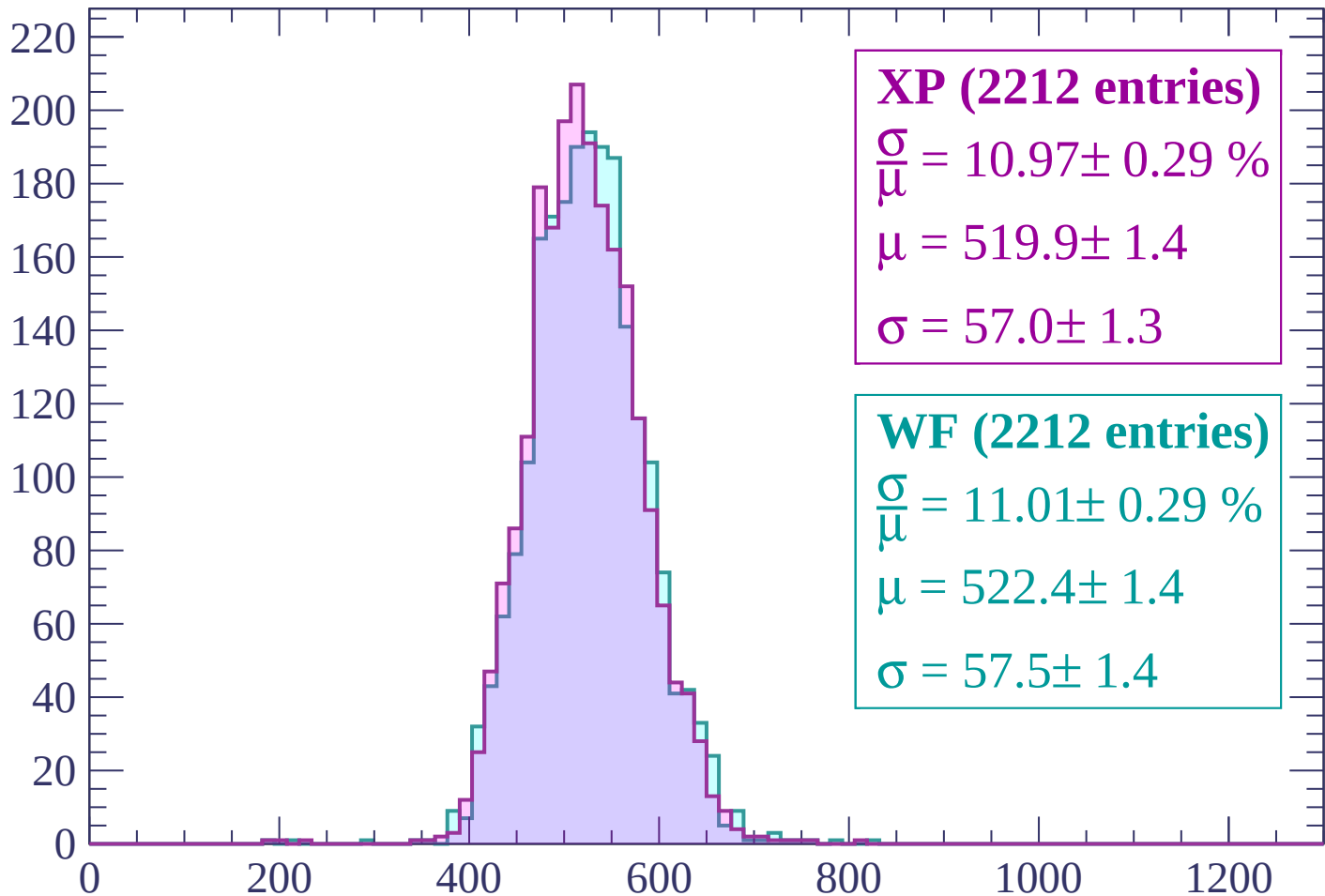
$$\mu = 526.8 \pm 1.5$$

$$\sigma = 57.7 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-2820 < p < -2760$

Count



XP (2212 entries)

$$\frac{\sigma}{\mu} = 10.97 \pm 0.29 \%$$

$$\mu = 519.9 \pm 1.4$$

$$\sigma = 57.0 \pm 1.3$$

WF (2212 entries)

$$\frac{\sigma}{\mu} = 11.01 \pm 0.29 \%$$

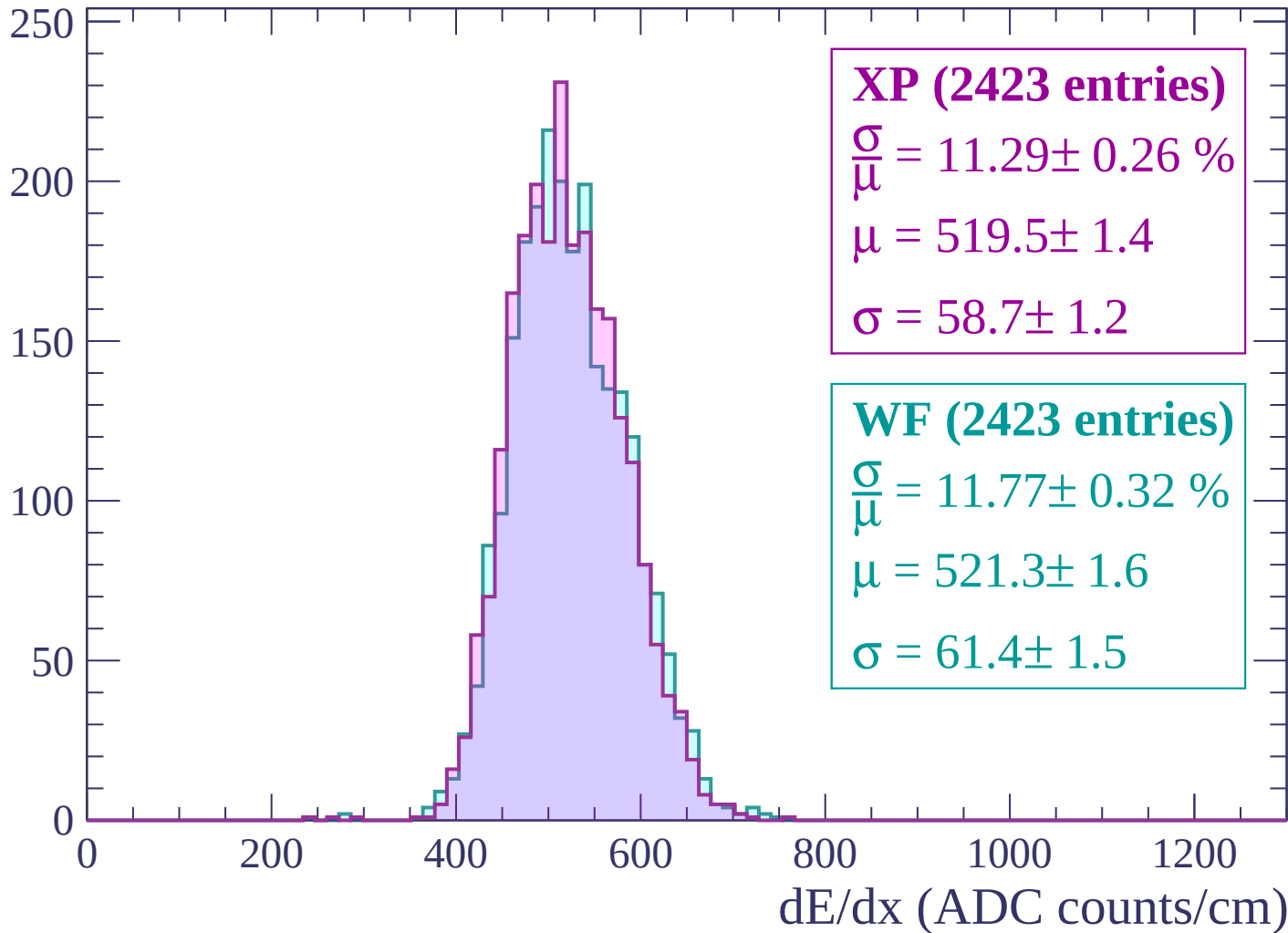
$$\mu = 522.4 \pm 1.4$$

$$\sigma = 57.5 \pm 1.4$$

dE/dx (ADC counts/cm)

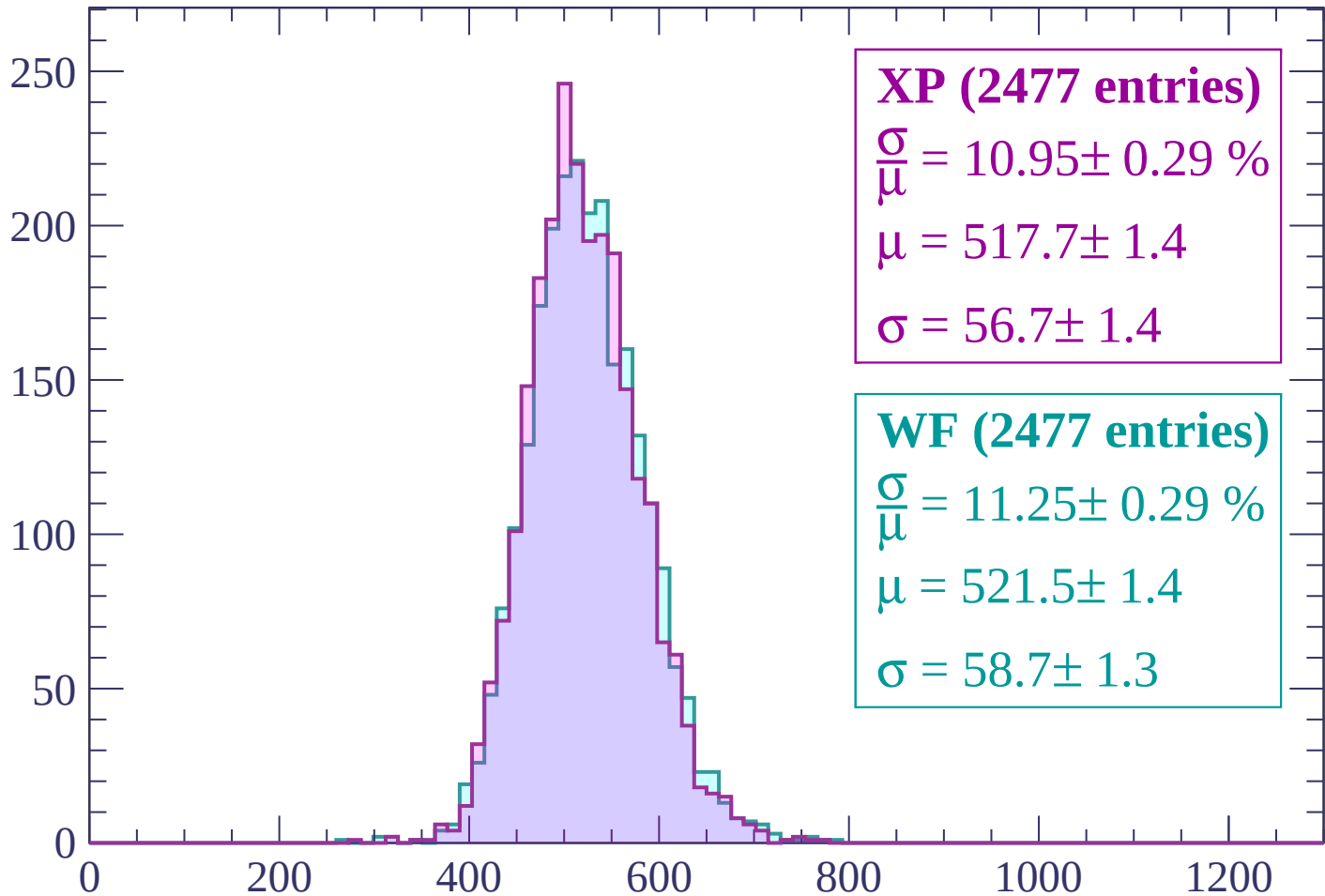
Energy loss | $-2760 < p < -2700$

Count



Energy loss | -2700 < p < -2640

Count



XP (2477 entries)

$\frac{\sigma}{\mu} = 10.95 \pm 0.29 \%$

$\mu = 517.7 \pm 1.4$

$\sigma = 56.7 \pm 1.4$

WF (2477 entries)

$\frac{\sigma}{\mu} = 11.25 \pm 0.29 \%$

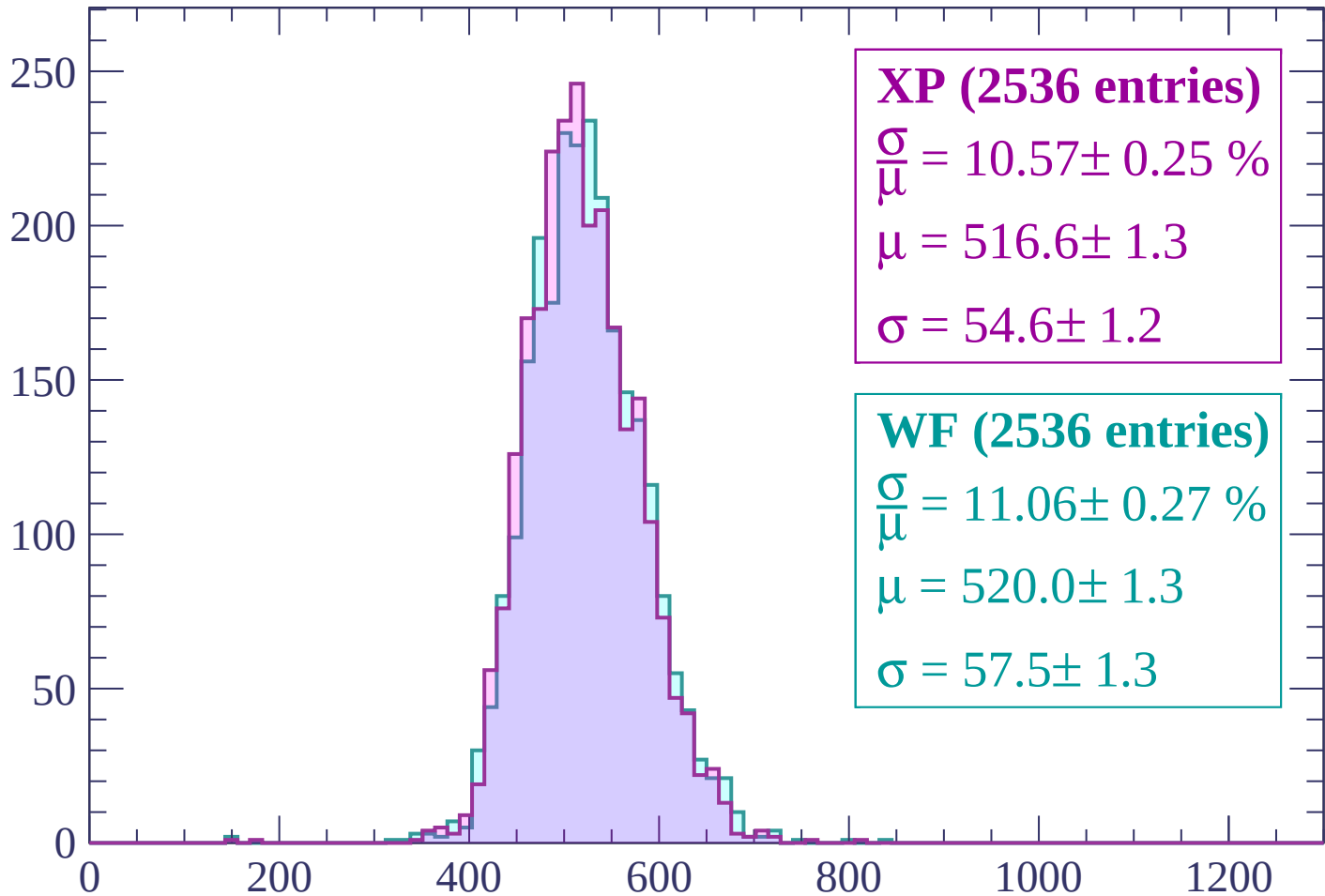
$\mu = 521.5 \pm 1.4$

$\sigma = 58.7 \pm 1.3$

dE/dx (ADC counts/cm)

Energy loss | $-2640 < p < -2580$

Count



XP (2536 entries)

$$\frac{\sigma}{\mu} = 10.57 \pm 0.25 \%$$

$$\mu = 516.6 \pm 1.3$$

$$\sigma = 54.6 \pm 1.2$$

WF (2536 entries)

$$\frac{\sigma}{\mu} = 11.06 \pm 0.27 \%$$

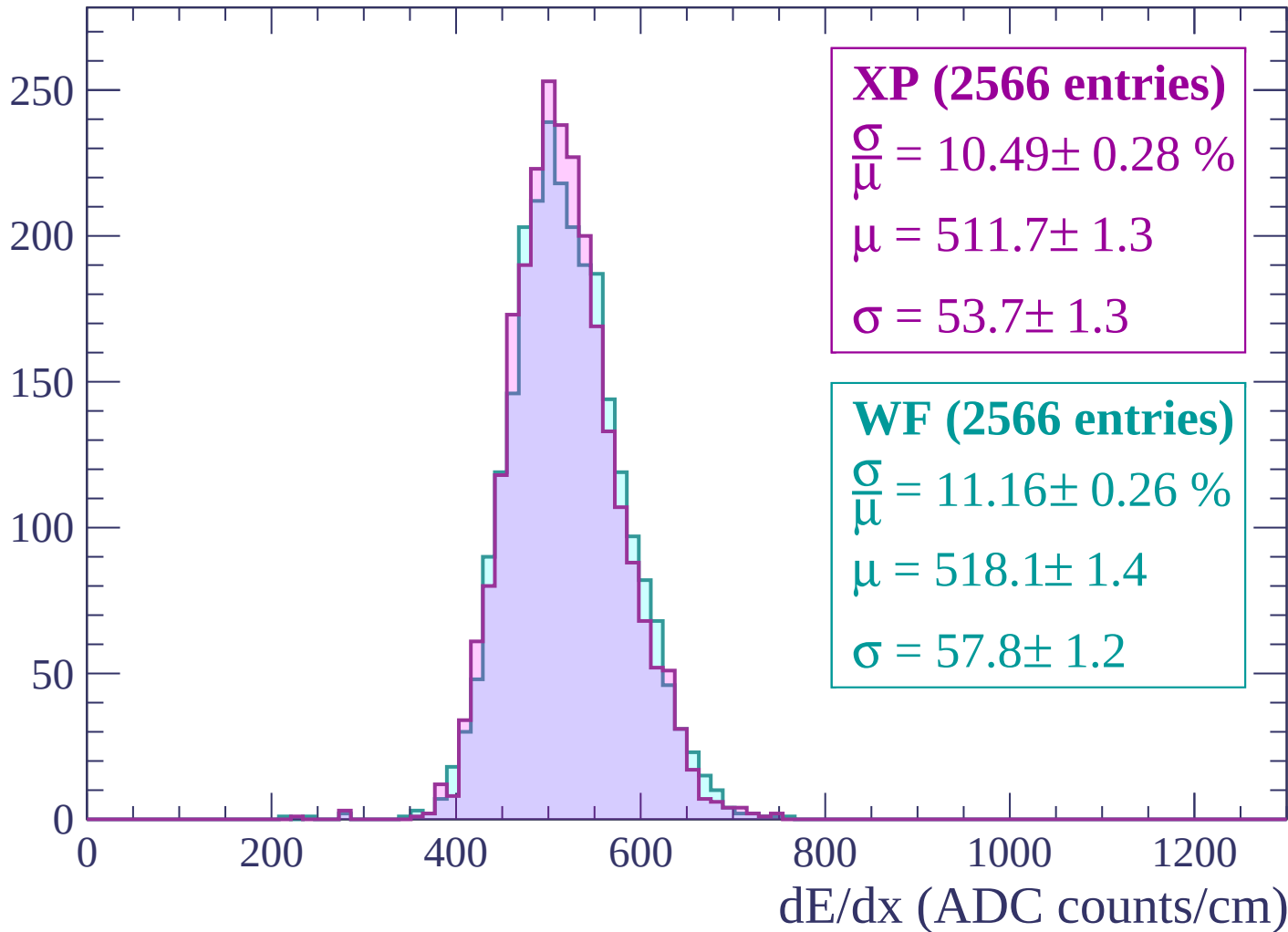
$$\mu = 520.0 \pm 1.3$$

$$\sigma = 57.5 \pm 1.3$$

dE/dx (ADC counts/cm)

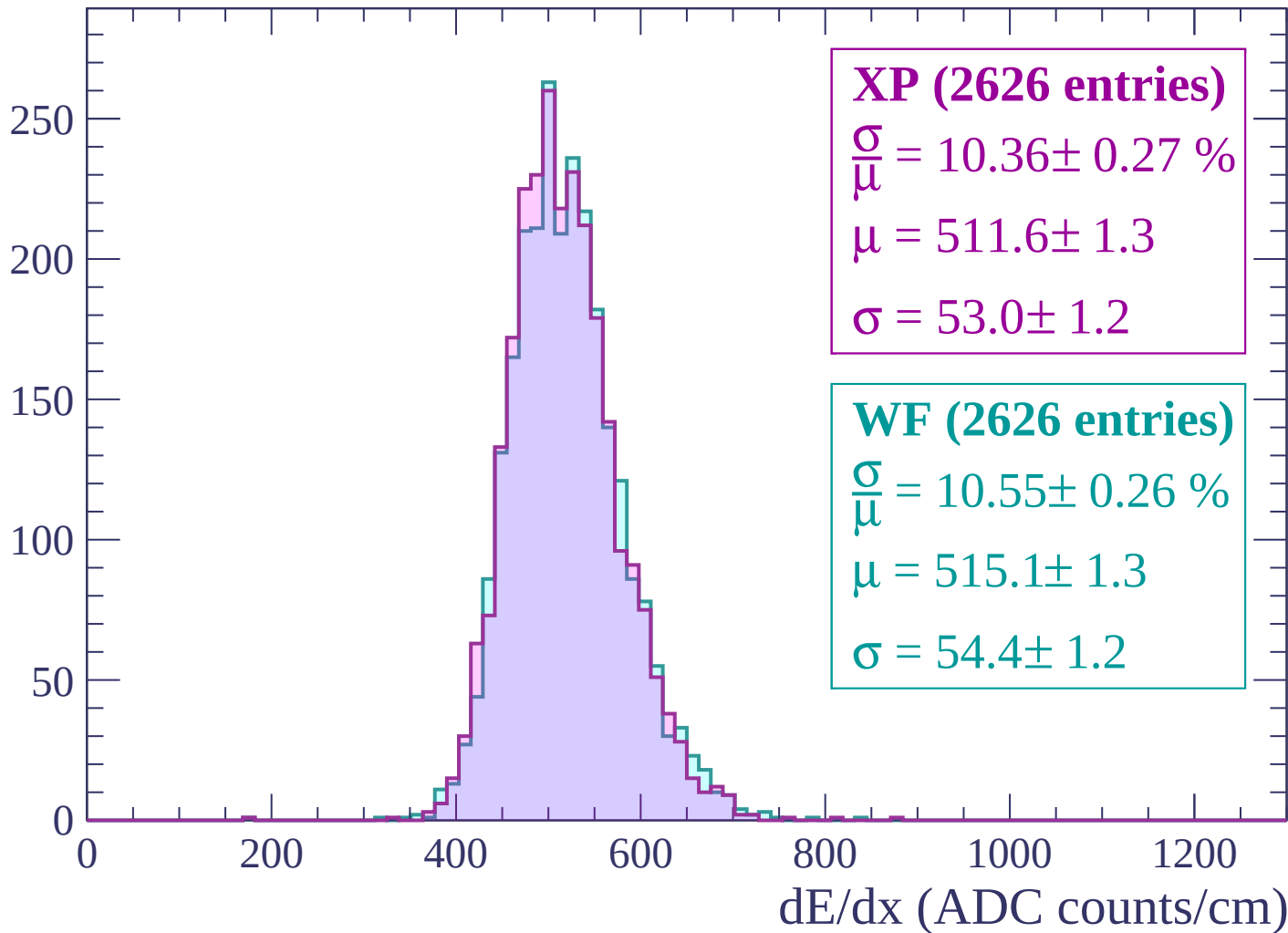
Energy loss | -2580 < p < -2520

Count



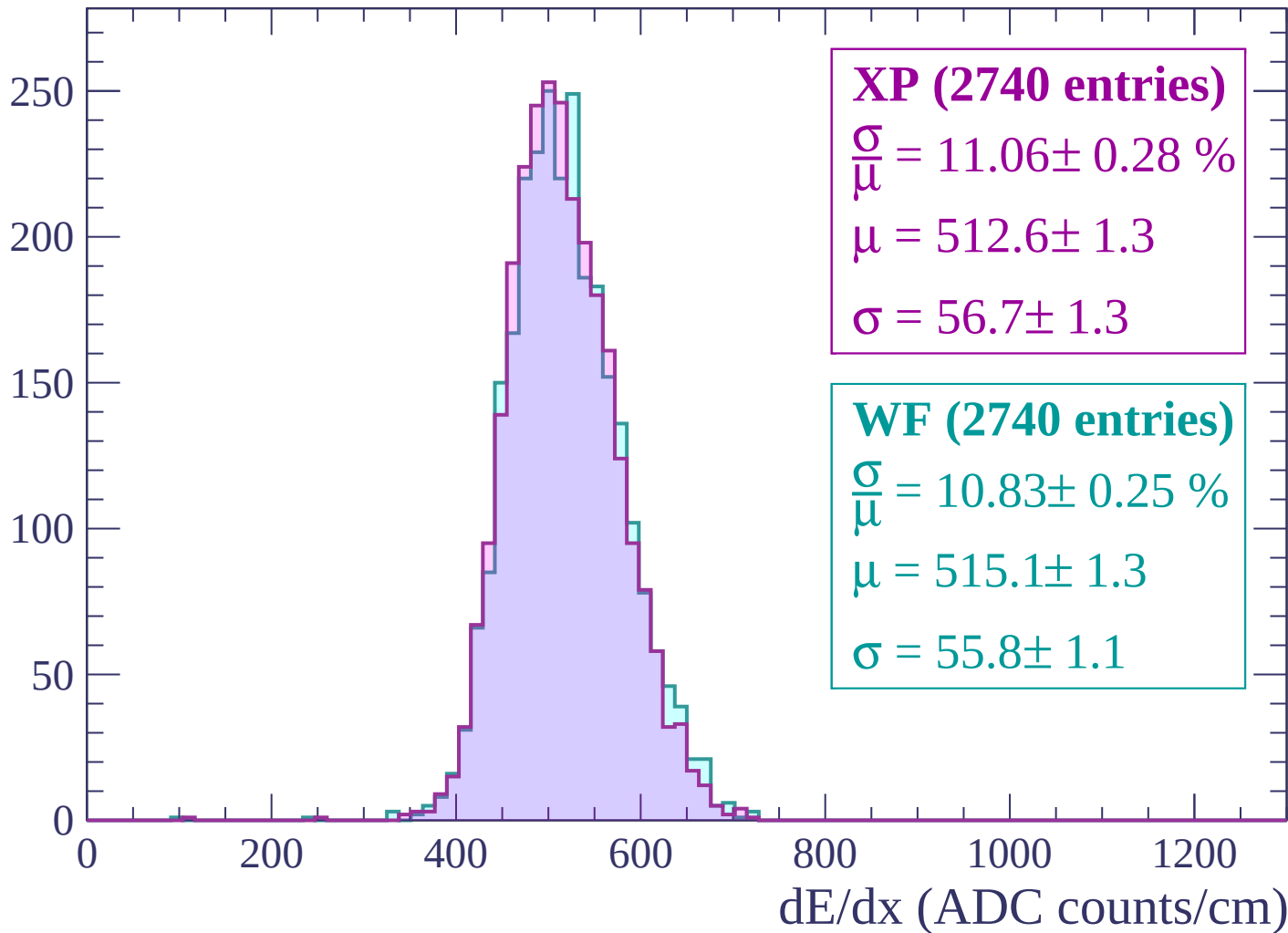
Energy loss | -2520 < p < -2460

Count



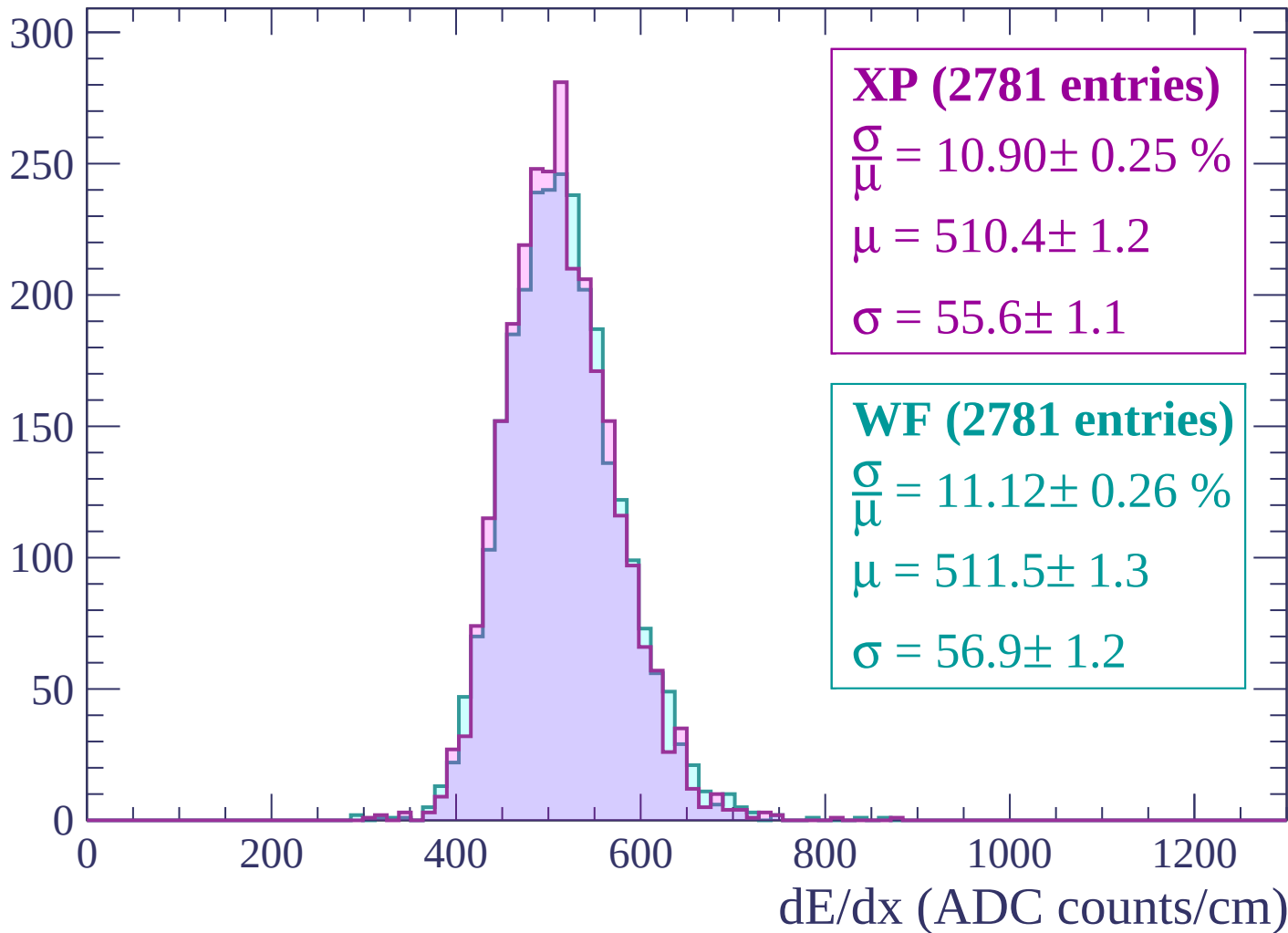
Energy loss | $-2460 < p < -2400$

Count



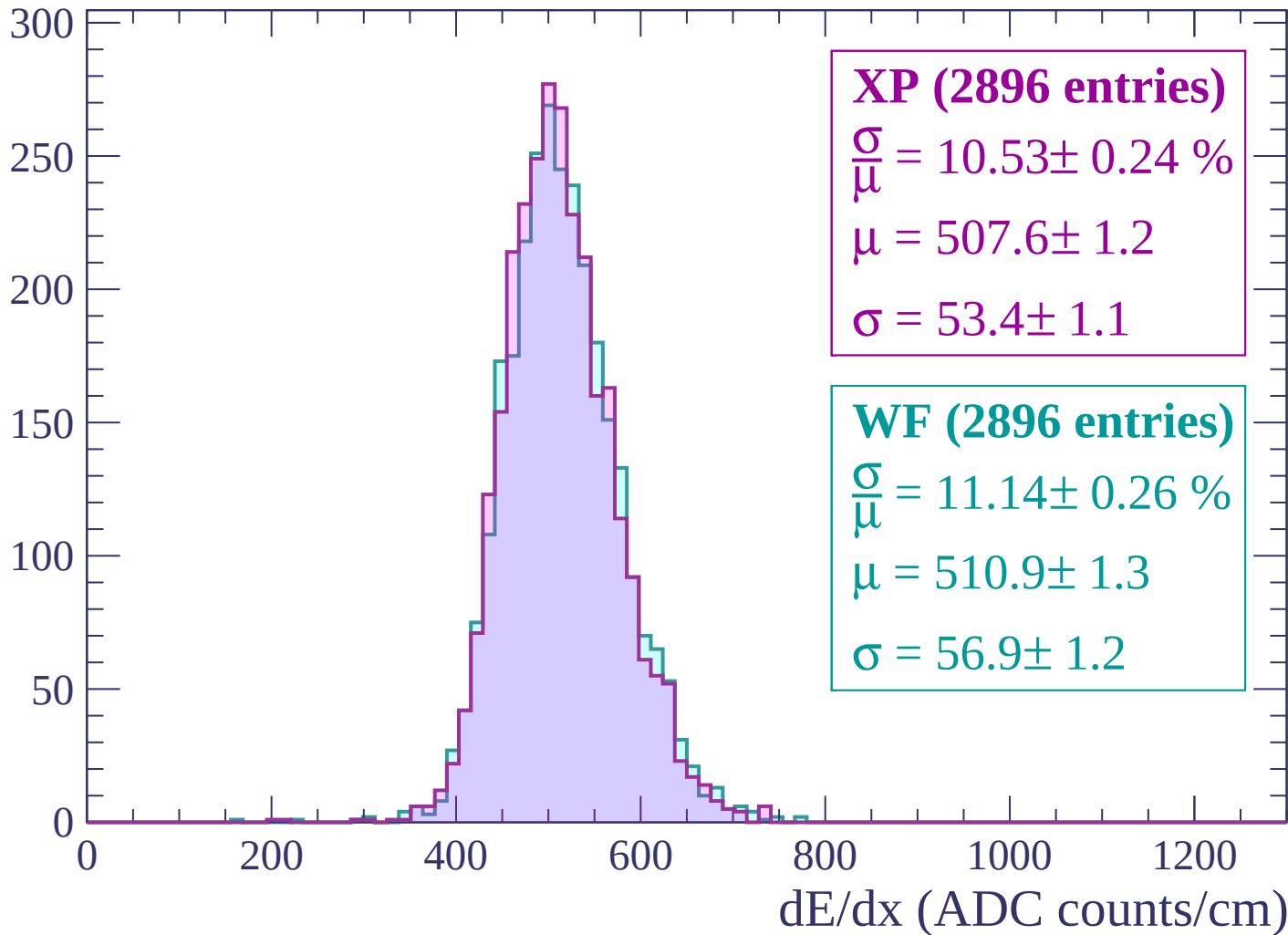
Energy loss | -2400 < p < -2340

Count



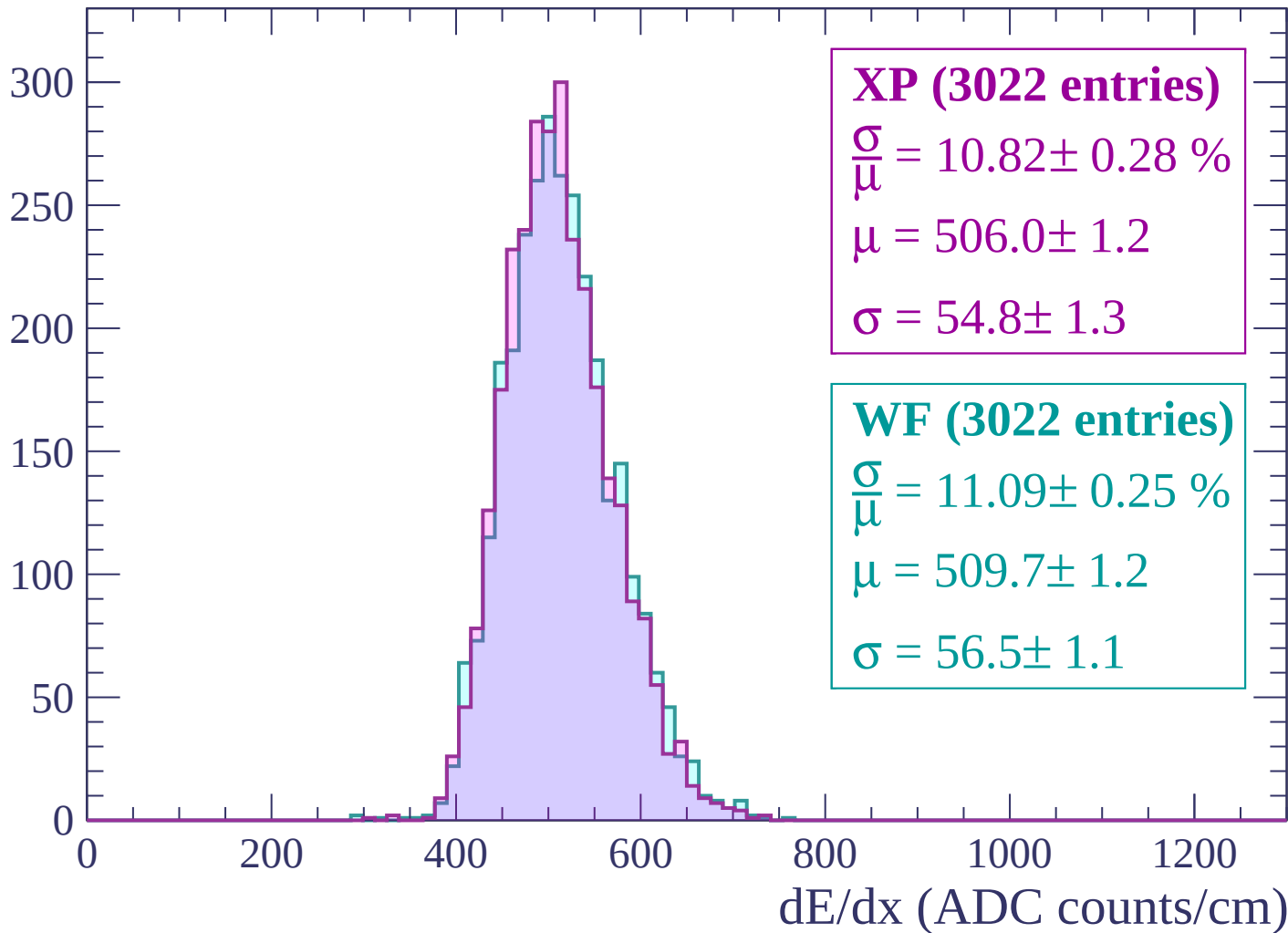
Energy loss | $-2340 < p < -2280$

Count



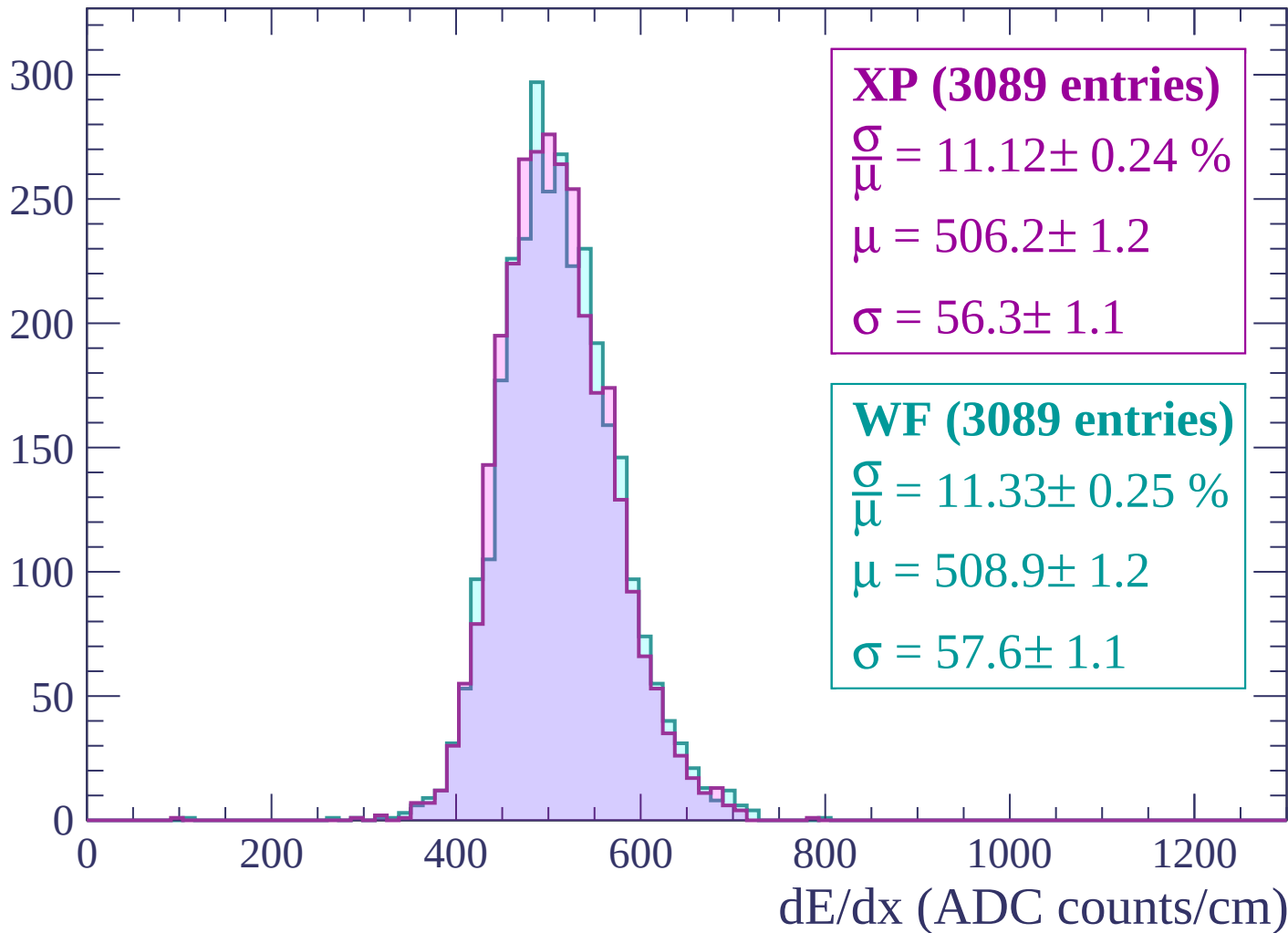
Energy loss | -2280 < p < -2220

Count

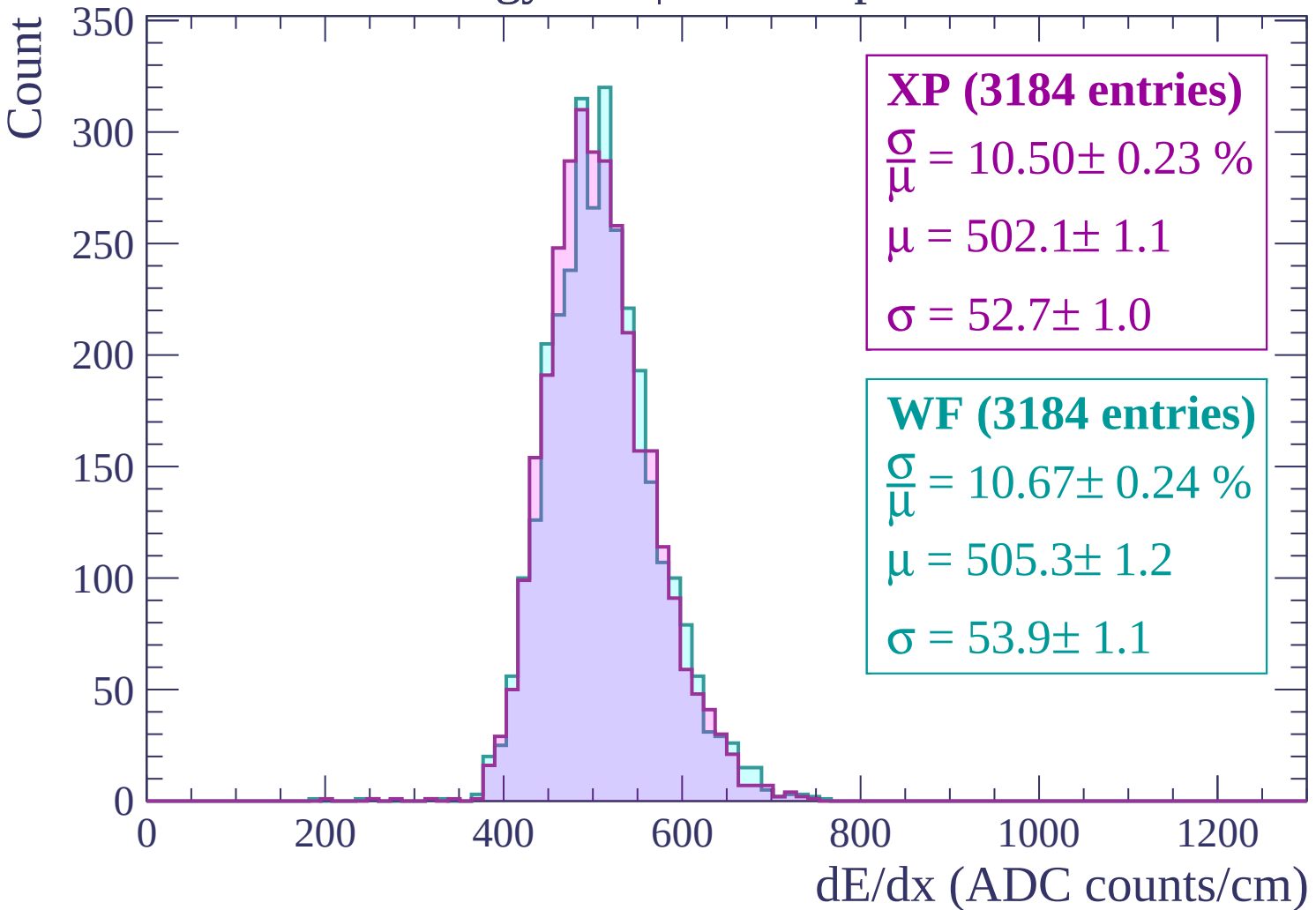


Energy loss | -2220 < p < -2160

Count

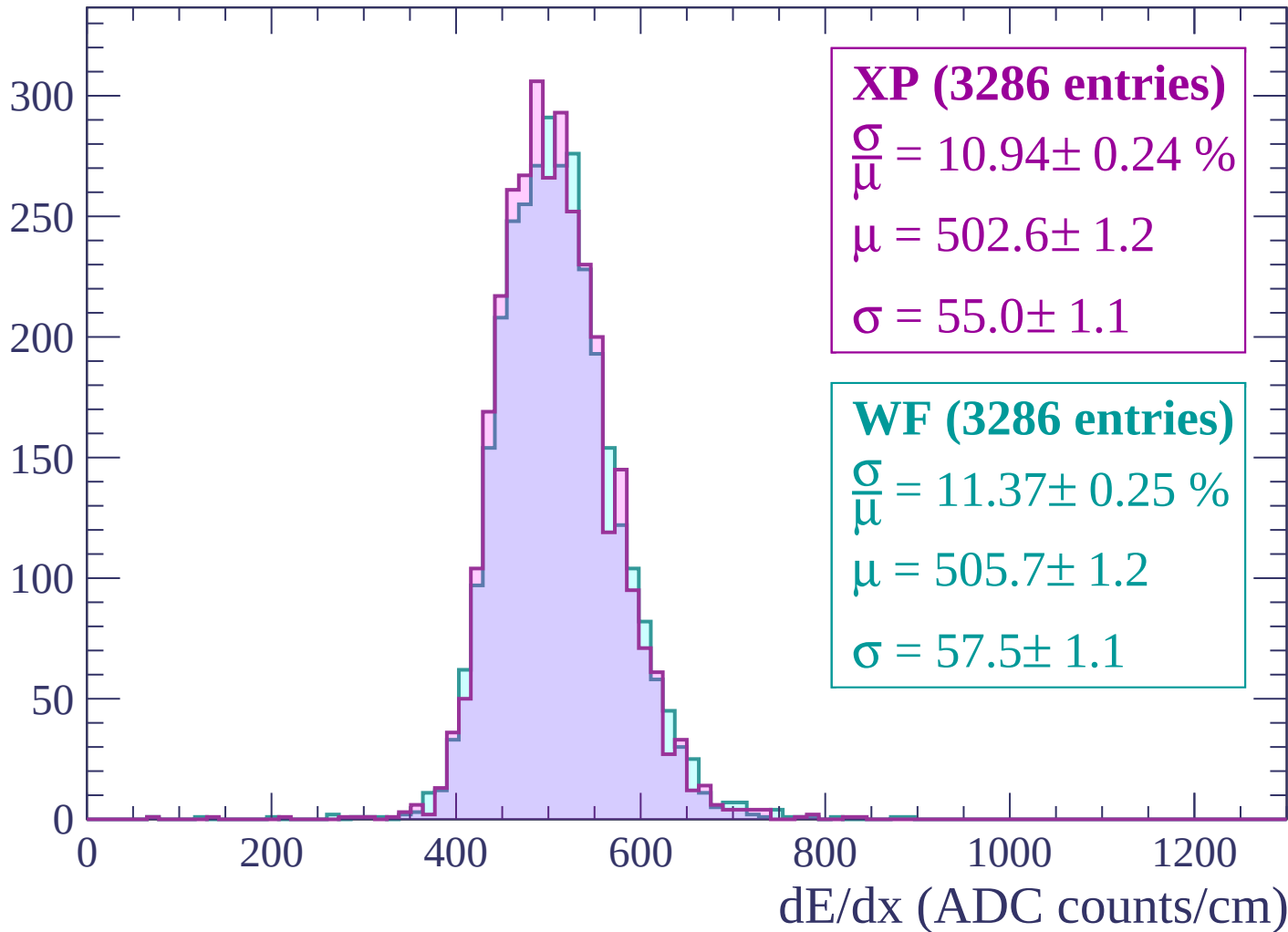


Energy loss | -2160 < p < -2100



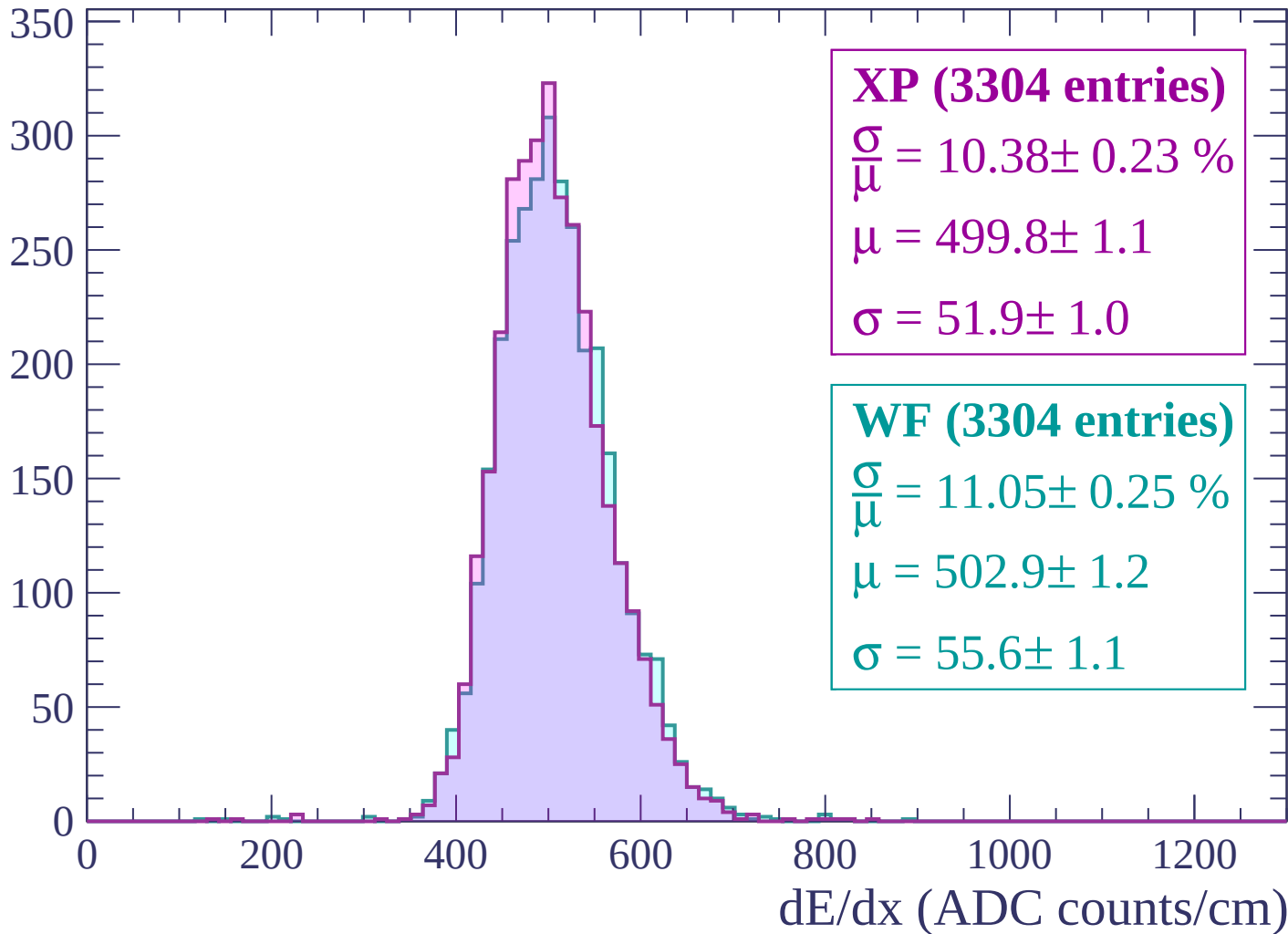
Energy loss | -2100 < p < -2040

Count



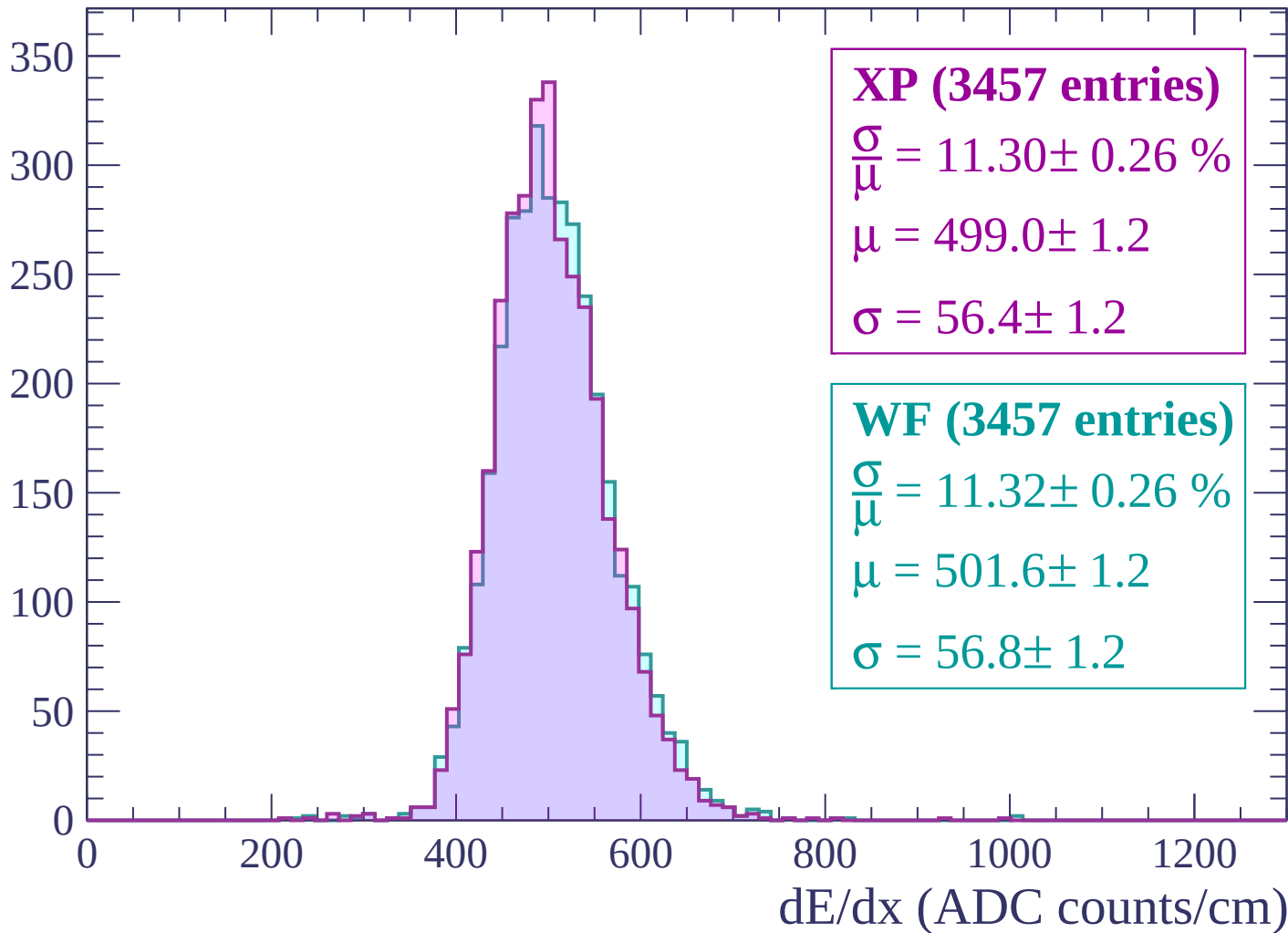
Energy loss | -2040 < p < -1980

Count



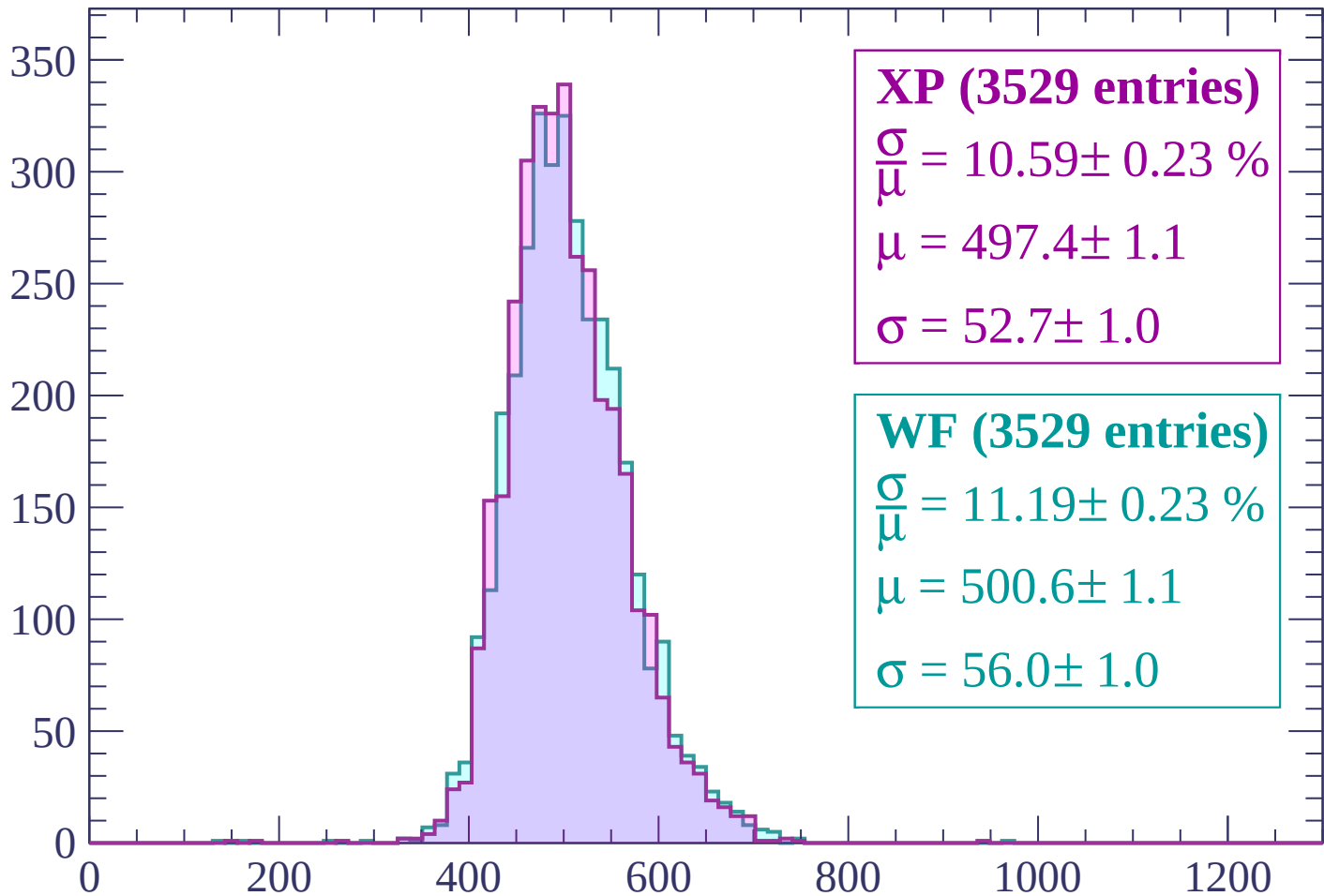
Energy loss | -1980 < p < -1920

Count



Energy loss | -1920 < p < -1860

Count



XP (3529 entries)

$$\frac{\sigma}{\mu} = 10.59 \pm 0.23 \%$$

$$\mu = 497.4 \pm 1.1$$

$$\sigma = 52.7 \pm 1.0$$

WF (3529 entries)

$$\frac{\sigma}{\mu} = 11.19 \pm 0.23 \%$$

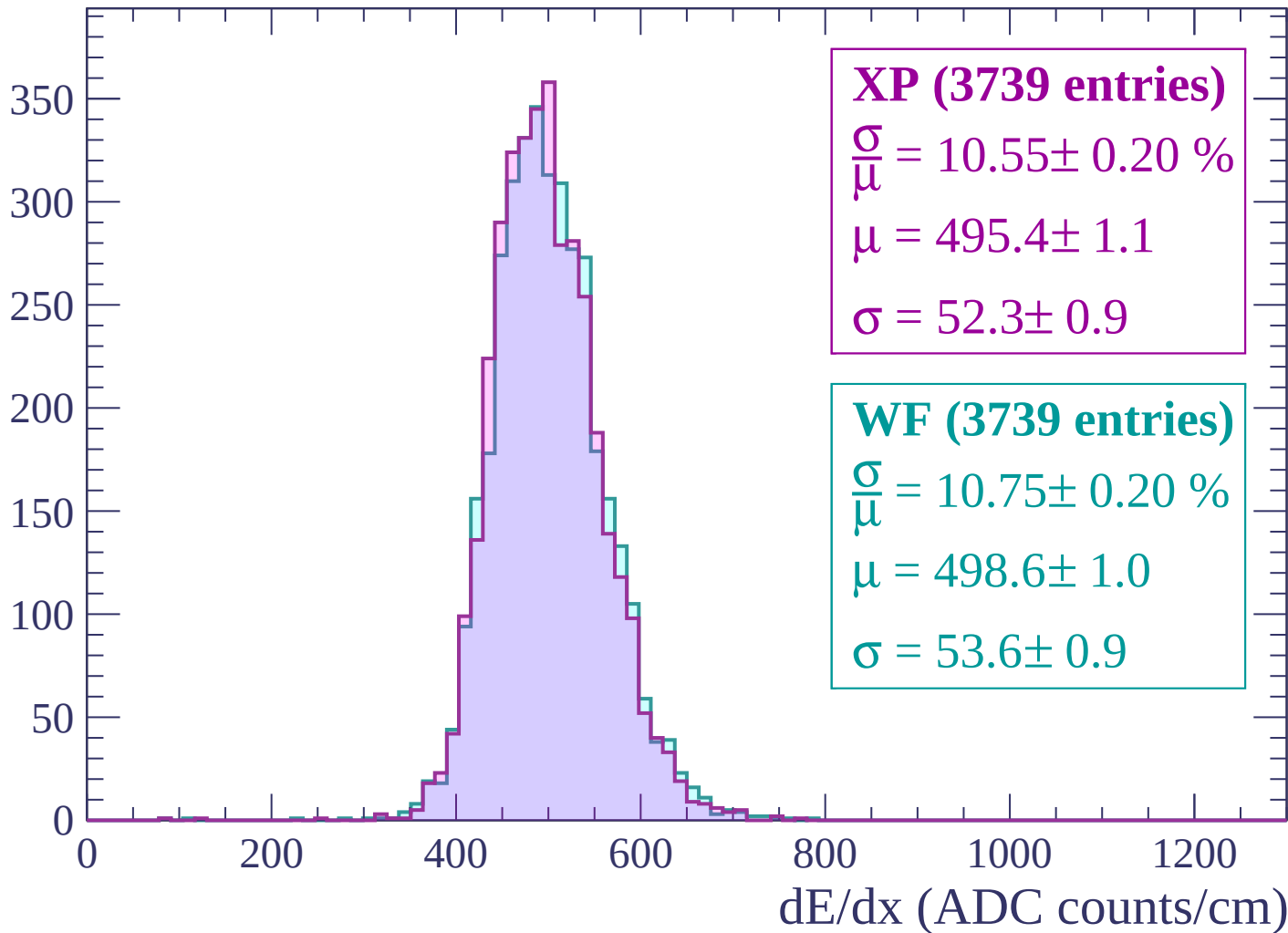
$$\mu = 500.6 \pm 1.1$$

$$\sigma = 56.0 \pm 1.0$$

dE/dx (ADC counts/cm)

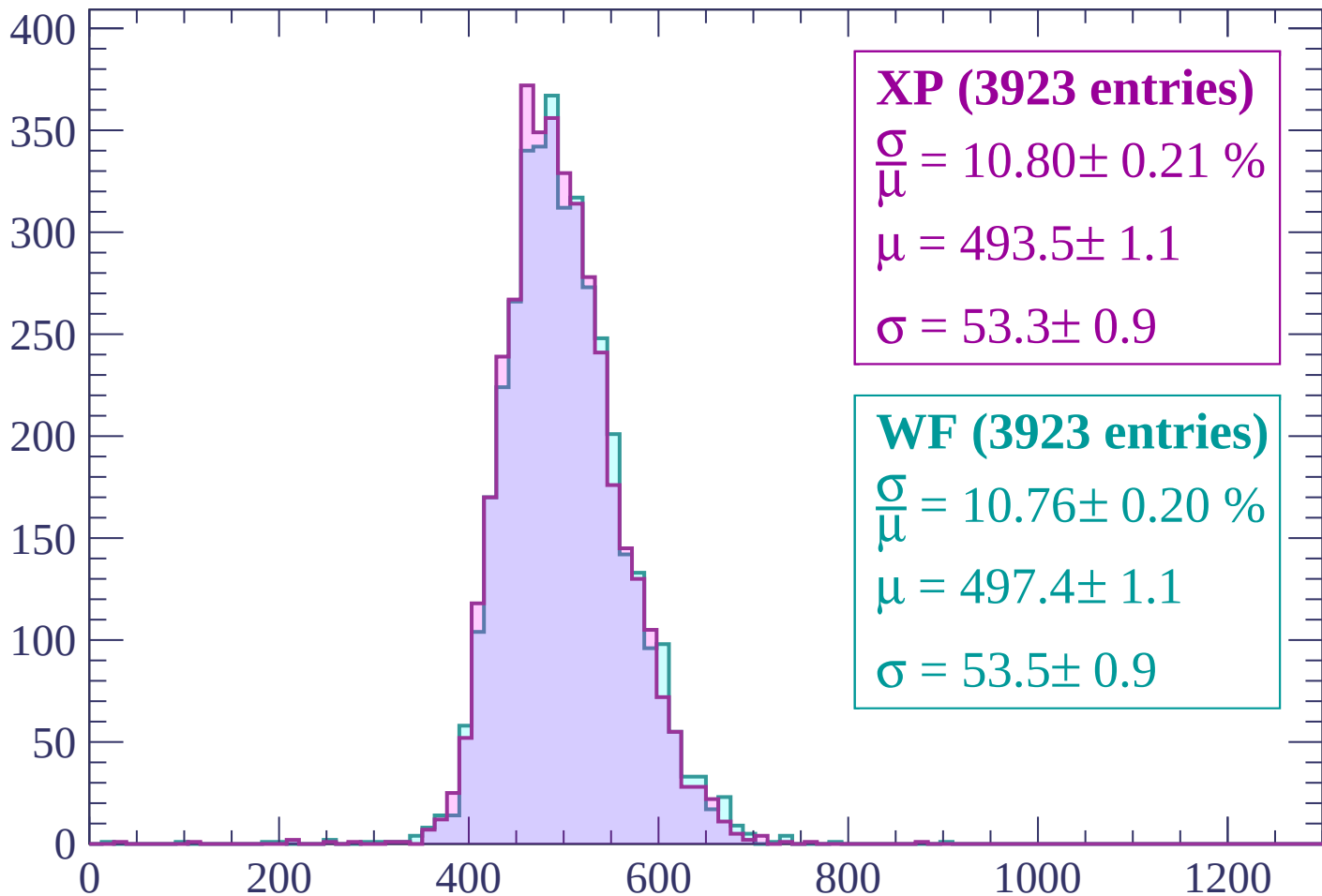
Energy loss | $-1860 < p < -1800$

Count



Energy loss | -1800 < p < -1740

Count



XP (3923 entries)

$$\frac{\sigma}{\mu} = 10.80 \pm 0.21 \%$$

$$\mu = 493.5 \pm 1.1$$

$$\sigma = 53.3 \pm 0.9$$

WF (3923 entries)

$$\frac{\sigma}{\mu} = 10.76 \pm 0.20 \%$$

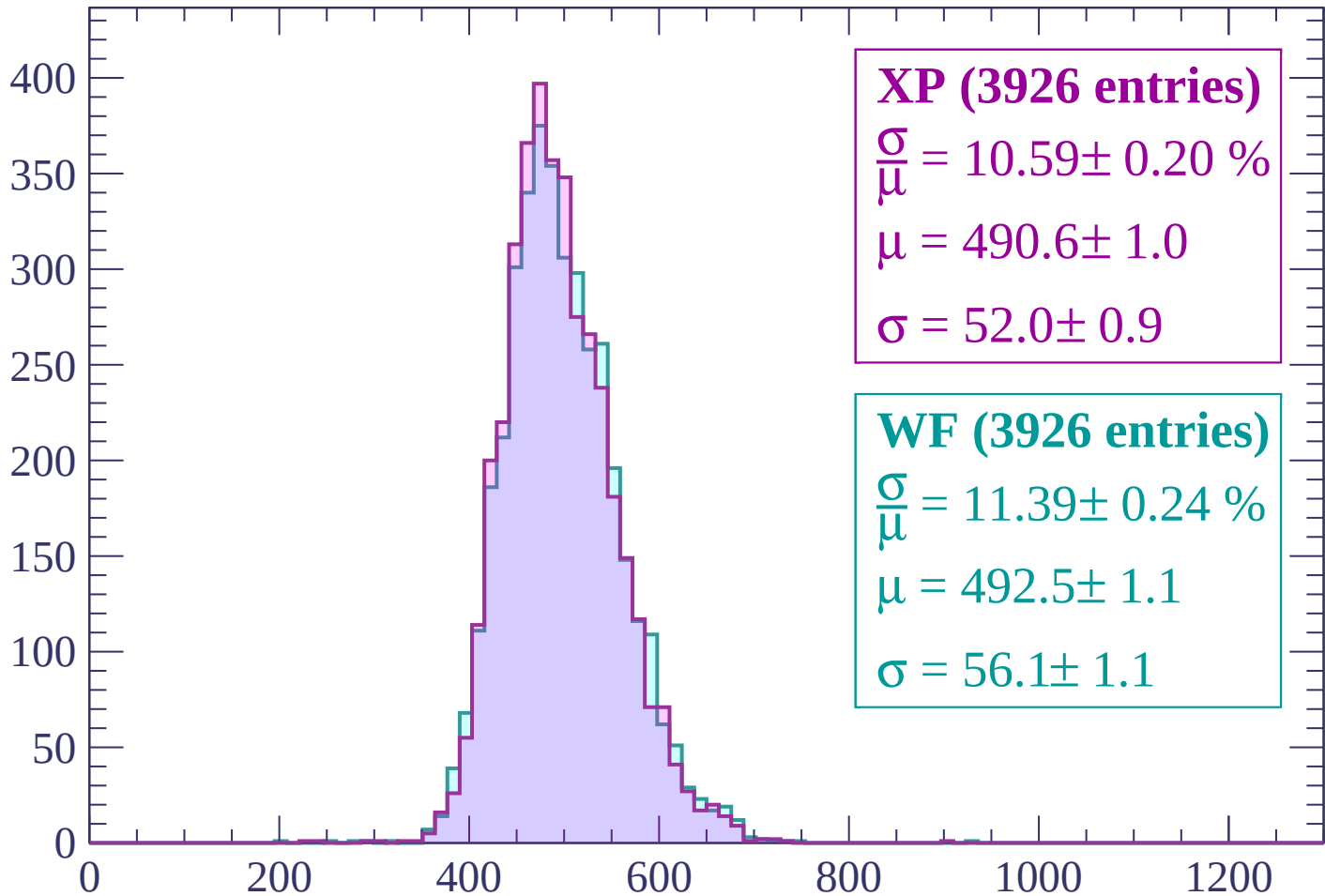
$$\mu = 497.4 \pm 1.1$$

$$\sigma = 53.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1740 < p < -1680$

Count



XP (3926 entries)

$$\frac{\sigma}{\mu} = 10.59 \pm 0.20 \%$$

$$\mu = 490.6 \pm 1.0$$

$$\sigma = 52.0 \pm 0.9$$

WF (3926 entries)

$$\frac{\sigma}{\mu} = 11.39 \pm 0.24 \%$$

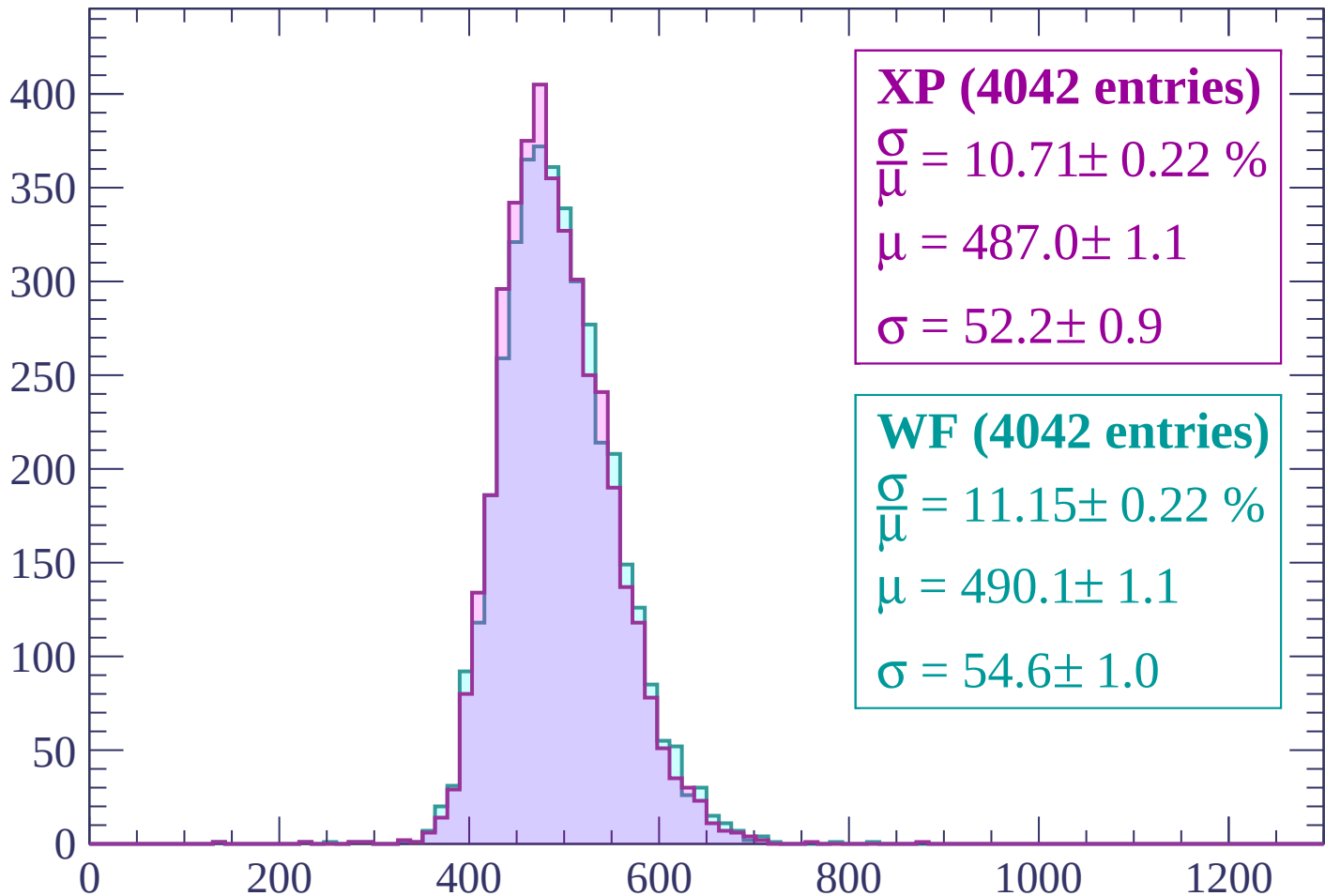
$$\mu = 492.5 \pm 1.1$$

$$\sigma = 56.1 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -1680 < p < -1620

Count



XP (4042 entries)

$\frac{\sigma}{\mu} = 10.71 \pm 0.22 \%$

$\mu = 487.0 \pm 1.1$

$\sigma = 52.2 \pm 0.9$

WF (4042 entries)

$\frac{\sigma}{\mu} = 11.15 \pm 0.22 \%$

$\mu = 490.1 \pm 1.1$

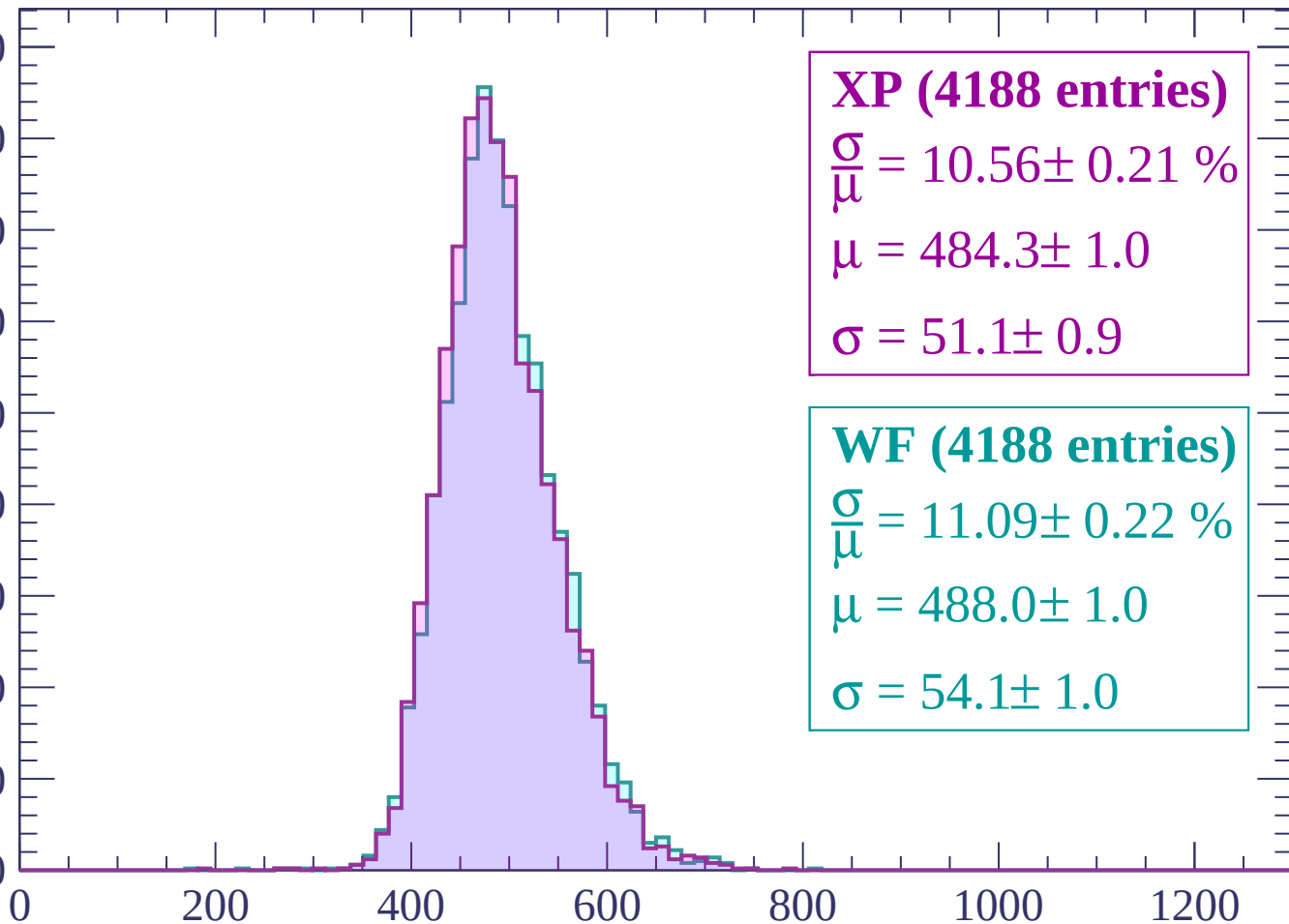
$\sigma = 54.6 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | -1620 < p < -1560

Count

450
400
350
300
250
200
150
100
50
0



XP (4188 entries)

$$\frac{\sigma}{\mu} = 10.56 \pm 0.21 \%$$

$$\mu = 484.3 \pm 1.0$$

$$\sigma = 51.1 \pm 0.9$$

WF (4188 entries)

$$\frac{\sigma}{\mu} = 11.09 \pm 0.22 \%$$

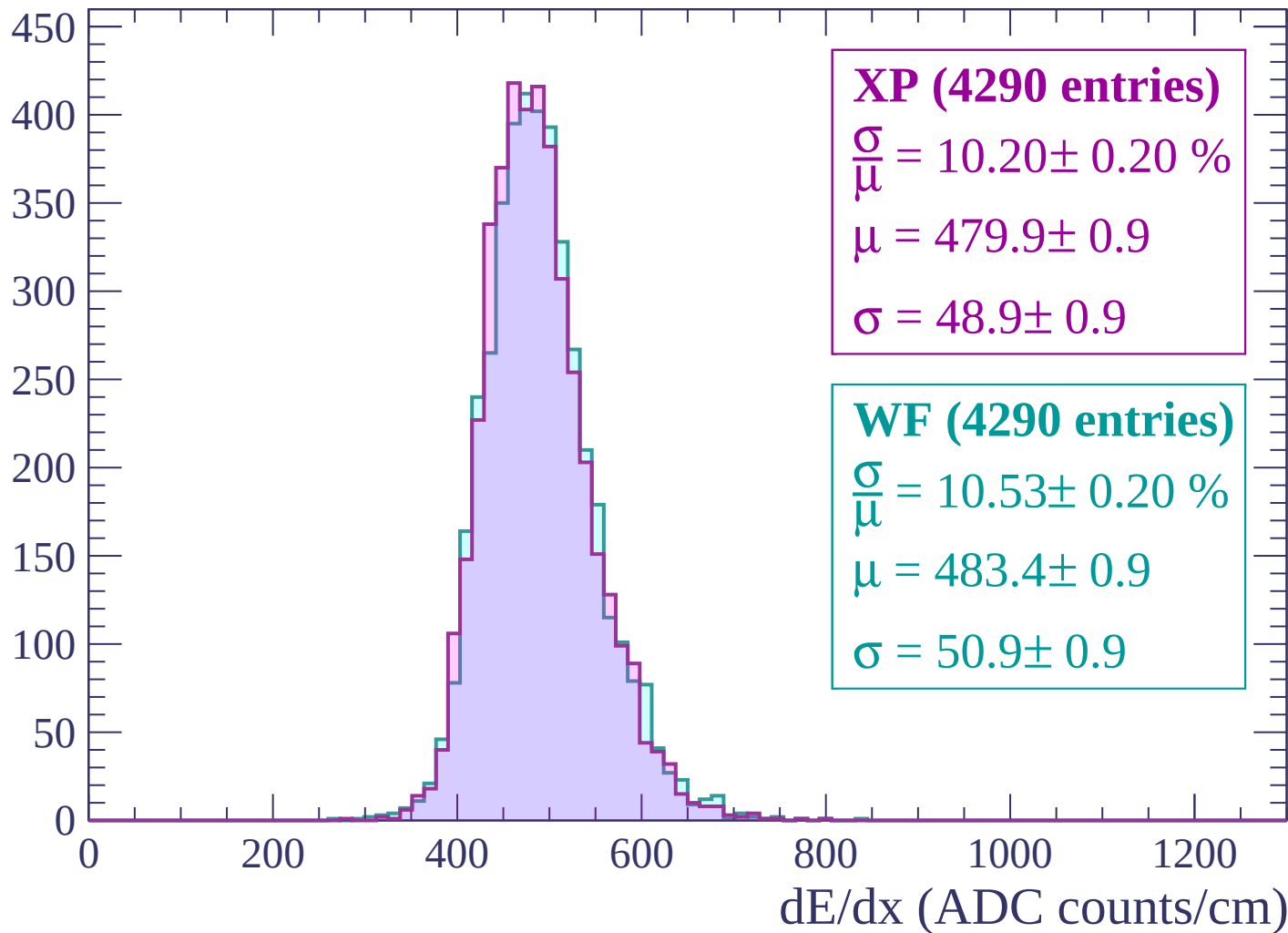
$$\mu = 488.0 \pm 1.0$$

$$\sigma = 54.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1500

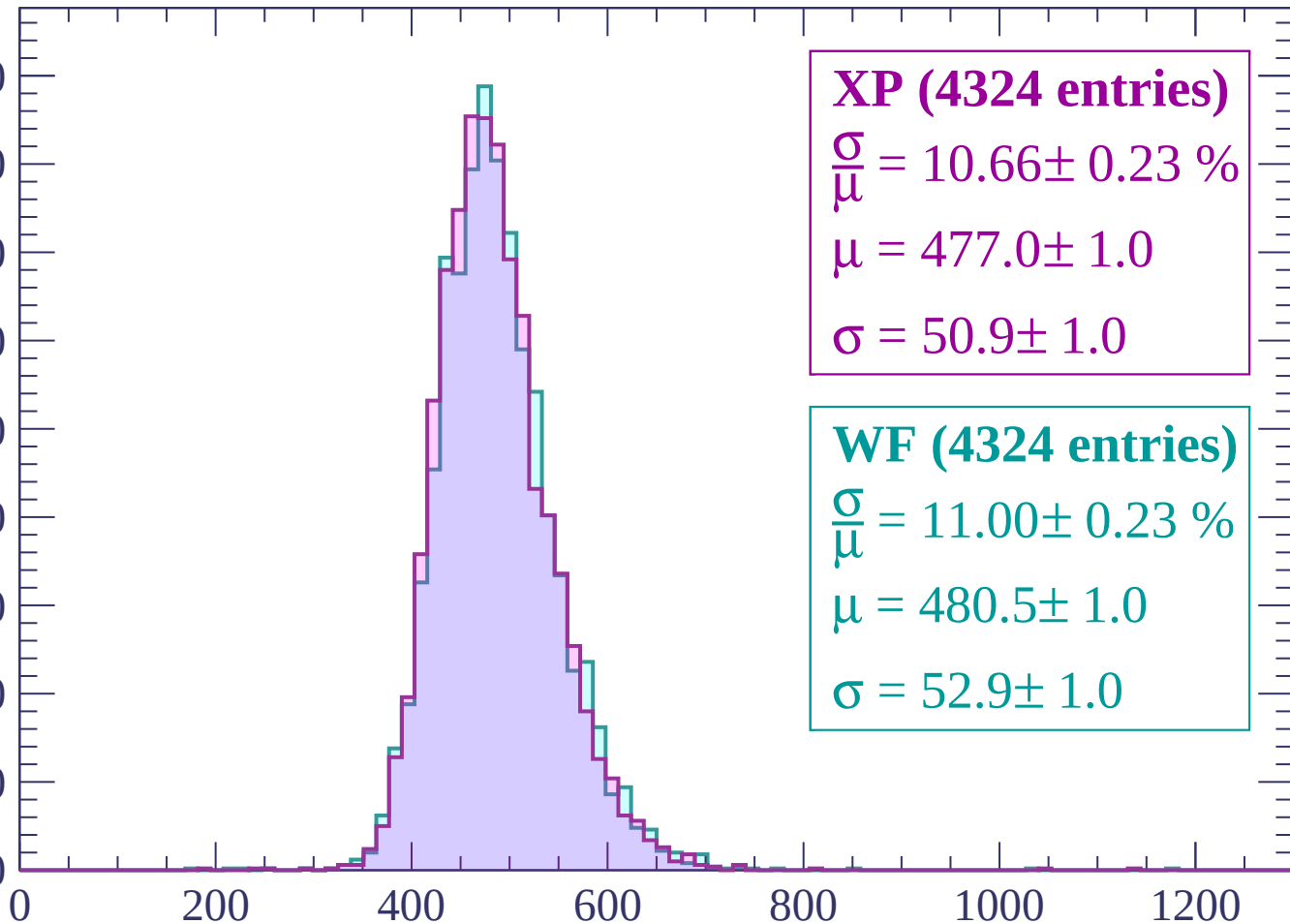
Count



Energy loss | $-1500 < p < -1440$

Count

450
400
350
300
250
200
150
100
50
0



XP (4324 entries)

$\frac{\sigma}{\mu} = 10.66 \pm 0.23 \%$

$\mu = 477.0 \pm 1.0$

$\sigma = 50.9 \pm 1.0$

WF (4324 entries)

$\frac{\sigma}{\mu} = 11.00 \pm 0.23 \%$

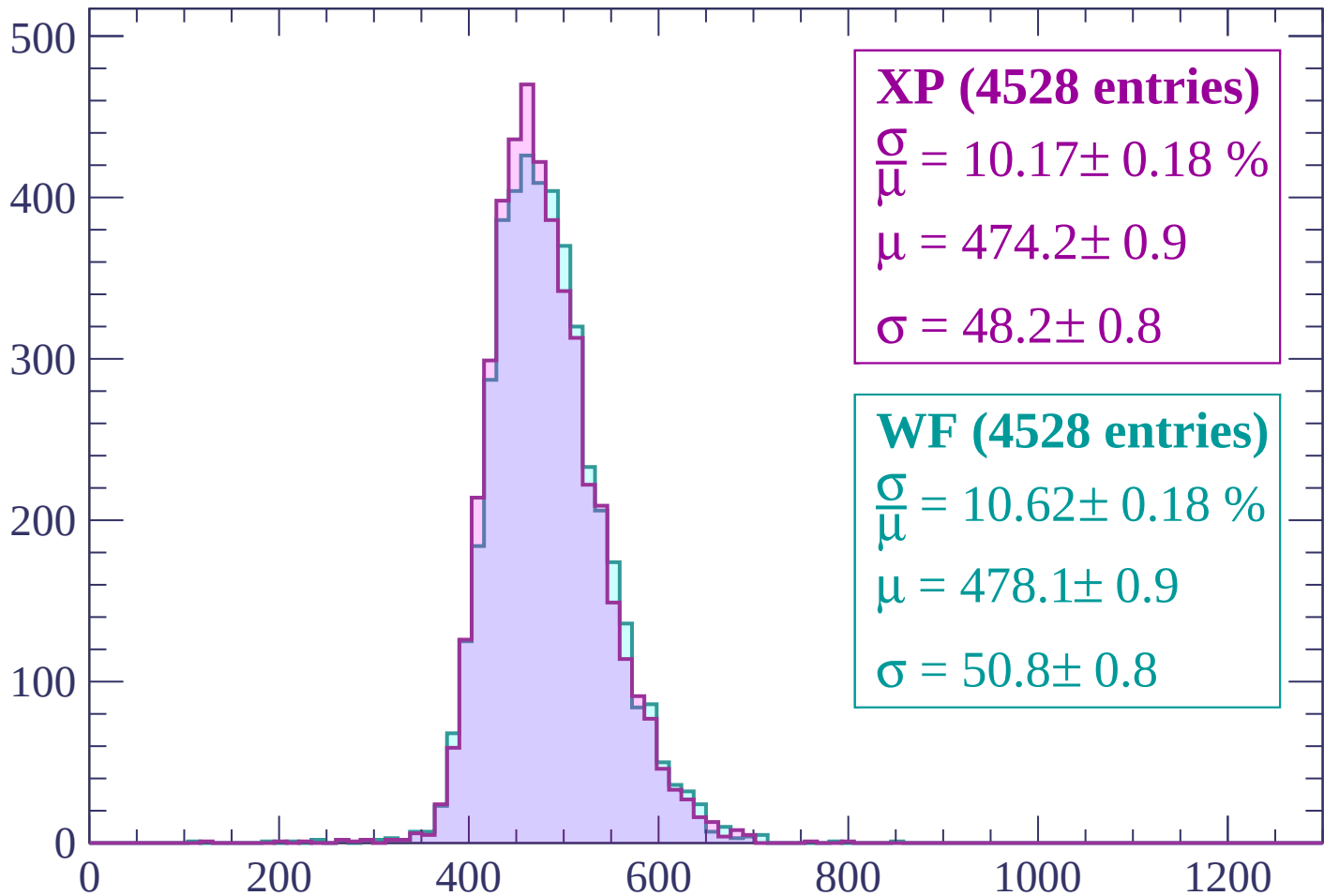
$\mu = 480.5 \pm 1.0$

$\sigma = 52.9 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1380$

Count



XP (4528 entries)

$\frac{\sigma}{\mu} = 10.17 \pm 0.18 \%$

$\mu = 474.2 \pm 0.9$

$\sigma = 48.2 \pm 0.8$

WF (4528 entries)

$\frac{\sigma}{\mu} = 10.62 \pm 0.18 \%$

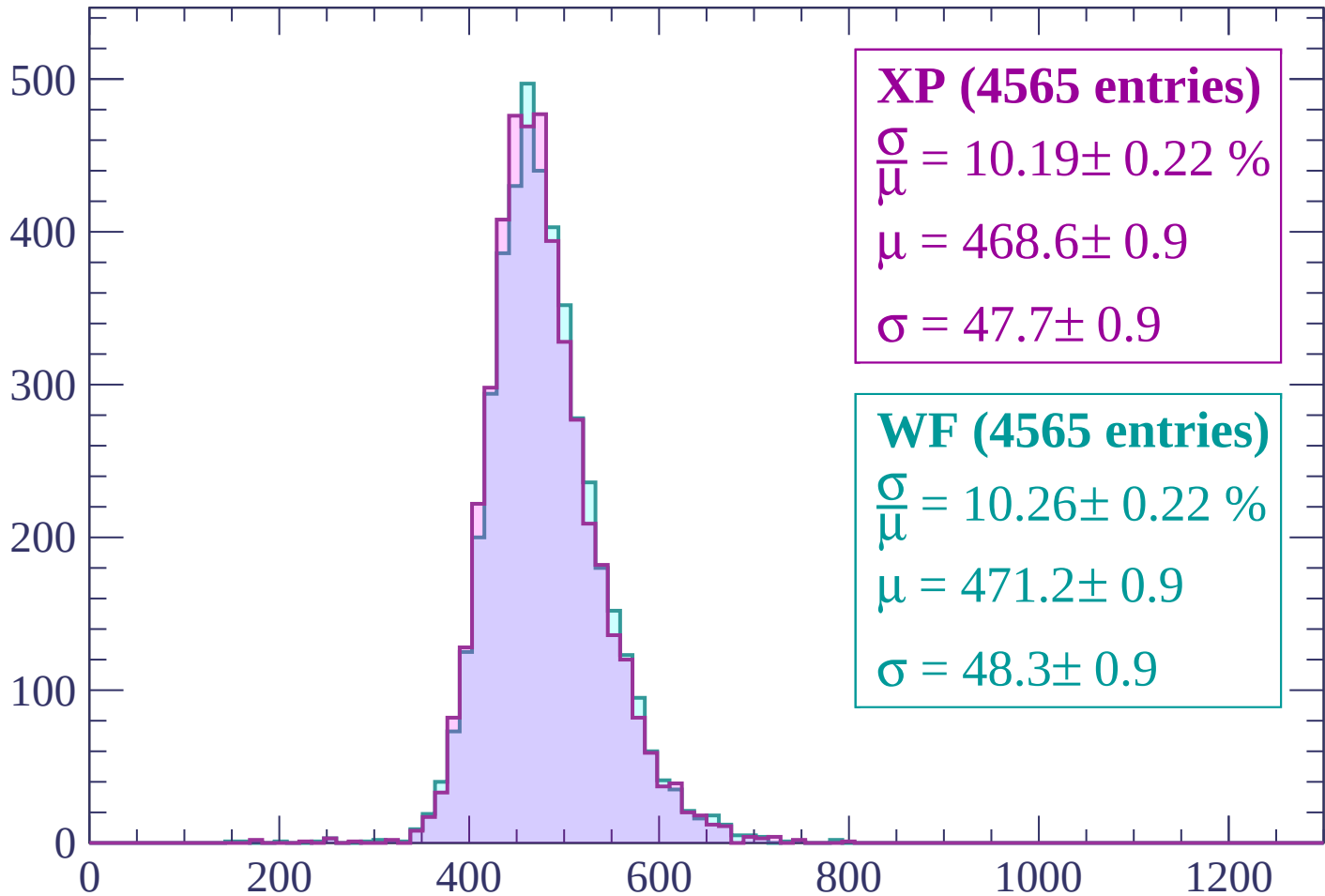
$\mu = 478.1 \pm 0.9$

$\sigma = 50.8 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | -1380 < p < -1320

Count



XP (4565 entries)

$\frac{\sigma}{\mu} = 10.19 \pm 0.22 \%$

$\mu = 468.6 \pm 0.9$

$\sigma = 47.7 \pm 0.9$

WF (4565 entries)

$\frac{\sigma}{\mu} = 10.26 \pm 0.22 \%$

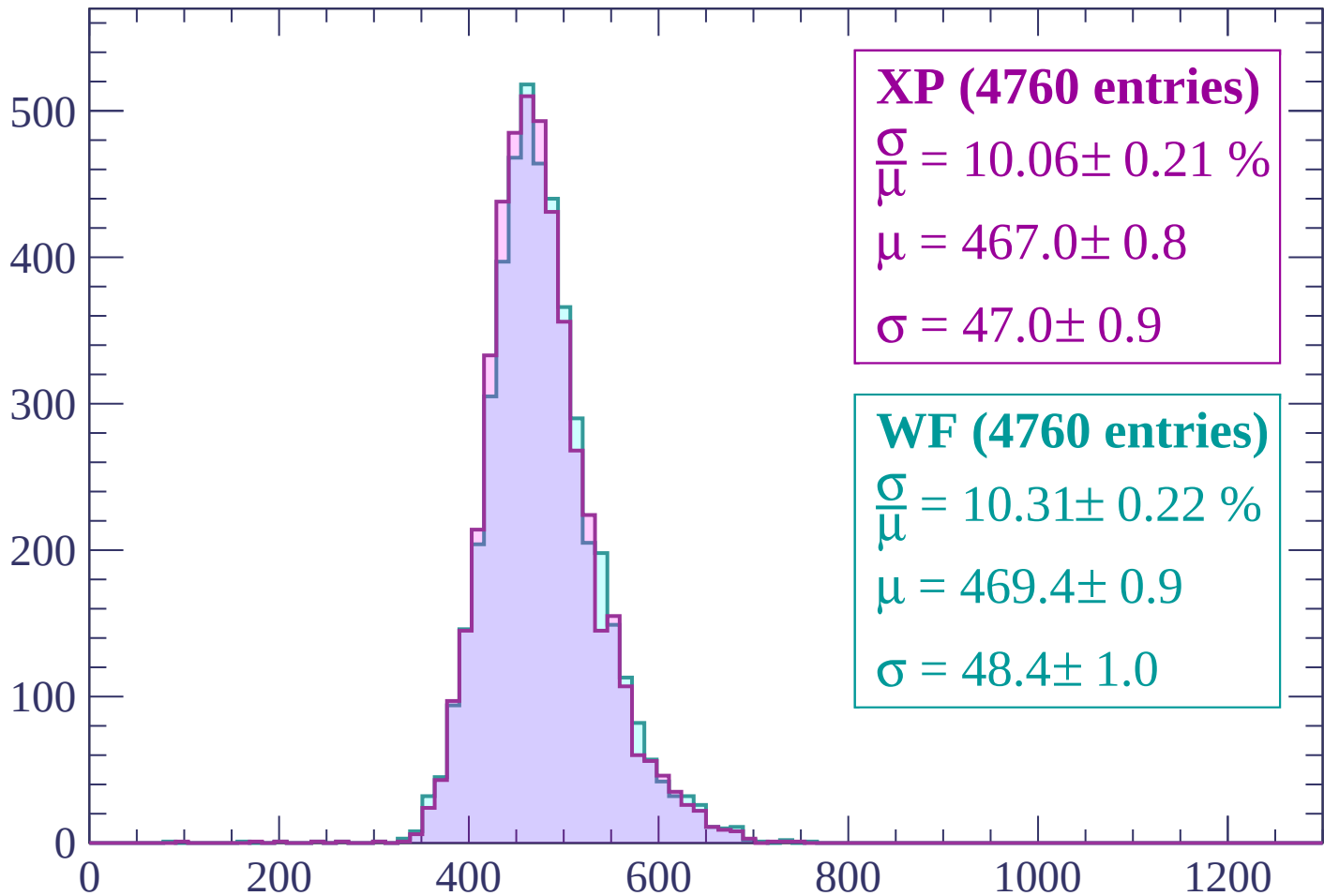
$\mu = 471.2 \pm 0.9$

$\sigma = 48.3 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1260

Count



XP (4760 entries)

$$\frac{\sigma}{\mu} = 10.06 \pm 0.21 \%$$

$$\mu = 467.0 \pm 0.8$$

$$\sigma = 47.0 \pm 0.9$$

WF (4760 entries)

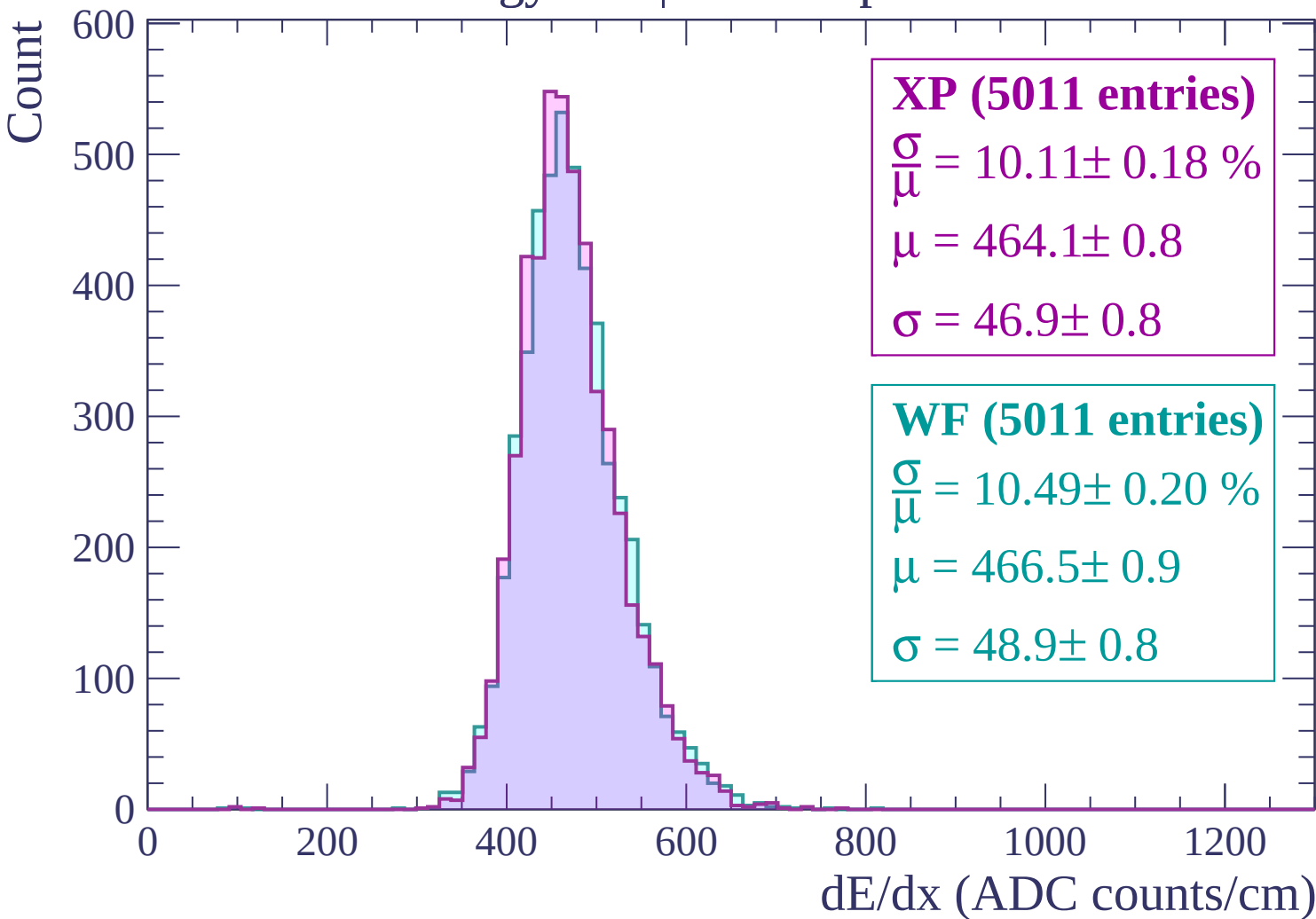
$$\frac{\sigma}{\mu} = 10.31 \pm 0.22 \%$$

$$\mu = 469.4 \pm 0.9$$

$$\sigma = 48.4 \pm 1.0$$

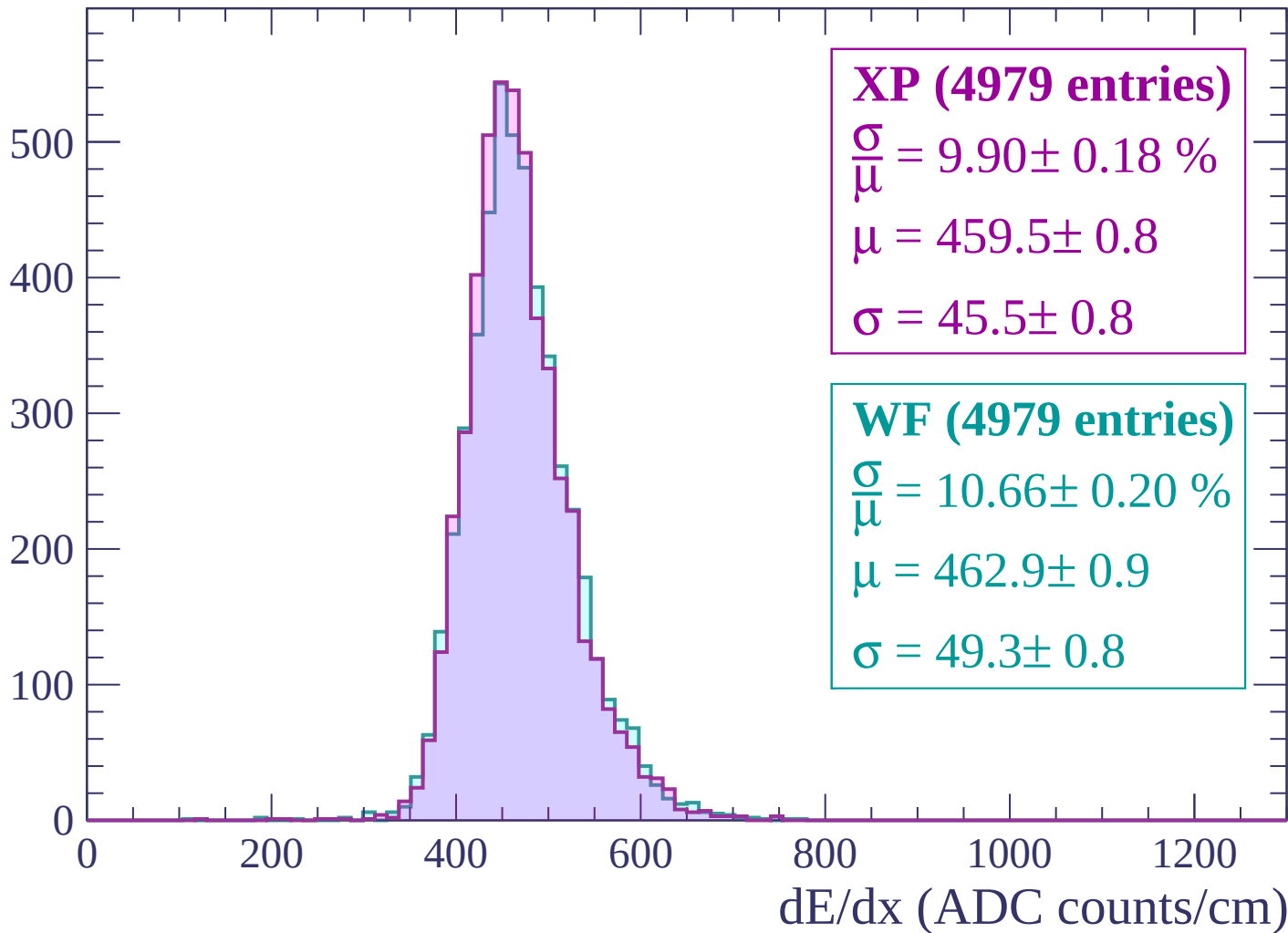
dE/dx (ADC counts/cm)

Energy loss | $-1260 < p < -1200$



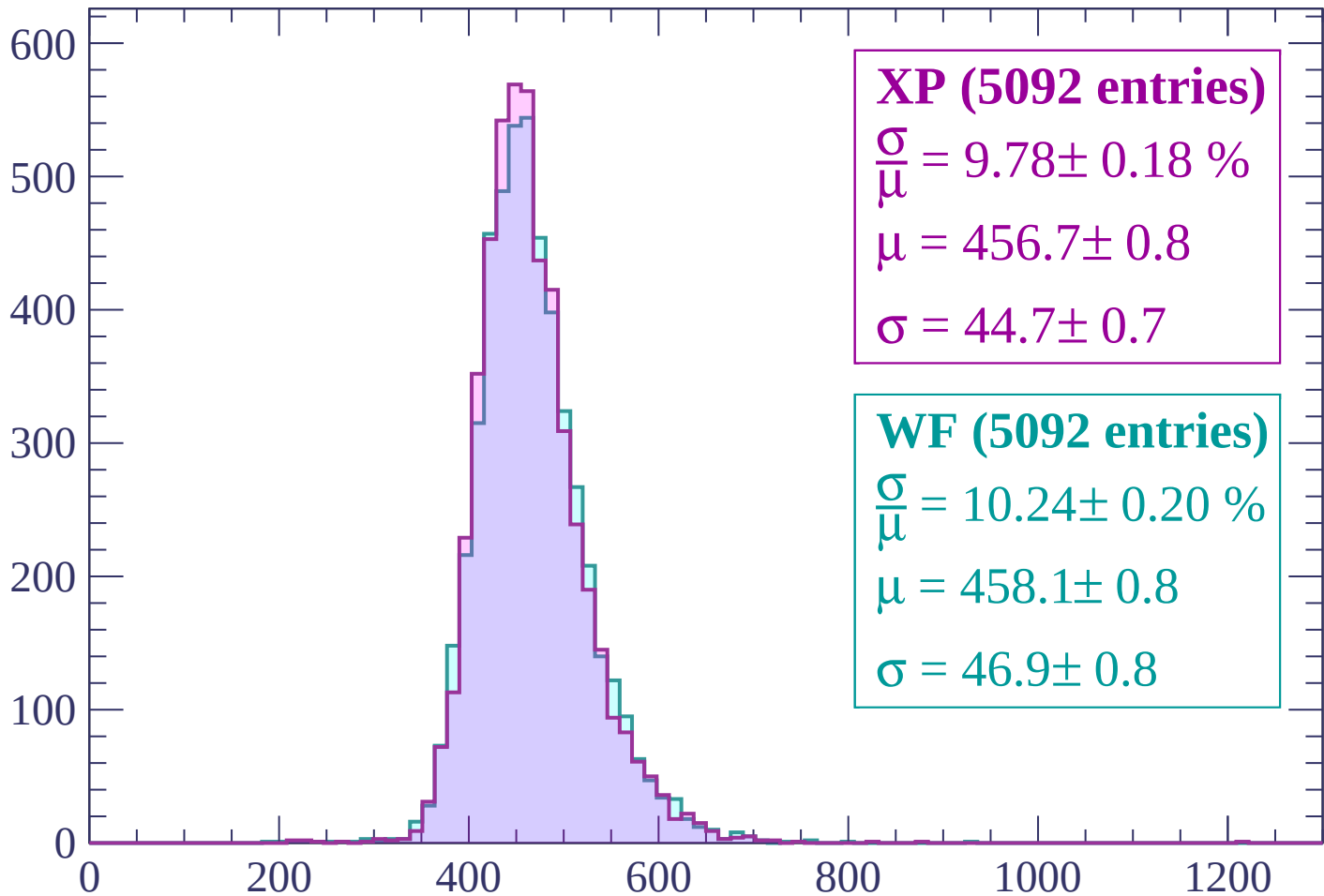
Energy loss | -1200 < p < -1140

Count



Energy loss | $-1140 < p < -1080$

Count



XP (5092 entries)

$$\frac{\sigma}{\mu} = 9.78 \pm 0.18 \%$$

$$\mu = 456.7 \pm 0.8$$

$$\sigma = 44.7 \pm 0.7$$

WF (5092 entries)

$$\frac{\sigma}{\mu} = 10.24 \pm 0.20 \%$$

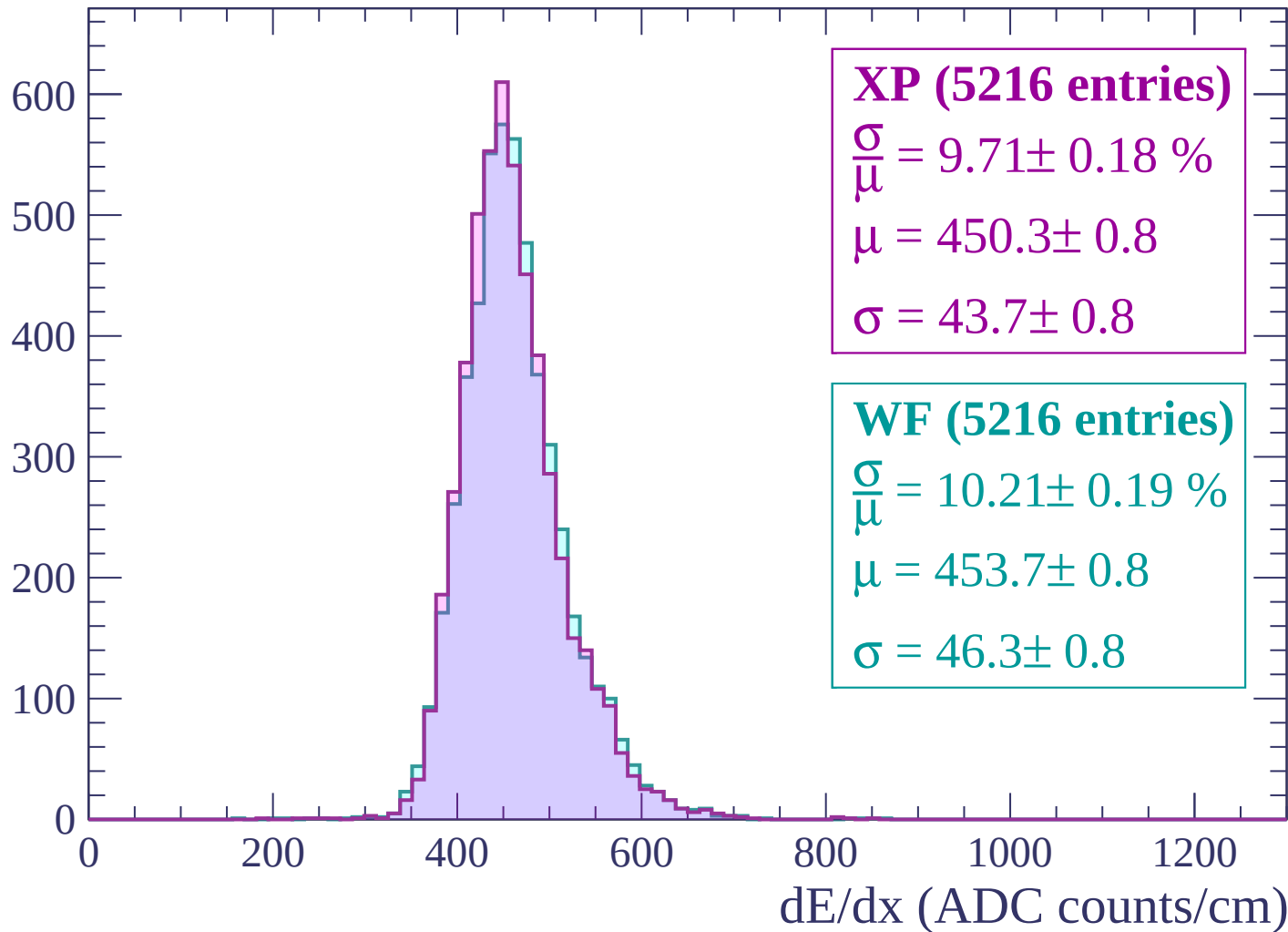
$$\mu = 458.1 \pm 0.8$$

$$\sigma = 46.9 \pm 0.8$$

dE/dx (ADC counts/cm)

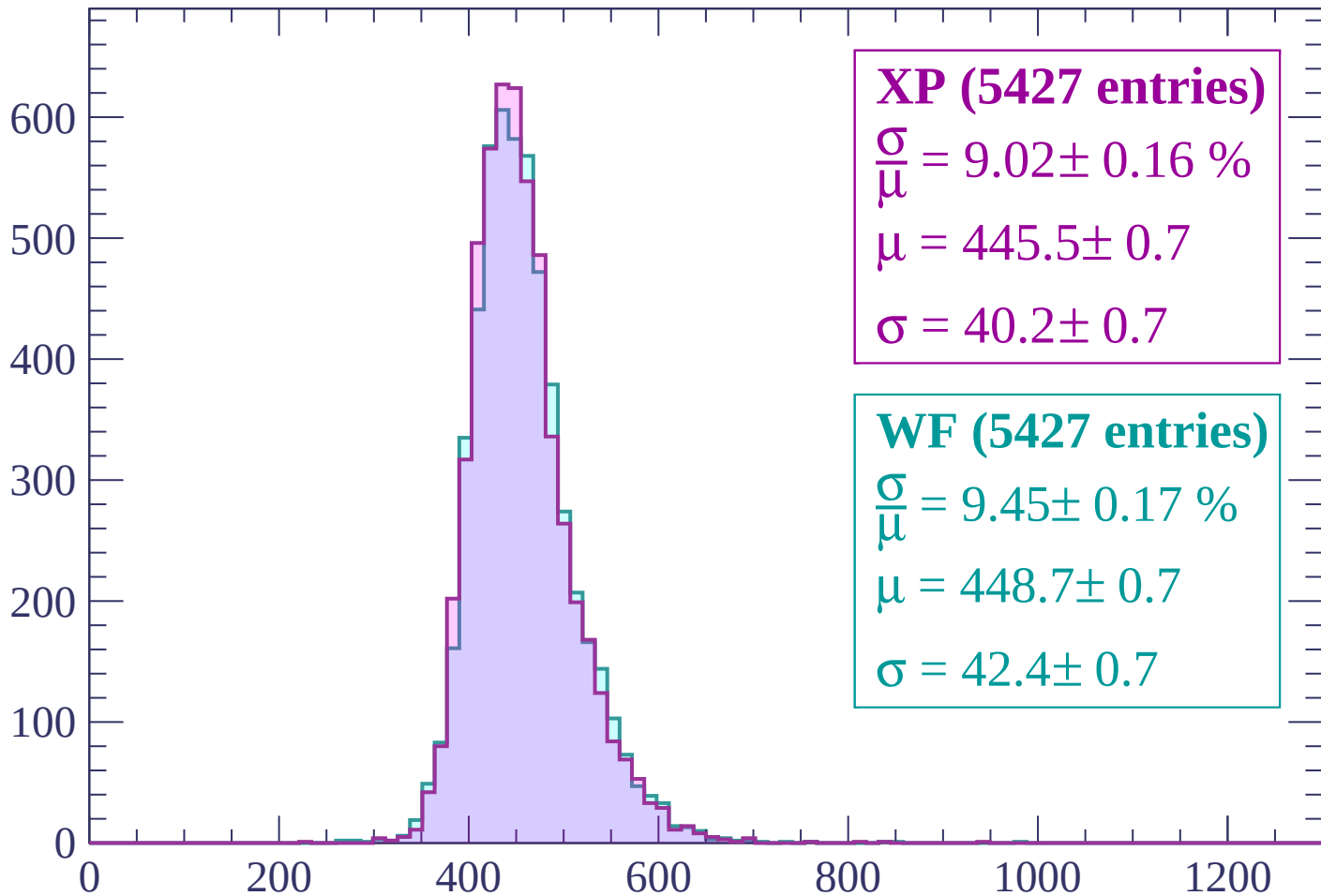
Energy loss | $-1080 < p < -1020$

Count



Energy loss | $-1020 < p < -960$

Count



XP (5427 entries)

$$\frac{\sigma}{\mu} = 9.02 \pm 0.16 \%$$

$$\mu = 445.5 \pm 0.7$$

$$\sigma = 40.2 \pm 0.7$$

WF (5427 entries)

$$\frac{\sigma}{\mu} = 9.45 \pm 0.17 \%$$

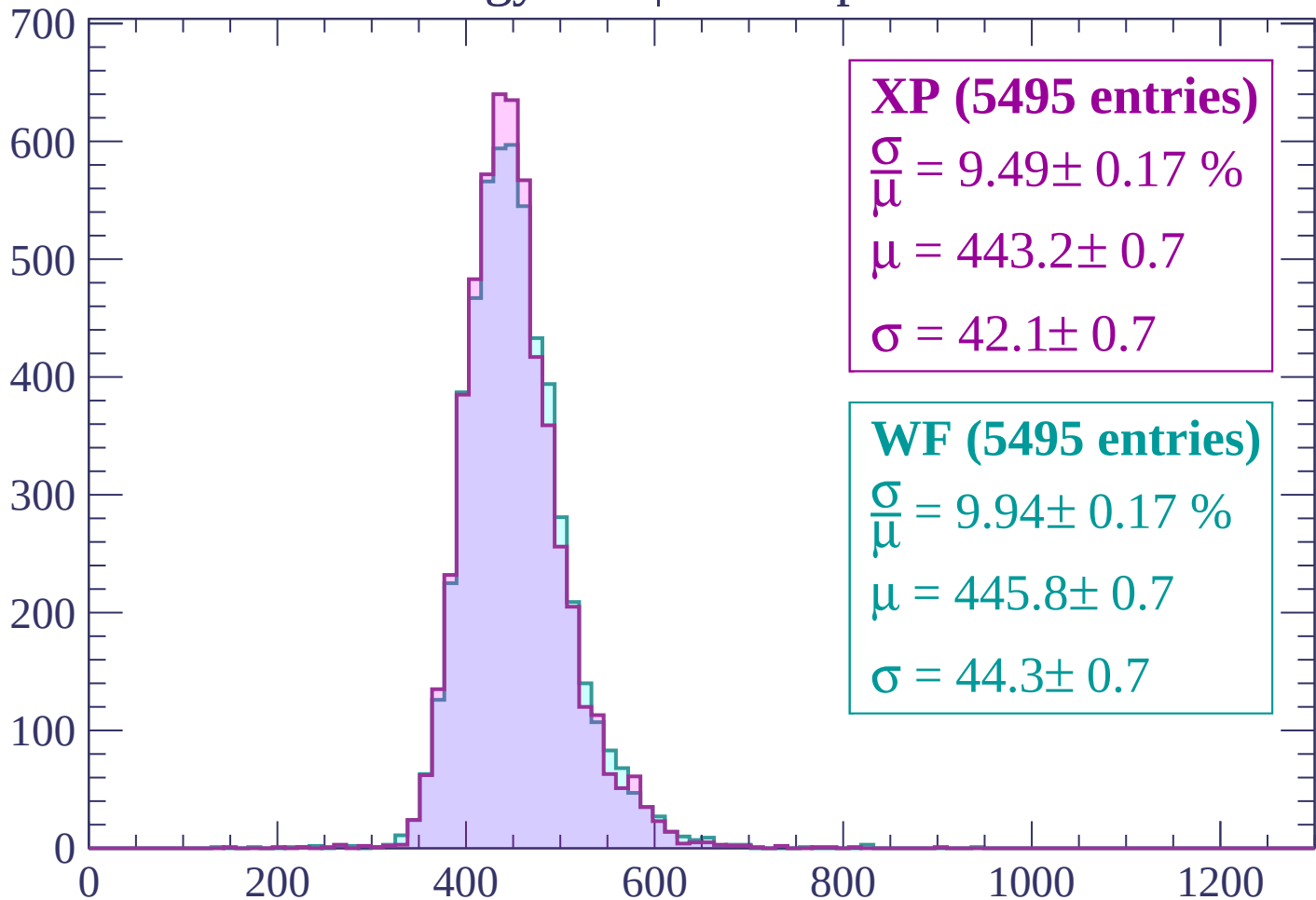
$$\mu = 448.7 \pm 0.7$$

$$\sigma = 42.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count



XP (5495 entries)

$$\frac{\sigma}{\mu} = 9.49 \pm 0.17 \%$$

$$\mu = 443.2 \pm 0.7$$

$$\sigma = 42.1 \pm 0.7$$

WF (5495 entries)

$$\frac{\sigma}{\mu} = 9.94 \pm 0.17 \%$$

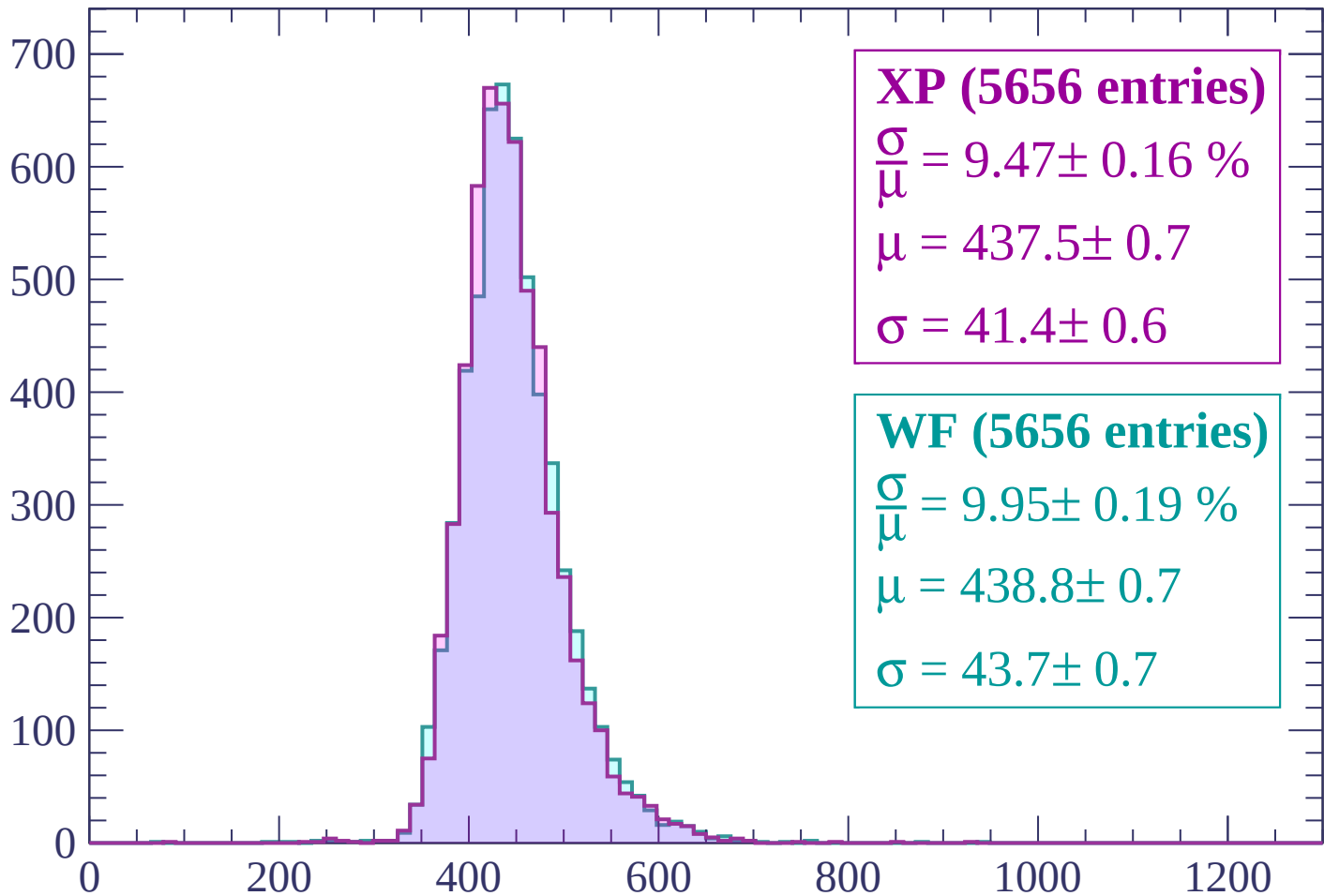
$$\mu = 445.8 \pm 0.7$$

$$\sigma = 44.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (5656 entries)

$$\frac{\sigma}{\mu} = 9.47 \pm 0.16 \%$$

$$\mu = 437.5 \pm 0.7$$

$$\sigma = 41.4 \pm 0.6$$

WF (5656 entries)

$$\frac{\sigma}{\mu} = 9.95 \pm 0.19 \%$$

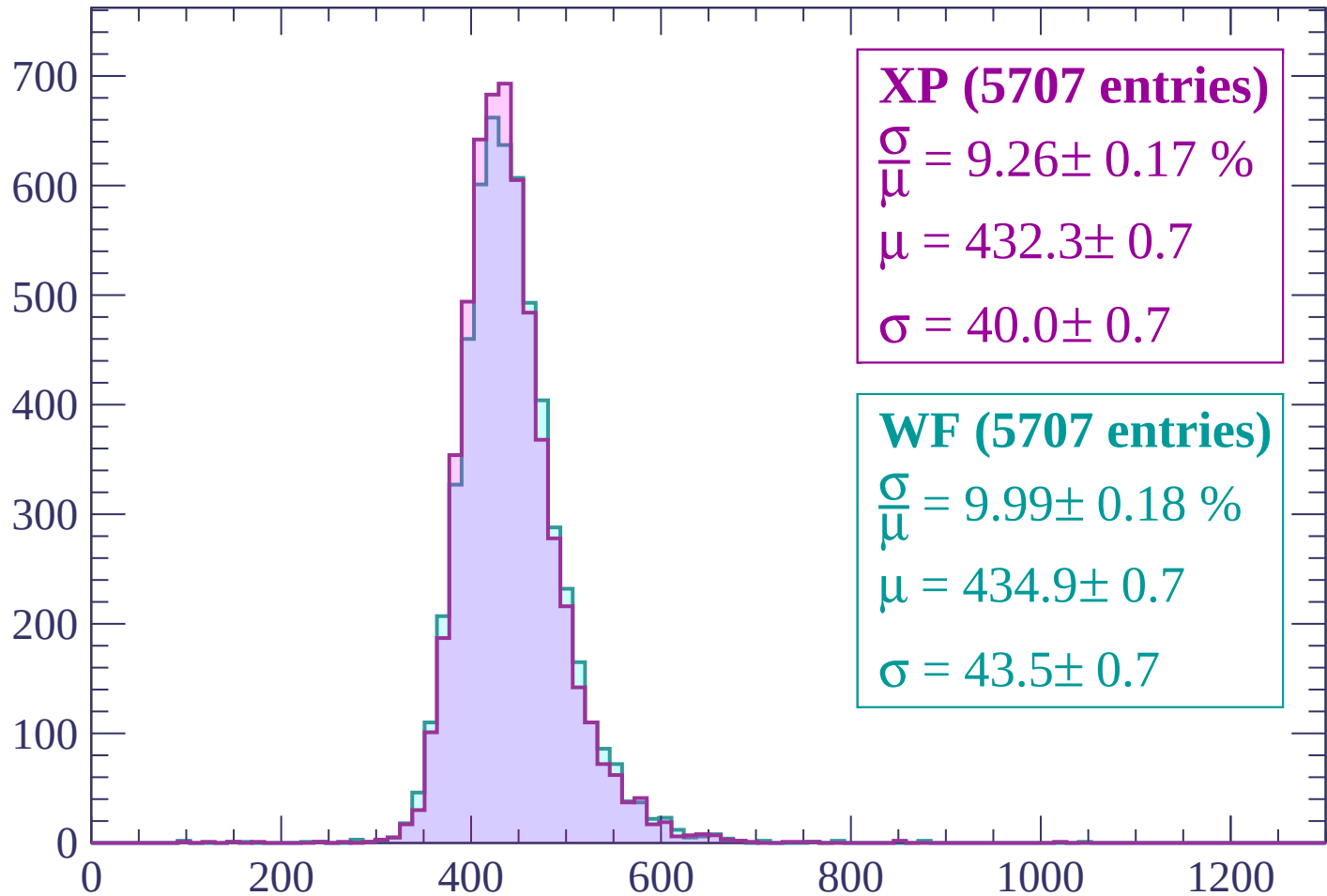
$$\mu = 438.8 \pm 0.7$$

$$\sigma = 43.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (5707 entries)

$$\frac{\sigma}{\mu} = 9.26 \pm 0.17 \%$$

$$\mu = 432.3 \pm 0.7$$

$$\sigma = 40.0 \pm 0.7$$

WF (5707 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.18 \%$$

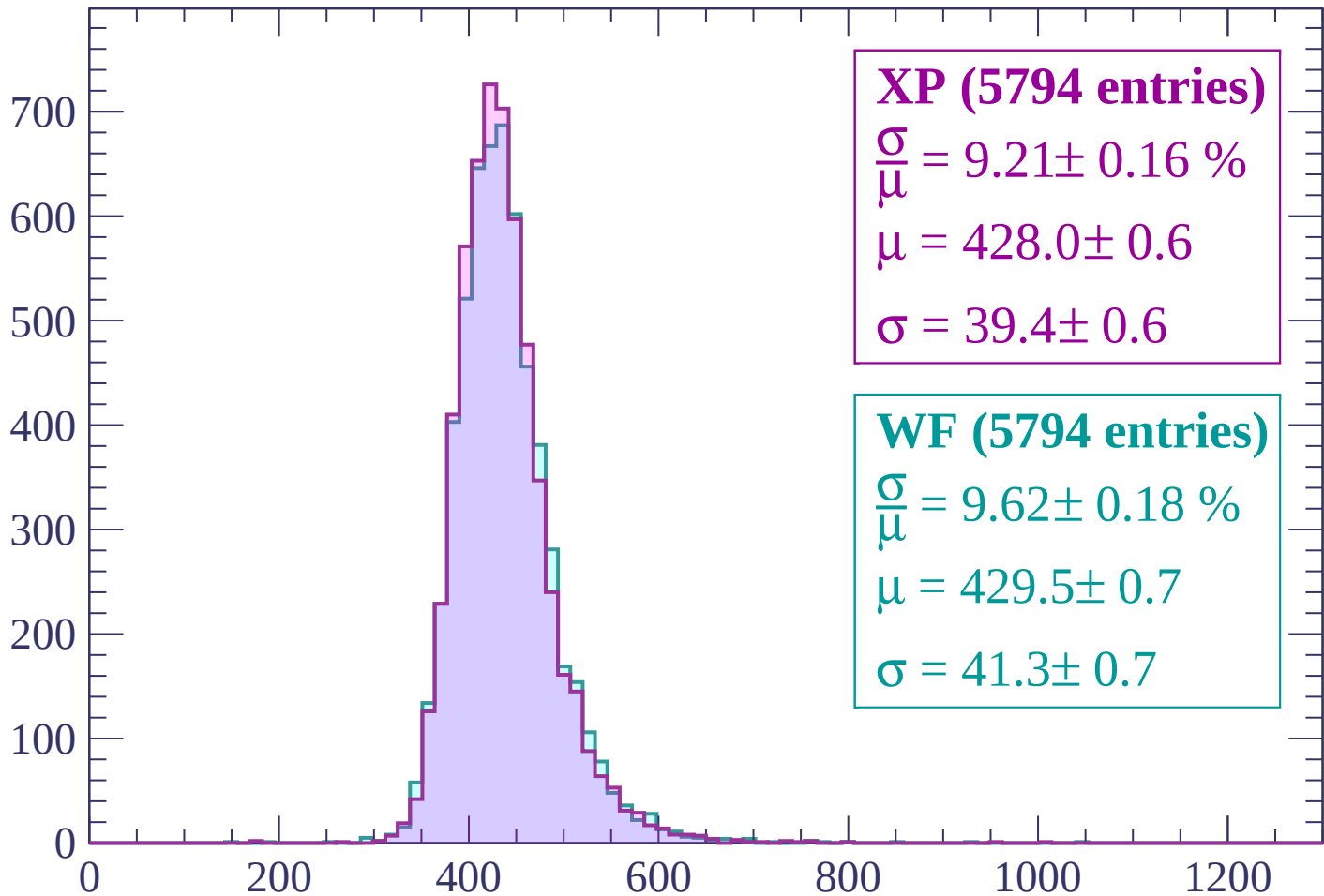
$$\mu = 434.9 \pm 0.7$$

$$\sigma = 43.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP (5794 entries)

$$\frac{\sigma}{\mu} = 9.21 \pm 0.16 \%$$

$$\mu = 428.0 \pm 0.6$$

$$\sigma = 39.4 \pm 0.6$$

WF (5794 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.18 \%$$

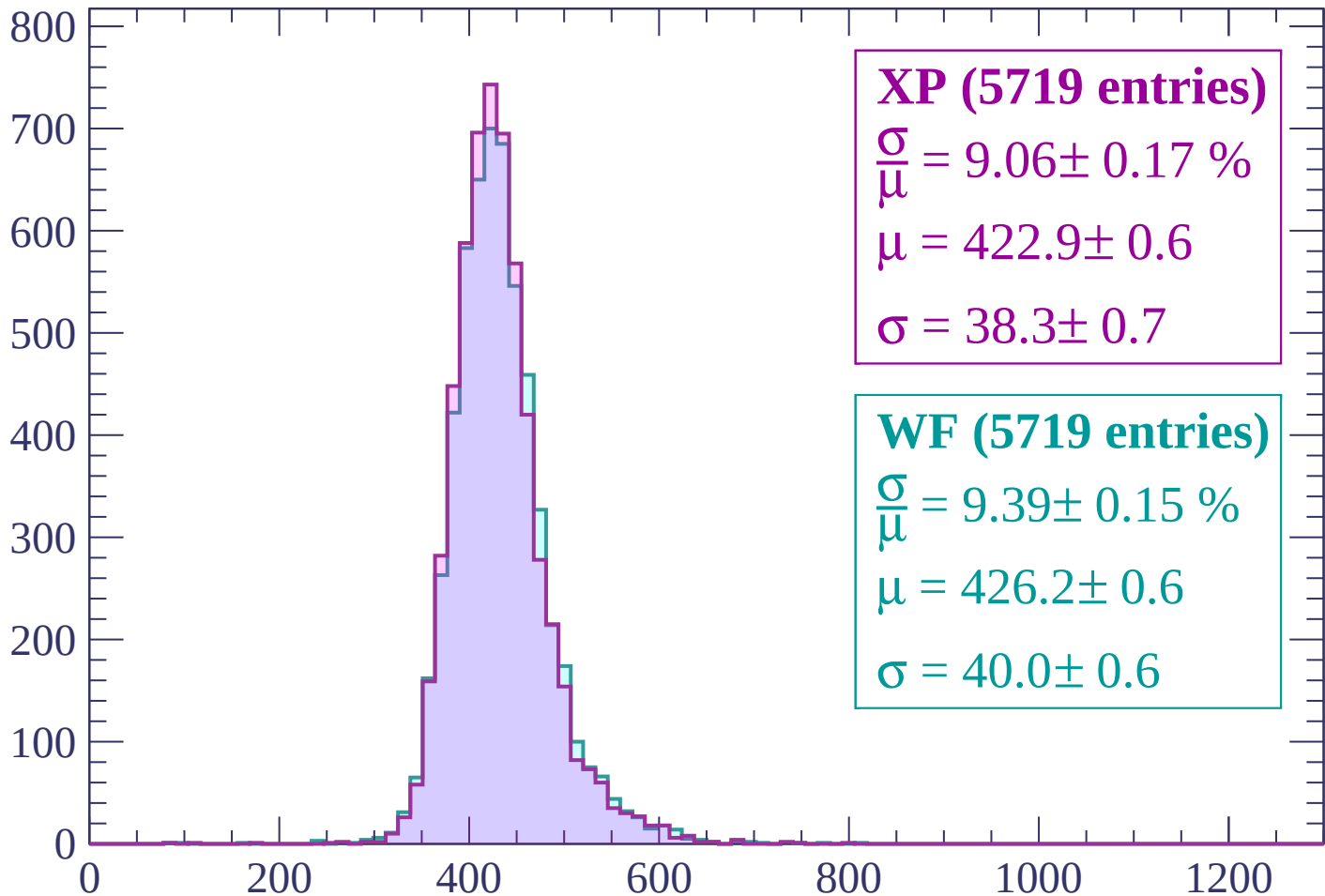
$$\mu = 429.5 \pm 0.7$$

$$\sigma = 41.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP (5719 entries)

$$\frac{\sigma}{\mu} = 9.06 \pm 0.17 \%$$

$$\mu = 422.9 \pm 0.6$$

$$\sigma = 38.3 \pm 0.7$$

WF (5719 entries)

$$\frac{\sigma}{\mu} = 9.39 \pm 0.15 \%$$

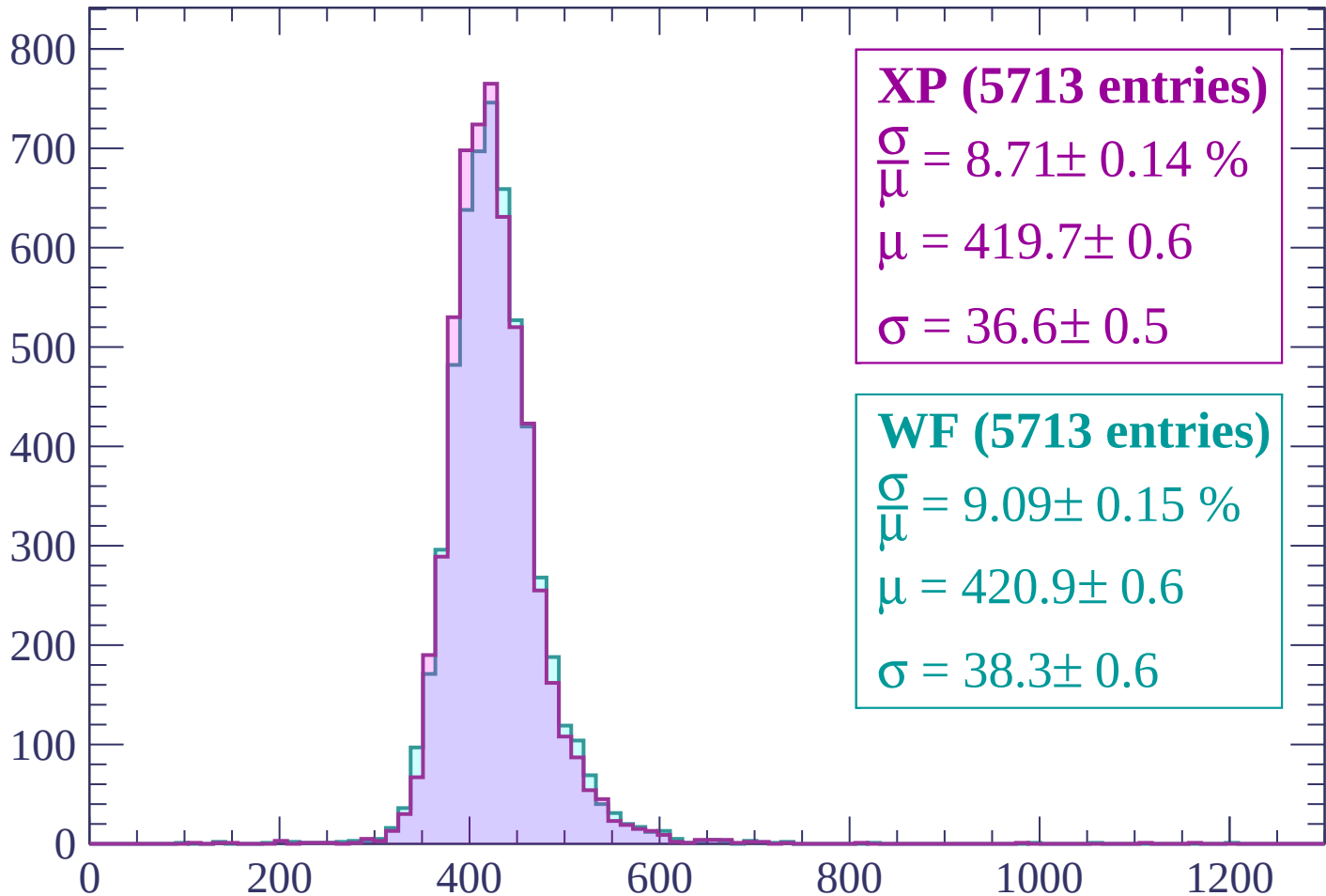
$$\mu = 426.2 \pm 0.6$$

$$\sigma = 40.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count



XP (5713 entries)

$$\frac{\sigma}{\mu} = 8.71 \pm 0.14 \%$$

$$\mu = 419.7 \pm 0.6$$

$$\sigma = 36.6 \pm 0.5$$

WF (5713 entries)

$$\frac{\sigma}{\mu} = 9.09 \pm 0.15 \%$$

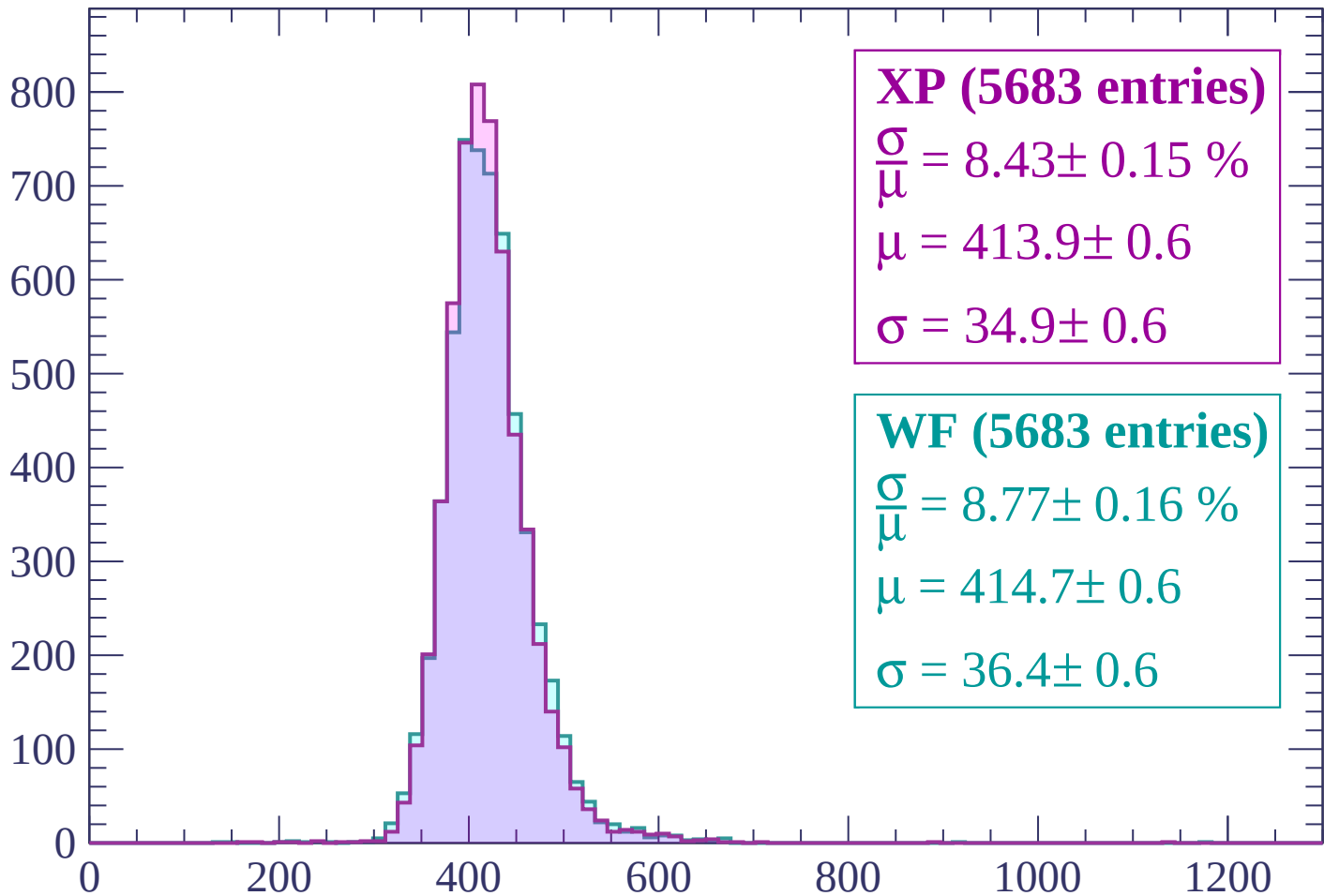
$$\mu = 420.9 \pm 0.6$$

$$\sigma = 38.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count



XP (5683 entries)

$\frac{\sigma}{\mu} = 8.43 \pm 0.15 \%$

$\mu = 413.9 \pm 0.6$

$\sigma = 34.9 \pm 0.6$

WF (5683 entries)

$\frac{\sigma}{\mu} = 8.77 \pm 0.16 \%$

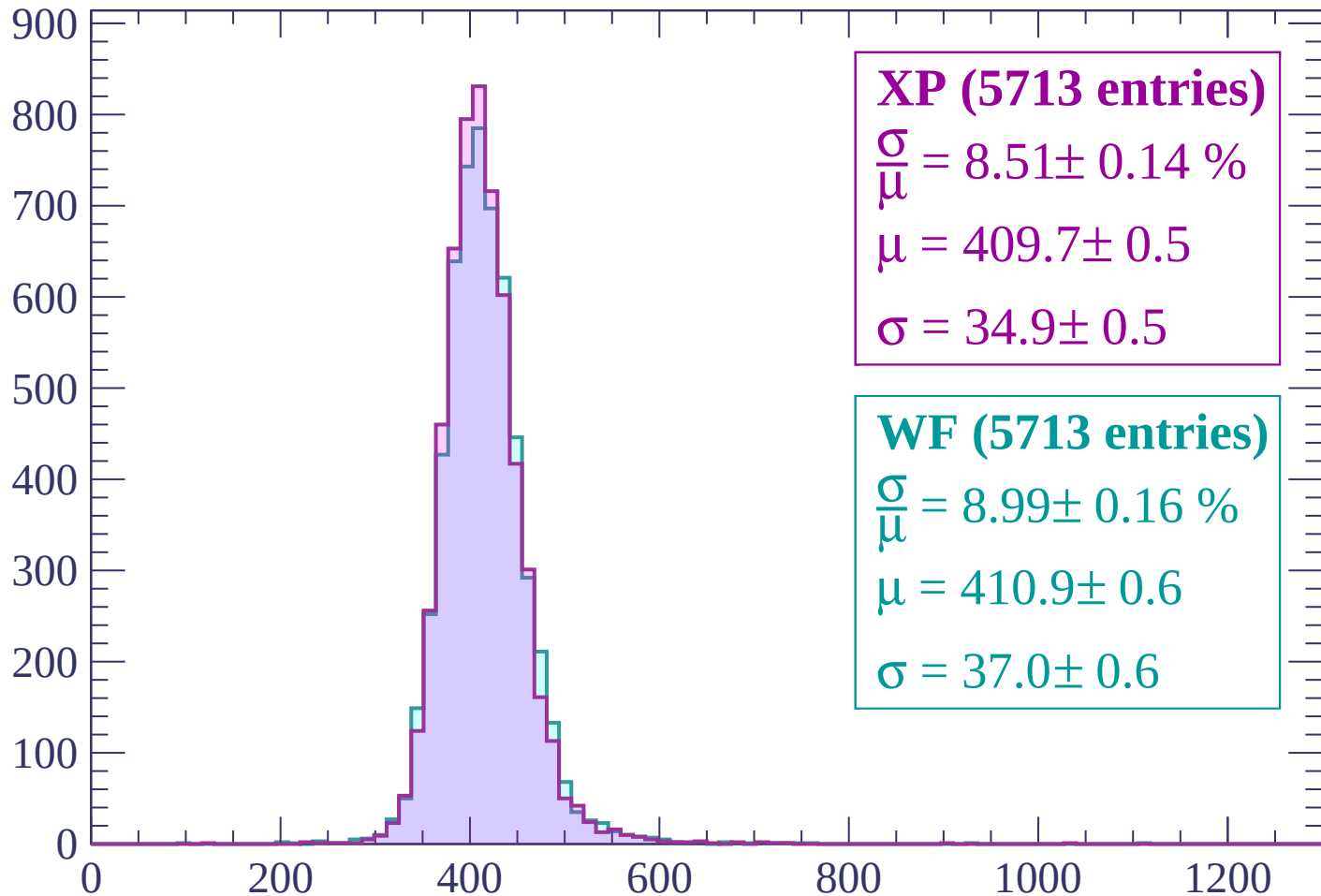
$\mu = 414.7 \pm 0.6$

$\sigma = 36.4 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count



XP (5713 entries)

$$\frac{\sigma}{\mu} = 8.51 \pm 0.14 \%$$

$$\mu = 409.7 \pm 0.5$$

$$\sigma = 34.9 \pm 0.5$$

WF (5713 entries)

$$\frac{\sigma}{\mu} = 8.99 \pm 0.16 \%$$

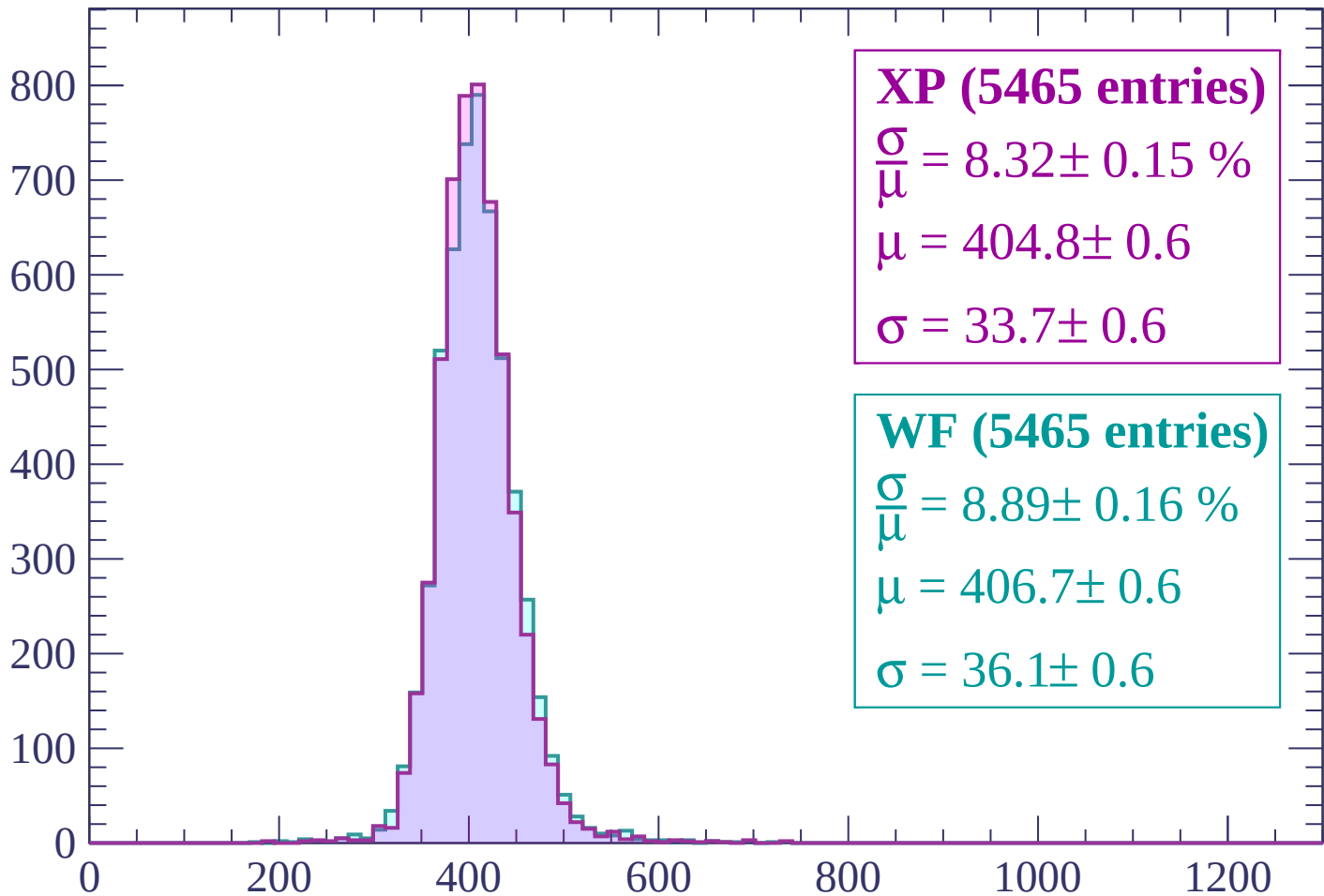
$$\mu = 410.9 \pm 0.6$$

$$\sigma = 37.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count



XP (5465 entries)

$\frac{\sigma}{\mu} = 8.32 \pm 0.15 \%$

$\mu = 404.8 \pm 0.6$

$\sigma = 33.7 \pm 0.6$

WF (5465 entries)

$\frac{\sigma}{\mu} = 8.89 \pm 0.16 \%$

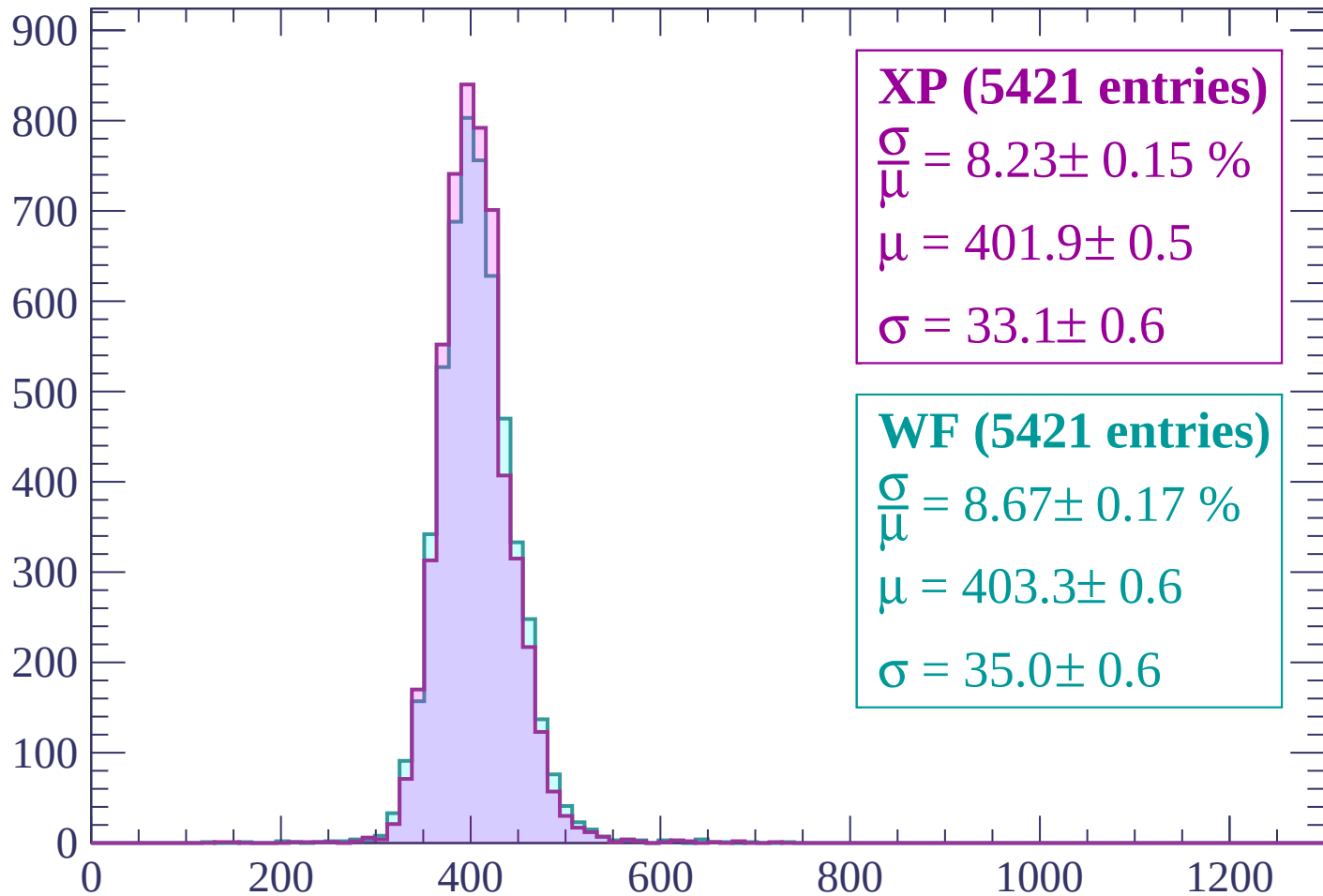
$\mu = 406.7 \pm 0.6$

$\sigma = 36.1 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count



XP (5421 entries)

$\frac{\sigma}{\mu} = 8.23 \pm 0.15 \%$

$\mu = 401.9 \pm 0.5$

$\sigma = 33.1 \pm 0.6$

WF (5421 entries)

$\frac{\sigma}{\mu} = 8.67 \pm 0.17 \%$

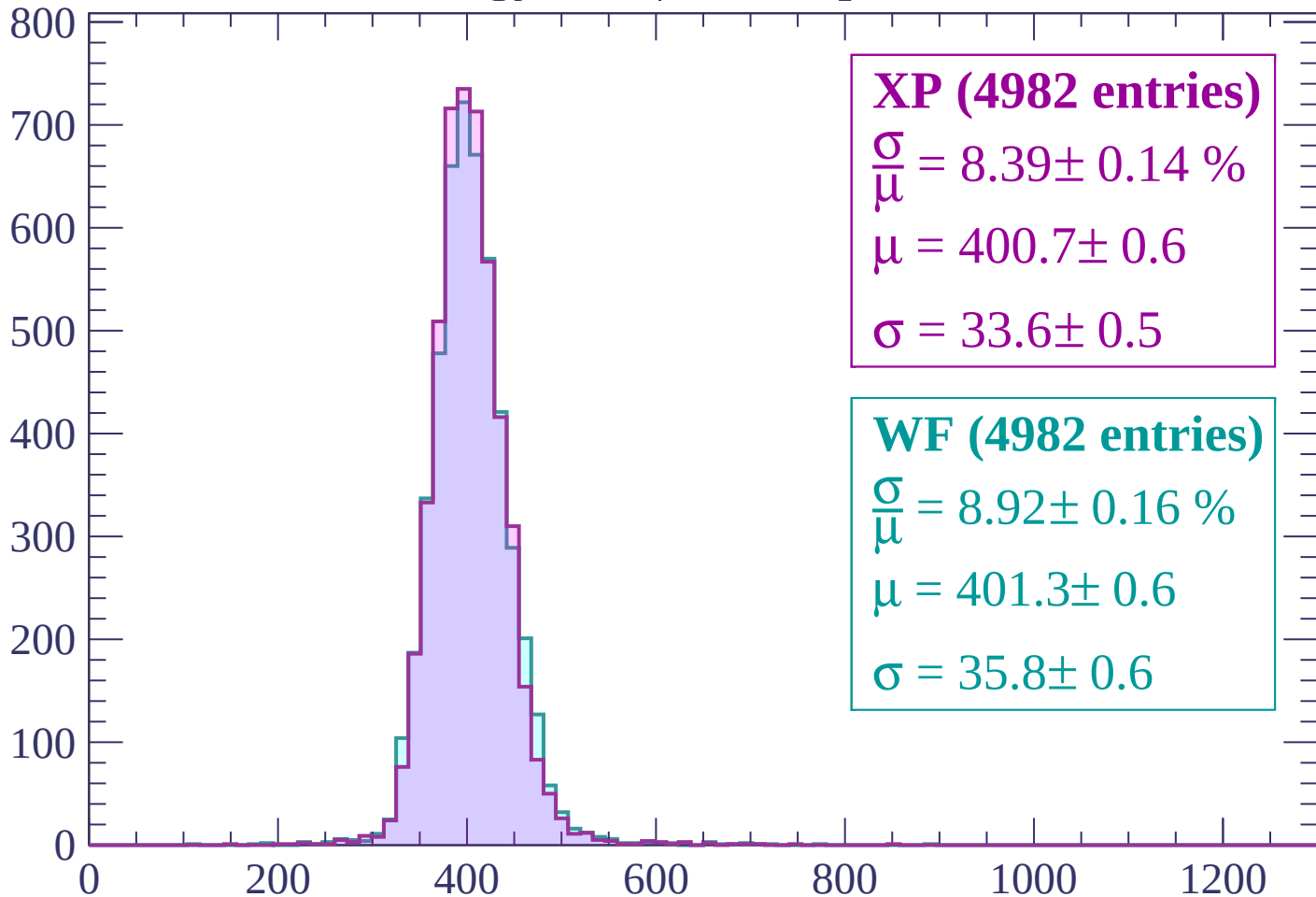
$\mu = 403.3 \pm 0.6$

$\sigma = 35.0 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$

Count



XP (4982 entries)

$$\frac{\sigma}{\mu} = 8.39 \pm 0.14 \%$$

$$\mu = 400.7 \pm 0.6$$

$$\sigma = 33.6 \pm 0.5$$

WF (4982 entries)

$$\frac{\sigma}{\mu} = 8.92 \pm 0.16 \%$$

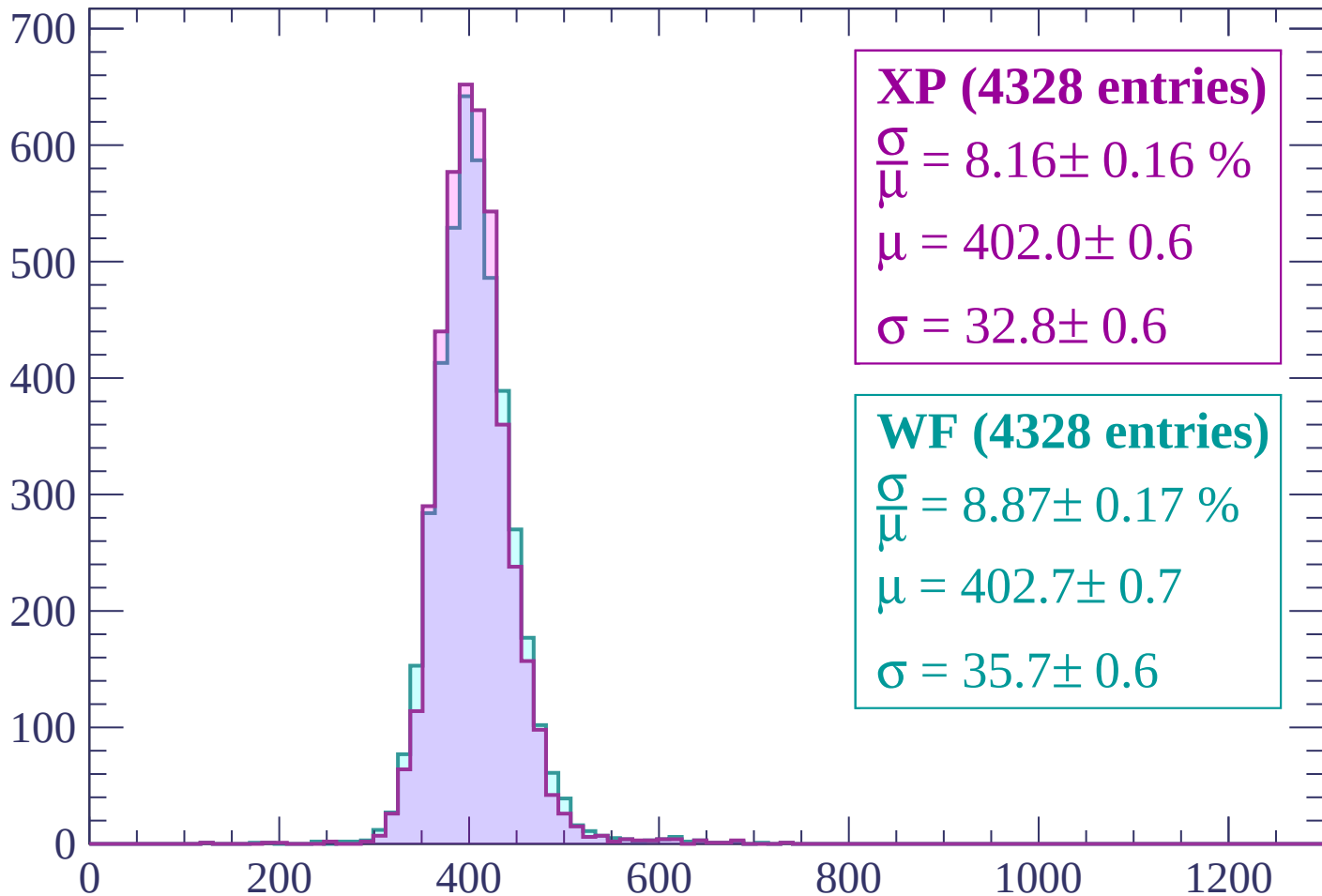
$$\mu = 401.3 \pm 0.6$$

$$\sigma = 35.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count



XP (4328 entries)

$$\frac{\sigma}{\mu} = 8.16 \pm 0.16 \%$$

$$\mu = 402.0 \pm 0.6$$

$$\sigma = 32.8 \pm 0.6$$

WF (4328 entries)

$$\frac{\sigma}{\mu} = 8.87 \pm 0.17 \%$$

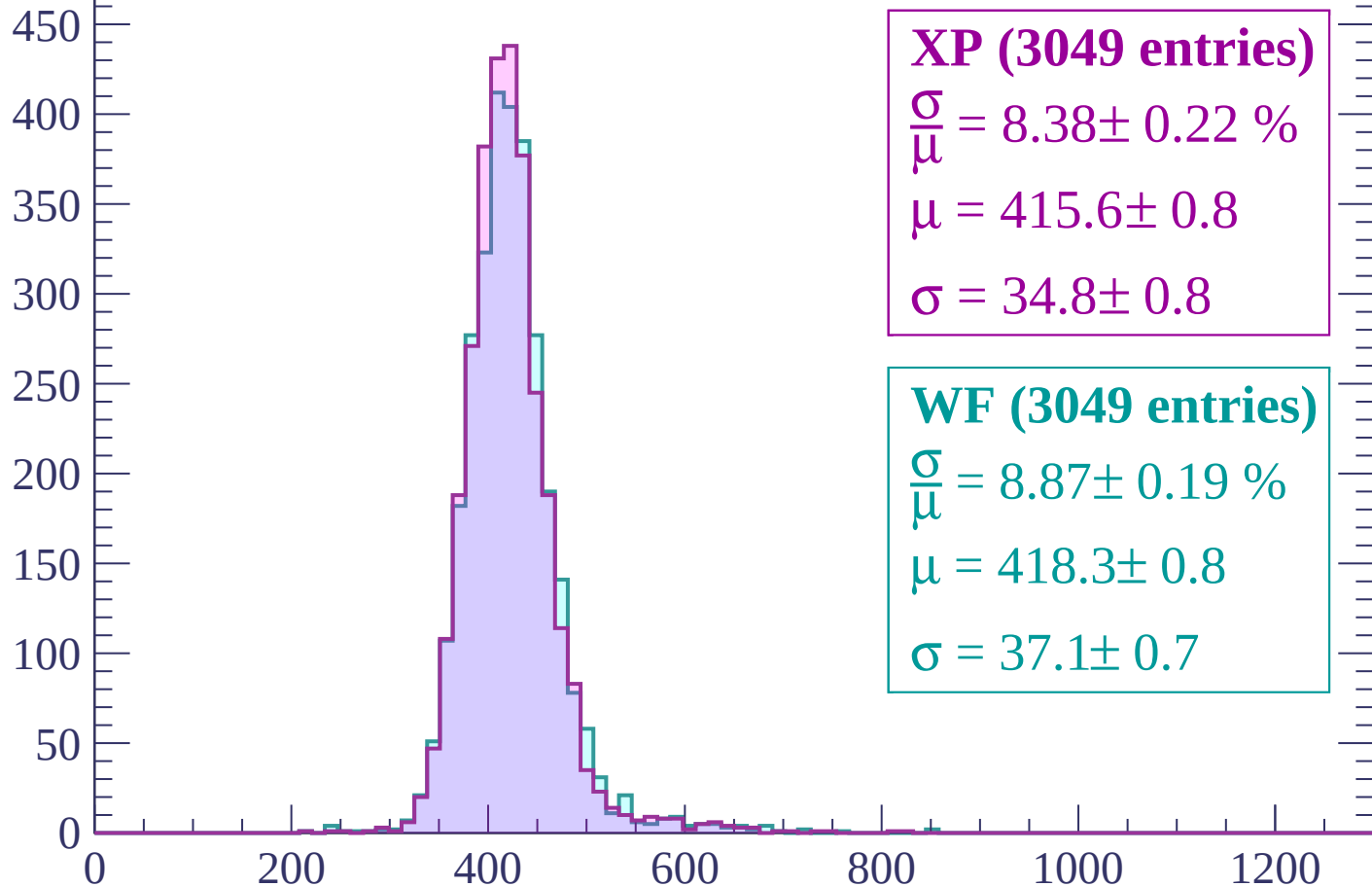
$$\mu = 402.7 \pm 0.7$$

$$\sigma = 35.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (3049 entries)

$\frac{\sigma}{\mu} = 8.38 \pm 0.22 \%$

$\mu = 415.6 \pm 0.8$

$\sigma = 34.8 \pm 0.8$

WF (3049 entries)

$\frac{\sigma}{\mu} = 8.87 \pm 0.19 \%$

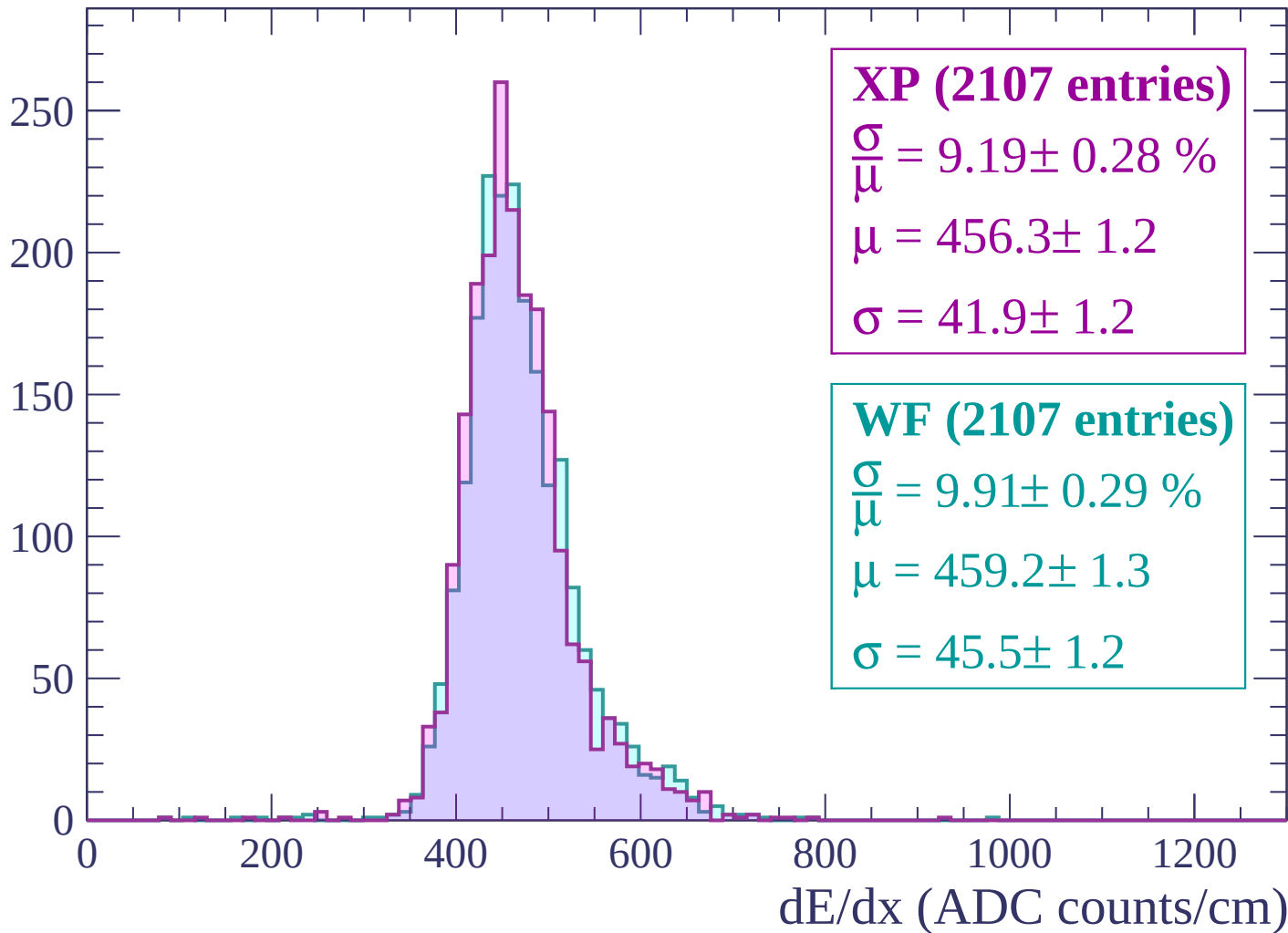
$\mu = 418.3 \pm 0.8$

$\sigma = 37.1 \pm 0.7$

dE/dx (ADC counts/cm)

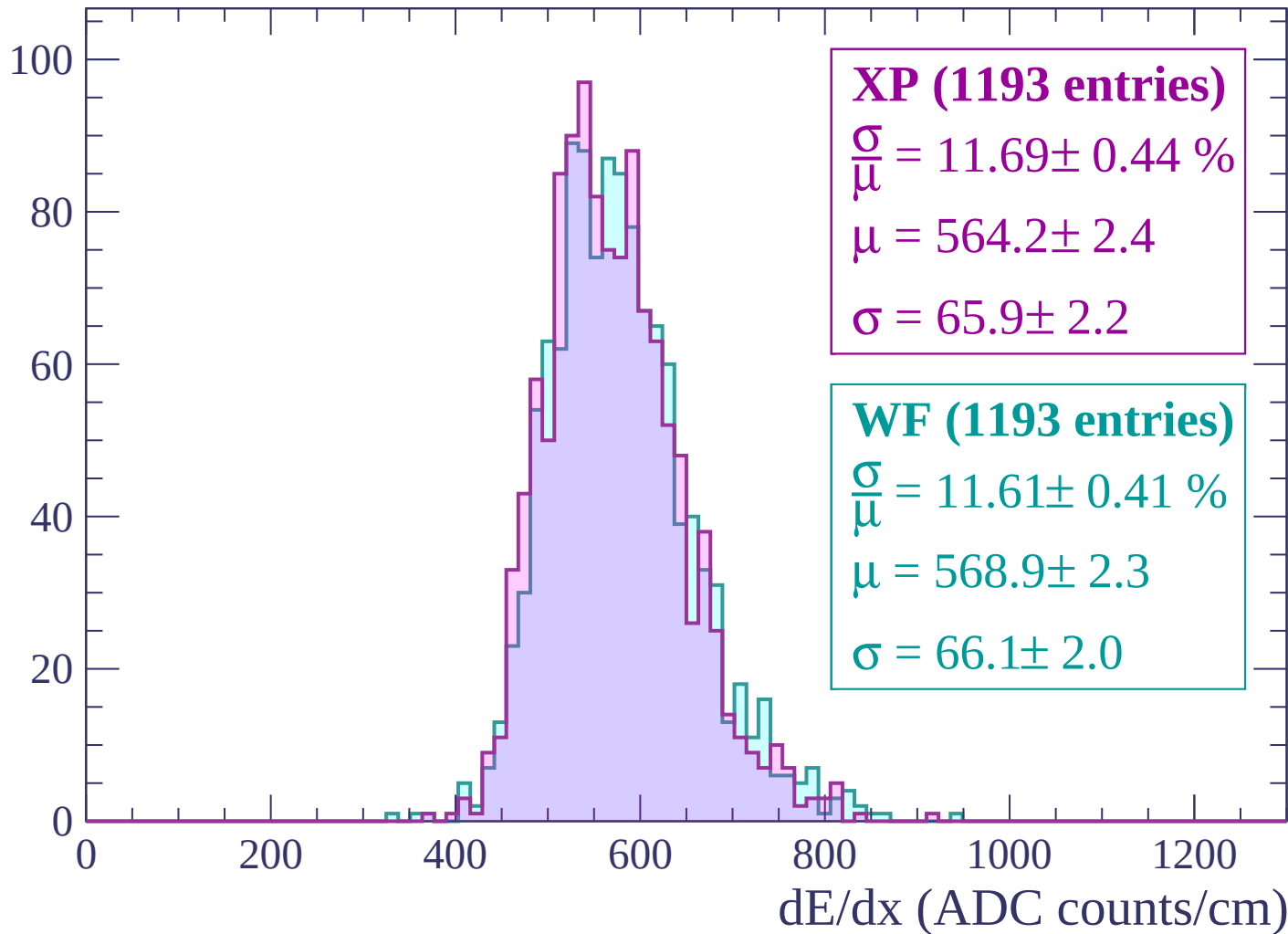
Energy loss | $-180 < p < -120$

Count



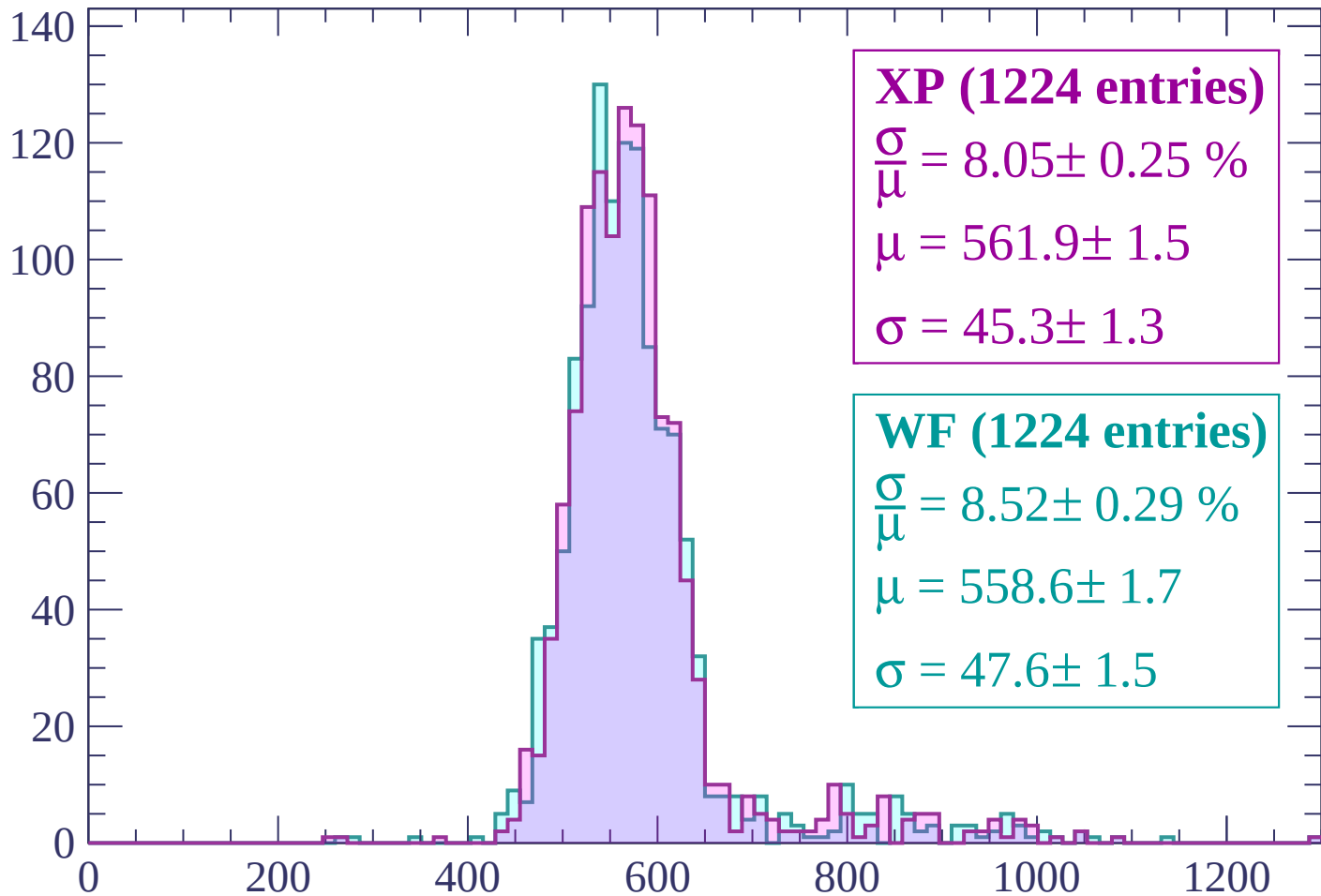
Energy loss | $-120 < p < -60$

Count



Energy loss | $-60 < p < 0$

Count



XP (1224 entries)

$$\frac{\sigma}{\mu} = 8.05 \pm 0.25 \%$$

$$\mu = 561.9 \pm 1.5$$

$$\sigma = 45.3 \pm 1.3$$

WF (1224 entries)

$$\frac{\sigma}{\mu} = 8.52 \pm 0.29 \%$$

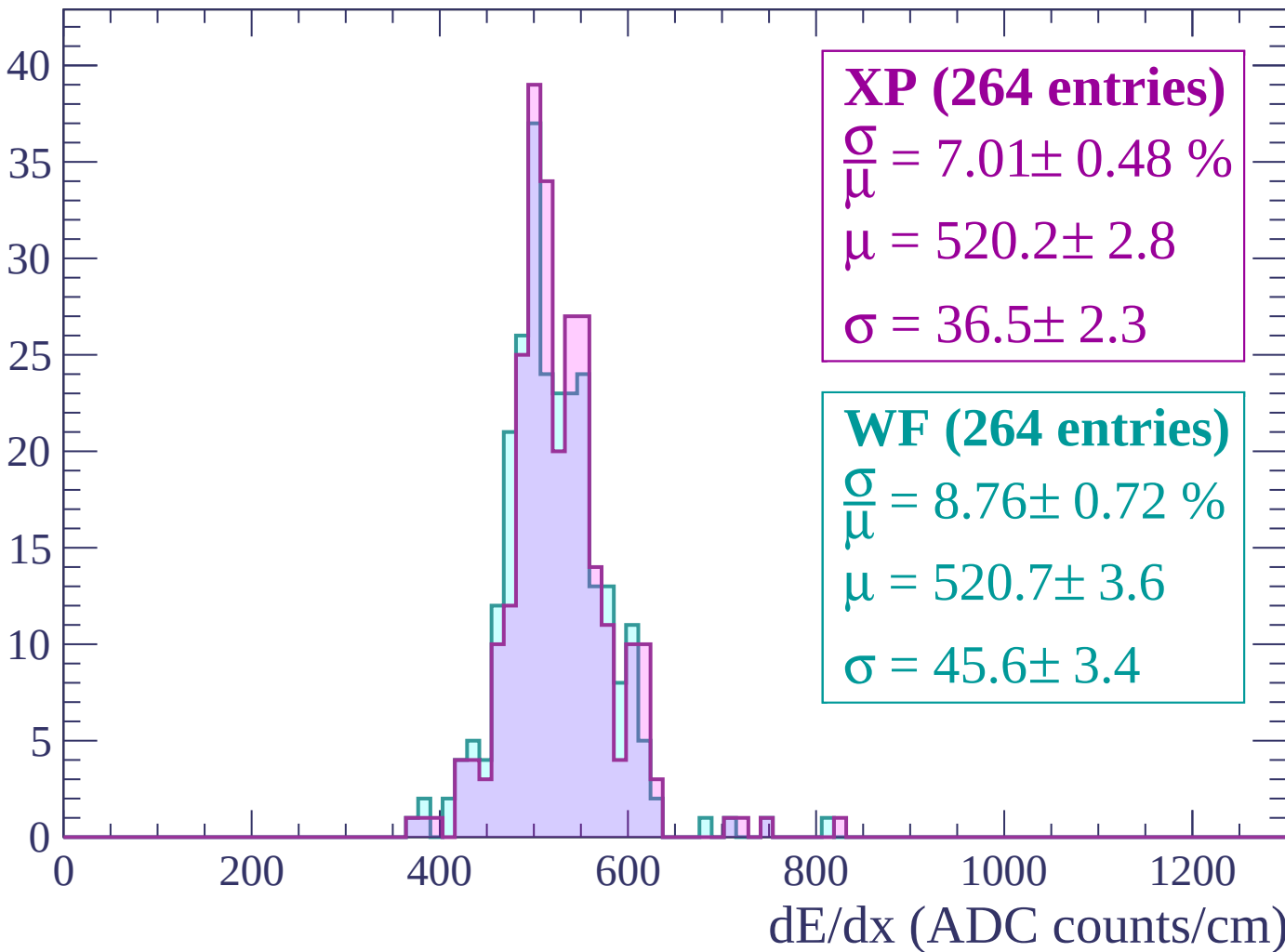
$$\mu = 558.6 \pm 1.7$$

$$\sigma = 47.6 \pm 1.5$$

dE/dx (ADC counts/cm)

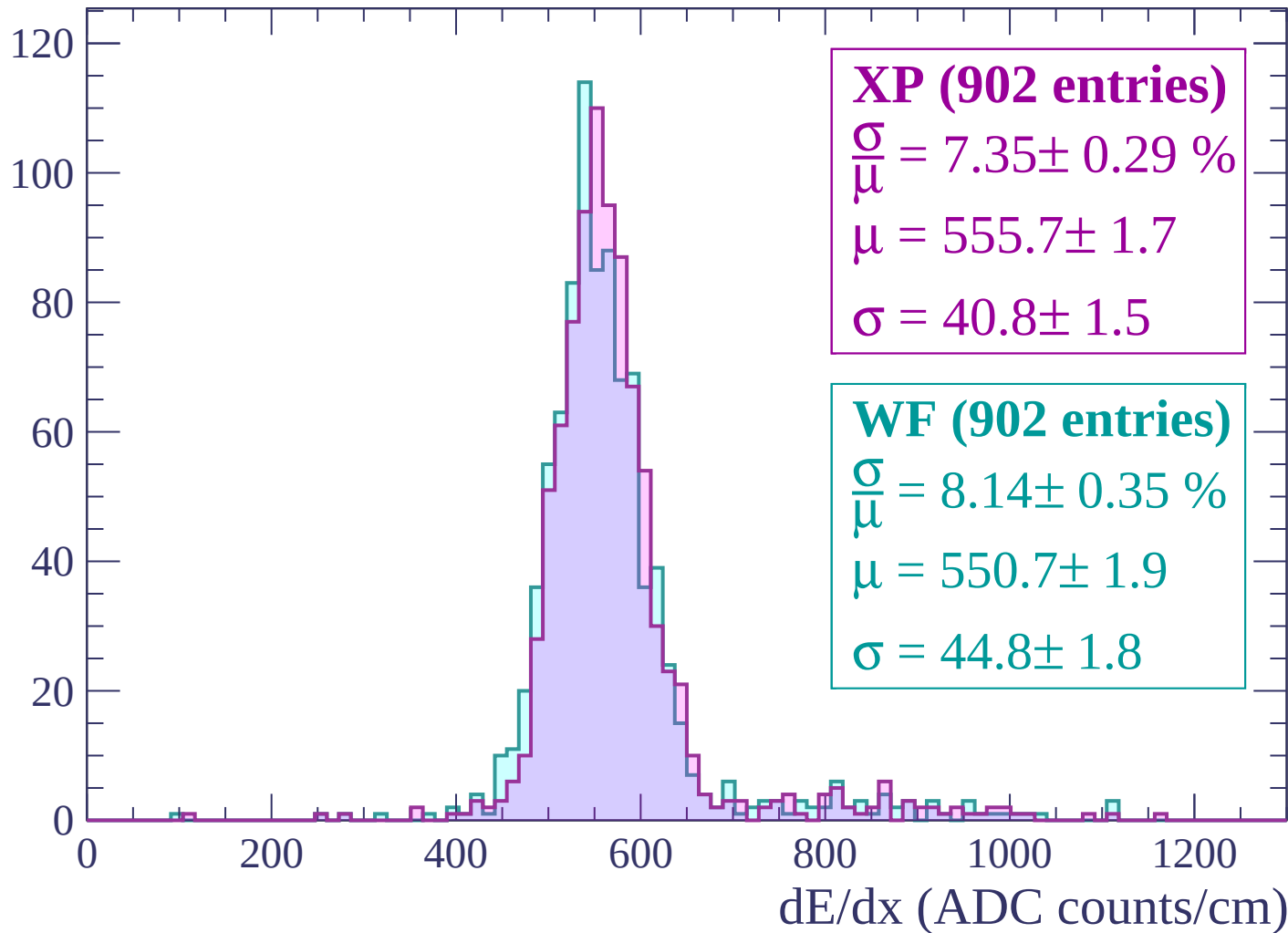
Energy loss | $0 < p < 60$

Count



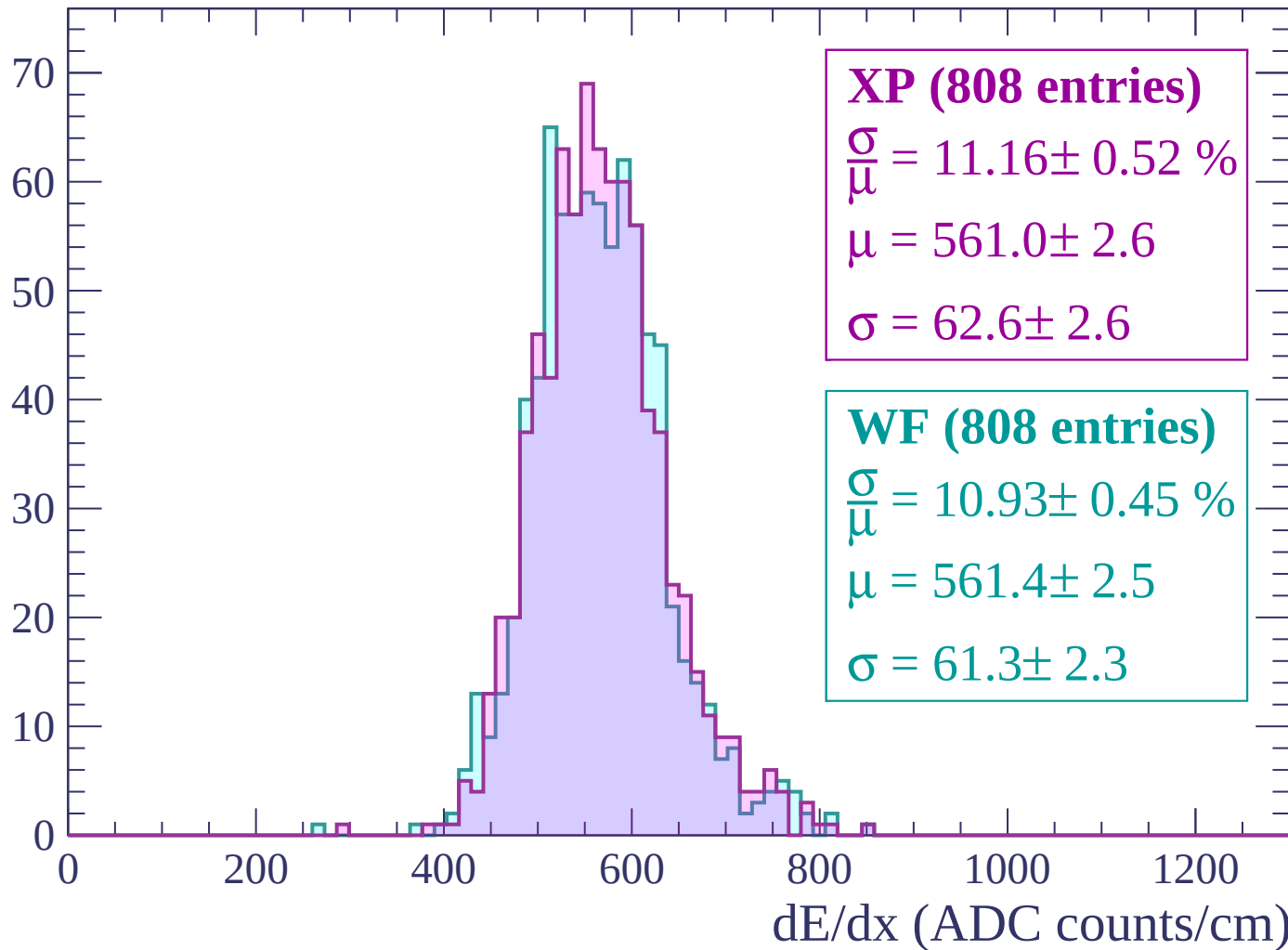
Energy loss | $60 < p < 120$

Count



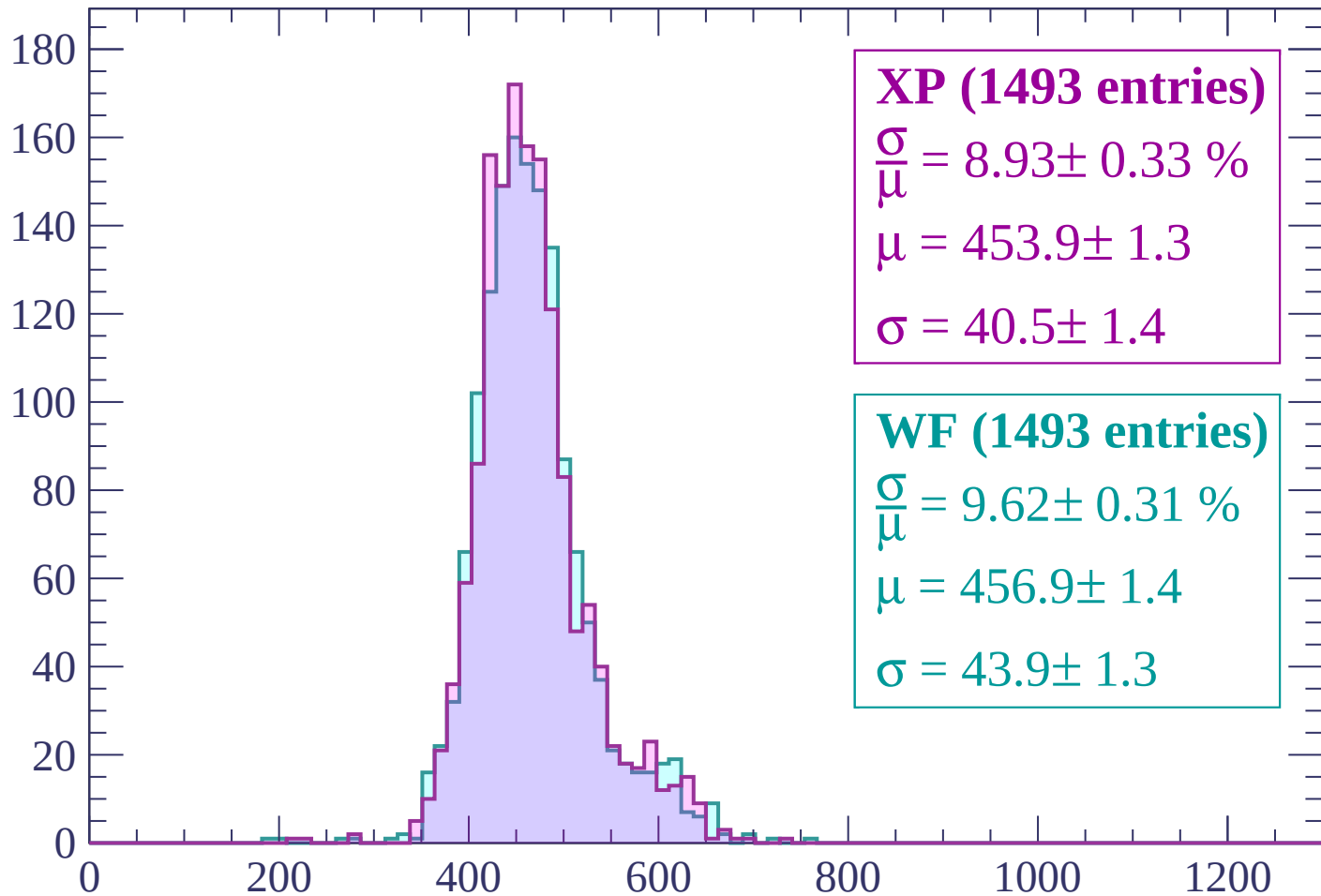
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (1493 entries)

$$\frac{\sigma}{\mu} = 8.93 \pm 0.33 \%$$

$$\mu = 453.9 \pm 1.3$$

$$\sigma = 40.5 \pm 1.4$$

WF (1493 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.31 \%$$

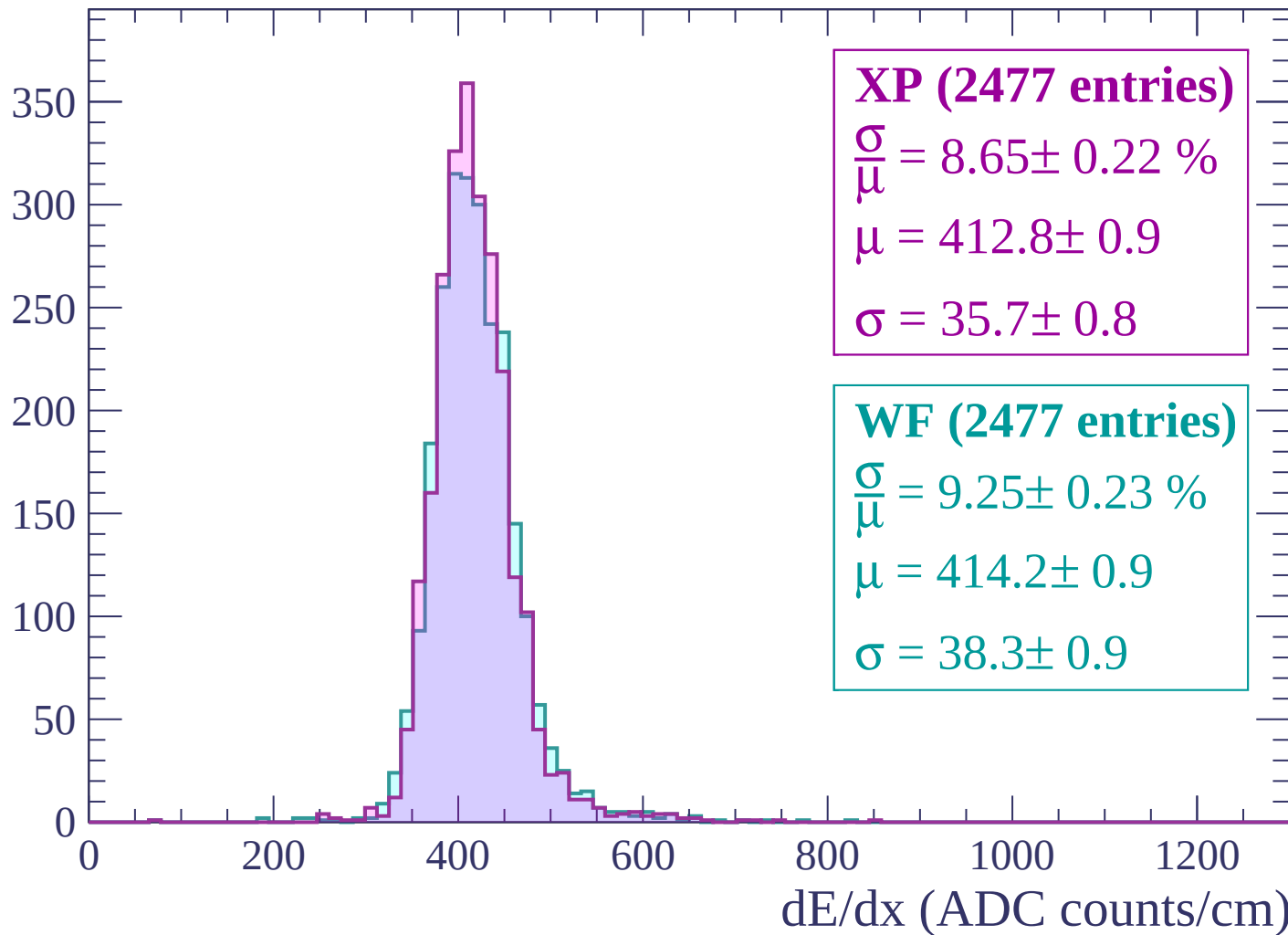
$$\mu = 456.9 \pm 1.4$$

$$\sigma = 43.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



Energy loss | $300 < p < 360$

Count

600
500
400
300
200
100
0

XP (4221 entries)

$$\frac{\sigma}{\mu} = 8.57 \pm 0.15 \%$$

$$\mu = 400.8 \pm 0.6$$

$$\sigma = 34.3 \pm 0.6$$

WF (4221 entries)

$$\frac{\sigma}{\mu} = 8.94 \pm 0.15 \%$$

$$\mu = 402.3 \pm 0.6$$

$$\sigma = 36.0 \pm 0.6$$

0

200

400

600

800

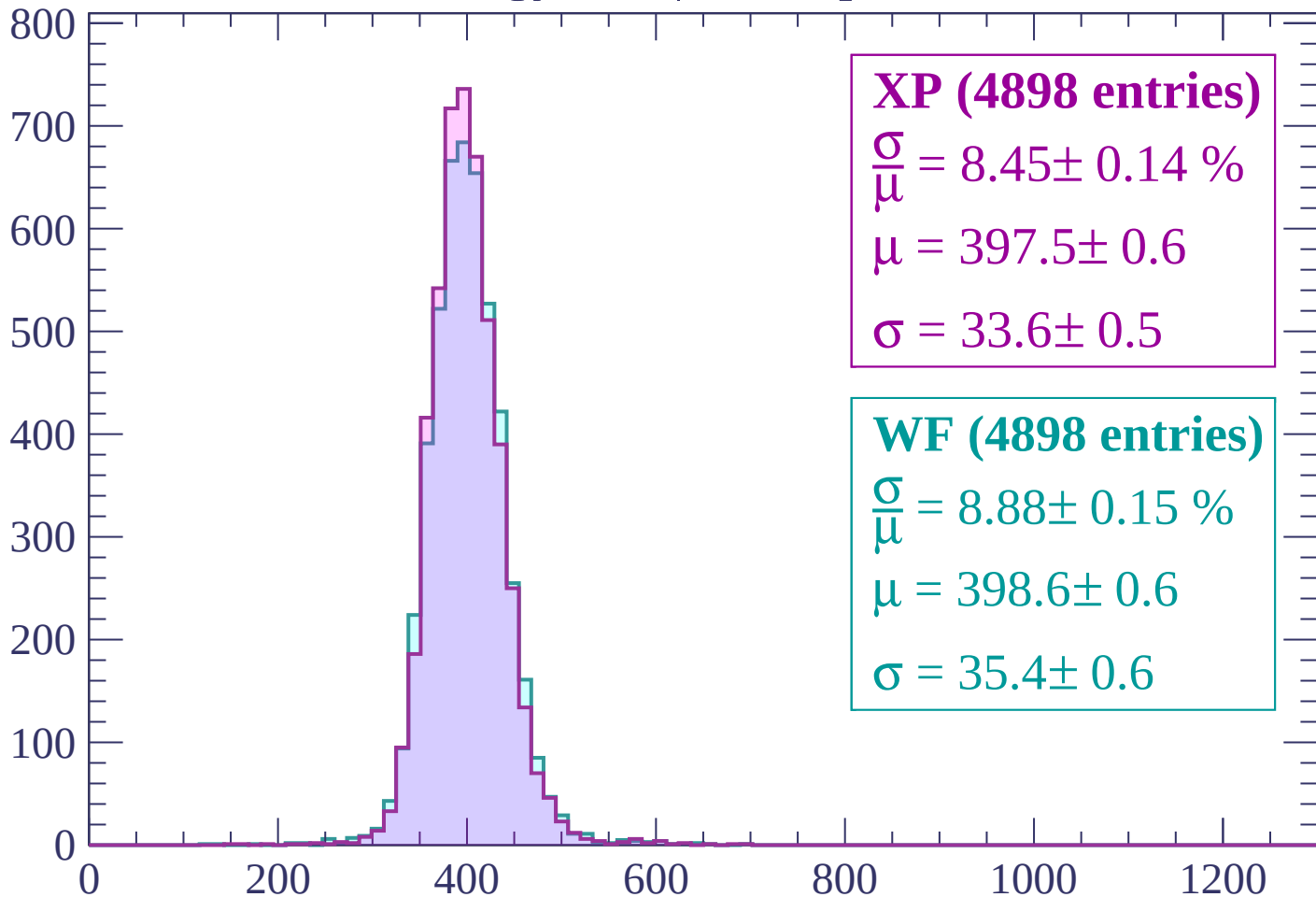
1000

1200

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (4898 entries)

$$\frac{\sigma}{\mu} = 8.45 \pm 0.14 \%$$

$$\mu = 397.5 \pm 0.6$$

$$\sigma = 33.6 \pm 0.5$$

WF (4898 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.15 \%$$

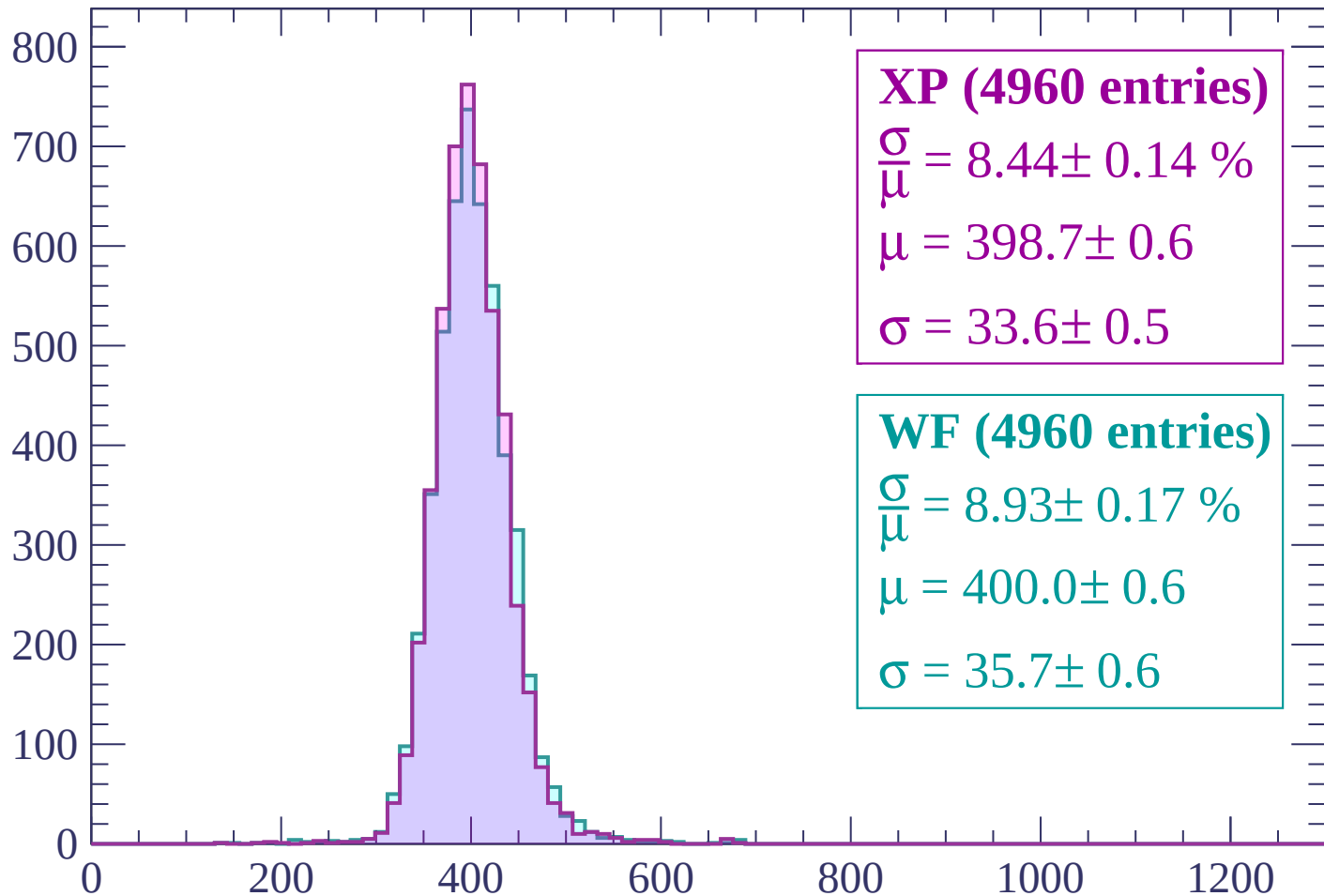
$$\mu = 398.6 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP (4960 entries)

$$\frac{\sigma}{\mu} = 8.44 \pm 0.14 \%$$

$$\mu = 398.7 \pm 0.6$$

$$\sigma = 33.6 \pm 0.5$$

WF (4960 entries)

$$\frac{\sigma}{\mu} = 8.93 \pm 0.17 \%$$

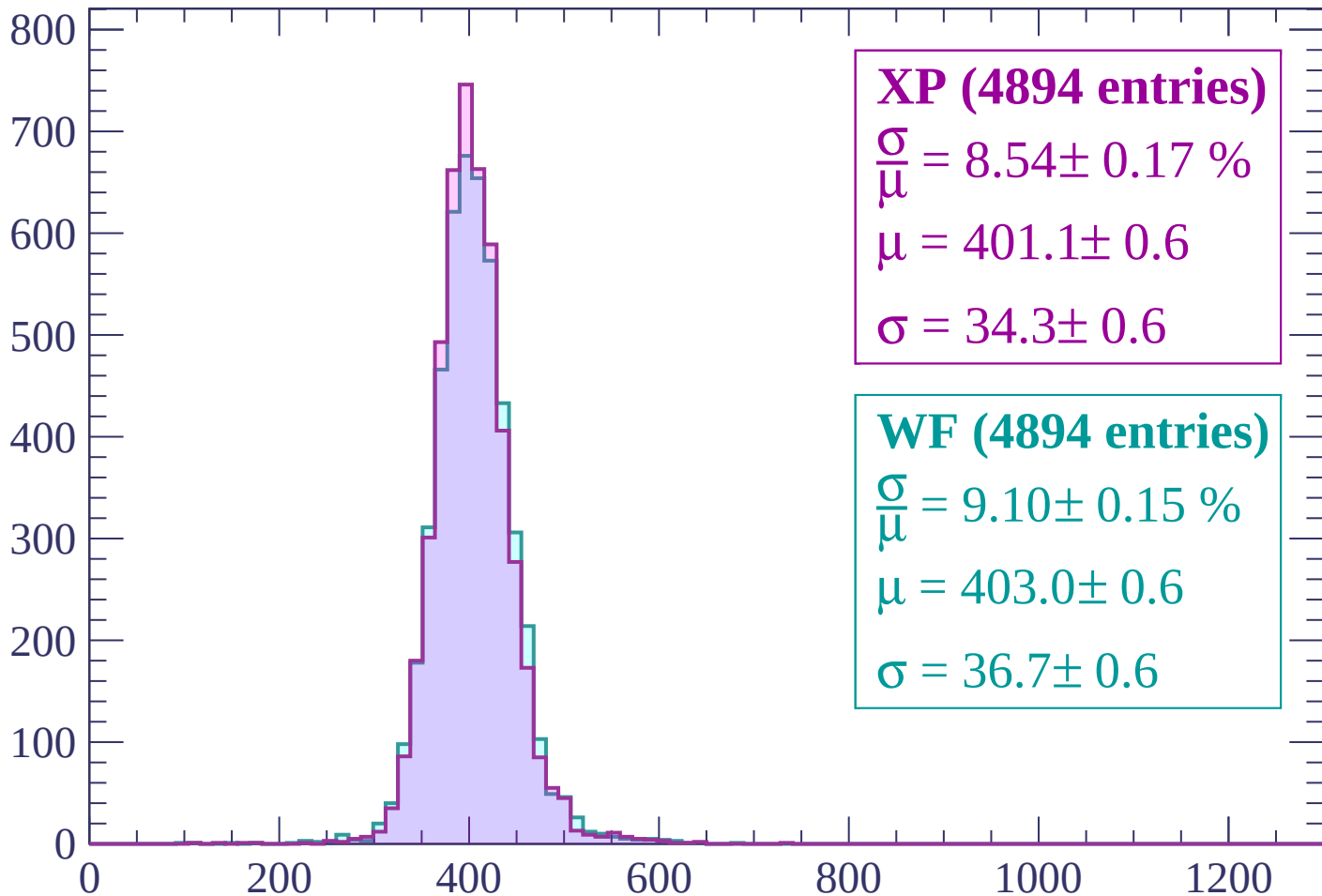
$$\mu = 400.0 \pm 0.6$$

$$\sigma = 35.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP (4894 entries)

$$\frac{\sigma}{\mu} = 8.54 \pm 0.17 \%$$

$$\mu = 401.1 \pm 0.6$$

$$\sigma = 34.3 \pm 0.6$$

WF (4894 entries)

$$\frac{\sigma}{\mu} = 9.10 \pm 0.15 \%$$

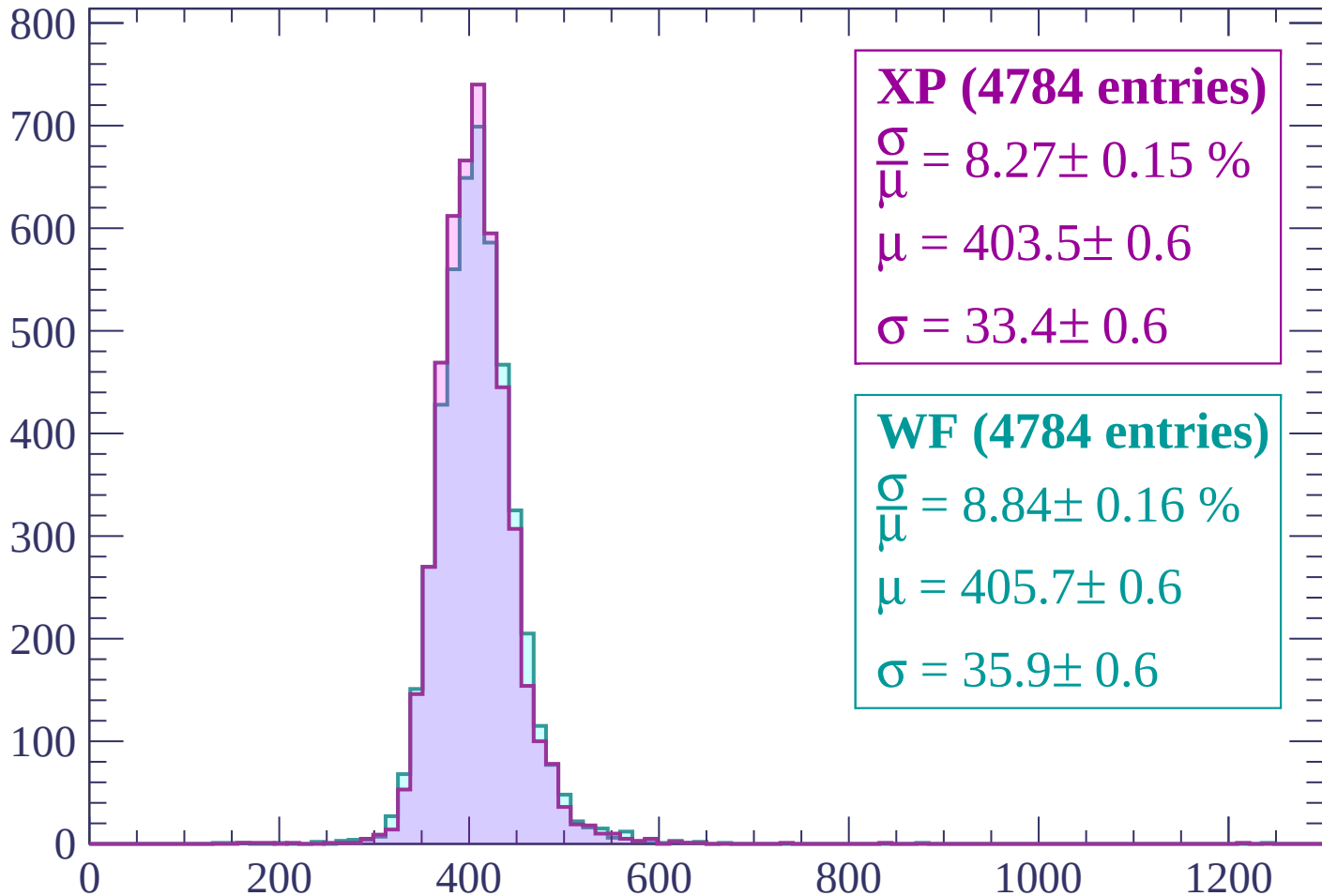
$$\mu = 403.0 \pm 0.6$$

$$\sigma = 36.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (4784 entries)

$\frac{\sigma}{\mu} = 8.27 \pm 0.15 \%$

$\mu = 403.5 \pm 0.6$

$\sigma = 33.4 \pm 0.6$

WF (4784 entries)

$\frac{\sigma}{\mu} = 8.84 \pm 0.16 \%$

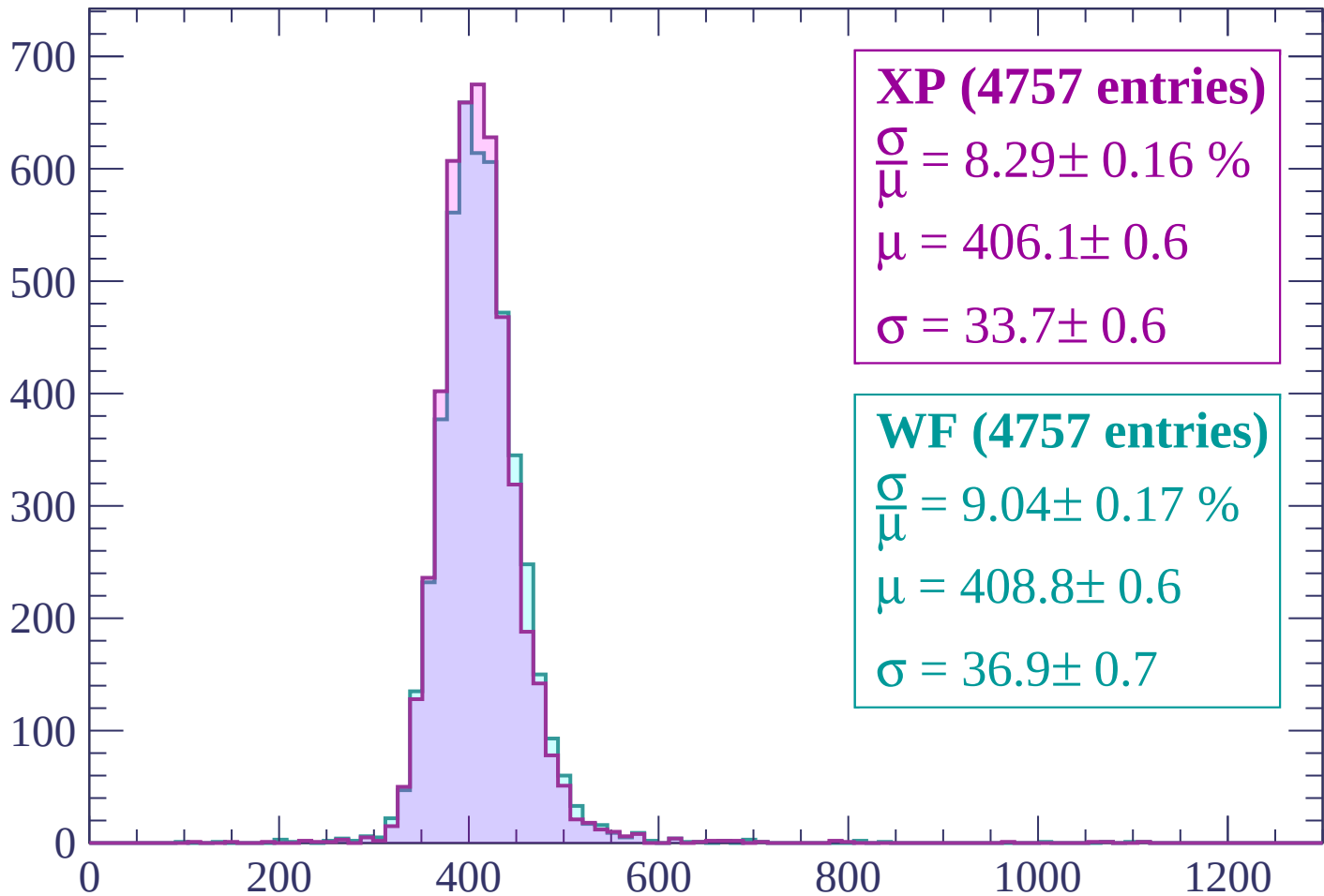
$\mu = 405.7 \pm 0.6$

$\sigma = 35.9 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (4757 entries)

$$\frac{\sigma}{\mu} = 8.29 \pm 0.16 \%$$

$$\mu = 406.1 \pm 0.6$$

$$\sigma = 33.7 \pm 0.6$$

WF (4757 entries)

$$\frac{\sigma}{\mu} = 9.04 \pm 0.17 \%$$

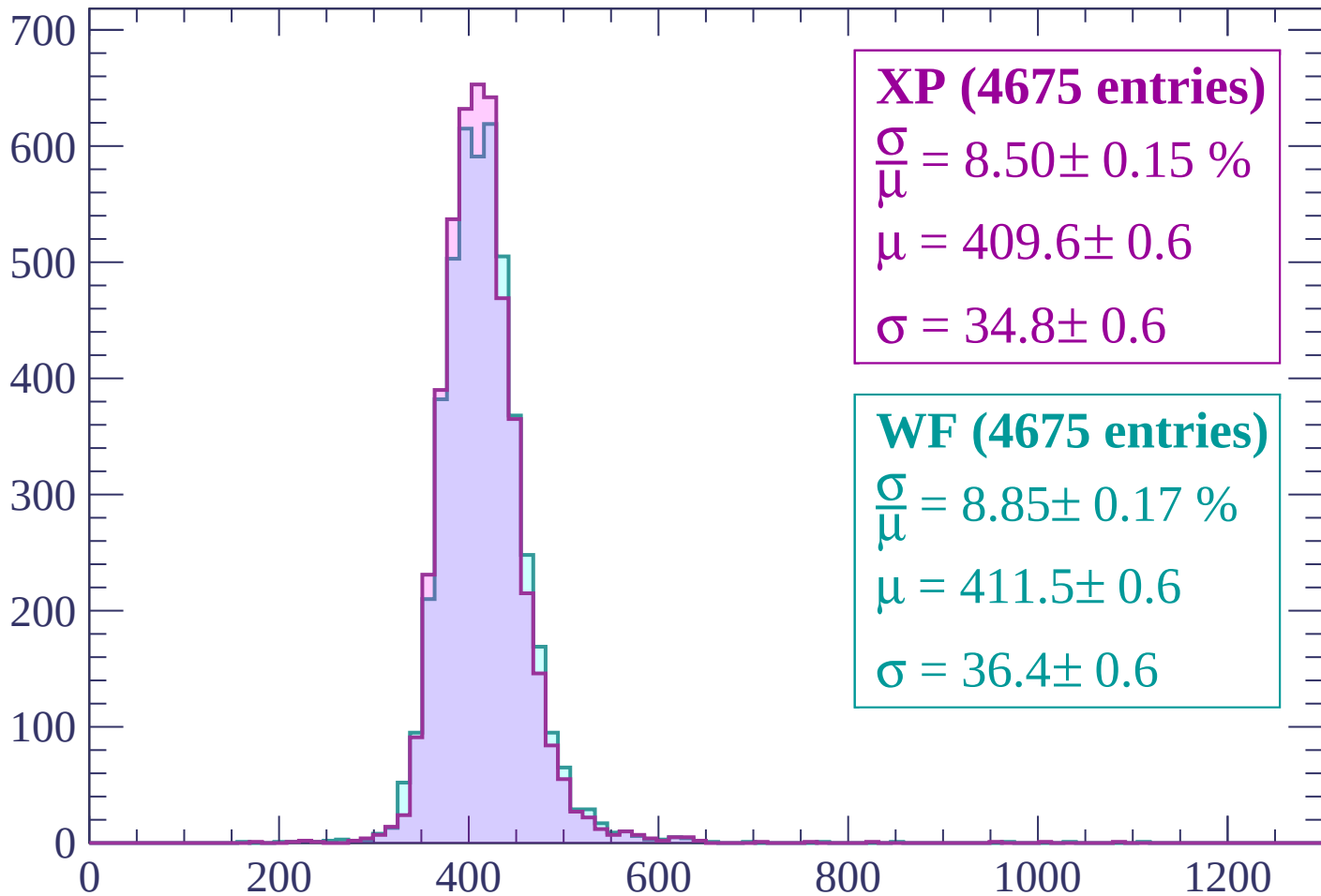
$$\mu = 408.8 \pm 0.6$$

$$\sigma = 36.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (4675 entries)

$$\frac{\sigma}{\mu} = 8.50 \pm 0.15 \%$$

$$\mu = 409.6 \pm 0.6$$

$$\sigma = 34.8 \pm 0.6$$

WF (4675 entries)

$$\frac{\sigma}{\mu} = 8.85 \pm 0.17 \%$$

$$\mu = 411.5 \pm 0.6$$

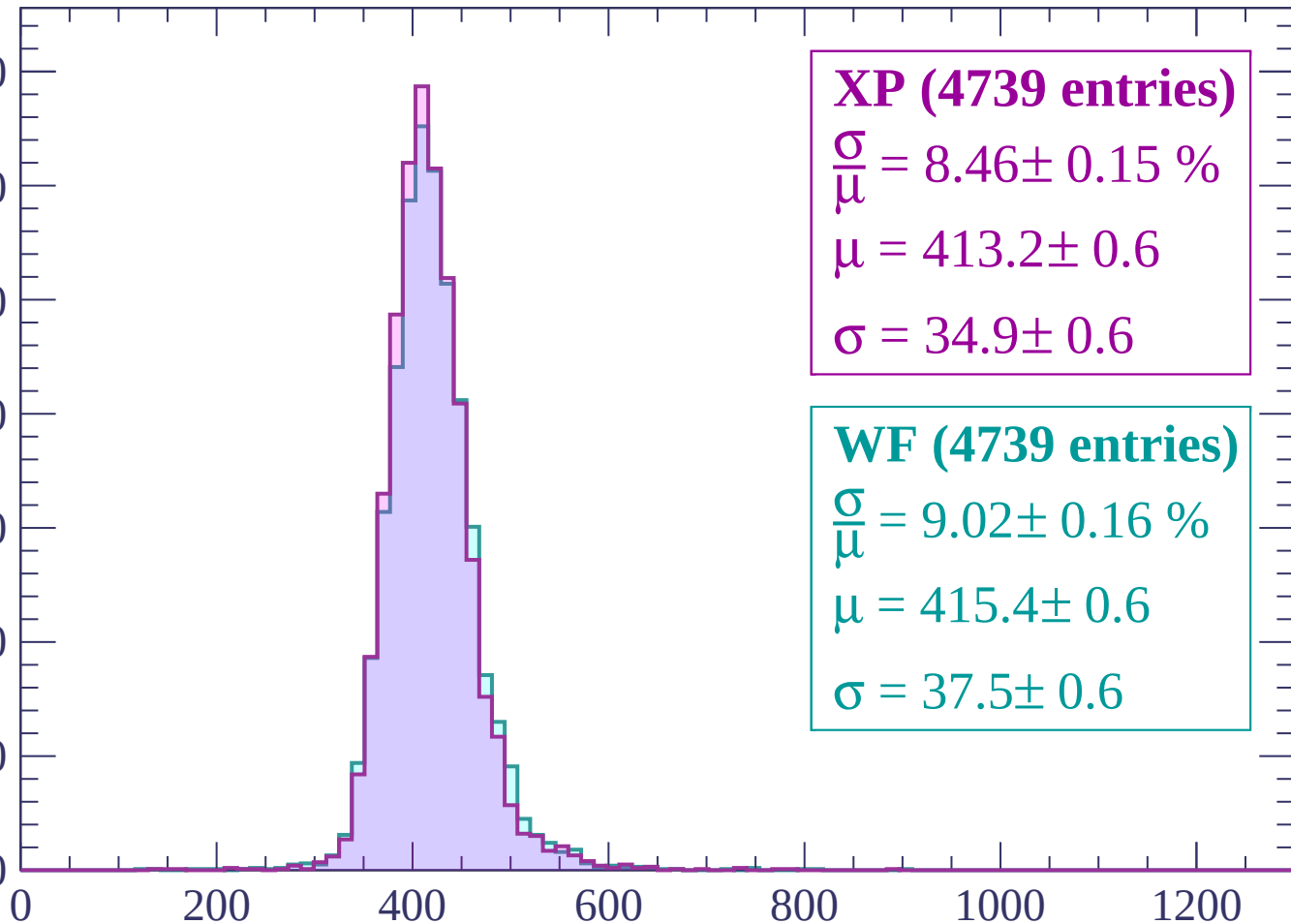
$$\sigma = 36.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count

700
600
500
400
300
200
100
0



XP (4739 entries)

$\frac{\sigma}{\mu} = 8.46 \pm 0.15 \%$

$\mu = 413.2 \pm 0.6$

$\sigma = 34.9 \pm 0.6$

WF (4739 entries)

$\frac{\sigma}{\mu} = 9.02 \pm 0.16 \%$

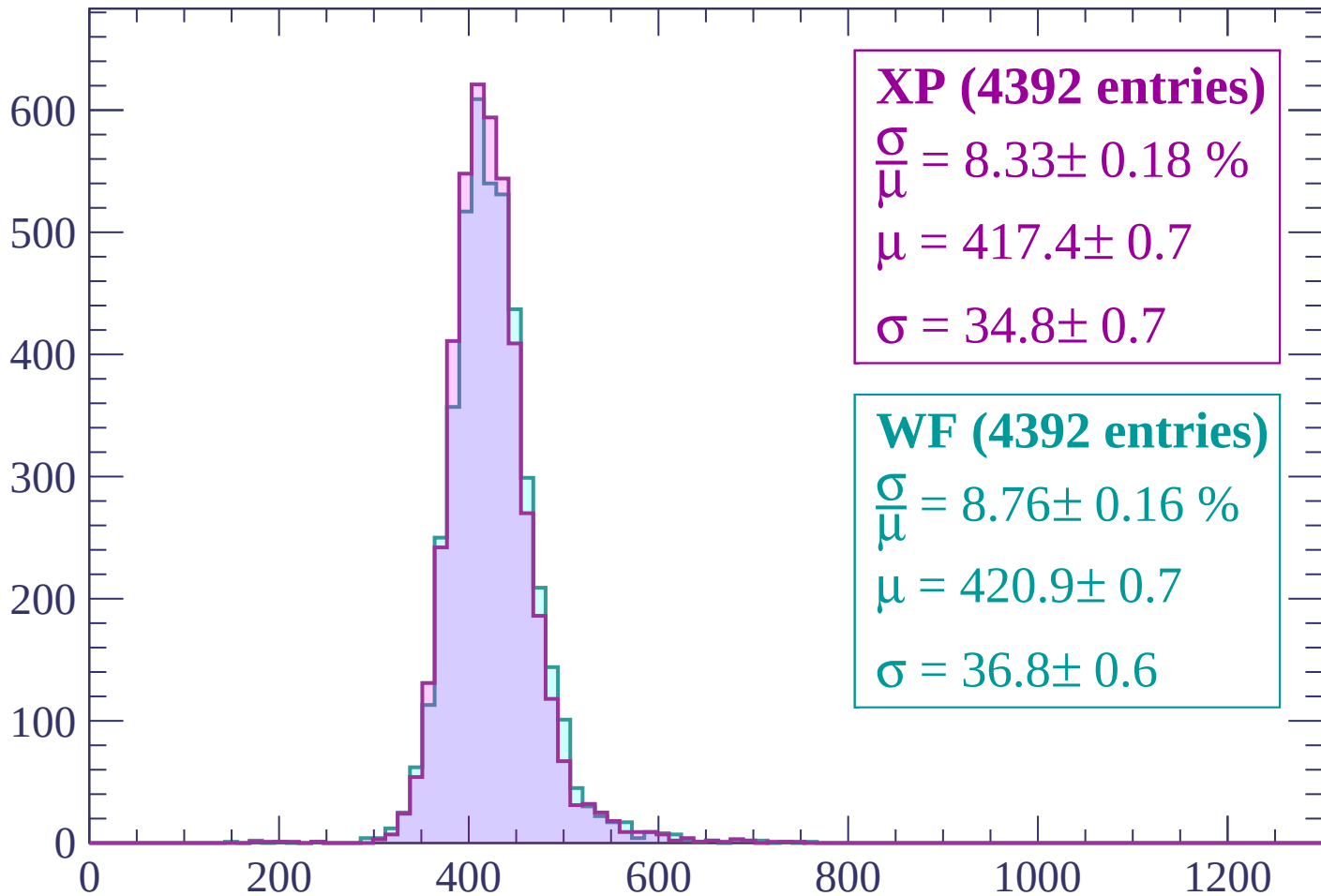
$\mu = 415.4 \pm 0.6$

$\sigma = 37.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP (4392 entries)

$$\frac{\sigma}{\mu} = 8.33 \pm 0.18 \%$$

$$\mu = 417.4 \pm 0.7$$

$$\sigma = 34.8 \pm 0.7$$

WF (4392 entries)

$$\frac{\sigma}{\mu} = 8.76 \pm 0.16 \%$$

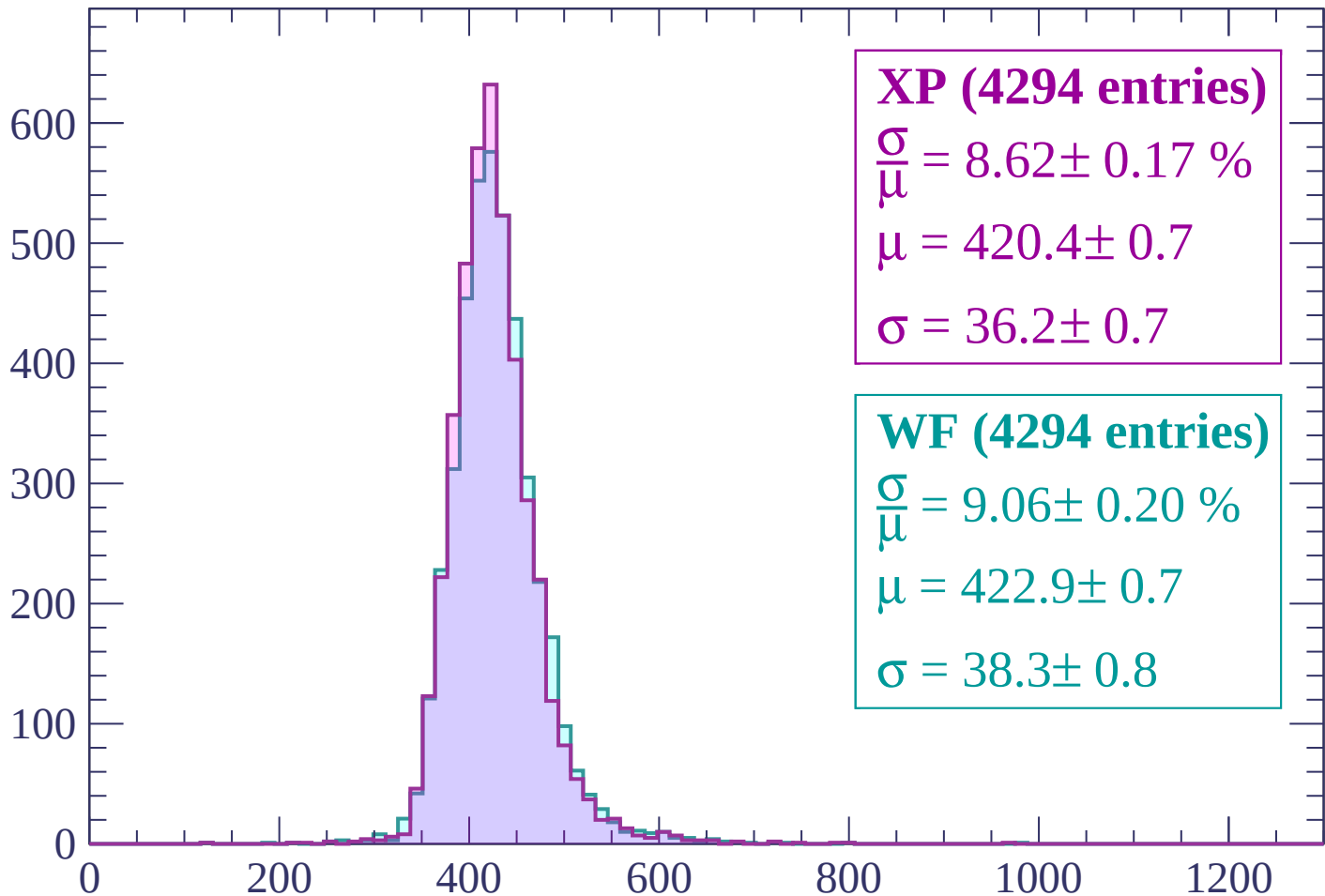
$$\mu = 420.9 \pm 0.7$$

$$\sigma = 36.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count



XP (4294 entries)

$$\frac{\sigma}{\mu} = 8.62 \pm 0.17 \%$$

$$\mu = 420.4 \pm 0.7$$

$$\sigma = 36.2 \pm 0.7$$

WF (4294 entries)

$$\frac{\sigma}{\mu} = 9.06 \pm 0.20 \%$$

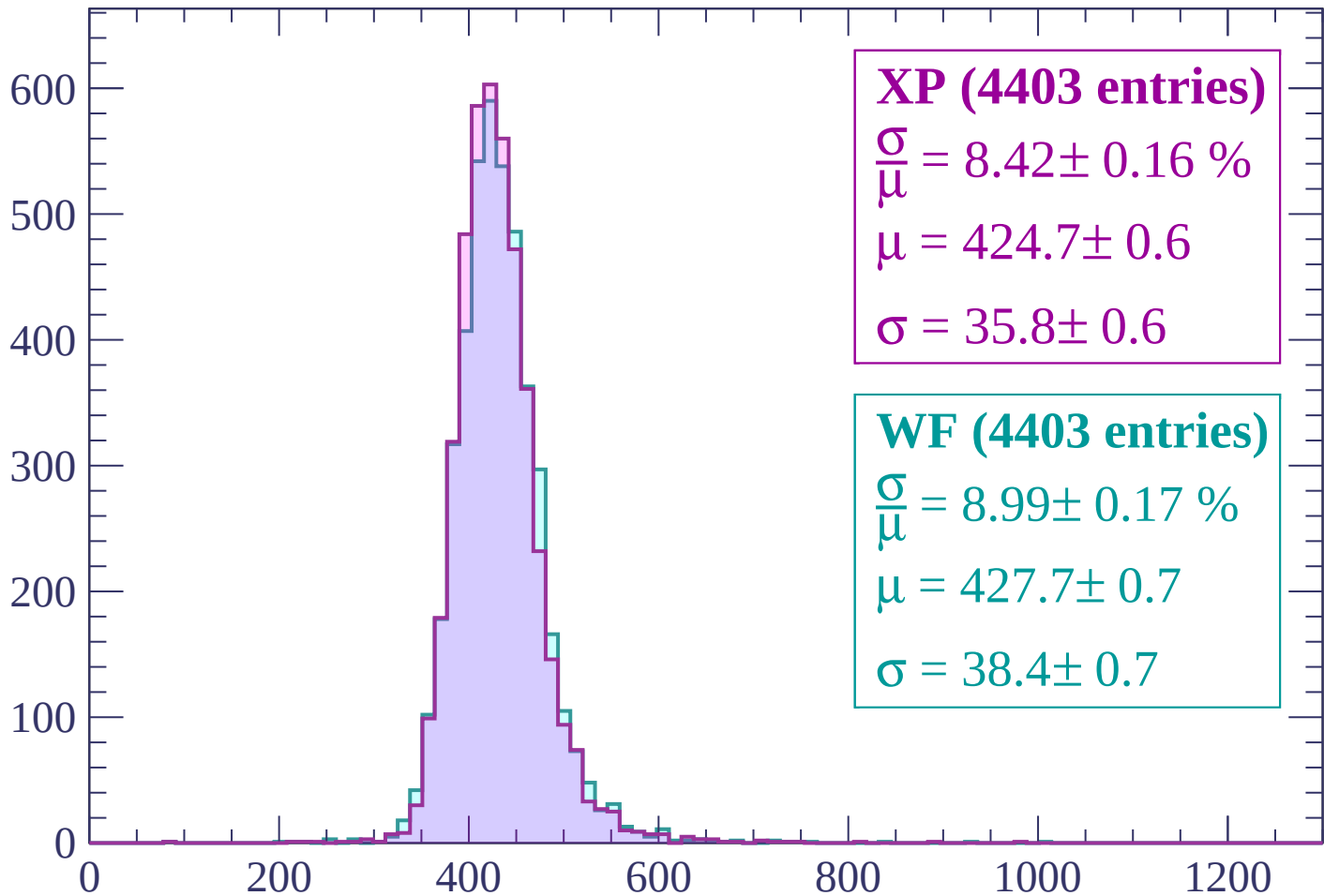
$$\mu = 422.9 \pm 0.7$$

$$\sigma = 38.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (4403 entries)

$$\frac{\sigma}{\mu} = 8.42 \pm 0.16 \%$$

$$\mu = 424.7 \pm 0.6$$

$$\sigma = 35.8 \pm 0.6$$

WF (4403 entries)

$$\frac{\sigma}{\mu} = 8.99 \pm 0.17 \%$$

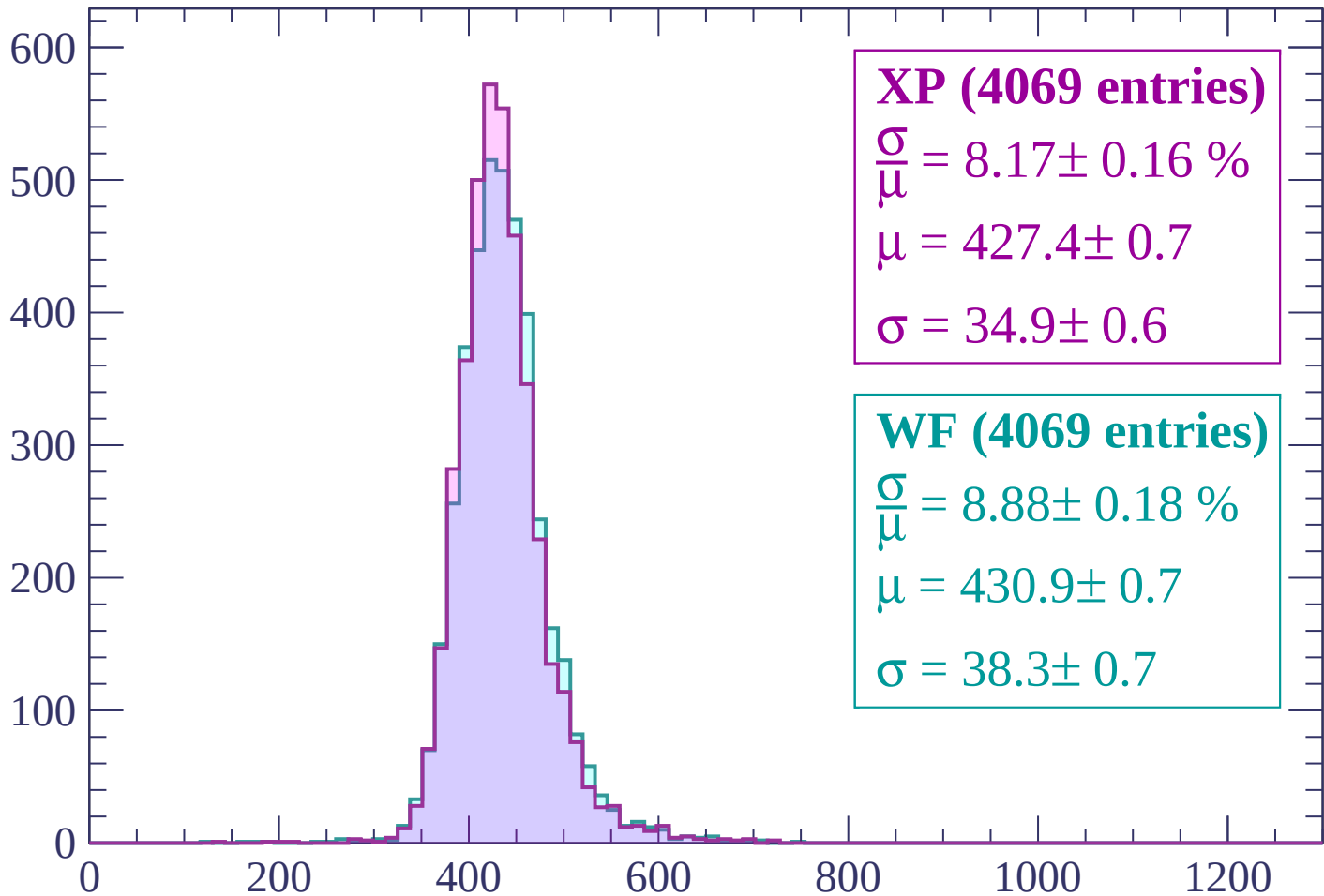
$$\mu = 427.7 \pm 0.7$$

$$\sigma = 38.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count



XP (4069 entries)

$$\frac{\sigma}{\mu} = 8.17 \pm 0.16 \%$$

$$\mu = 427.4 \pm 0.7$$

$$\sigma = 34.9 \pm 0.6$$

WF (4069 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.18 \%$$

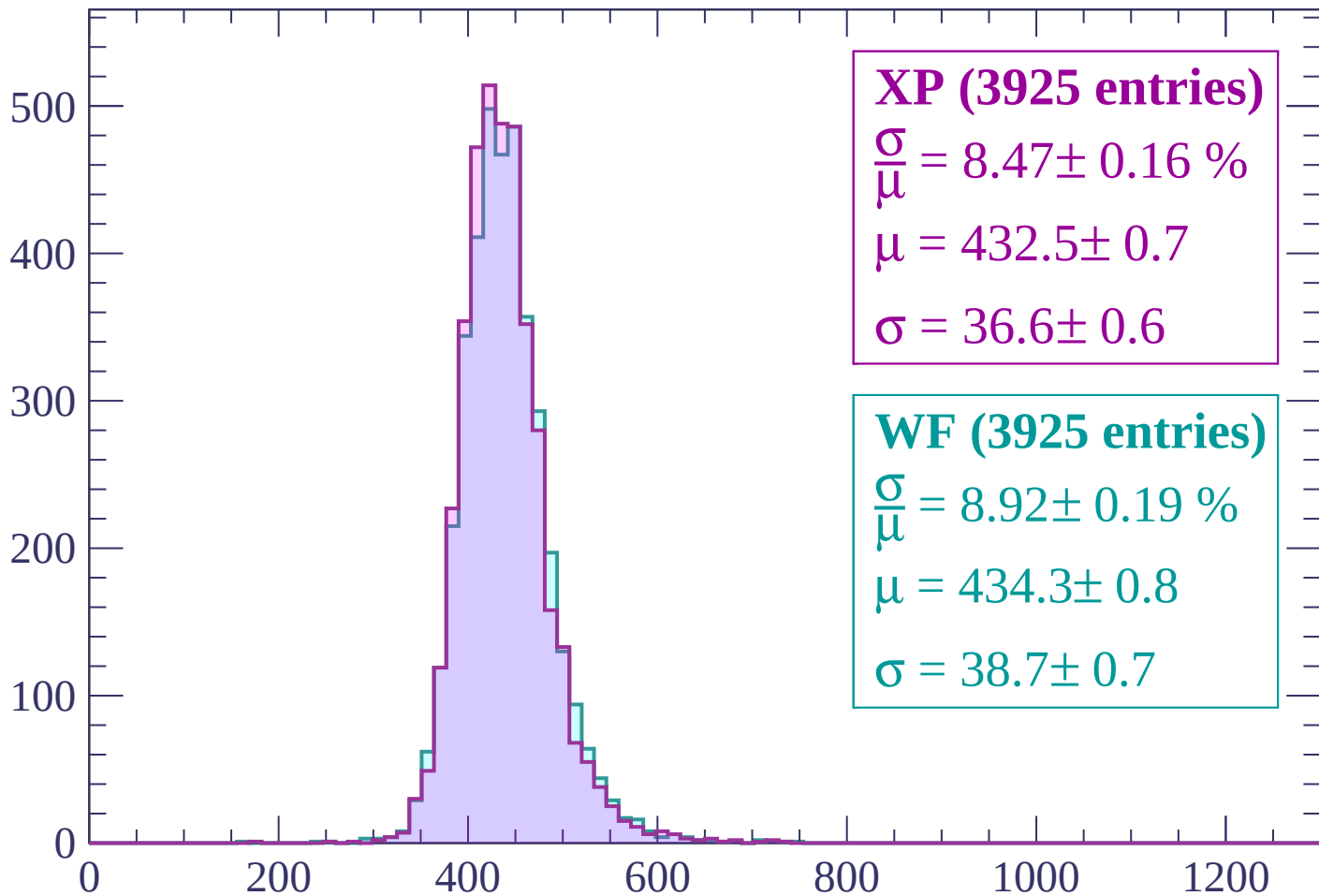
$$\mu = 430.9 \pm 0.7$$

$$\sigma = 38.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count



XP (3925 entries)

$$\frac{\sigma}{\mu} = 8.47 \pm 0.16 \%$$

$$\mu = 432.5 \pm 0.7$$

$$\sigma = 36.6 \pm 0.6$$

WF (3925 entries)

$$\frac{\sigma}{\mu} = 8.92 \pm 0.19 \%$$

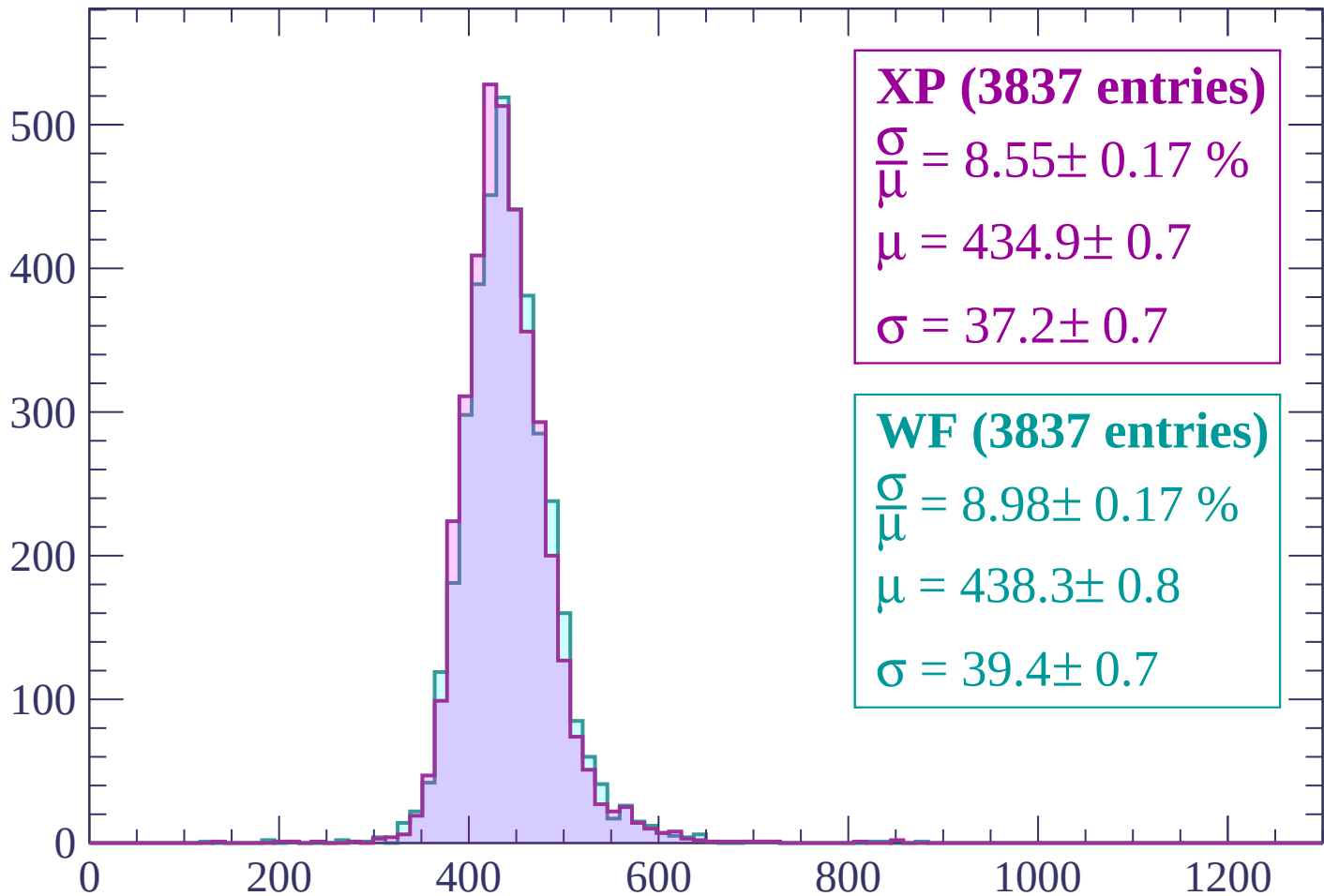
$$\mu = 434.3 \pm 0.8$$

$$\sigma = 38.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP (3837 entries)

$$\frac{\sigma}{\mu} = 8.55 \pm 0.17 \%$$

$$\mu = 434.9 \pm 0.7$$

$$\sigma = 37.2 \pm 0.7$$

WF (3837 entries)

$$\frac{\sigma}{\mu} = 8.98 \pm 0.17 \%$$

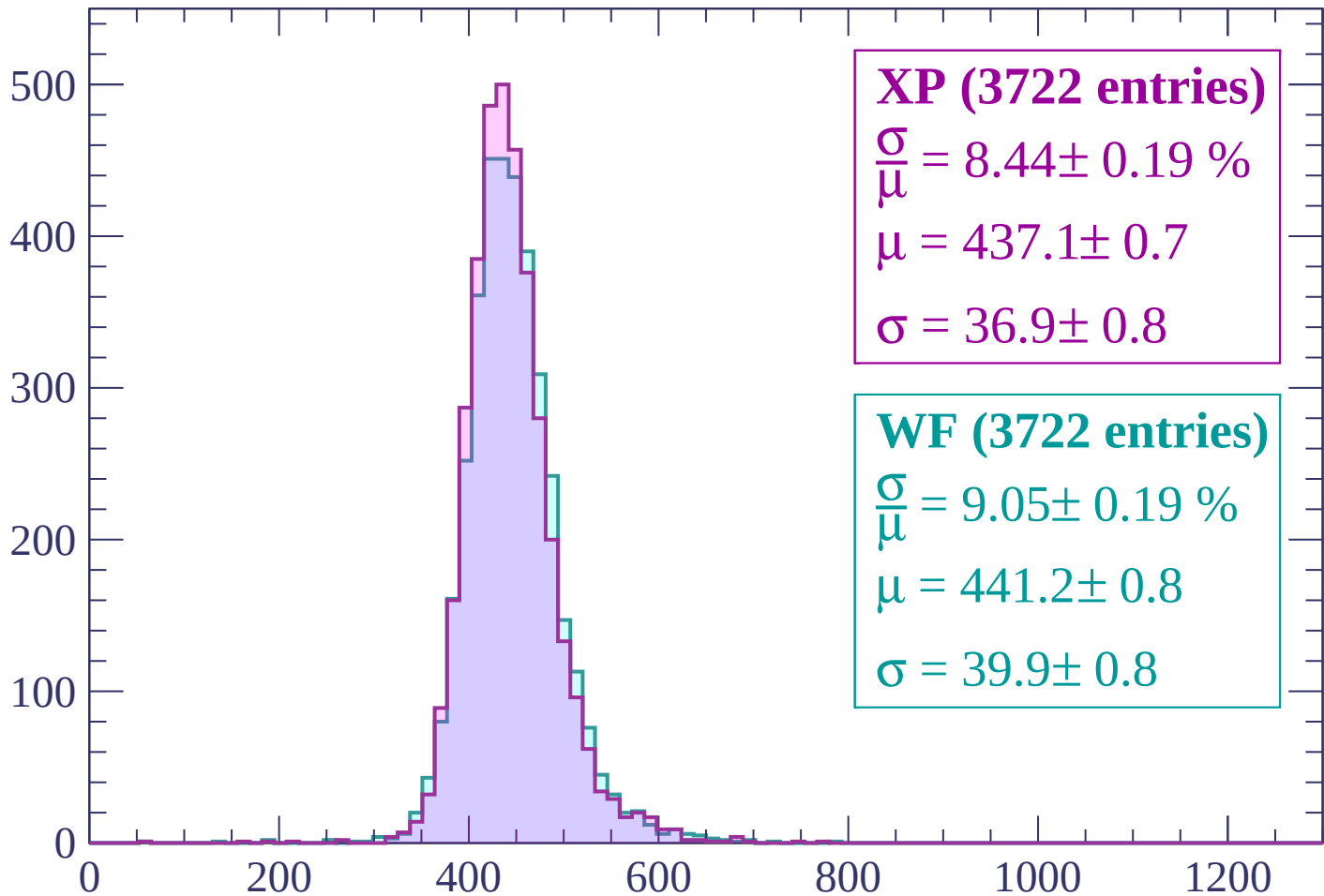
$$\mu = 438.3 \pm 0.8$$

$$\sigma = 39.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$

Count



XP (3722 entries)

$$\frac{\sigma}{\mu} = 8.44 \pm 0.19 \%$$

$$\mu = 437.1 \pm 0.7$$

$$\sigma = 36.9 \pm 0.8$$

WF (3722 entries)

$$\frac{\sigma}{\mu} = 9.05 \pm 0.19 \%$$

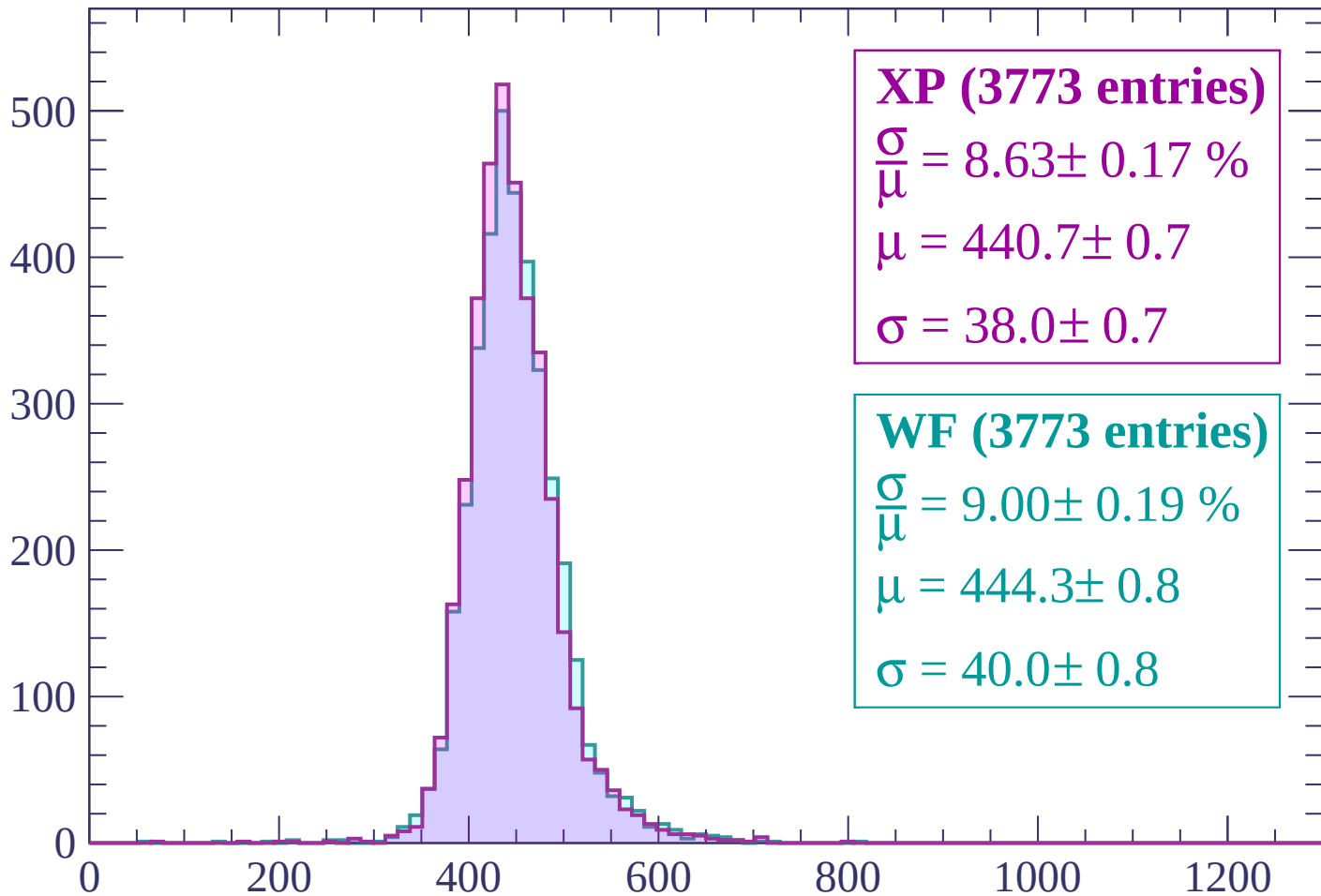
$$\mu = 441.2 \pm 0.8$$

$$\sigma = 39.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$

Count



XP (3773 entries)

$$\frac{\sigma}{\mu} = 8.63 \pm 0.17 \%$$

$$\mu = 440.7 \pm 0.7$$

$$\sigma = 38.0 \pm 0.7$$

WF (3773 entries)

$$\frac{\sigma}{\mu} = 9.00 \pm 0.19 \%$$

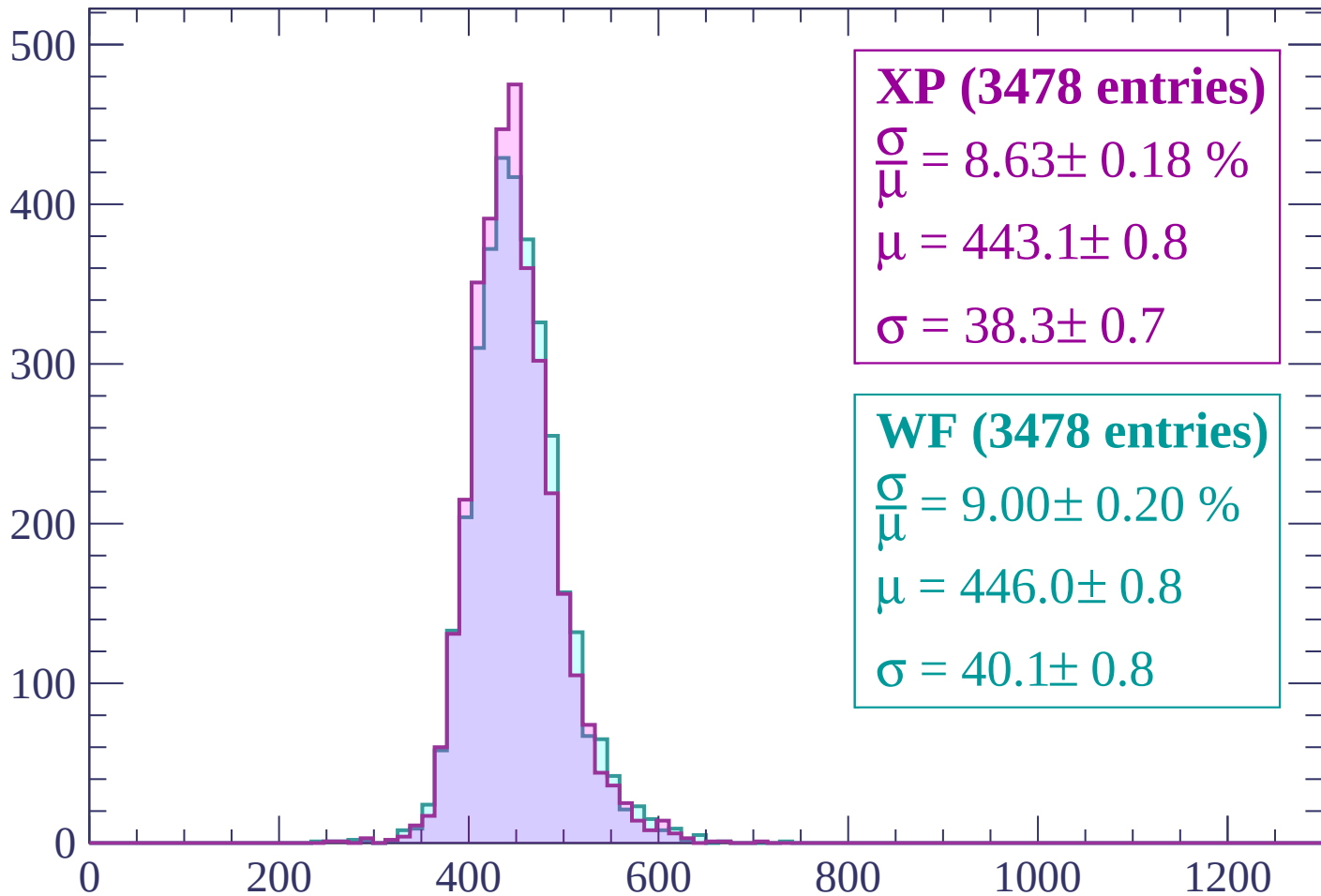
$$\mu = 444.3 \pm 0.8$$

$$\sigma = 40.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP (3478 entries)

$\frac{\sigma}{\mu} = 8.63 \pm 0.18 \%$

$\mu = 443.1 \pm 0.8$

$\sigma = 38.3 \pm 0.7$

WF (3478 entries)

$\frac{\sigma}{\mu} = 9.00 \pm 0.20 \%$

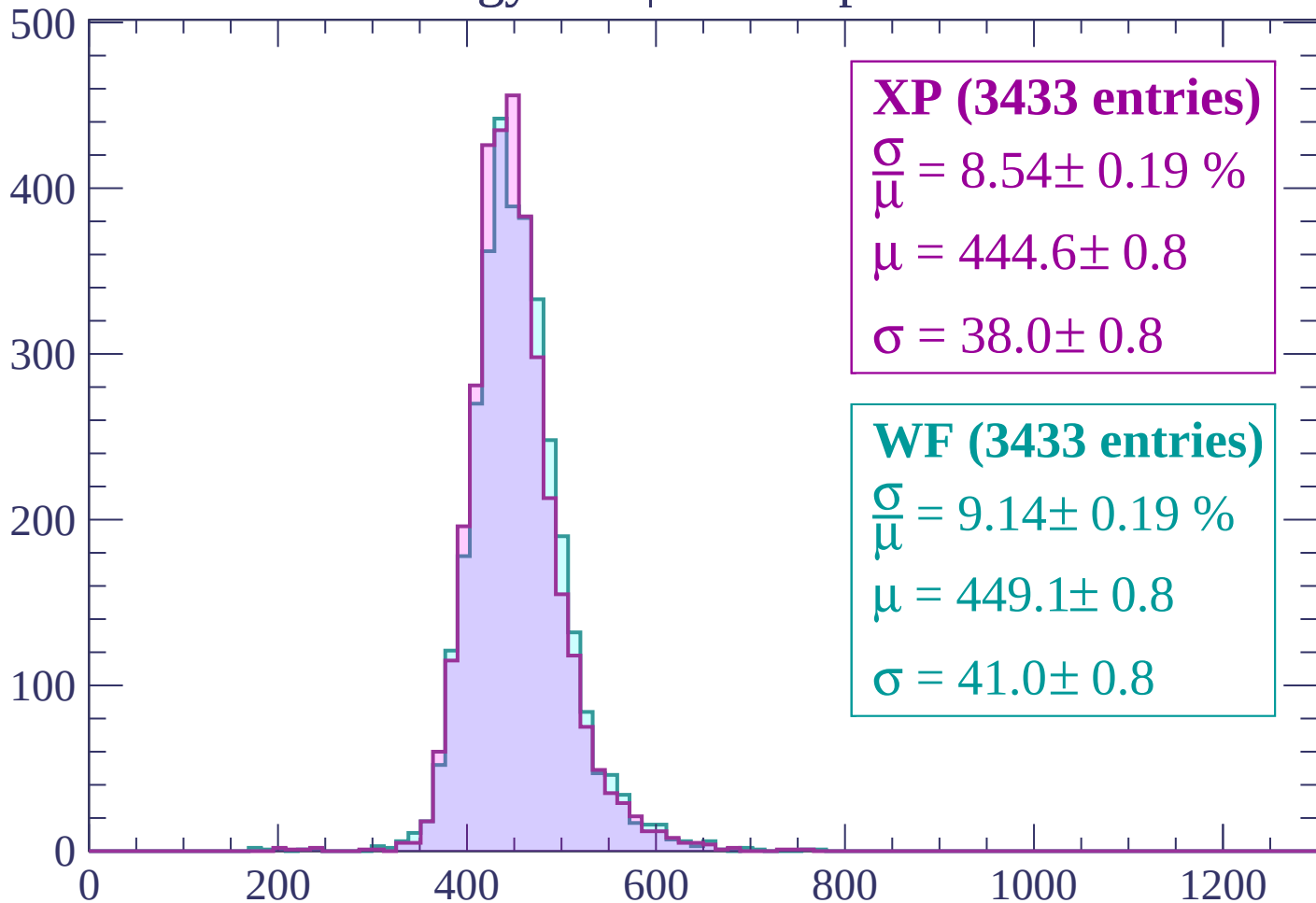
$\mu = 446.0 \pm 0.8$

$\sigma = 40.1 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (3433 entries)

$$\frac{\sigma}{\mu} = 8.54 \pm 0.19 \%$$

$$\mu = 444.6 \pm 0.8$$

$$\sigma = 38.0 \pm 0.8$$

WF (3433 entries)

$$\frac{\sigma}{\mu} = 9.14 \pm 0.19 \%$$

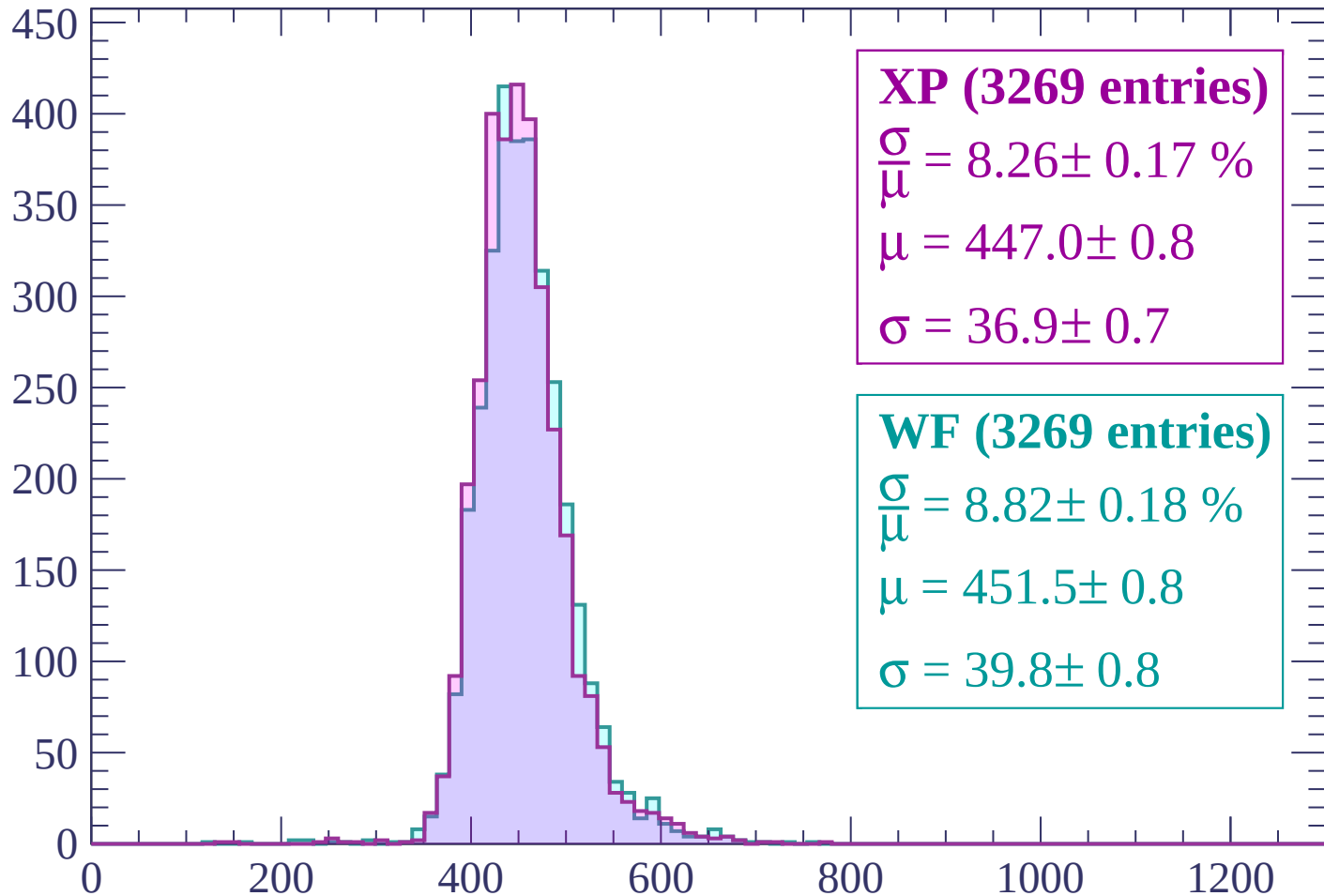
$$\mu = 449.1 \pm 0.8$$

$$\sigma = 41.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP (3269 entries)

$\frac{\sigma}{\mu} = 8.26 \pm 0.17 \%$

$\mu = 447.0 \pm 0.8$

$\sigma = 36.9 \pm 0.7$

WF (3269 entries)

$\frac{\sigma}{\mu} = 8.82 \pm 0.18 \%$

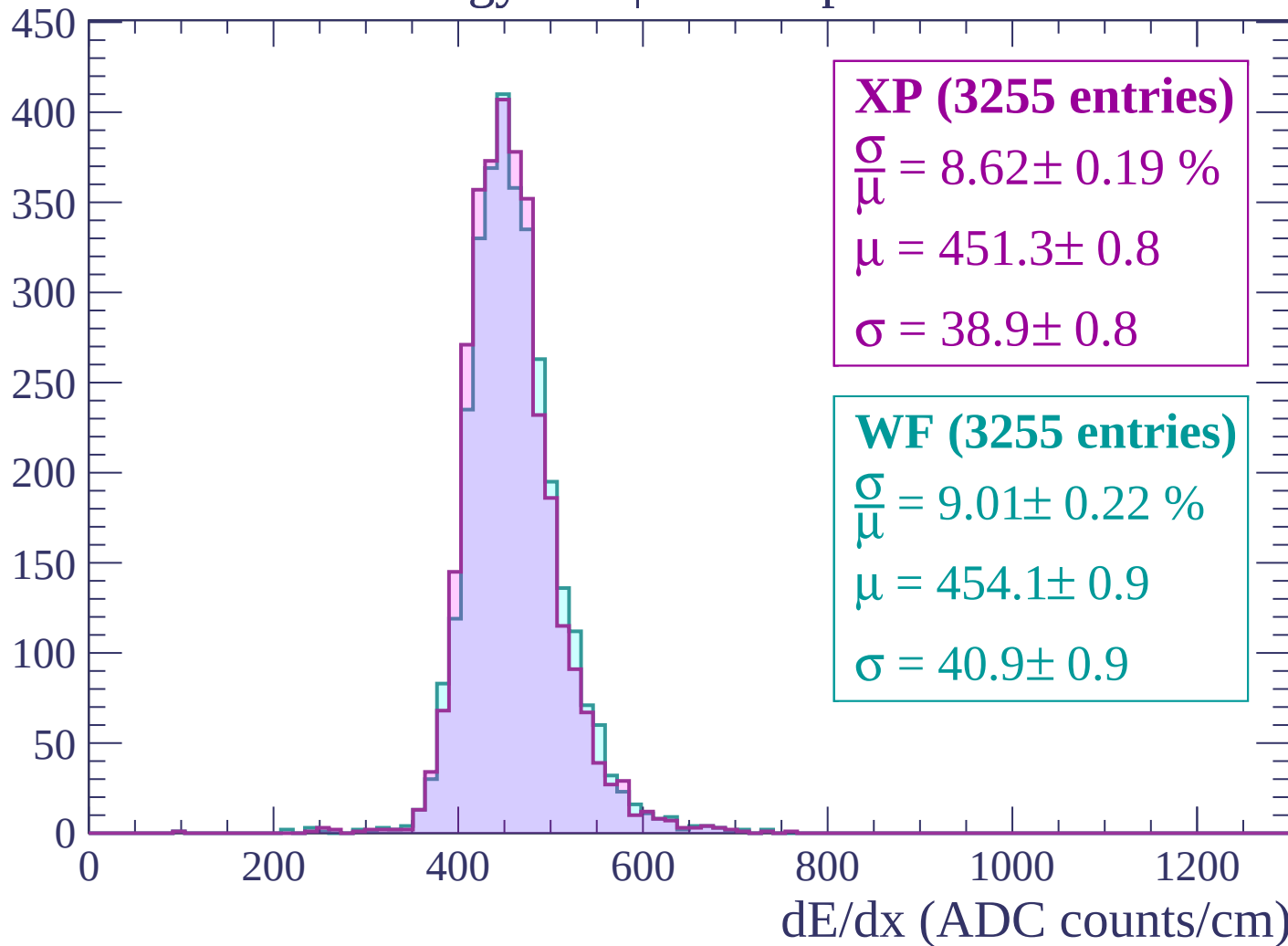
$\mu = 451.5 \pm 0.8$

$\sigma = 39.8 \pm 0.8$

dE/dx (ADC counts/cm)

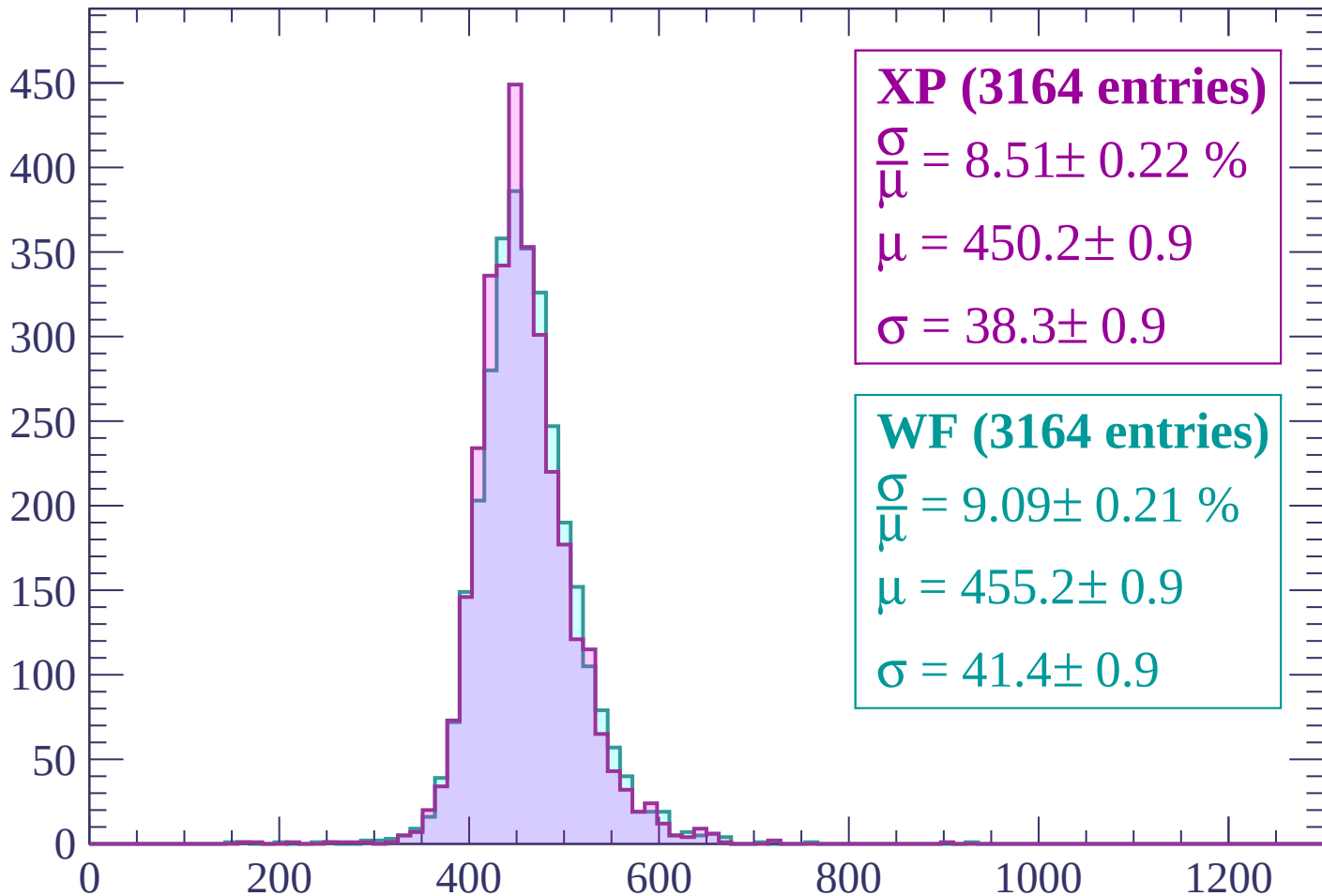
Energy loss | $1440 < p < 1500$

Count



Energy loss | $1500 < p < 1560$

Count



XP (3164 entries)

$$\frac{\sigma}{\mu} = 8.51 \pm 0.22 \%$$

$$\mu = 450.2 \pm 0.9$$

$$\sigma = 38.3 \pm 0.9$$

WF (3164 entries)

$$\frac{\sigma}{\mu} = 9.09 \pm 0.21 \%$$

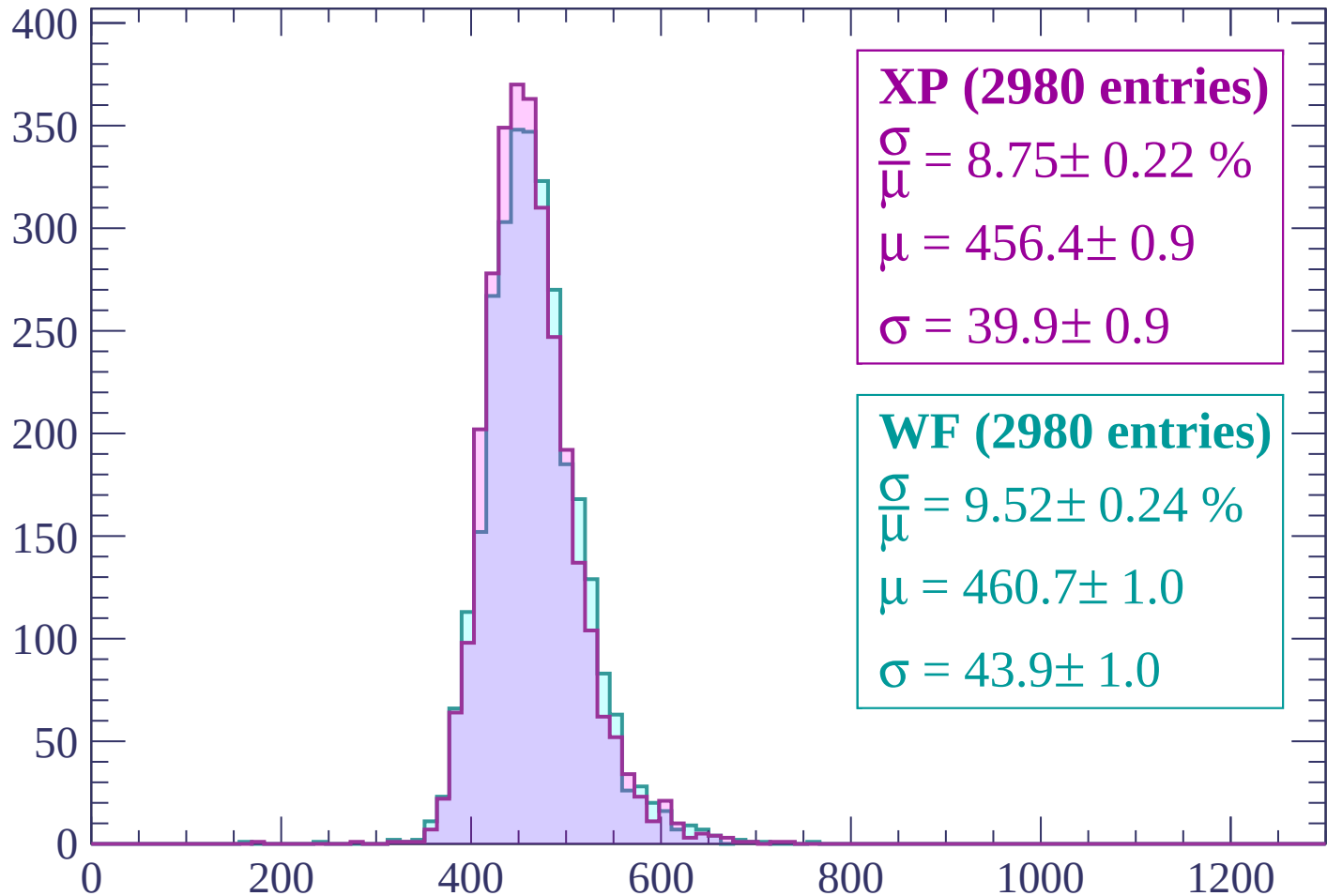
$$\mu = 455.2 \pm 0.9$$

$$\sigma = 41.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1620$

Count



XP (2980 entries)

$$\frac{\sigma}{\mu} = 8.75 \pm 0.22 \%$$

$$\mu = 456.4 \pm 0.9$$

$$\sigma = 39.9 \pm 0.9$$

WF (2980 entries)

$$\frac{\sigma}{\mu} = 9.52 \pm 0.24 \%$$

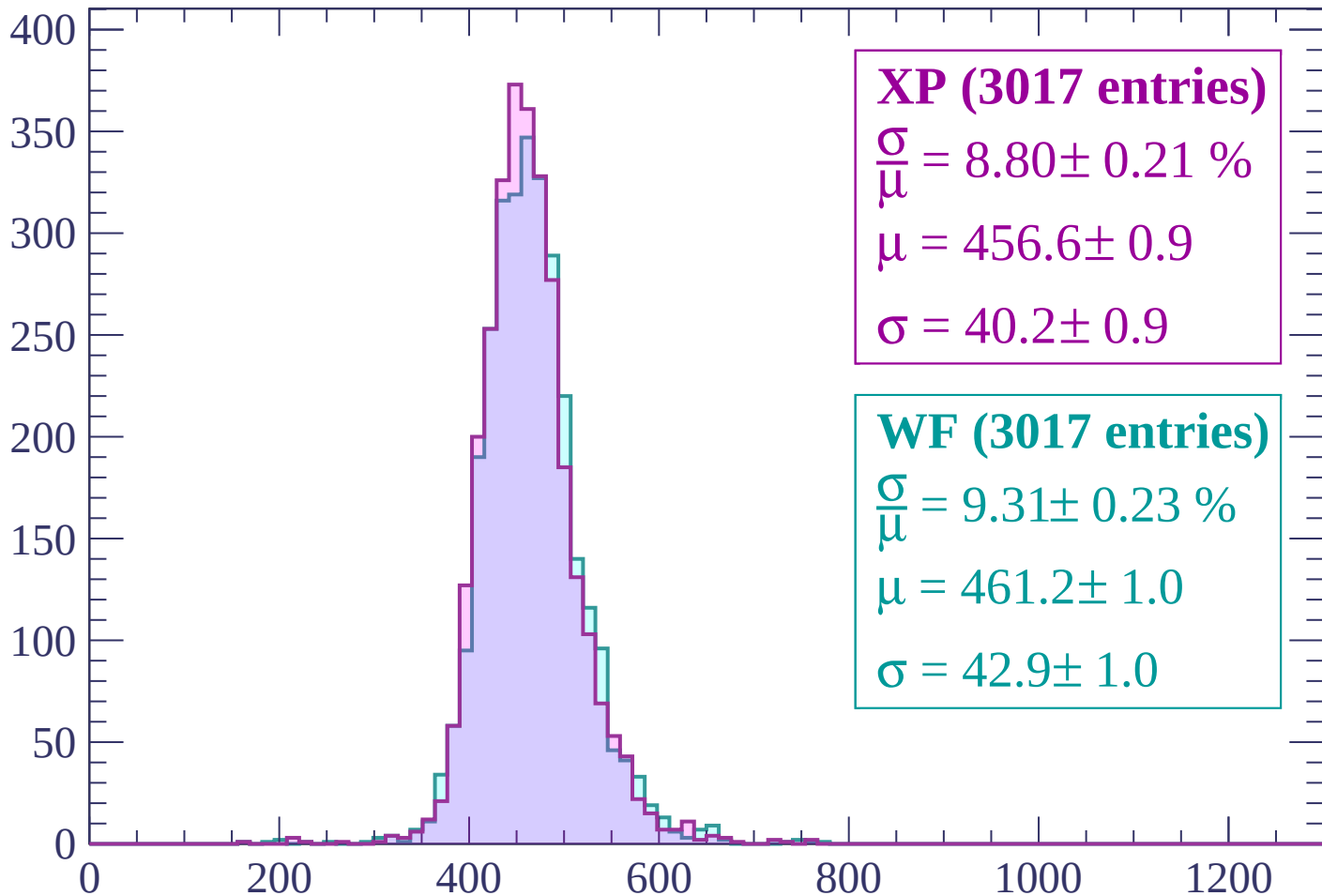
$$\mu = 460.7 \pm 1.0$$

$$\sigma = 43.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP (3017 entries)

$$\frac{\sigma}{\mu} = 8.80 \pm 0.21 \%$$

$$\mu = 456.6 \pm 0.9$$

$$\sigma = 40.2 \pm 0.9$$

WF (3017 entries)

$$\frac{\sigma}{\mu} = 9.31 \pm 0.23 \%$$

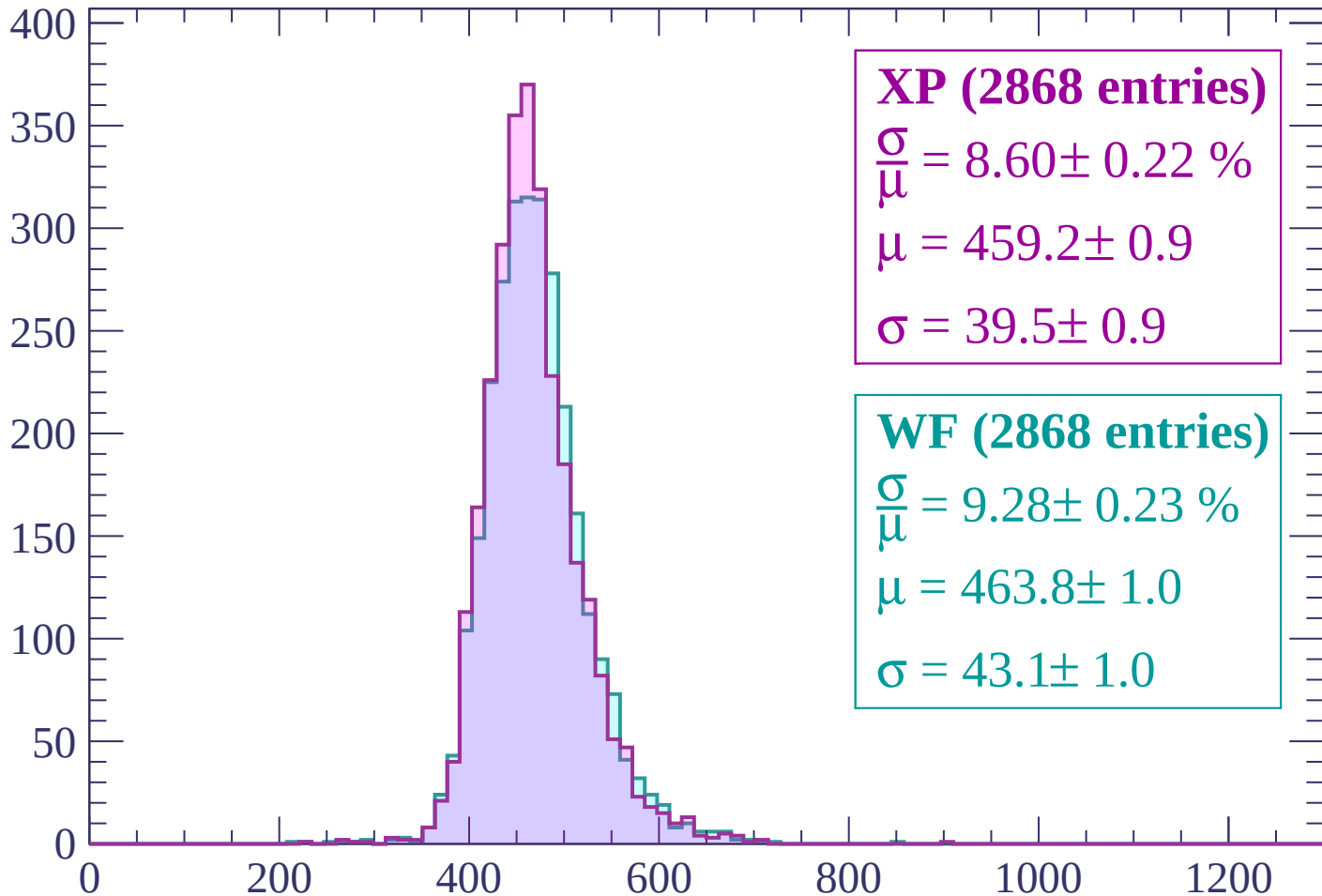
$$\mu = 461.2 \pm 1.0$$

$$\sigma = 42.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (2868 entries)

$$\frac{\sigma}{\mu} = 8.60 \pm 0.22 \%$$

$$\mu = 459.2 \pm 0.9$$

$$\sigma = 39.5 \pm 0.9$$

WF (2868 entries)

$$\frac{\sigma}{\mu} = 9.28 \pm 0.23 \%$$

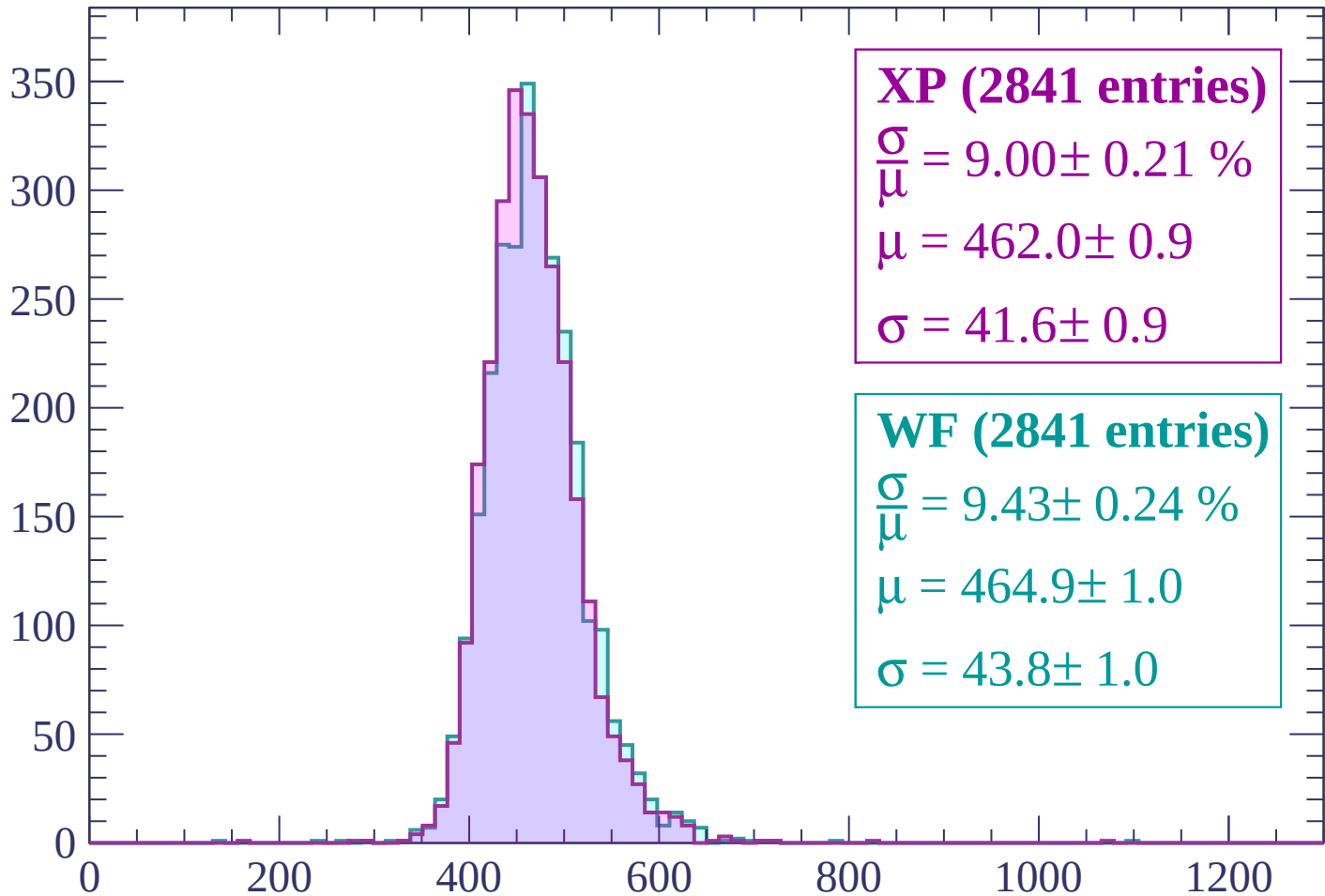
$$\mu = 463.8 \pm 1.0$$

$$\sigma = 43.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (2841 entries)

$\frac{\sigma}{\mu} = 9.00 \pm 0.21 \%$

$\mu = 462.0 \pm 0.9$

$\sigma = 41.6 \pm 0.9$

WF (2841 entries)

$\frac{\sigma}{\mu} = 9.43 \pm 0.24 \%$

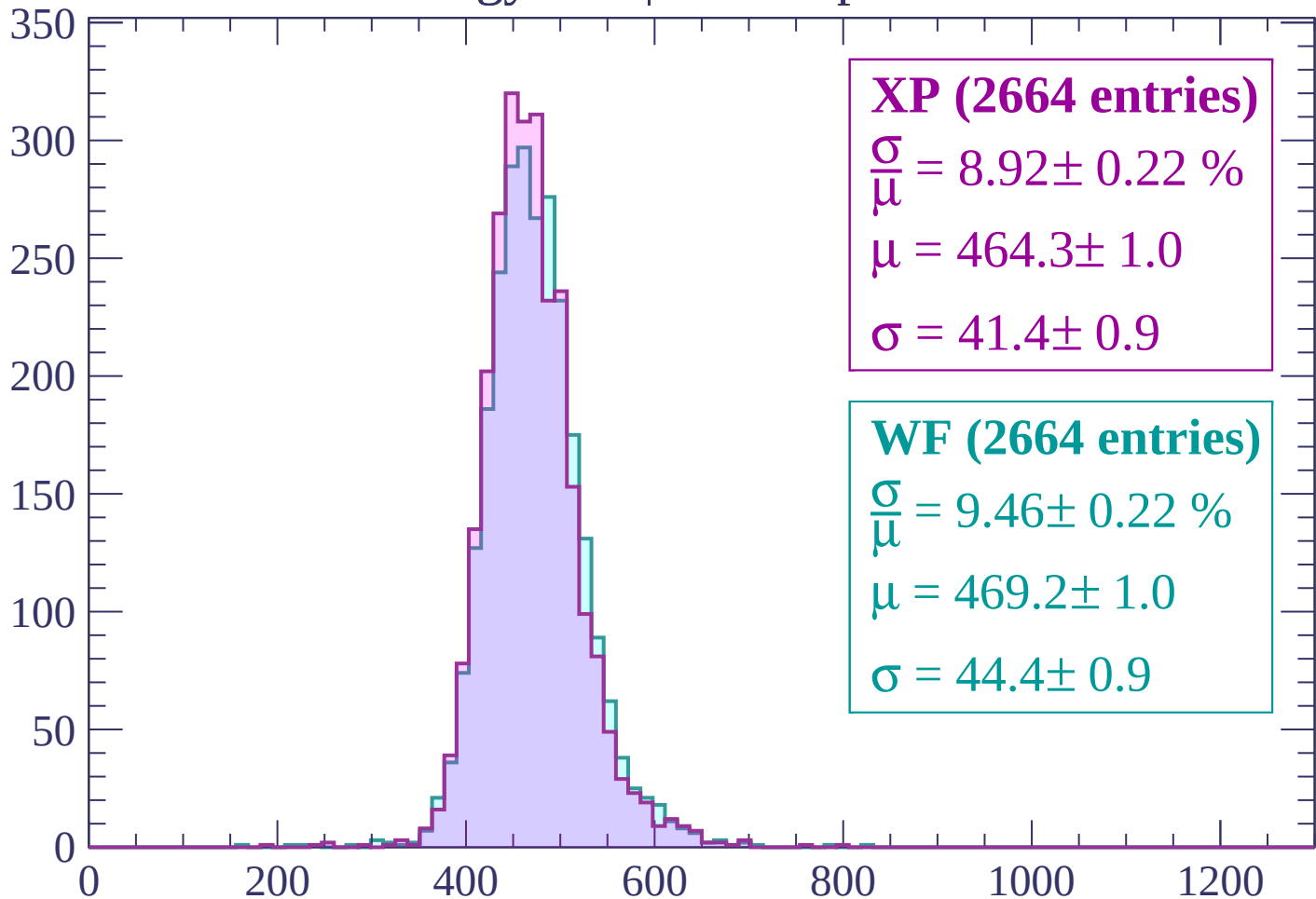
$\mu = 464.9 \pm 1.0$

$\sigma = 43.8 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1860$

Count



XP (2664 entries)

$$\frac{\sigma}{\mu} = 8.92 \pm 0.22 \%$$

$$\mu = 464.3 \pm 1.0$$

$$\sigma = 41.4 \pm 0.9$$

WF (2664 entries)

$$\frac{\sigma}{\mu} = 9.46 \pm 0.22 \%$$

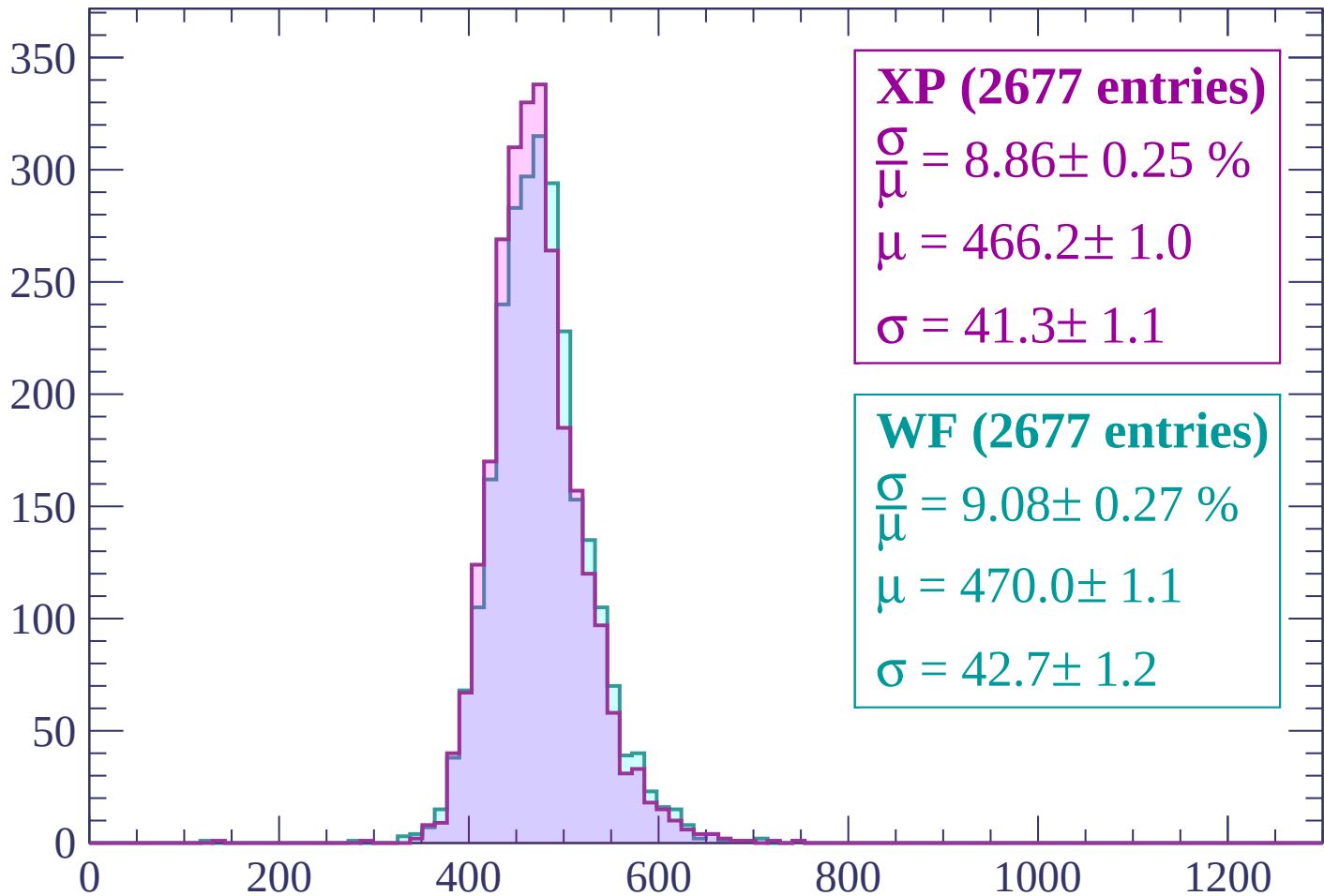
$$\mu = 469.2 \pm 1.0$$

$$\sigma = 44.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1860 < p < 1920$

Count



XP (2677 entries)

$$\frac{\sigma}{\mu} = 8.86 \pm 0.25 \%$$

$$\mu = 466.2 \pm 1.0$$

$$\sigma = 41.3 \pm 1.1$$

WF (2677 entries)

$$\frac{\sigma}{\mu} = 9.08 \pm 0.27 \%$$

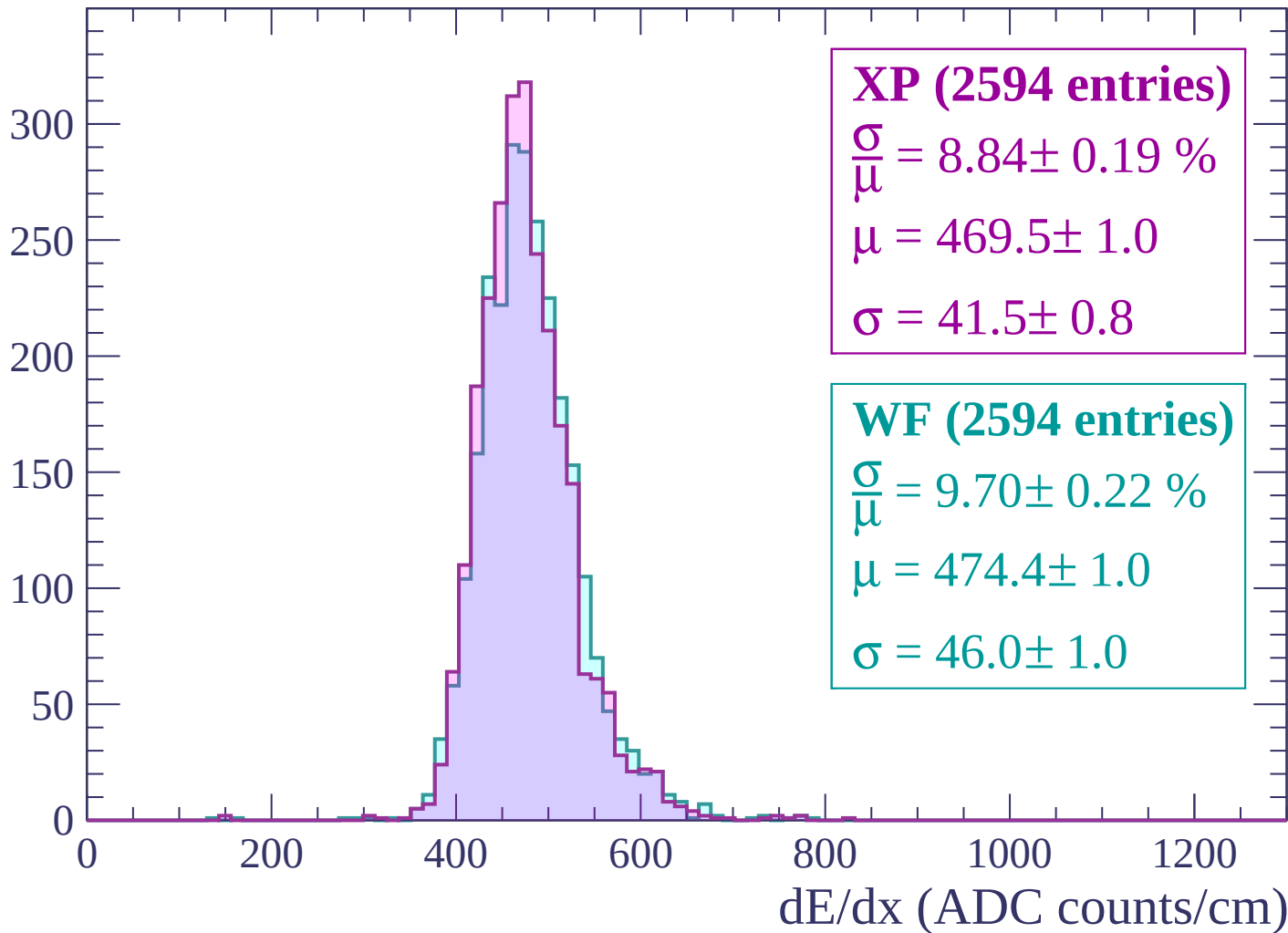
$$\mu = 470.0 \pm 1.1$$

$$\sigma = 42.7 \pm 1.2$$

dE/dx (ADC counts/cm)

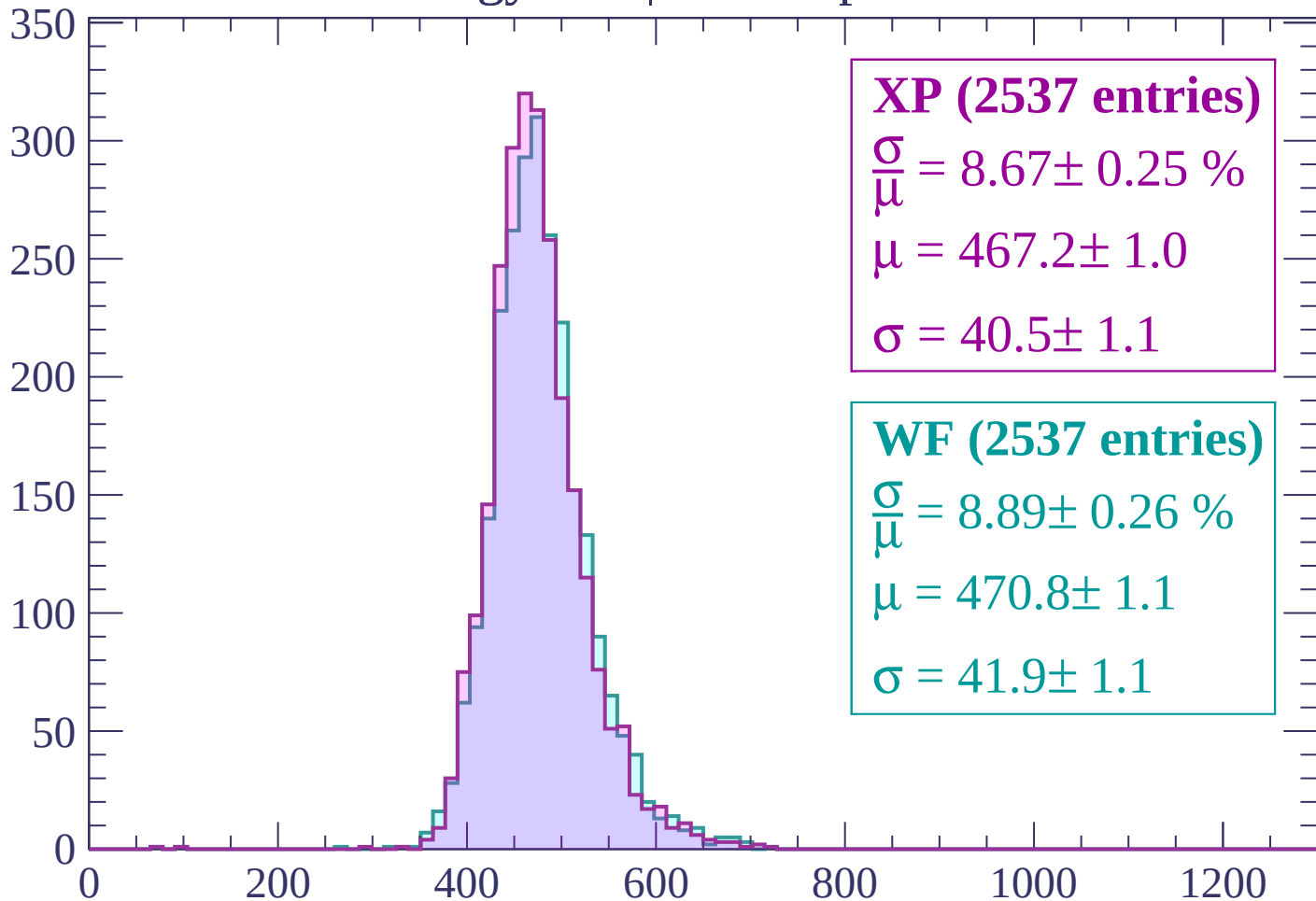
Energy loss | $1920 < p < 1980$

Count



Energy loss | $1980 < p < 2040$

Count



XP (2537 entries)

$$\frac{\sigma}{\mu} = 8.67 \pm 0.25 \%$$

$$\mu = 467.2 \pm 1.0$$

$$\sigma = 40.5 \pm 1.1$$

WF (2537 entries)

$$\frac{\sigma}{\mu} = 8.89 \pm 0.26 \%$$

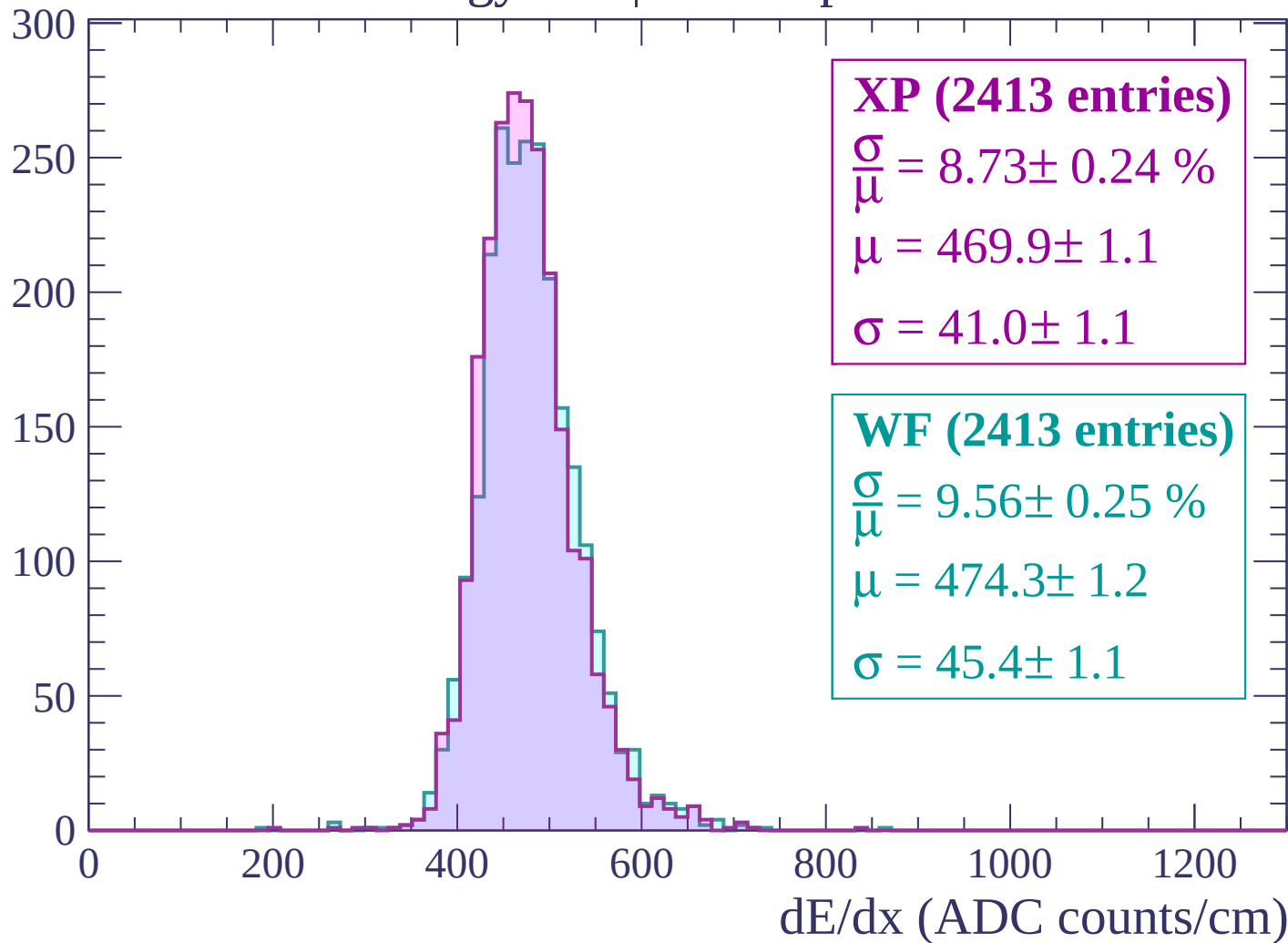
$$\mu = 470.8 \pm 1.1$$

$$\sigma = 41.9 \pm 1.1$$

dE/dx (ADC counts/cm)

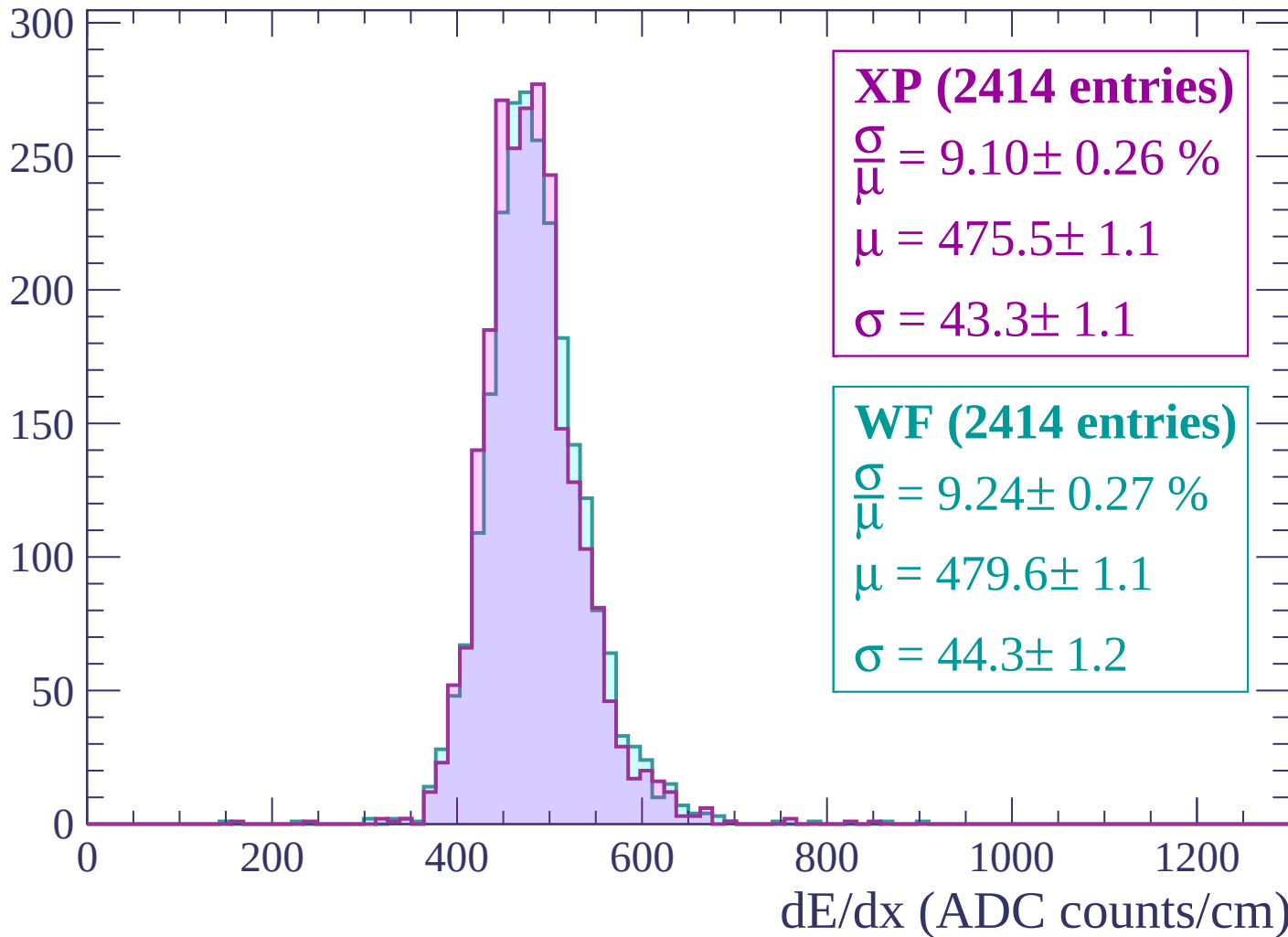
Energy loss | $2040 < p < 2100$

Count



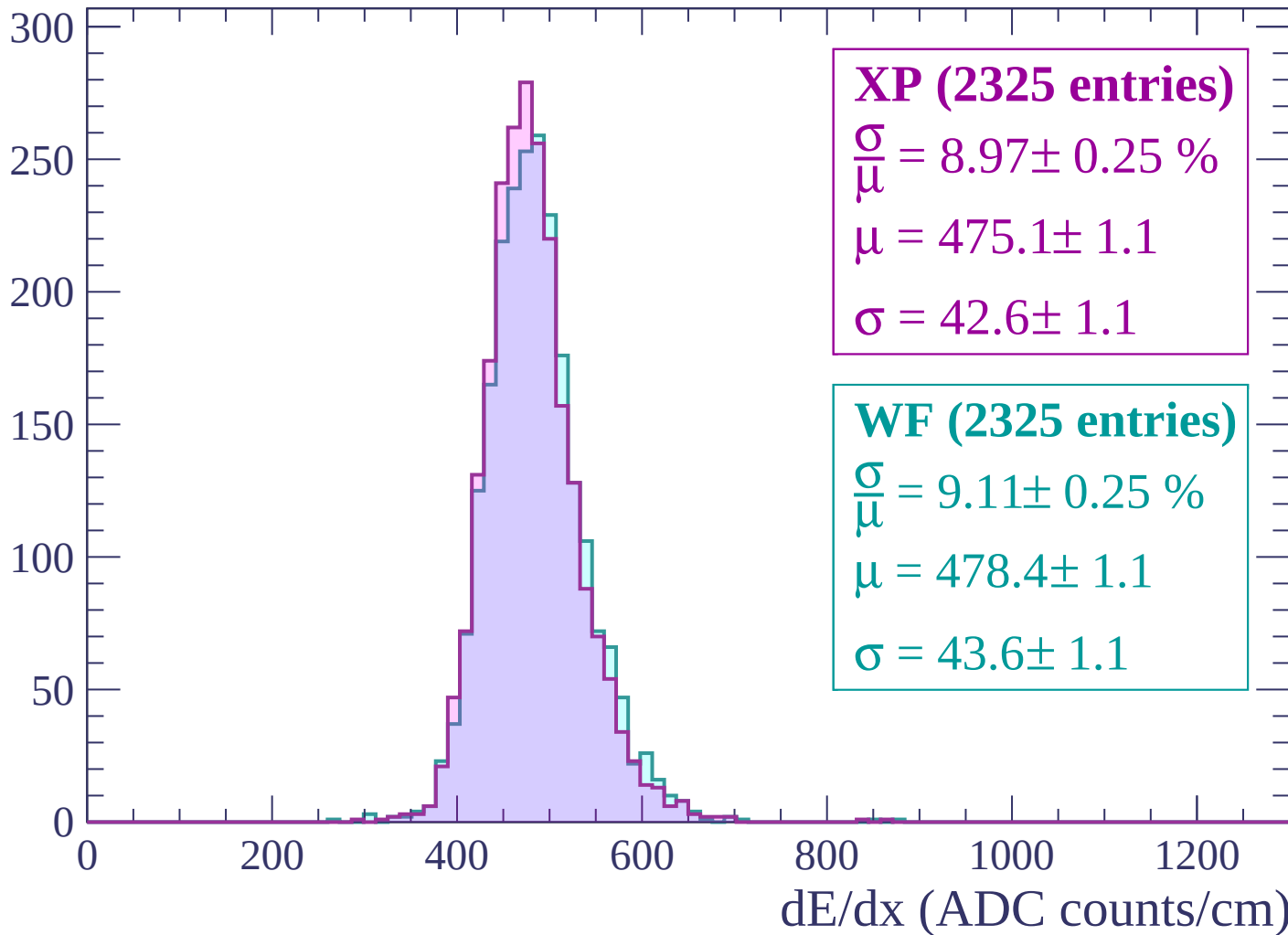
Energy loss | $2100 < p < 2160$

Count



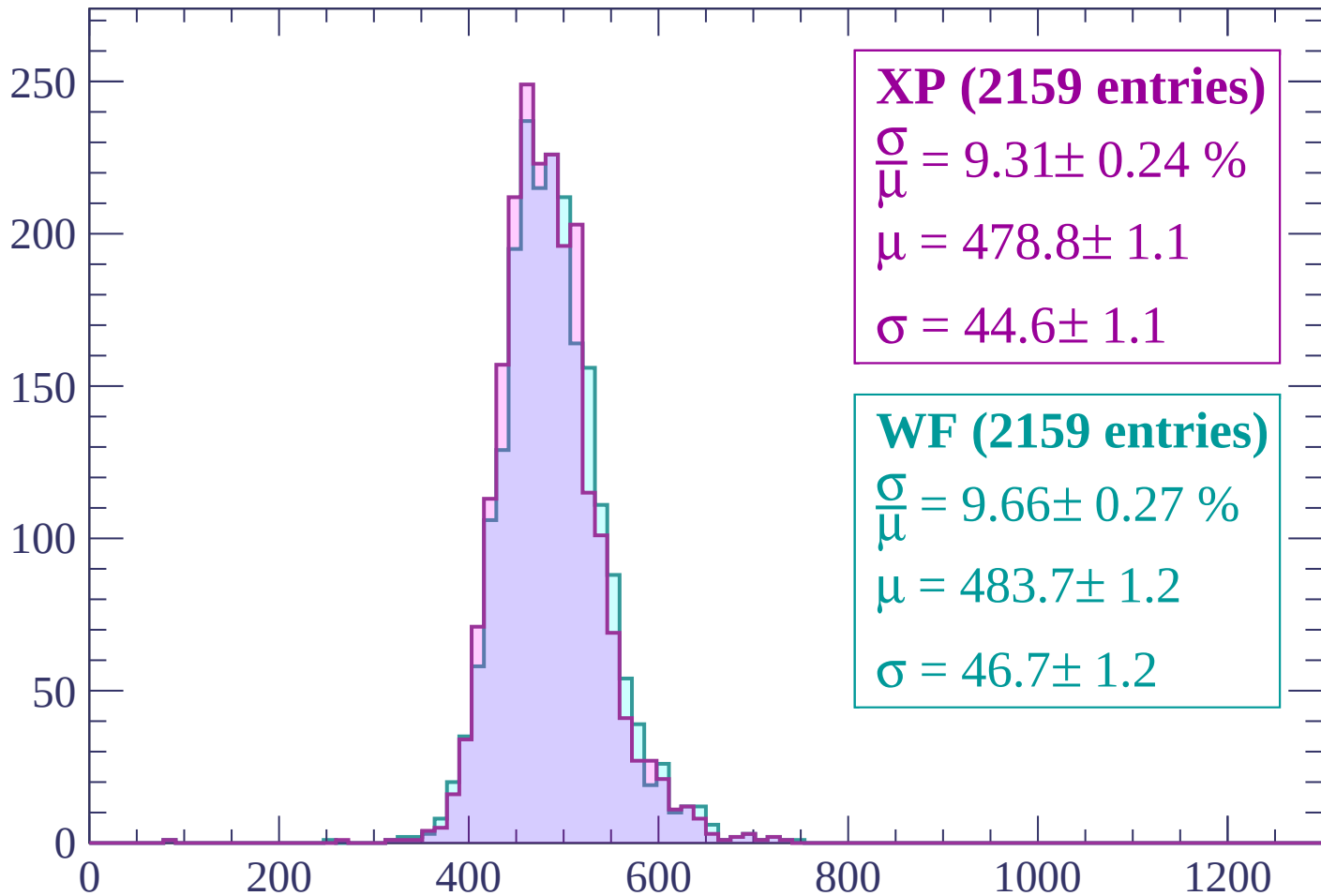
Energy loss | $2160 < p < 2220$

Count



Energy loss | $2220 < p < 2280$

Count



XP (2159 entries)

$\frac{\sigma}{\mu} = 9.31 \pm 0.24 \%$

$\mu = 478.8 \pm 1.1$

$\sigma = 44.6 \pm 1.1$

WF (2159 entries)

$\frac{\sigma}{\mu} = 9.66 \pm 0.27 \%$

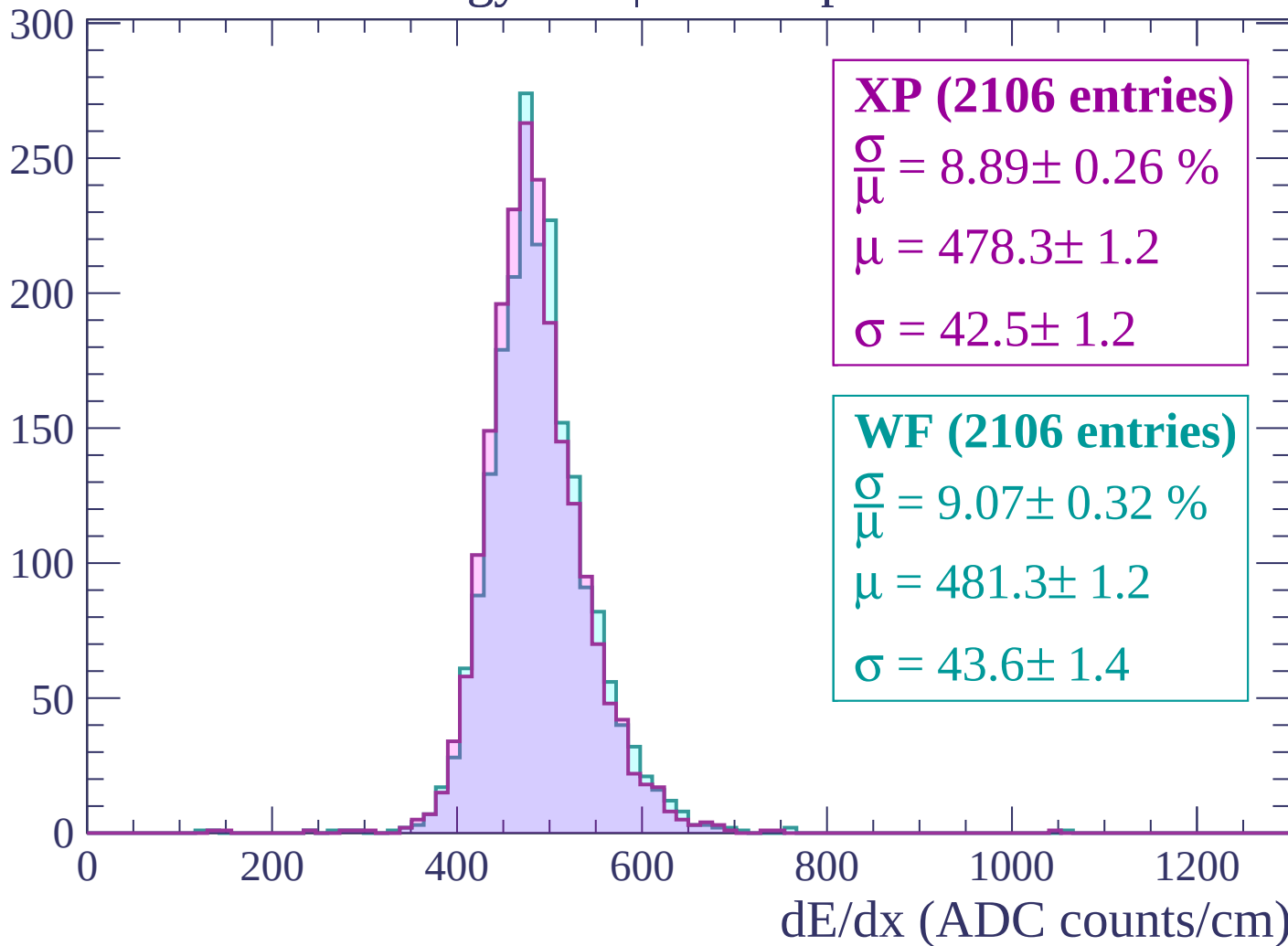
$\mu = 483.7 \pm 1.2$

$\sigma = 46.7 \pm 1.2$

dE/dx (ADC counts/cm)

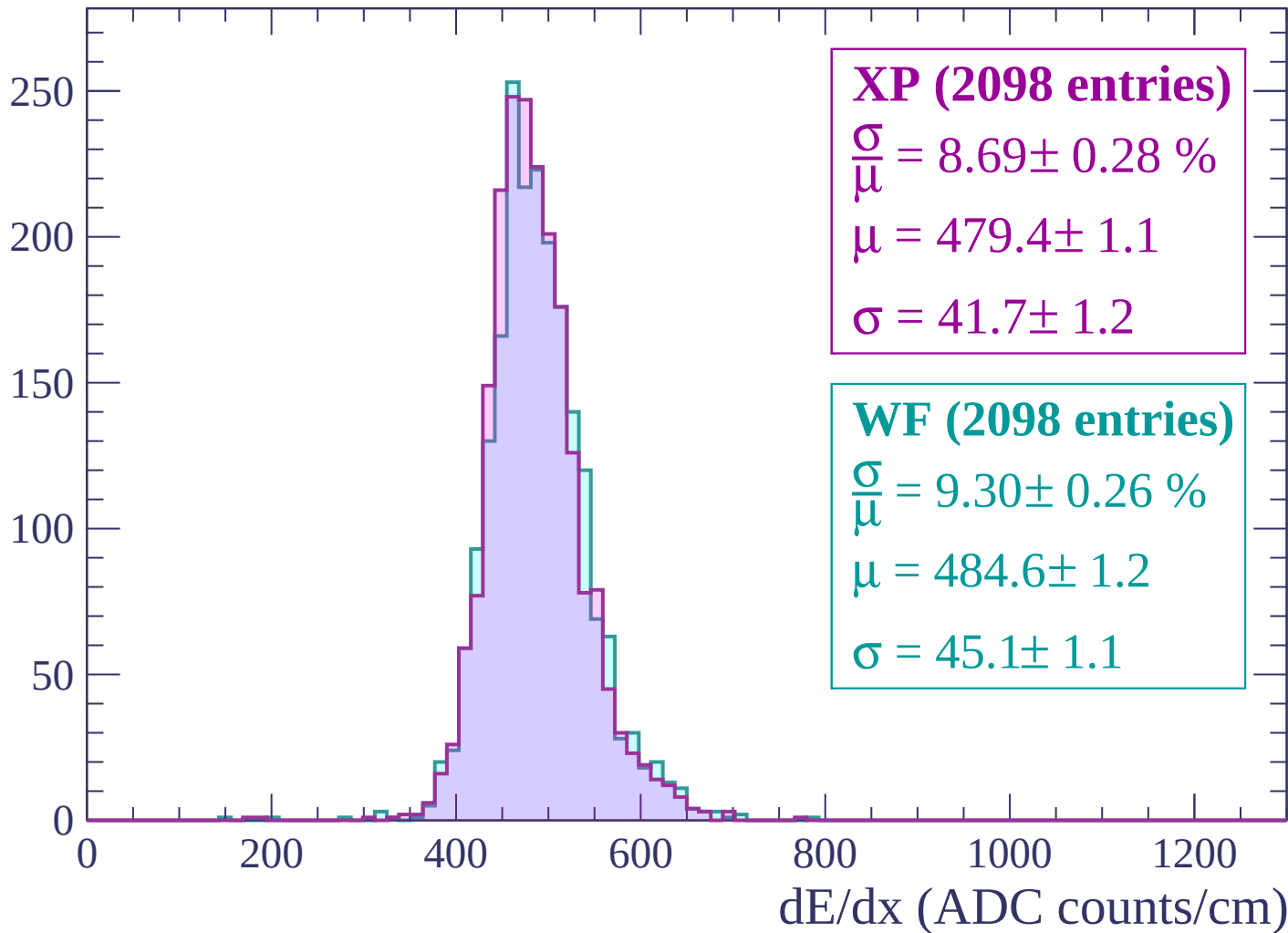
Energy loss | $2280 < p < 2340$

Count



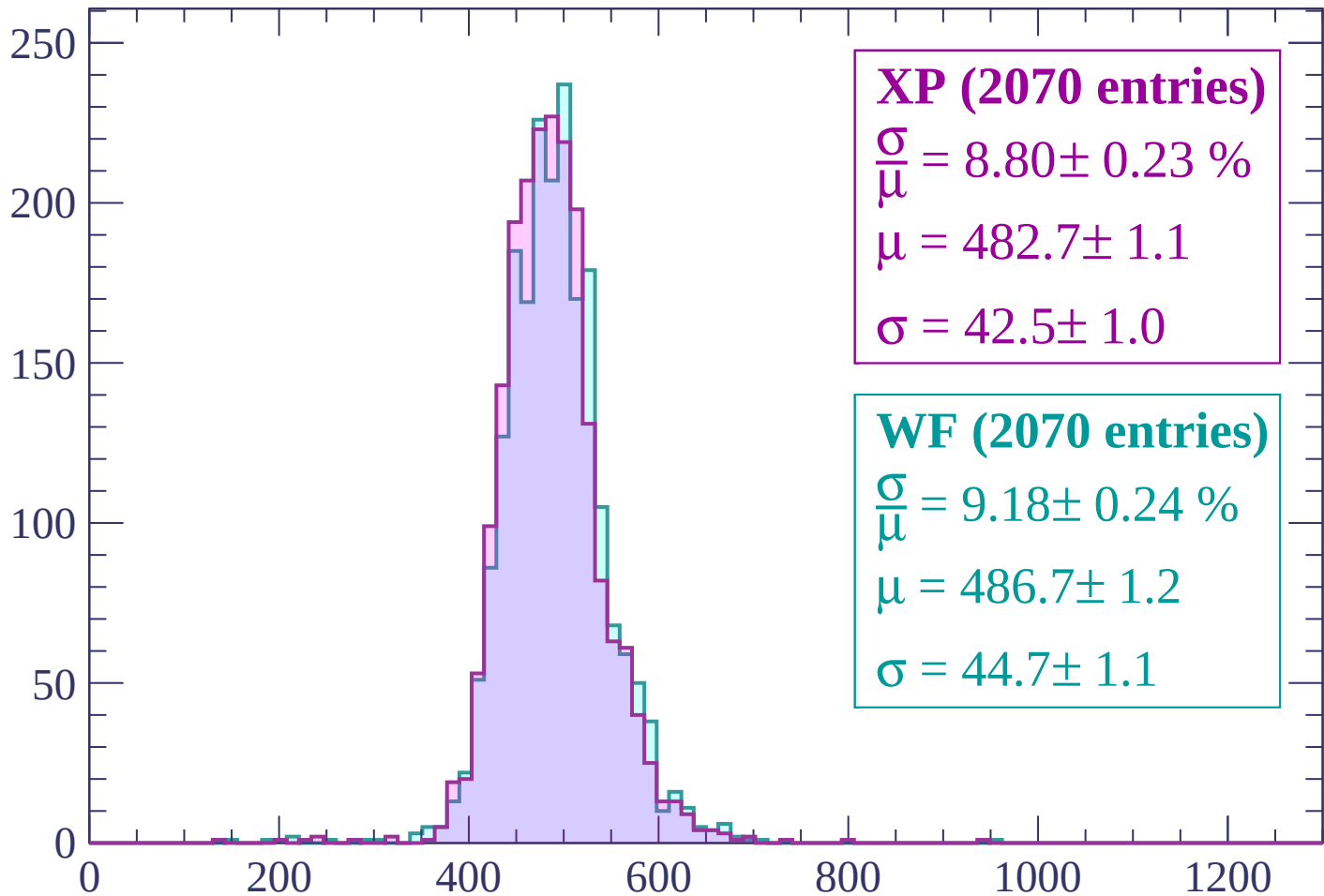
Energy loss | $2340 < p < 2400$

Count



Energy loss | $2400 < p < 2460$

Count



XP (2070 entries)

$$\frac{\sigma}{\mu} = 8.80 \pm 0.23 \%$$

$$\mu = 482.7 \pm 1.1$$

$$\sigma = 42.5 \pm 1.0$$

WF (2070 entries)

$$\frac{\sigma}{\mu} = 9.18 \pm 0.24 \%$$

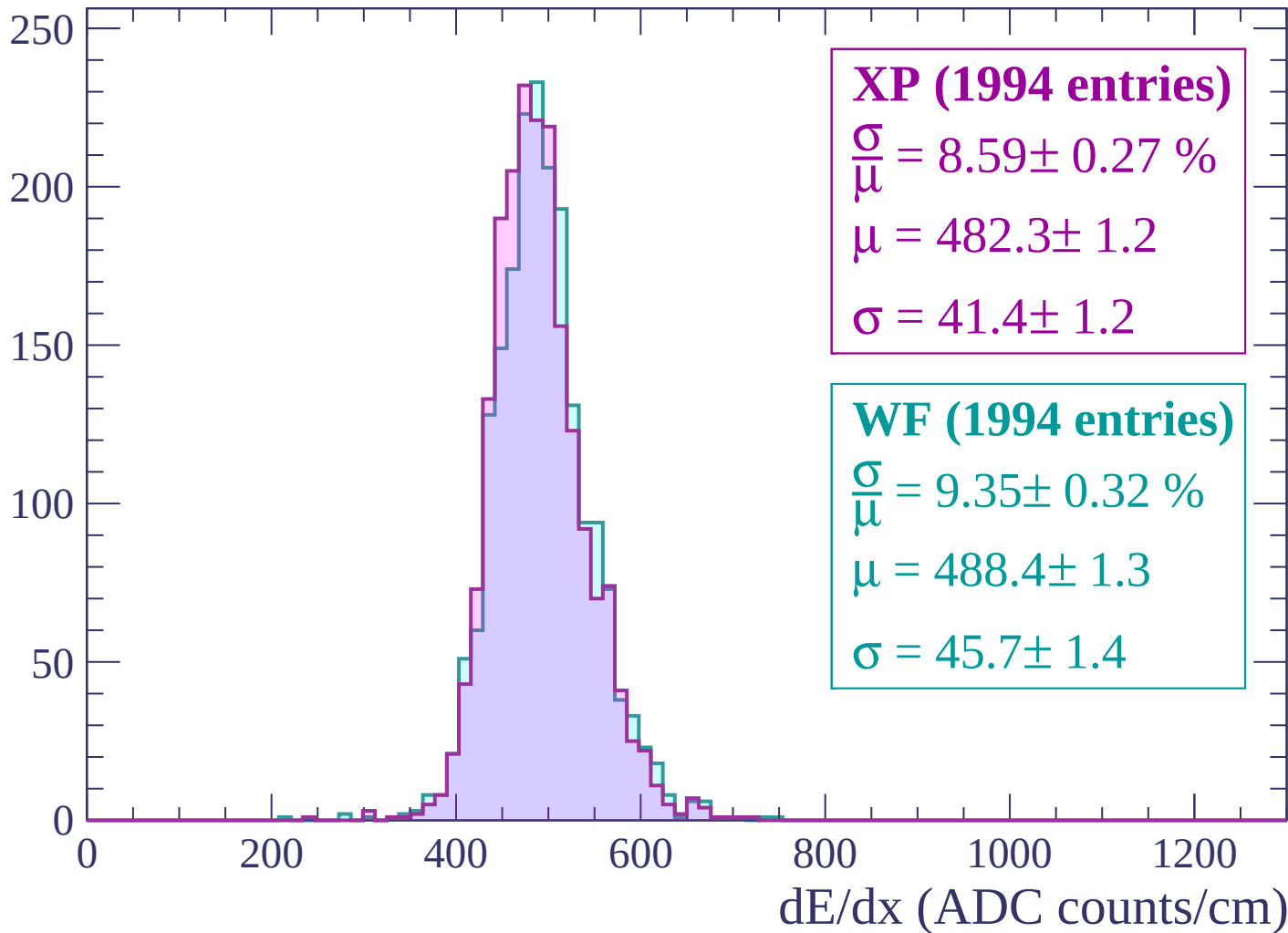
$$\mu = 486.7 \pm 1.2$$

$$\sigma = 44.7 \pm 1.1$$

dE/dx (ADC counts/cm)

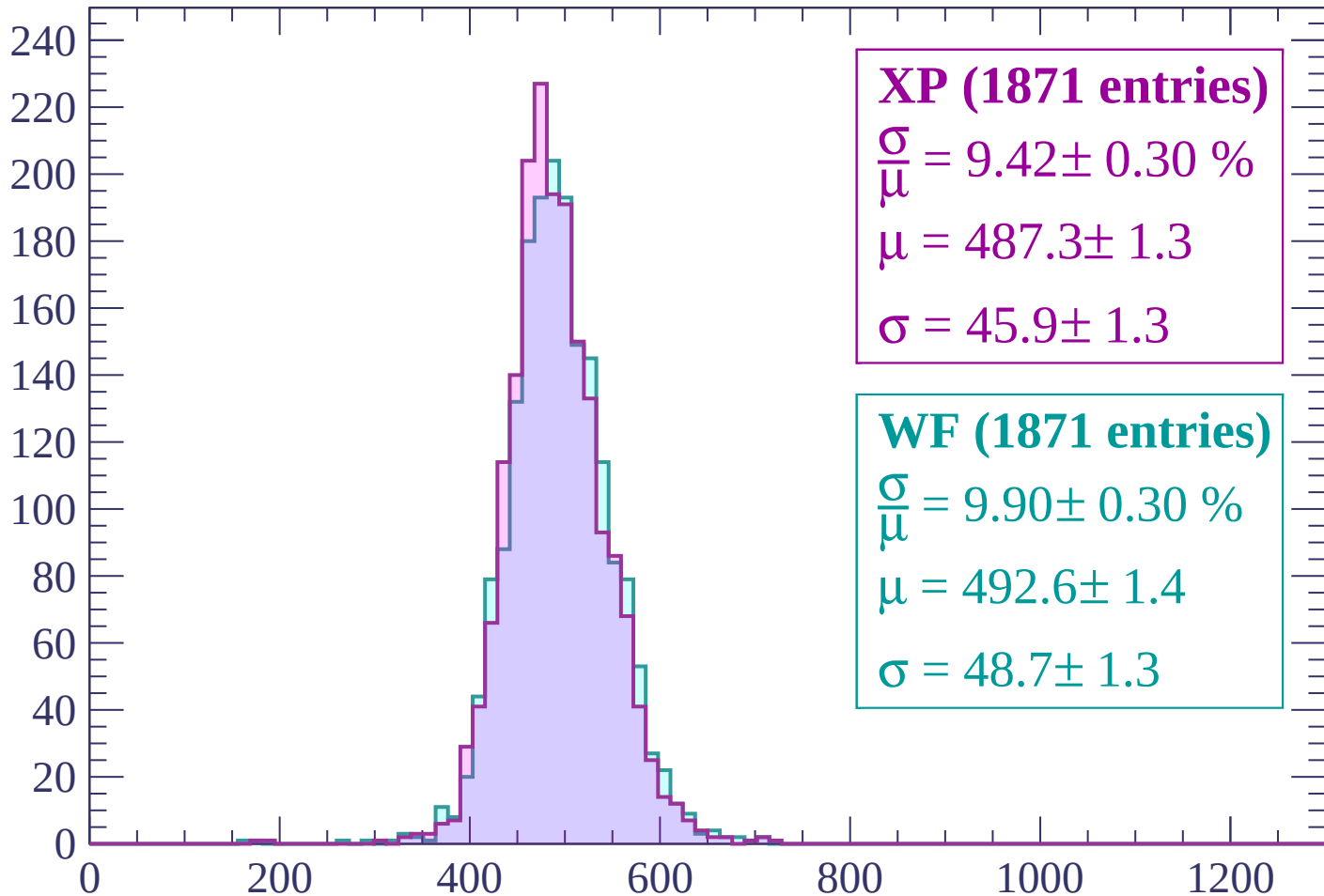
Energy loss | $2460 < p < 2520$

Count



Energy loss | $2520 < p < 2580$

Count



XP (1871 entries)

$$\frac{\sigma}{\mu} = 9.42 \pm 0.30 \%$$

$$\mu = 487.3 \pm 1.3$$

$$\sigma = 45.9 \pm 1.3$$

WF (1871 entries)

$$\frac{\sigma}{\mu} = 9.90 \pm 0.30 \%$$

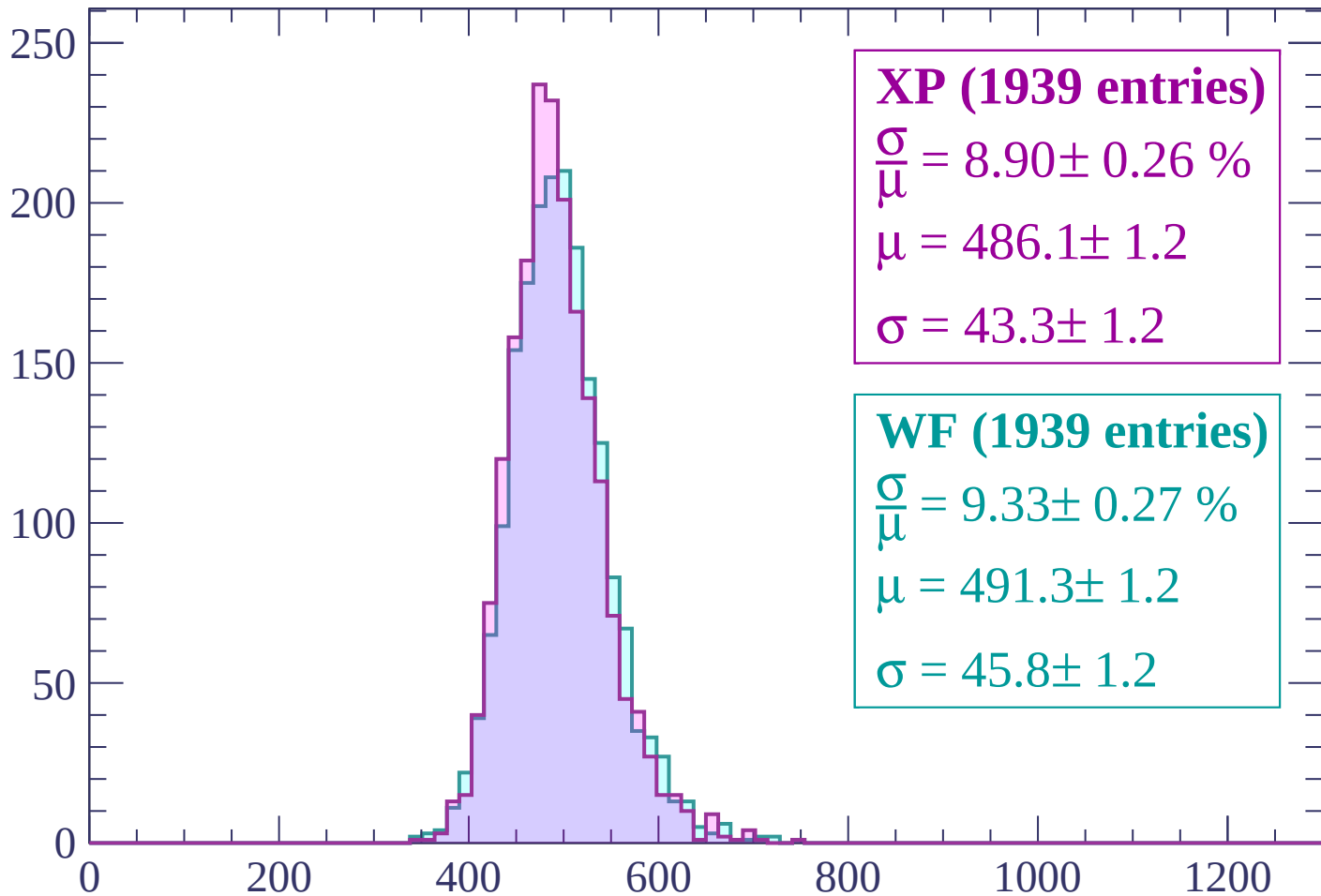
$$\mu = 492.6 \pm 1.4$$

$$\sigma = 48.7 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count



XP (1939 entries)

$$\frac{\sigma}{\mu} = 8.90 \pm 0.26 \%$$

$$\mu = 486.1 \pm 1.2$$

$$\sigma = 43.3 \pm 1.2$$

WF (1939 entries)

$$\frac{\sigma}{\mu} = 9.33 \pm 0.27 \%$$

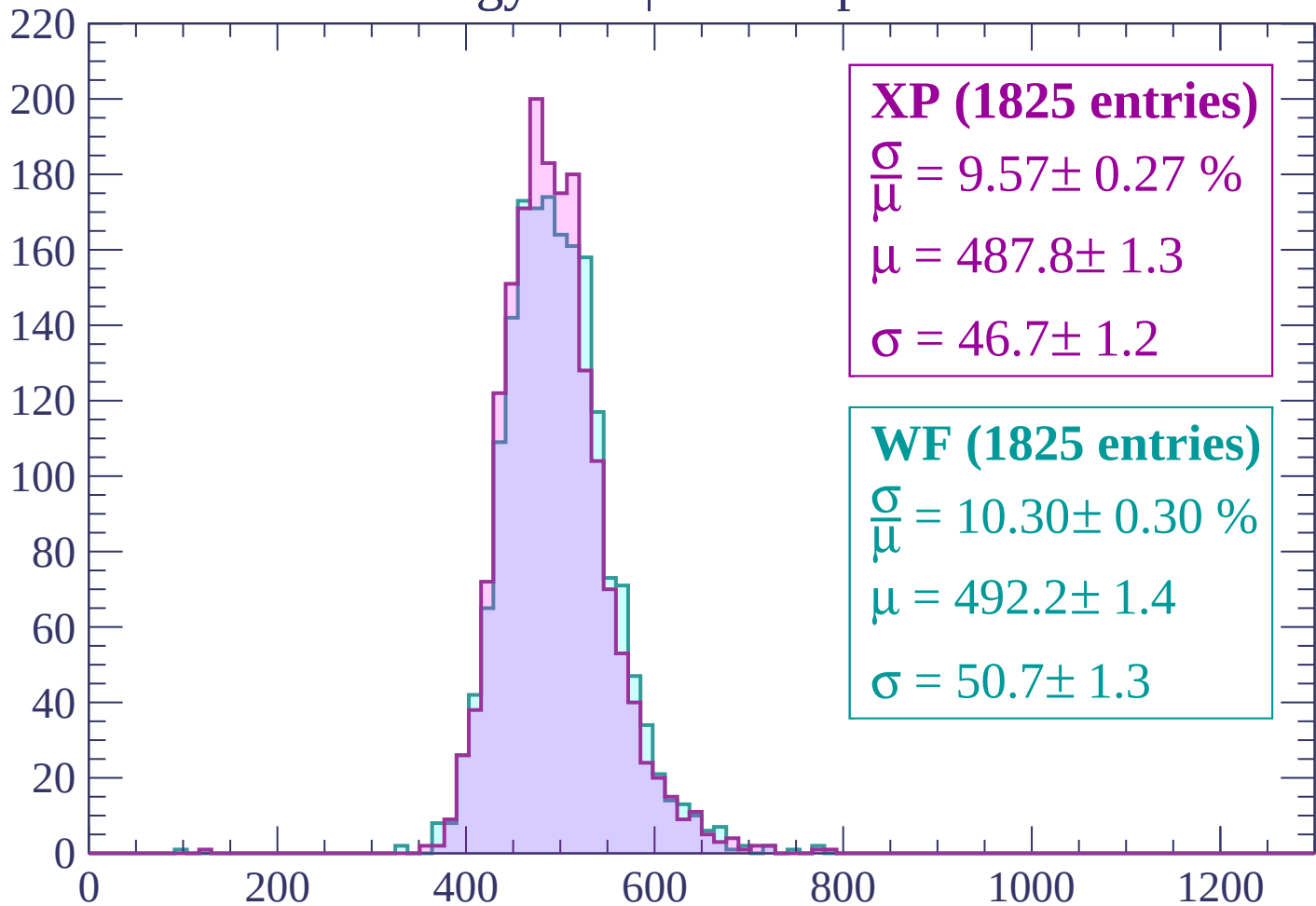
$$\mu = 491.3 \pm 1.2$$

$$\sigma = 45.8 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $2640 < p < 2700$

Count



XP (1825 entries)

$$\frac{\sigma}{\mu} = 9.57 \pm 0.27 \%$$

$$\mu = 487.8 \pm 1.3$$

$$\sigma = 46.7 \pm 1.2$$

WF (1825 entries)

$$\frac{\sigma}{\mu} = 10.30 \pm 0.30 \%$$

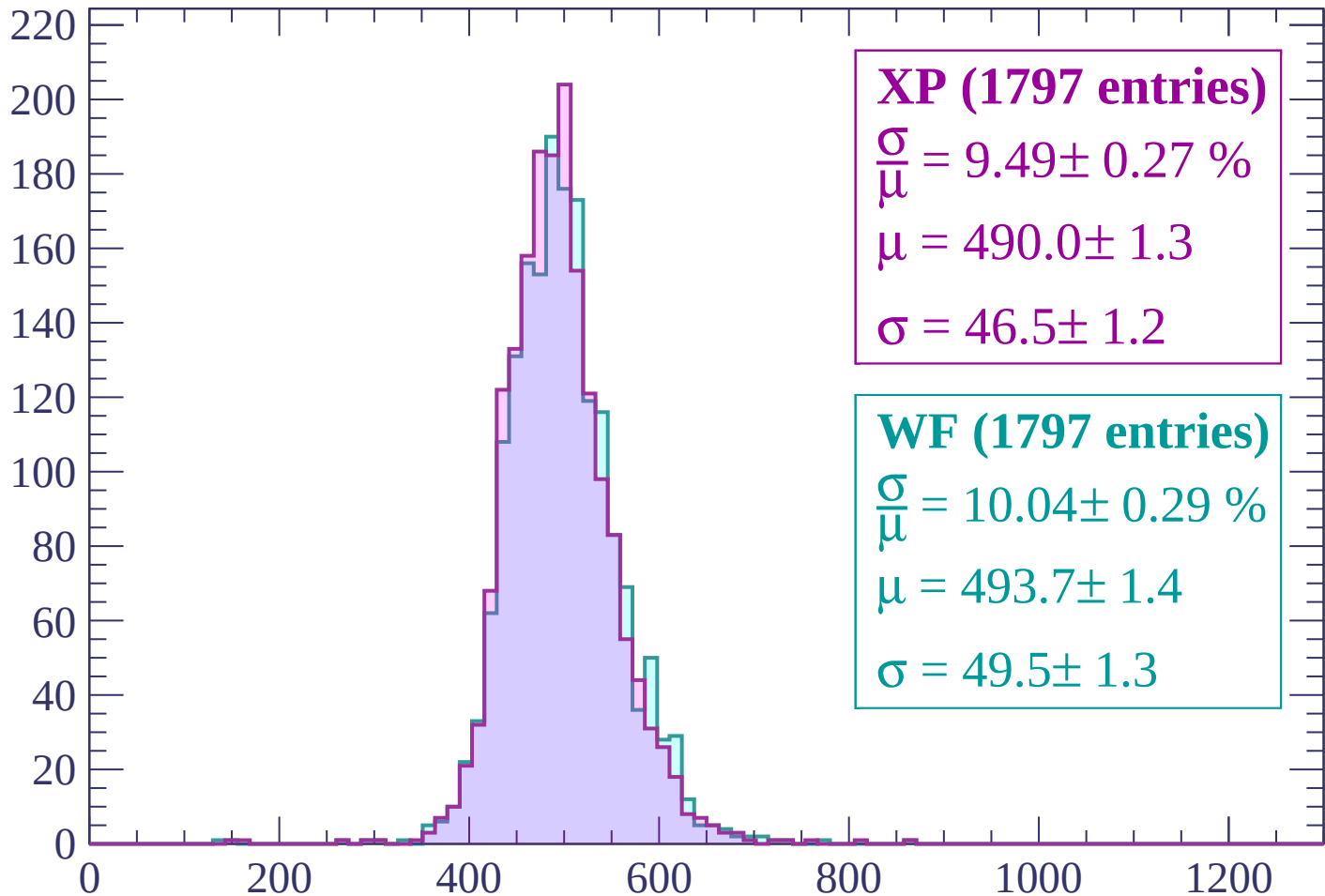
$$\mu = 492.2 \pm 1.4$$

$$\sigma = 50.7 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $2700 < p < 2760$

Count



XP (1797 entries)

$$\frac{\sigma}{\mu} = 9.49 \pm 0.27 \%$$

$$\mu = 490.0 \pm 1.3$$

$$\sigma = 46.5 \pm 1.2$$

WF (1797 entries)

$$\frac{\sigma}{\mu} = 10.04 \pm 0.29 \%$$

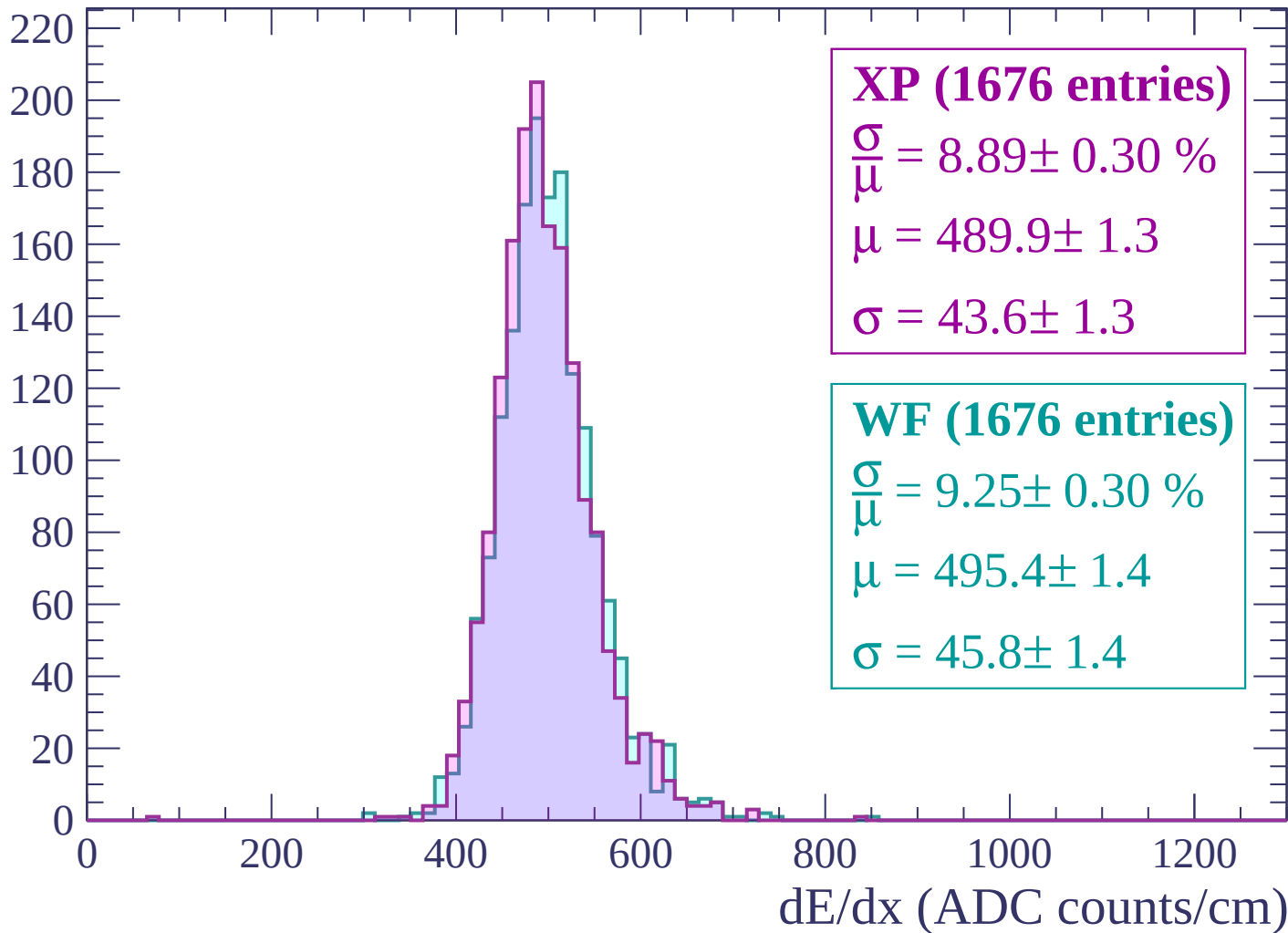
$$\mu = 493.7 \pm 1.4$$

$$\sigma = 49.5 \pm 1.3$$

dE/dx (ADC counts/cm)

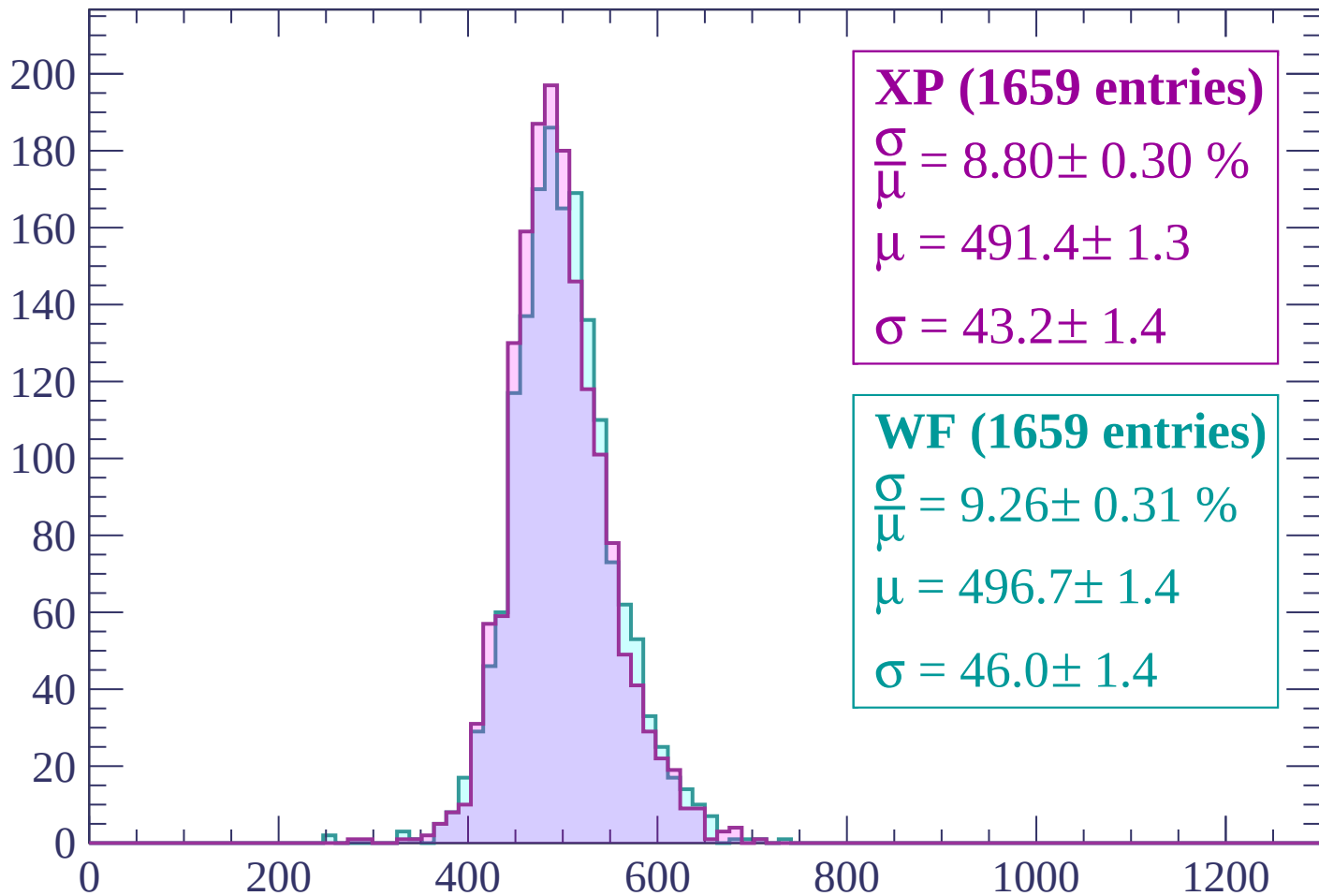
Energy loss | $2760 < p < 2820$

Count



Energy loss | $2820 < p < 2880$

Count



XP (1659 entries)

$$\frac{\sigma}{\mu} = 8.80 \pm 0.30 \%$$

$$\mu = 491.4 \pm 1.3$$

$$\sigma = 43.2 \pm 1.4$$

WF (1659 entries)

$$\frac{\sigma}{\mu} = 9.26 \pm 0.31 \%$$

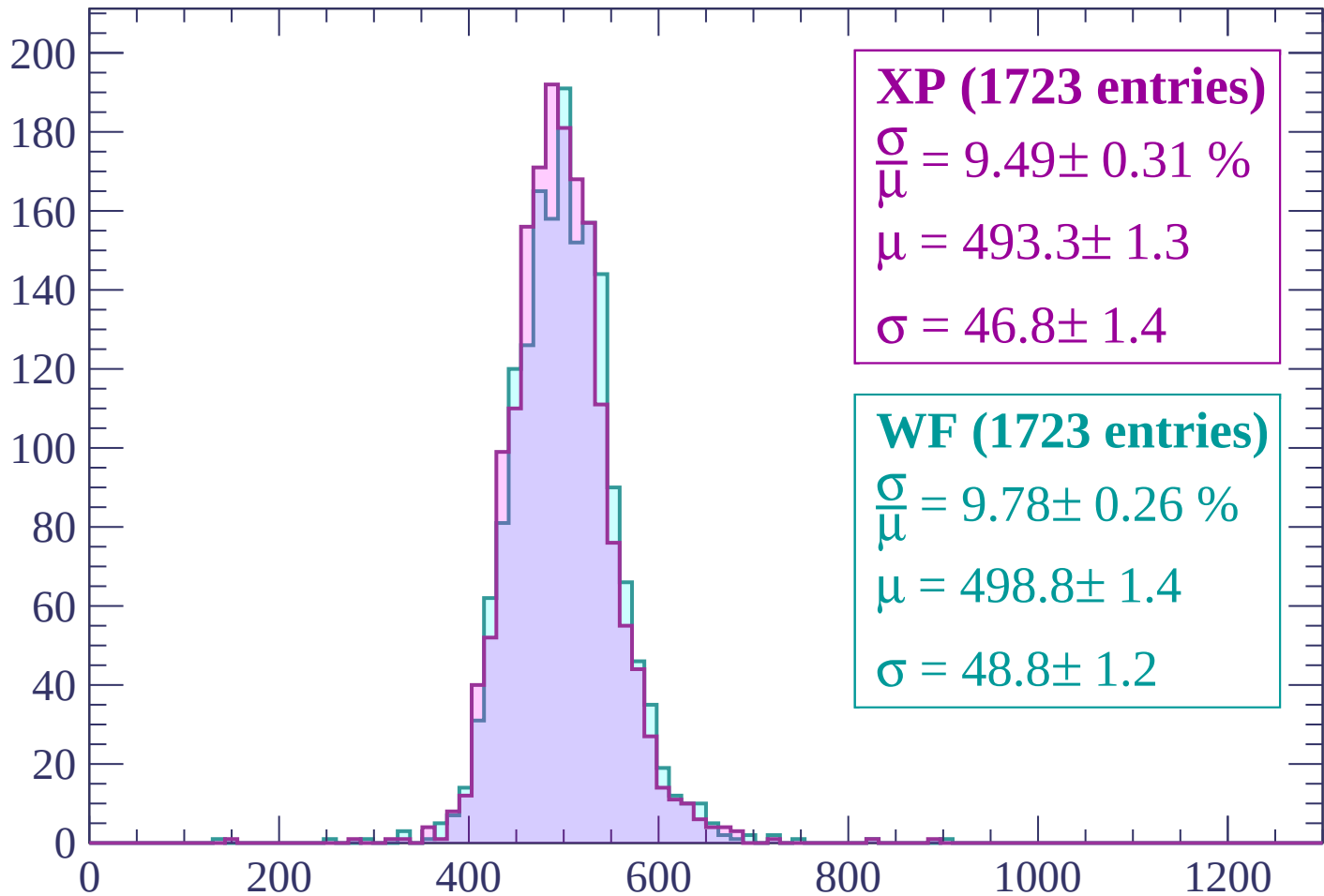
$$\mu = 496.7 \pm 1.4$$

$$\sigma = 46.0 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $2880 < p < 2940$

Count



XP (1723 entries)

$$\frac{\sigma}{\mu} = 9.49 \pm 0.31 \%$$

$$\mu = 493.3 \pm 1.3$$

$$\sigma = 46.8 \pm 1.4$$

WF (1723 entries)

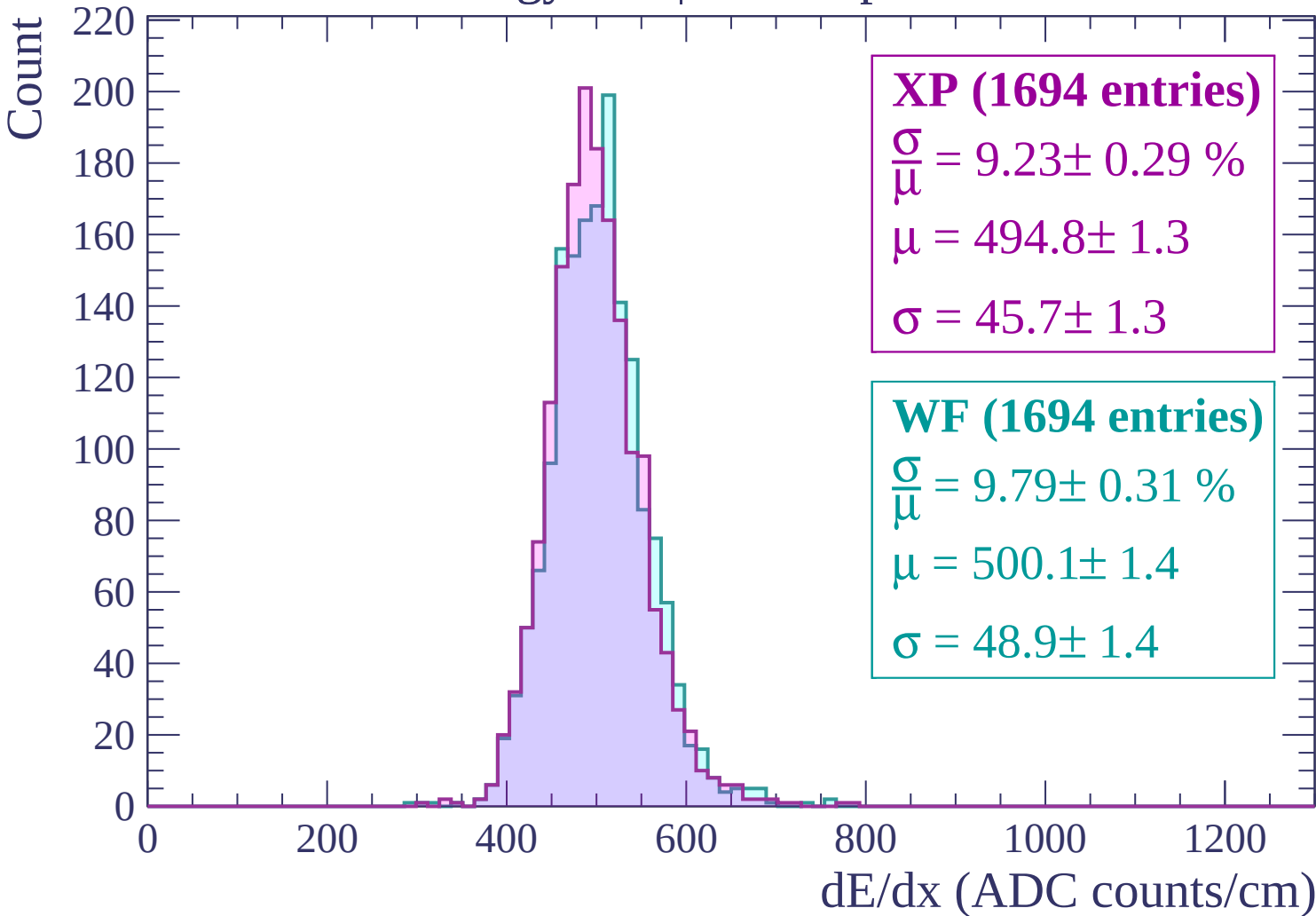
$$\frac{\sigma}{\mu} = 9.78 \pm 0.26 \%$$

$$\mu = 498.8 \pm 1.4$$

$$\sigma = 48.8 \pm 1.2$$

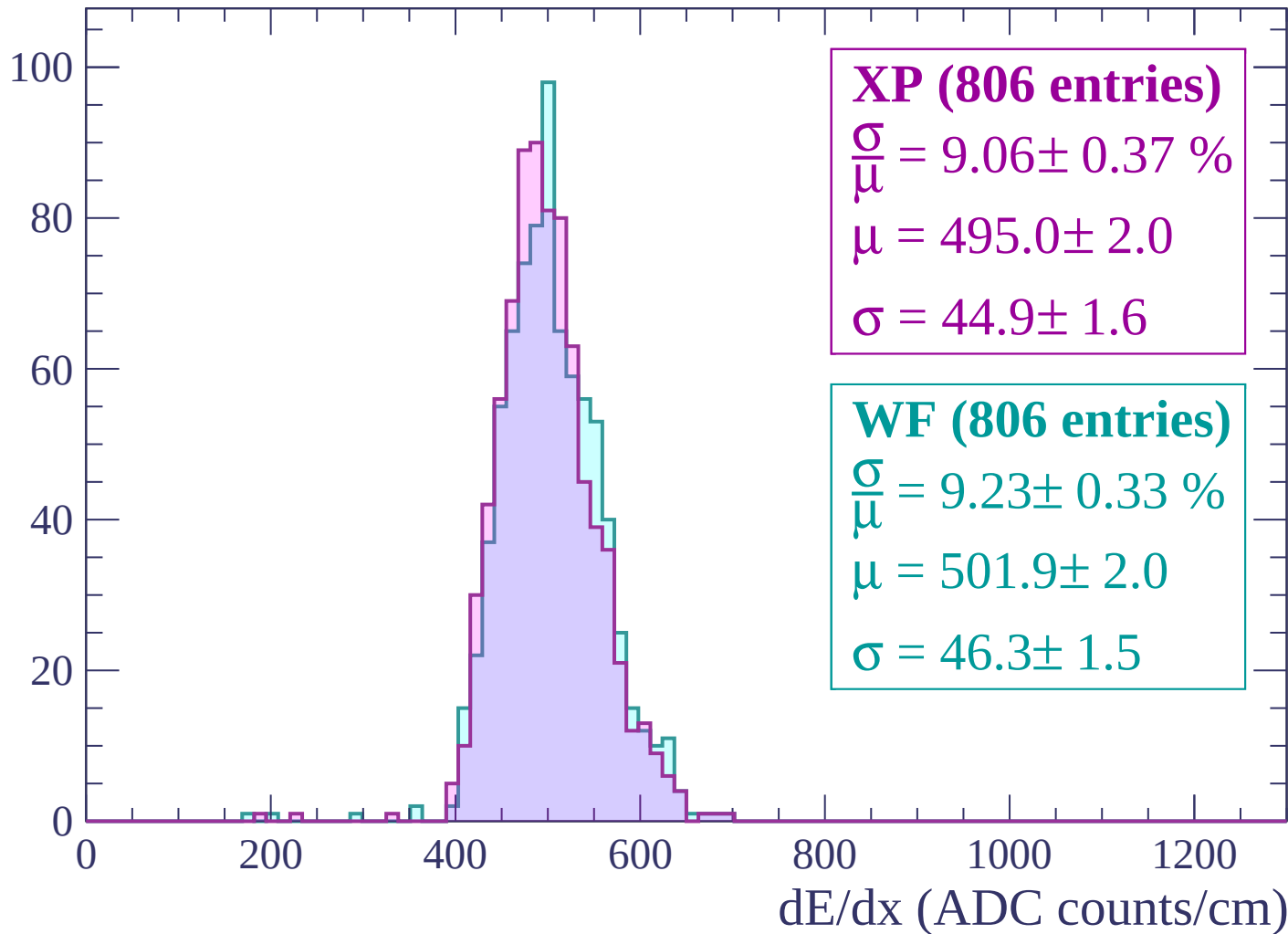
dE/dx (ADC counts/cm)

Energy loss | $2940 < p < 3000$



Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-82 < \theta < -80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

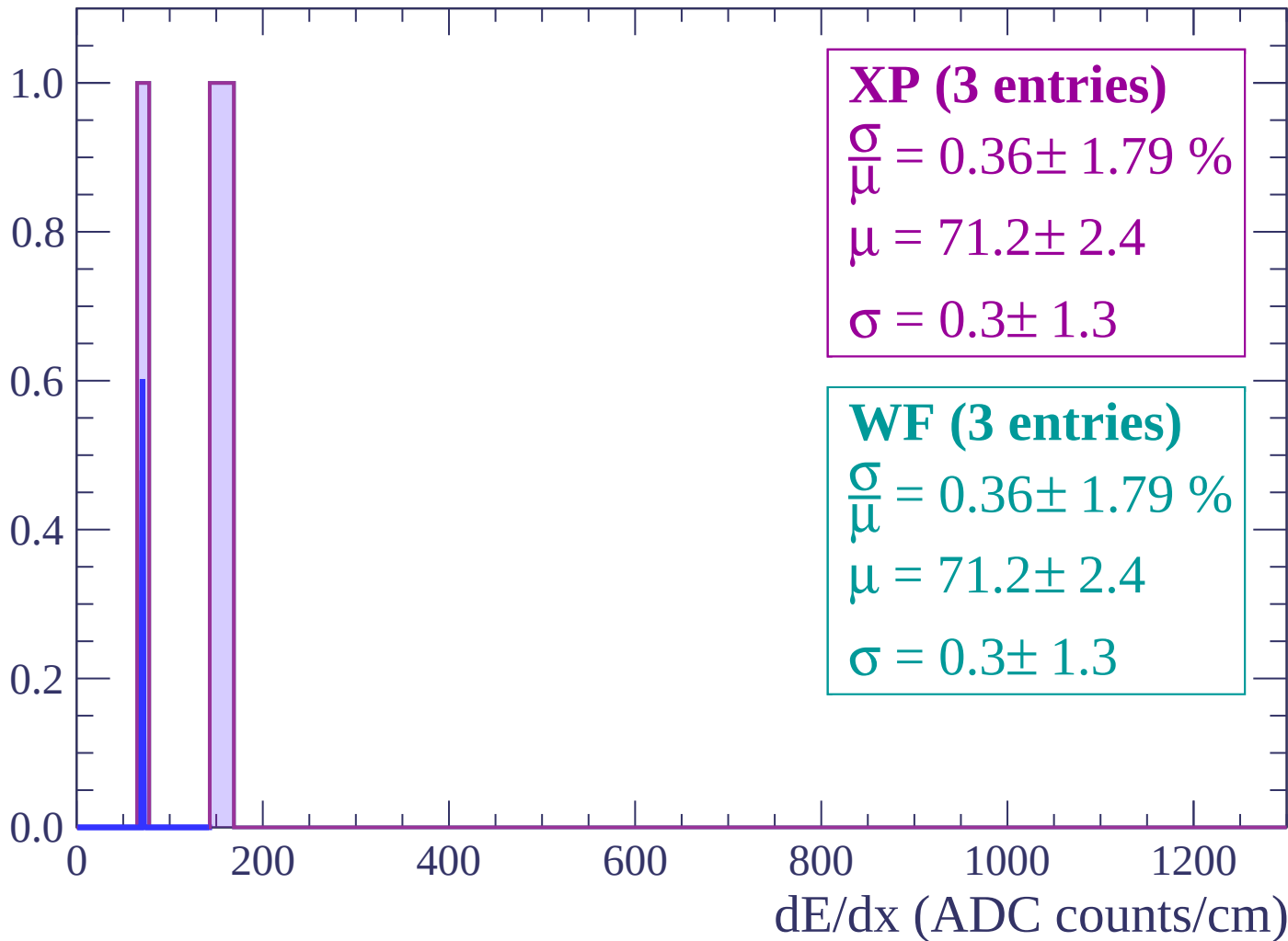
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-78 < \theta < -76$

Count



Energy loss | $-76 < \theta < -74$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

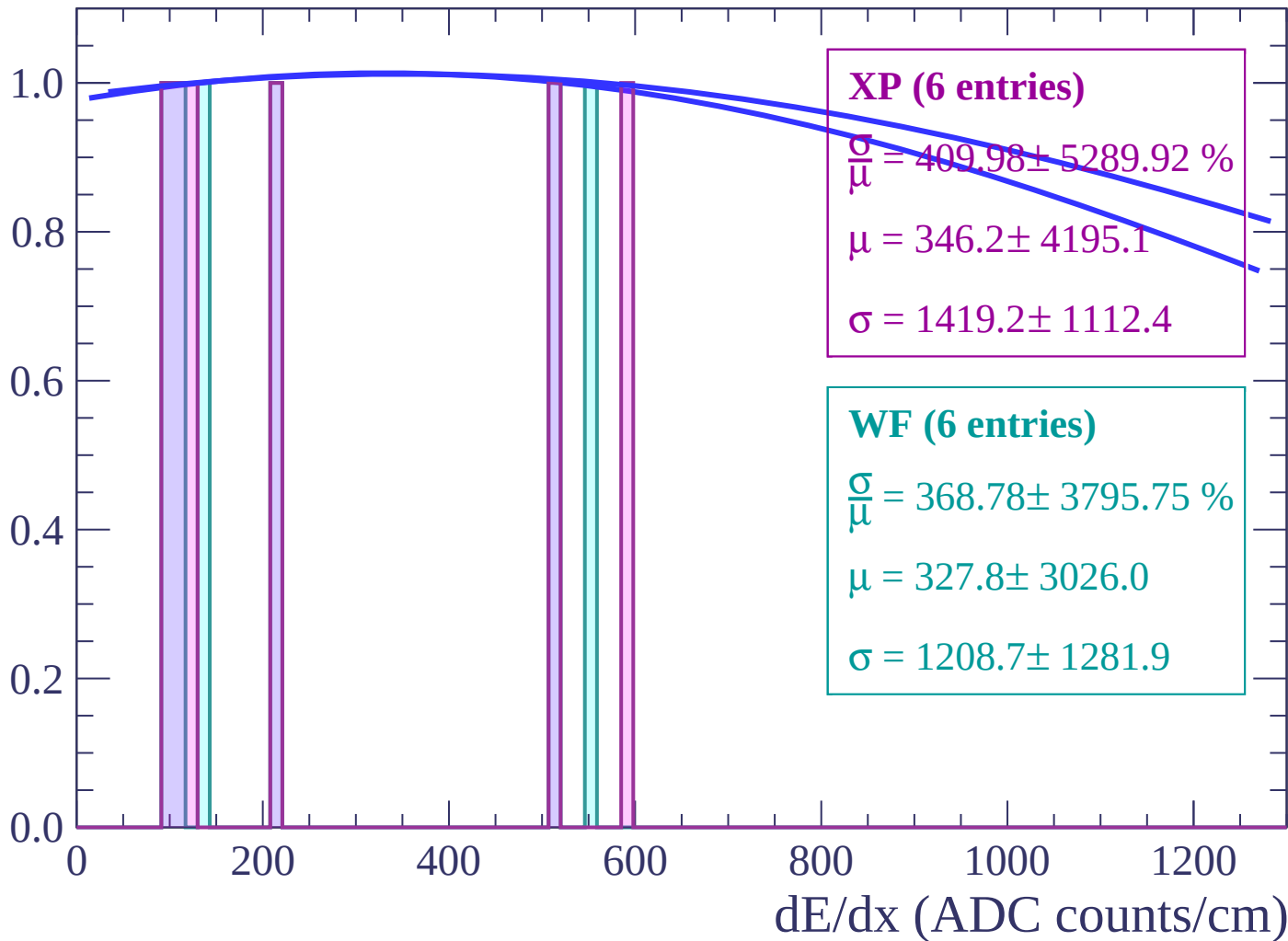
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

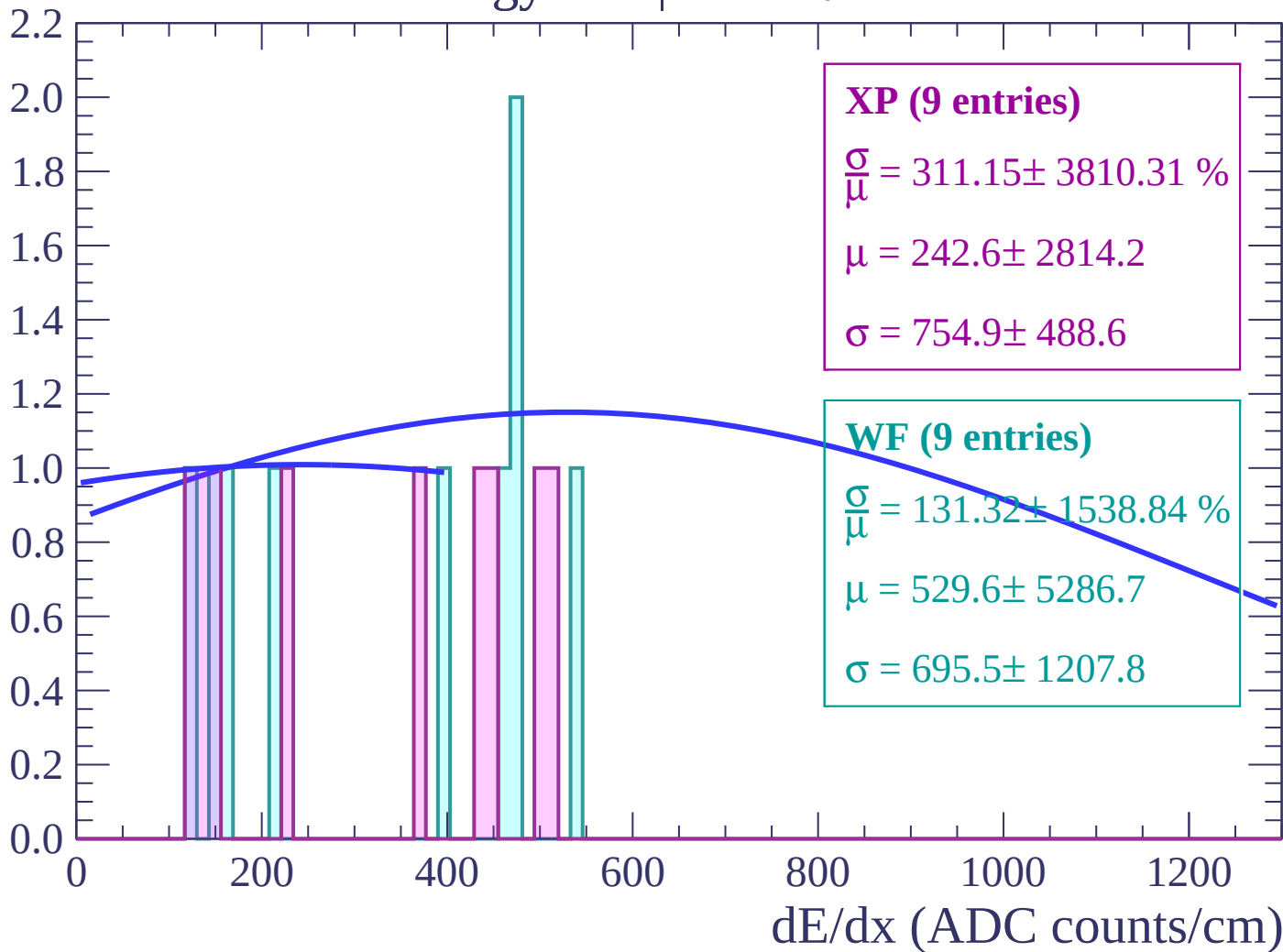
Energy loss | $-74 < \theta < -72$

Count



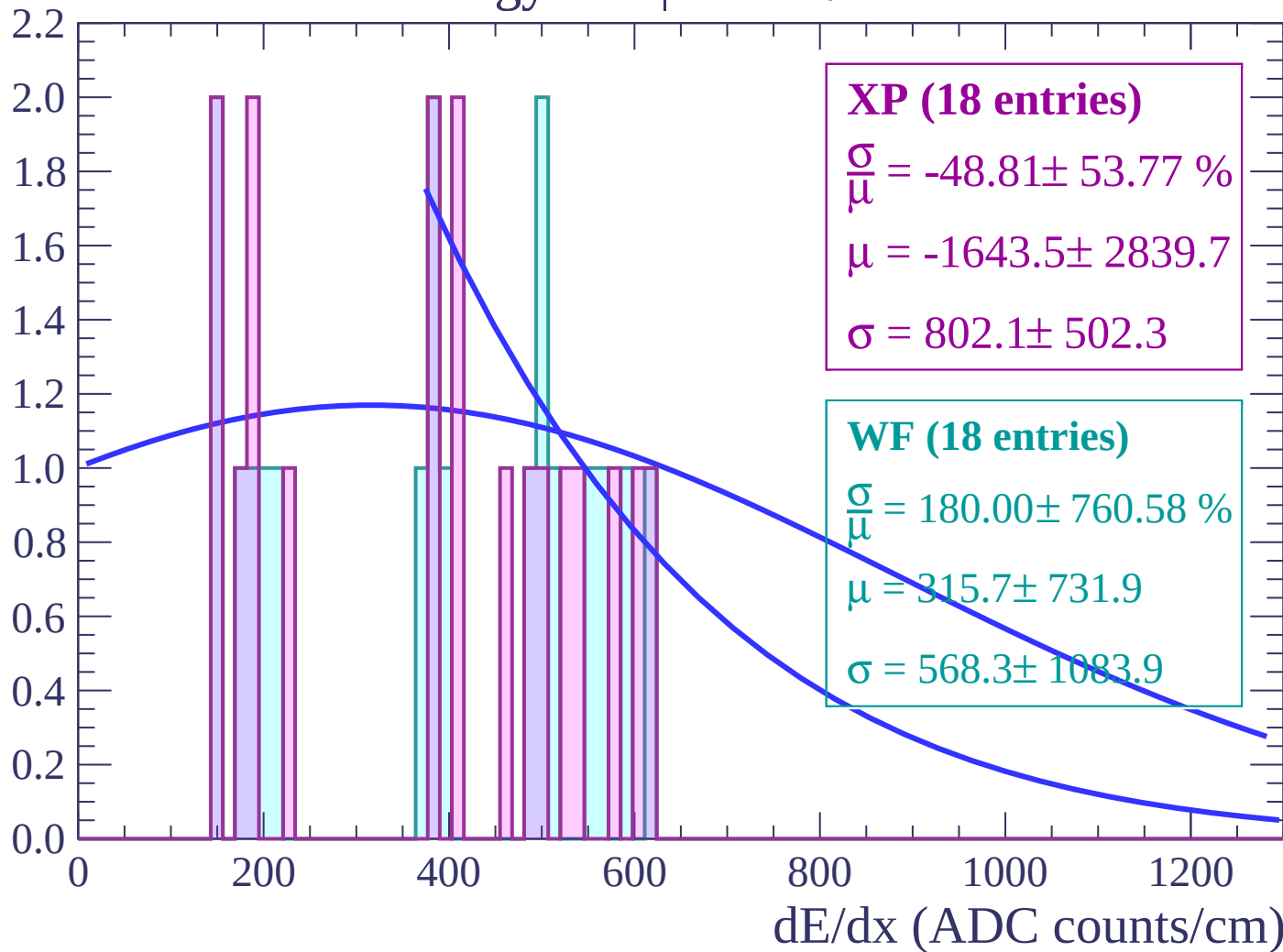
Energy loss | $-72 < \theta < -70$

Count



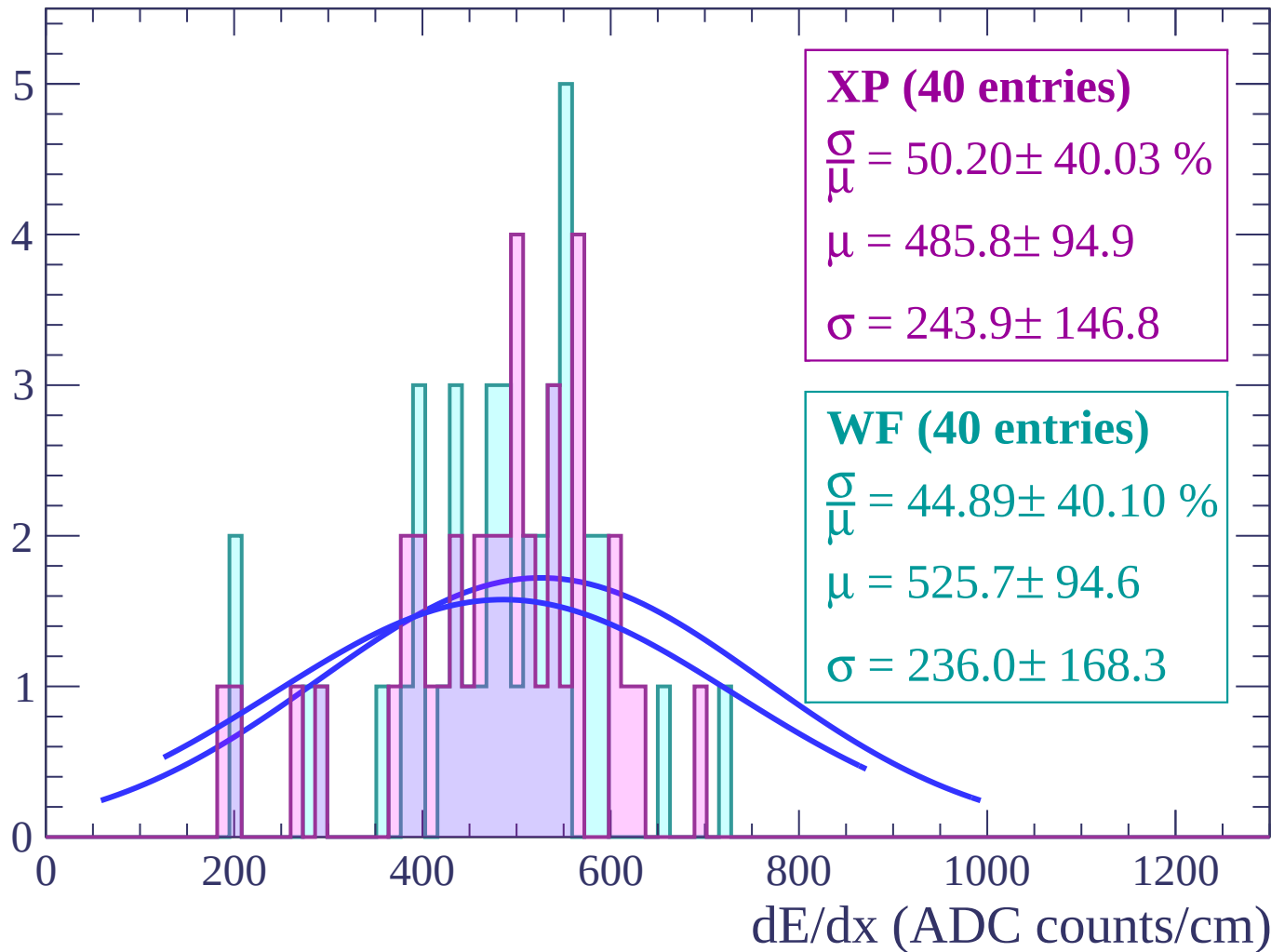
Energy loss | $-70 < \theta < -68$

Count



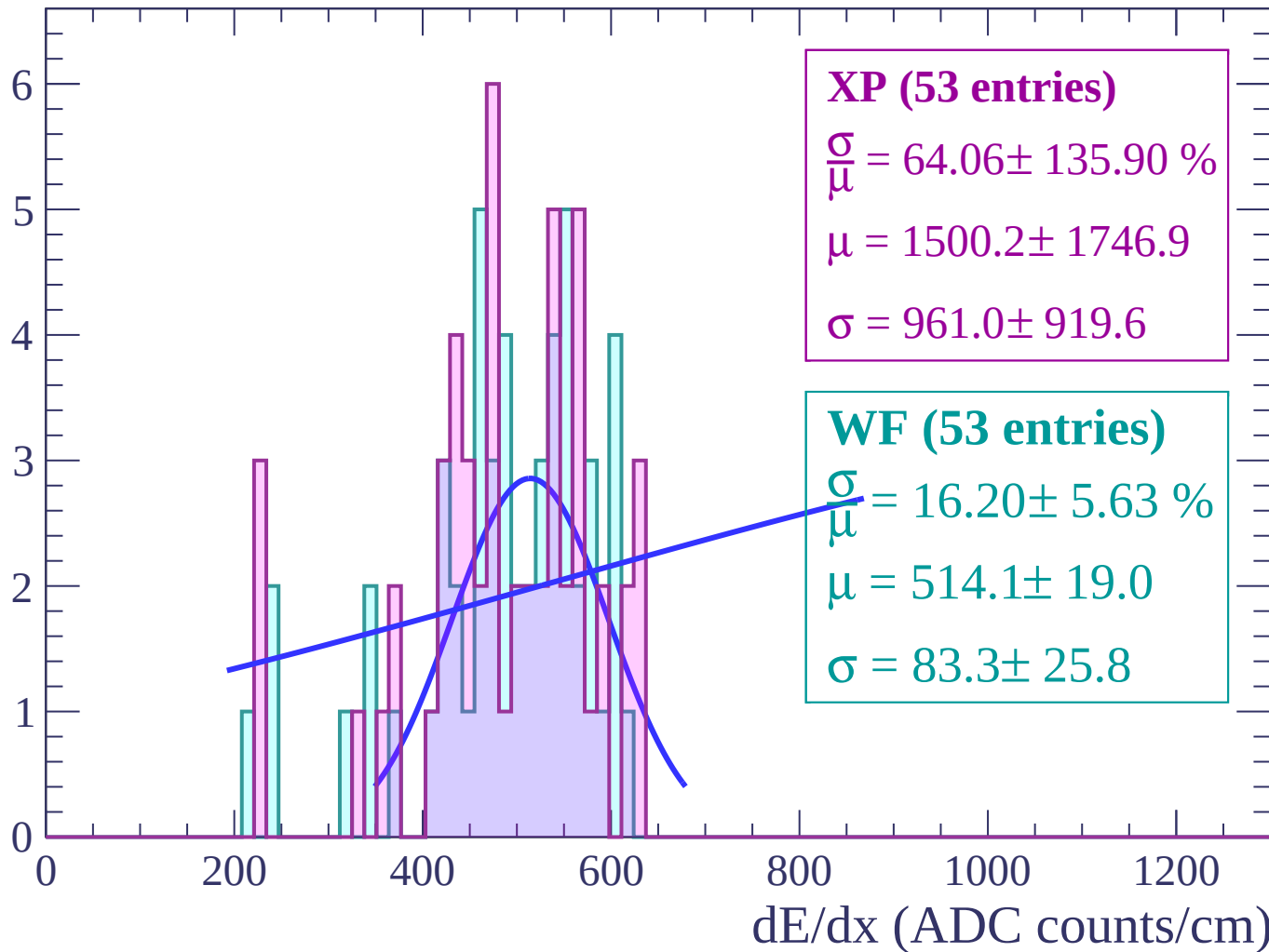
Energy loss | $-68 < \theta < -66$

Count



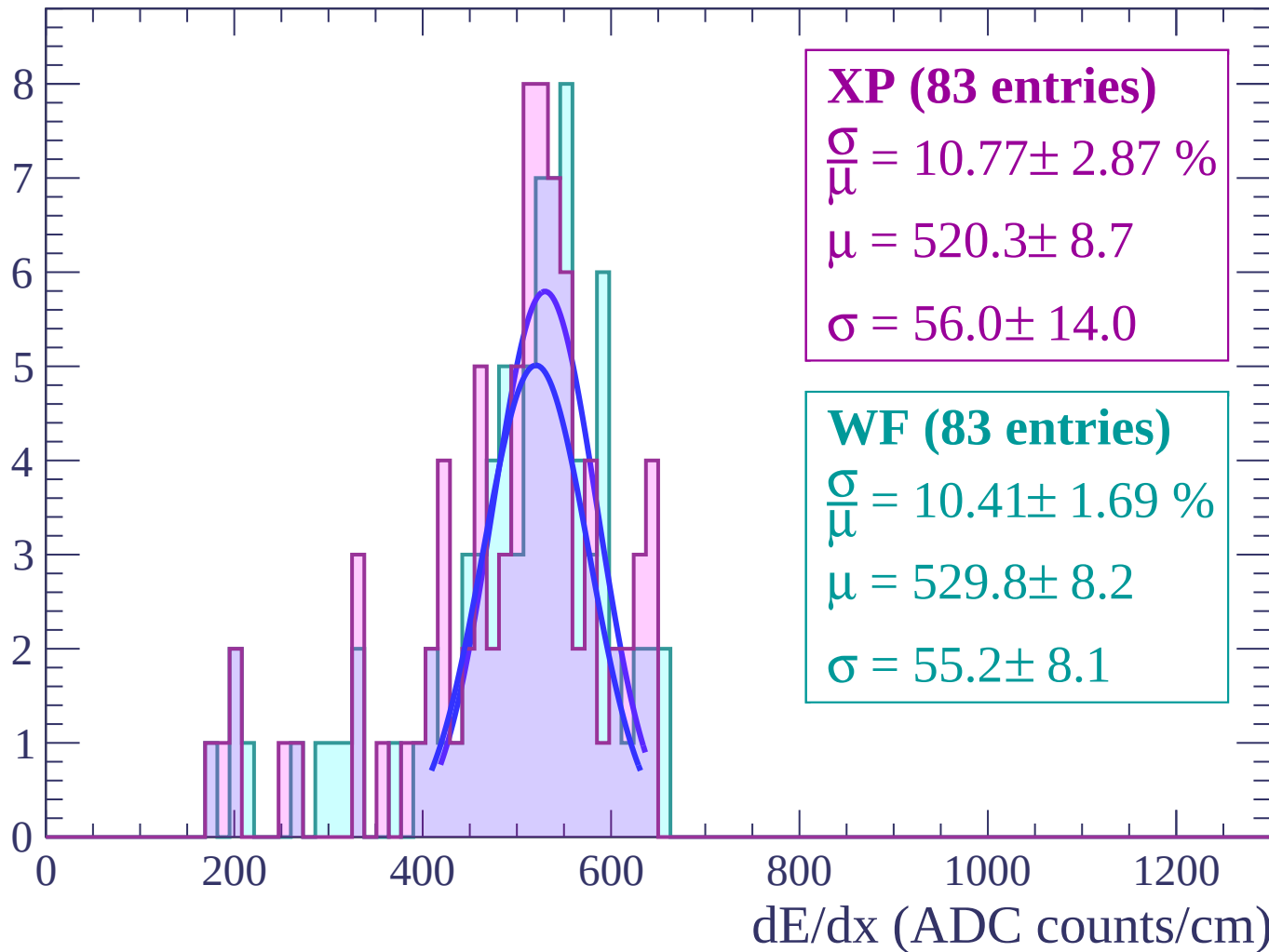
Energy loss | $-66 < \theta < -64$

Count



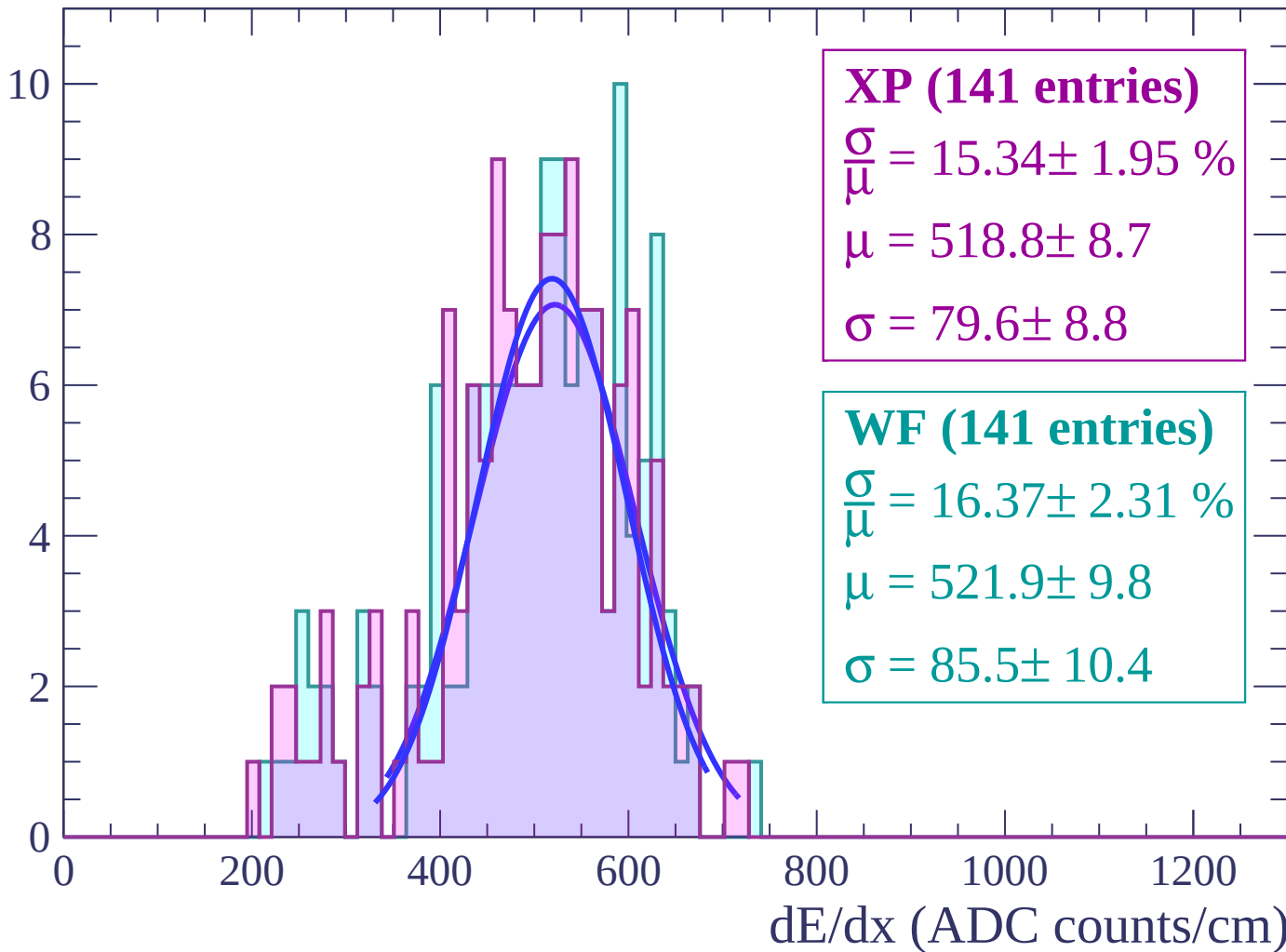
Energy loss | $-64 < \theta < -62$

Count



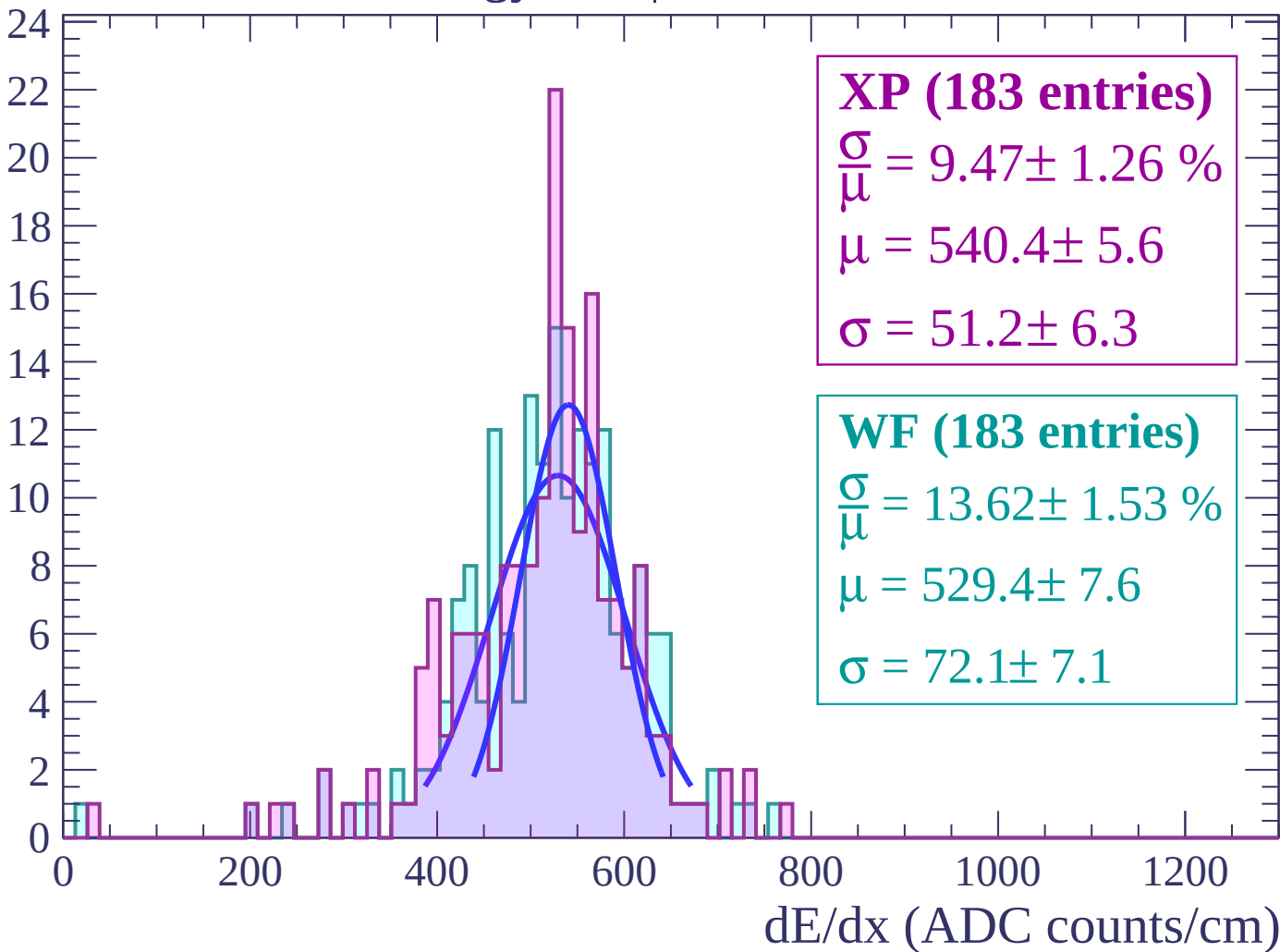
Energy loss | $-62 < \theta < -60$

Count



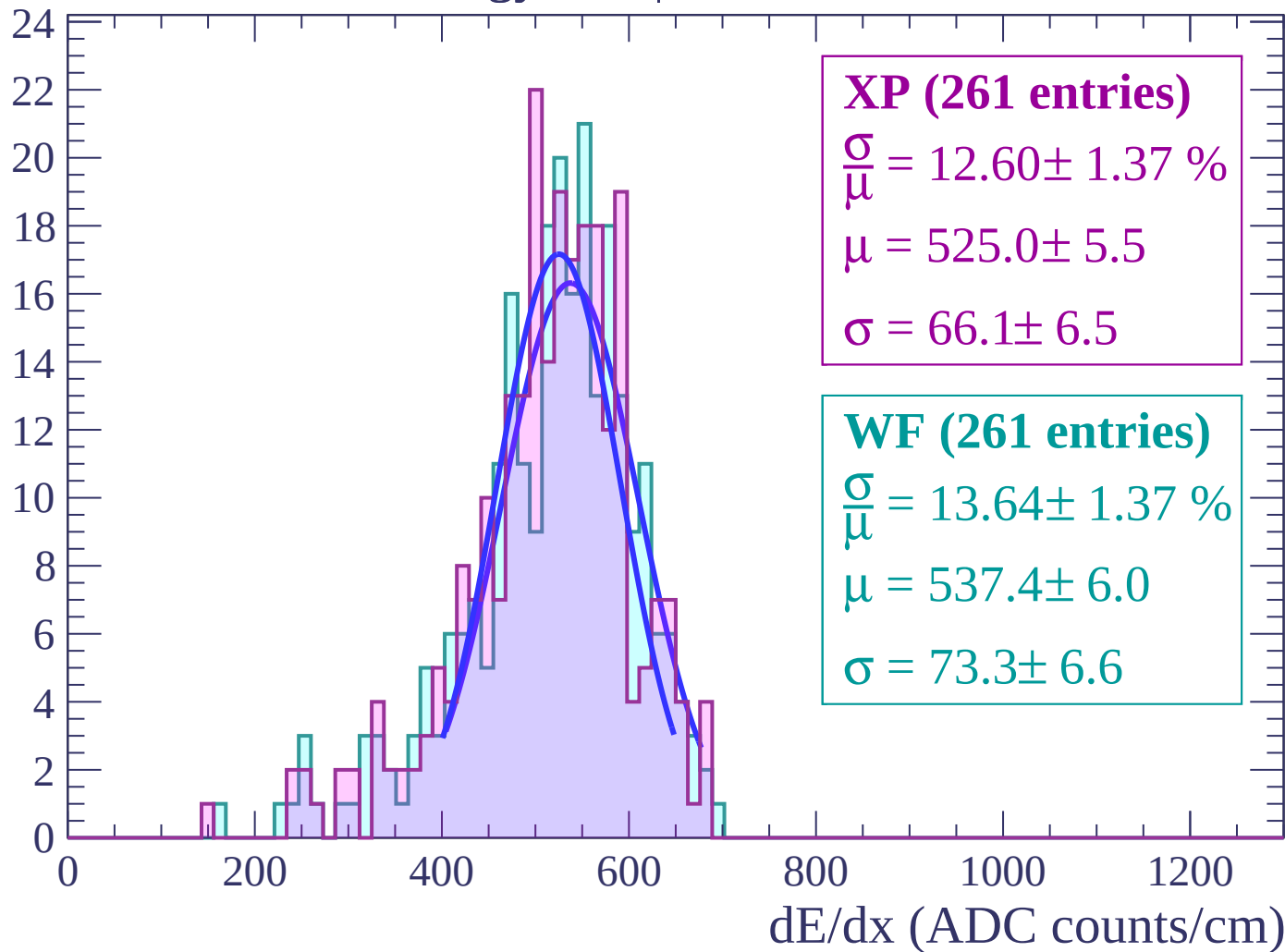
Energy loss | $-60 < \theta < -58$

Count



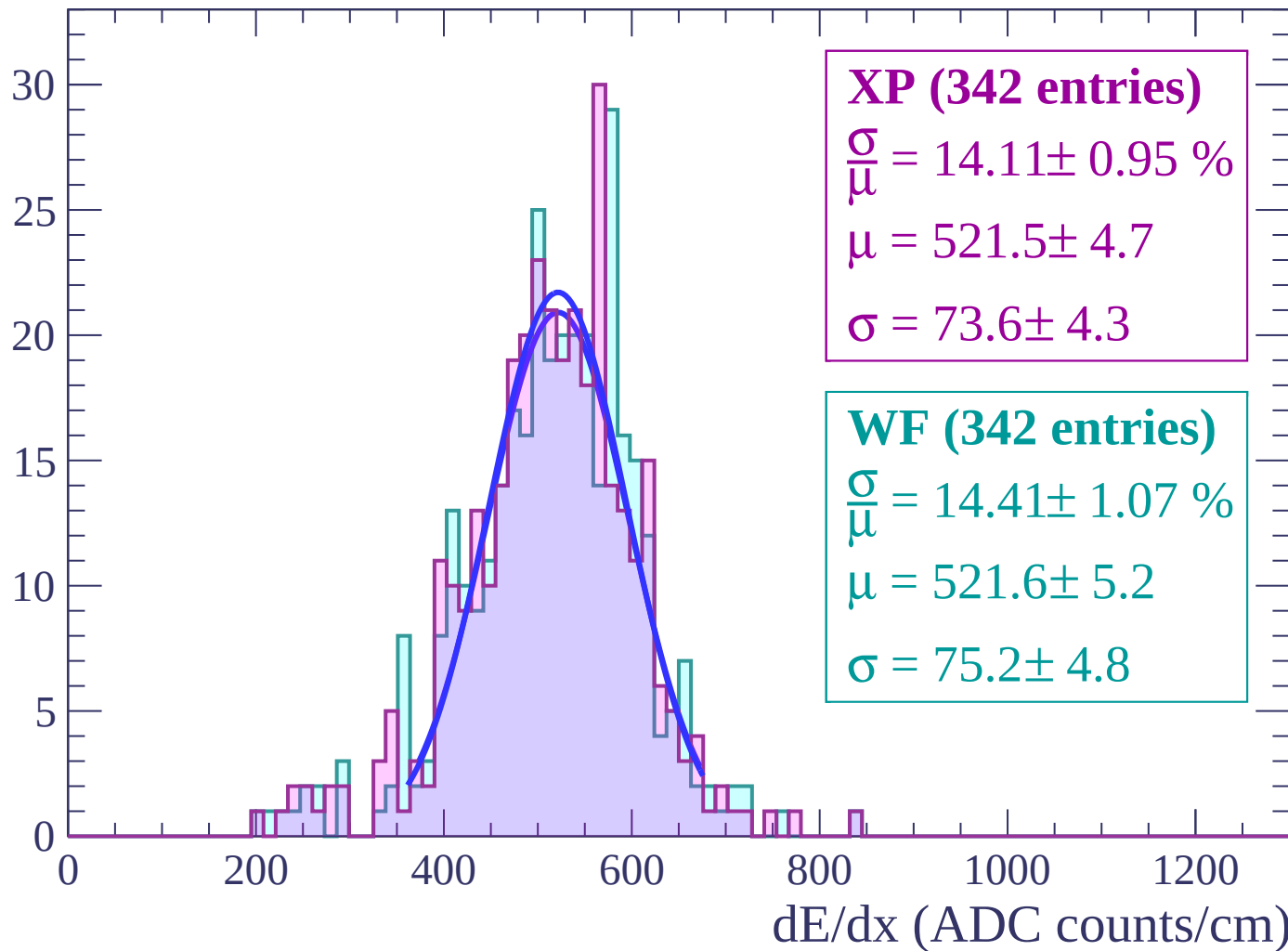
Energy loss | $-58 < \theta < -56$

Count



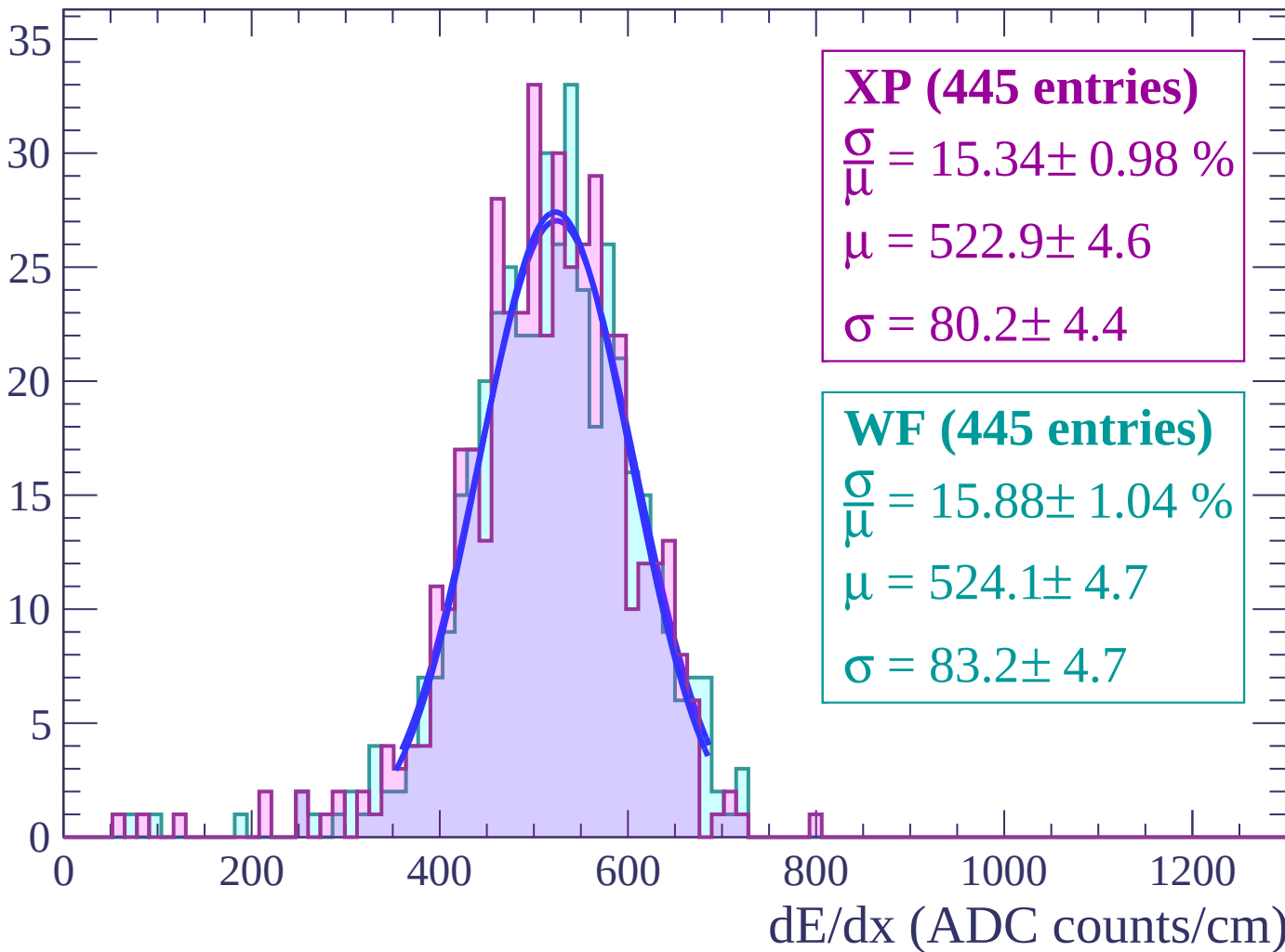
Energy loss | $-56 < \theta < -54$

Count



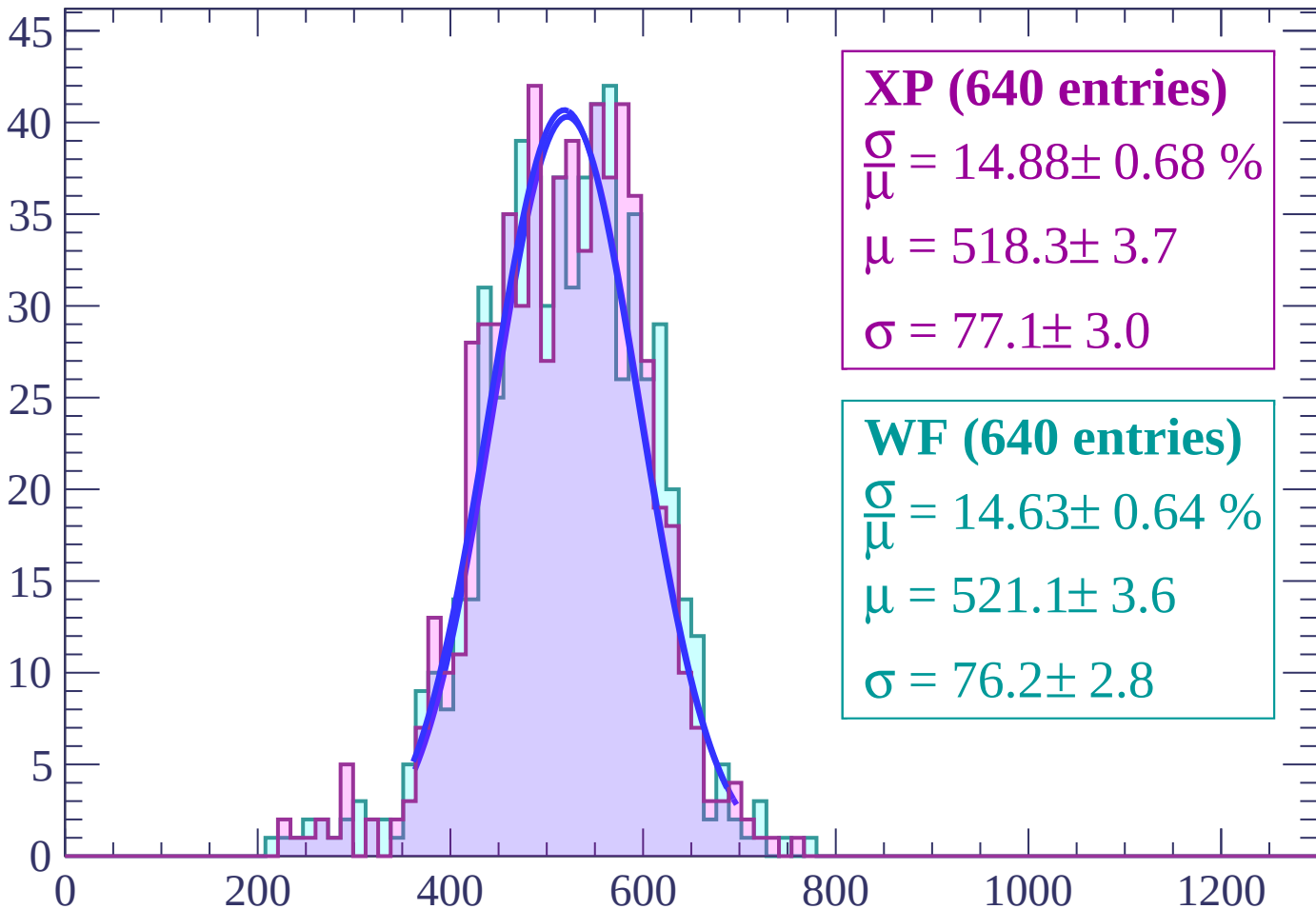
Energy loss | $-54 < \theta < -52$

Count



Energy loss | $-52 < \theta < -50$

Count



XP (640 entries)

$$\frac{\sigma}{\mu} = 14.88 \pm 0.68 \%$$

$$\mu = 518.3 \pm 3.7$$

$$\sigma = 77.1 \pm 3.0$$

WF (640 entries)

$$\frac{\sigma}{\mu} = 14.63 \pm 0.64 \%$$

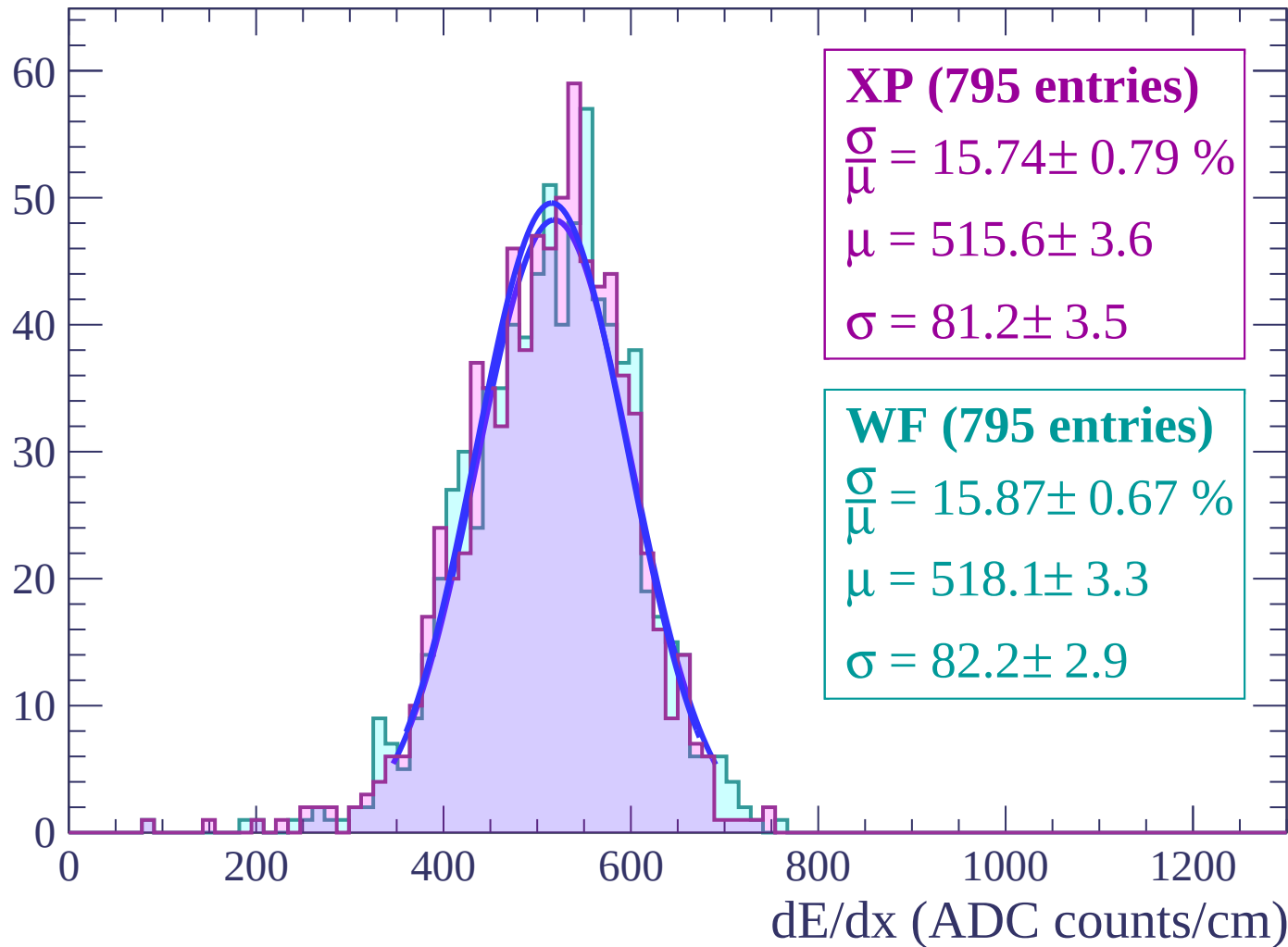
$$\mu = 521.1 \pm 3.6$$

$$\sigma = 76.2 \pm 2.8$$

dE/dx (ADC counts/cm)

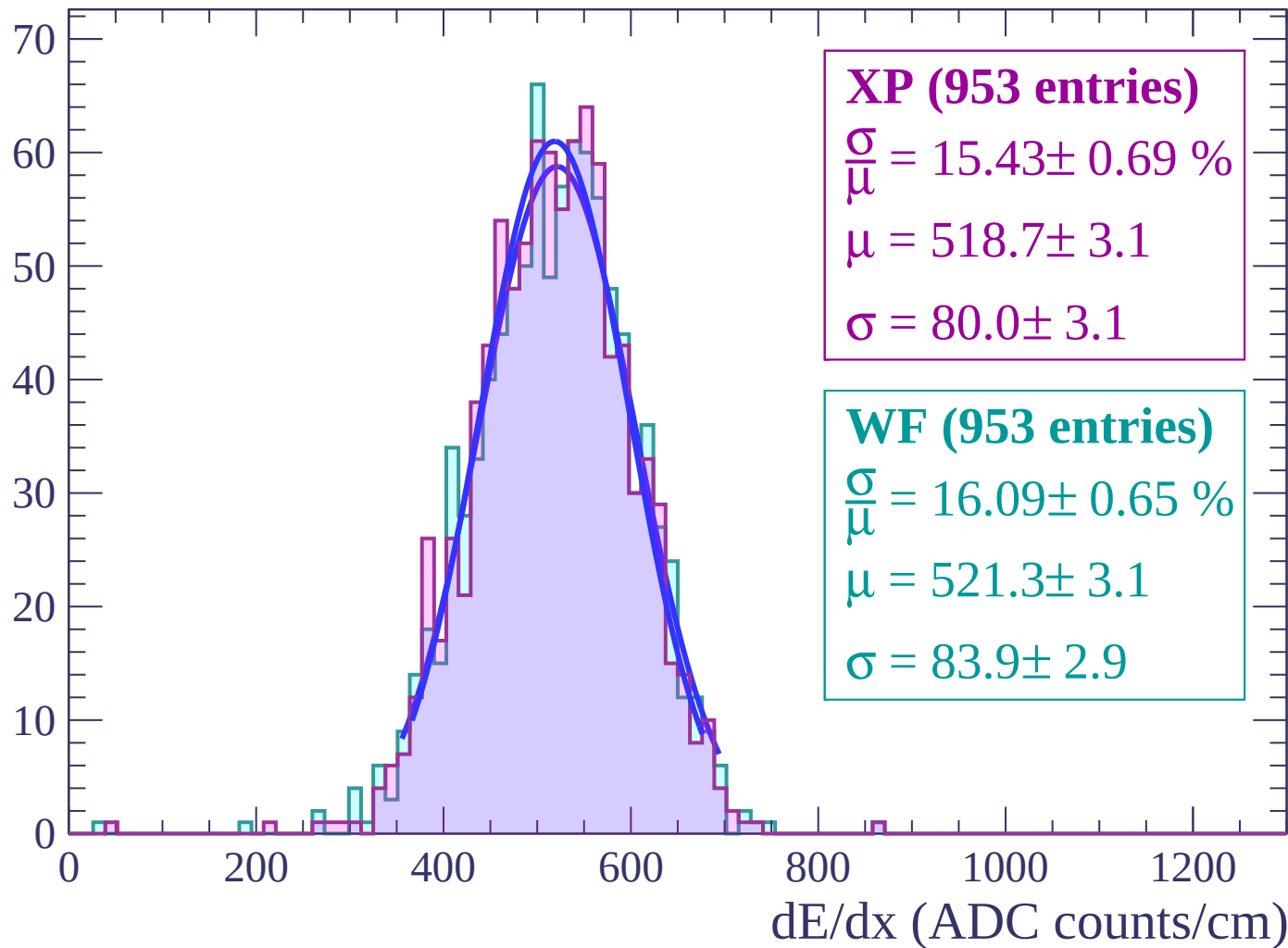
Energy loss | $-50 < \theta < -48$

Count



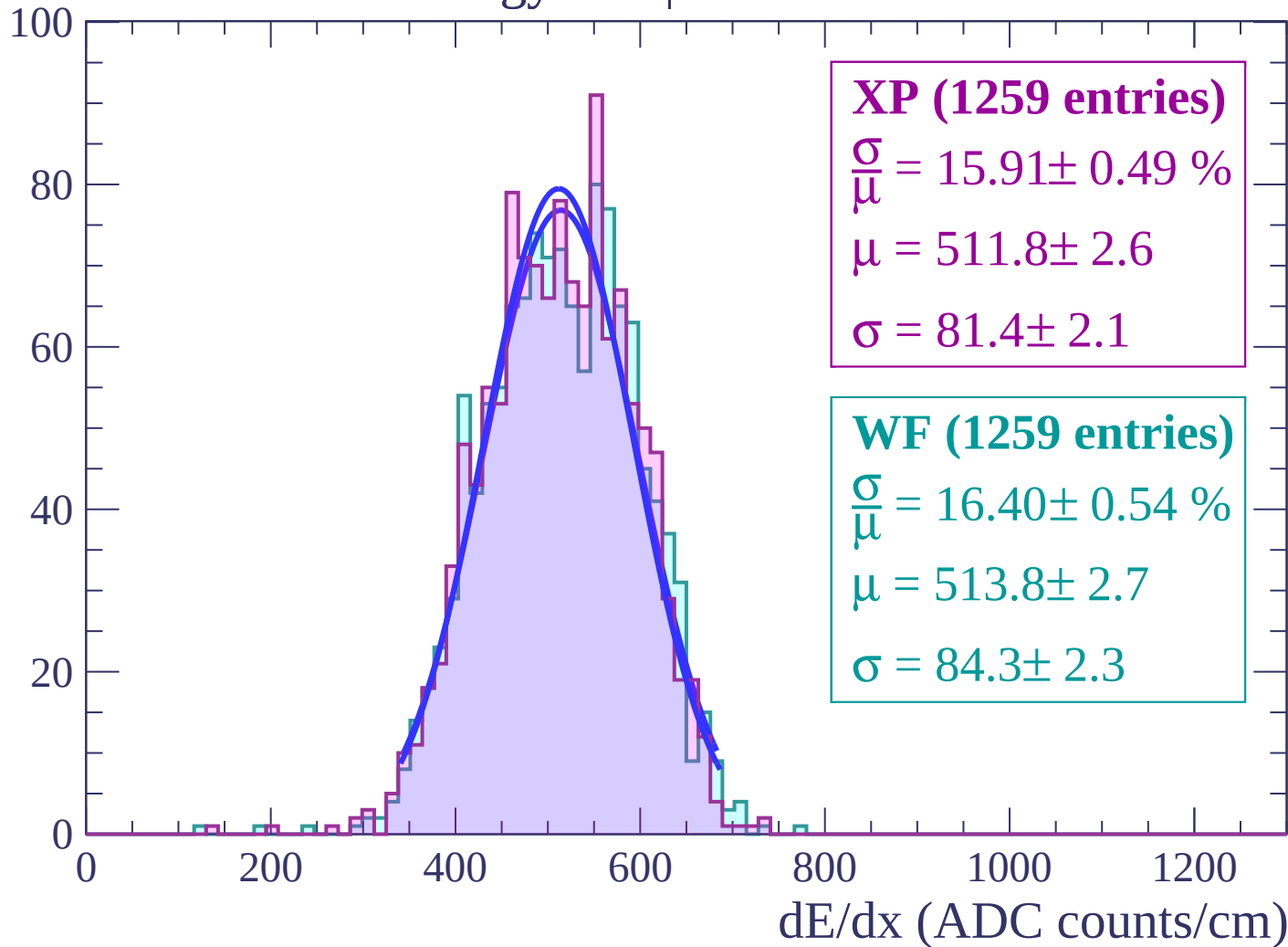
Energy loss | $-48 < \theta < -46$

Count



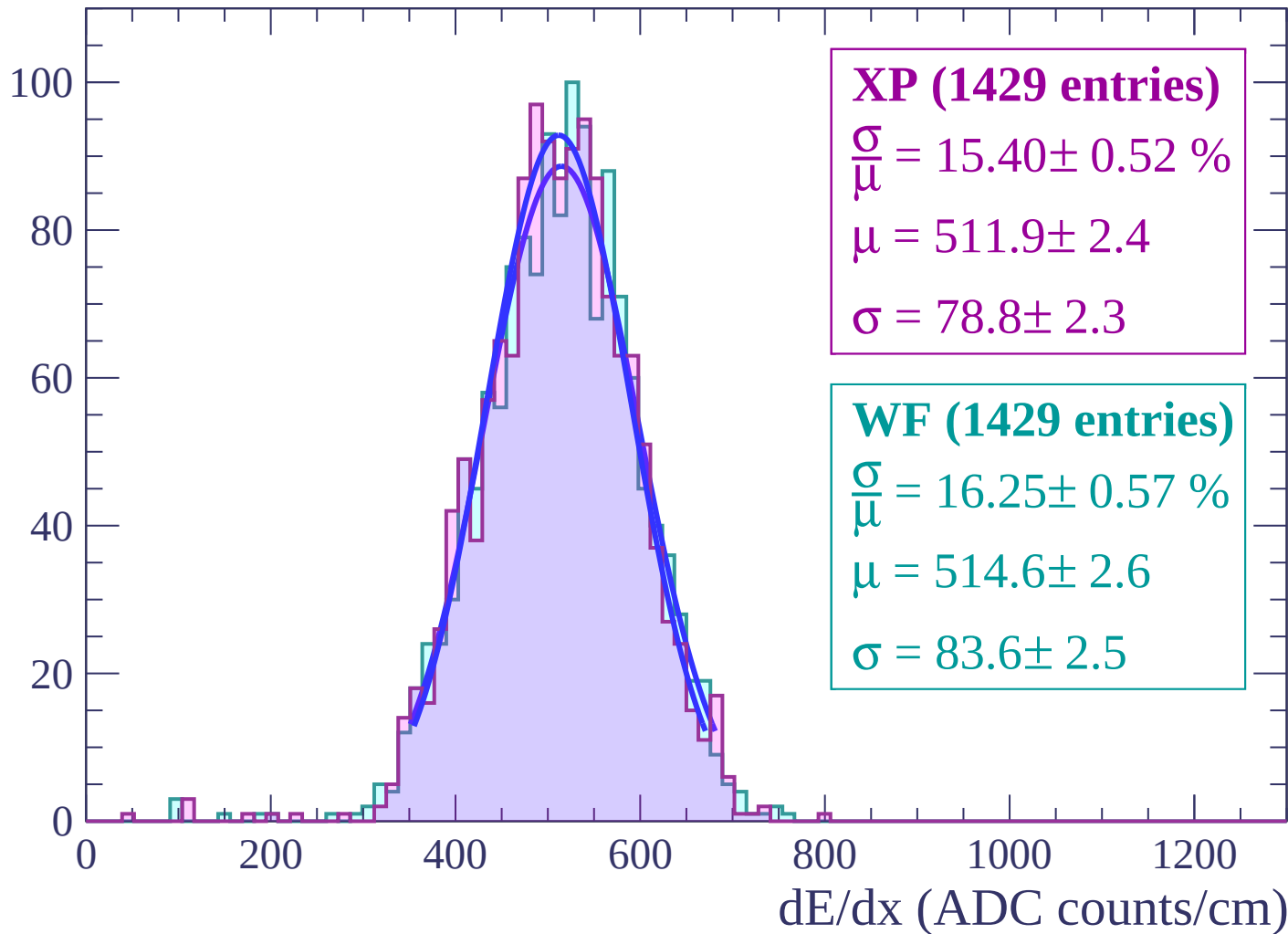
Energy loss | $-46 < \theta < -44$

Count



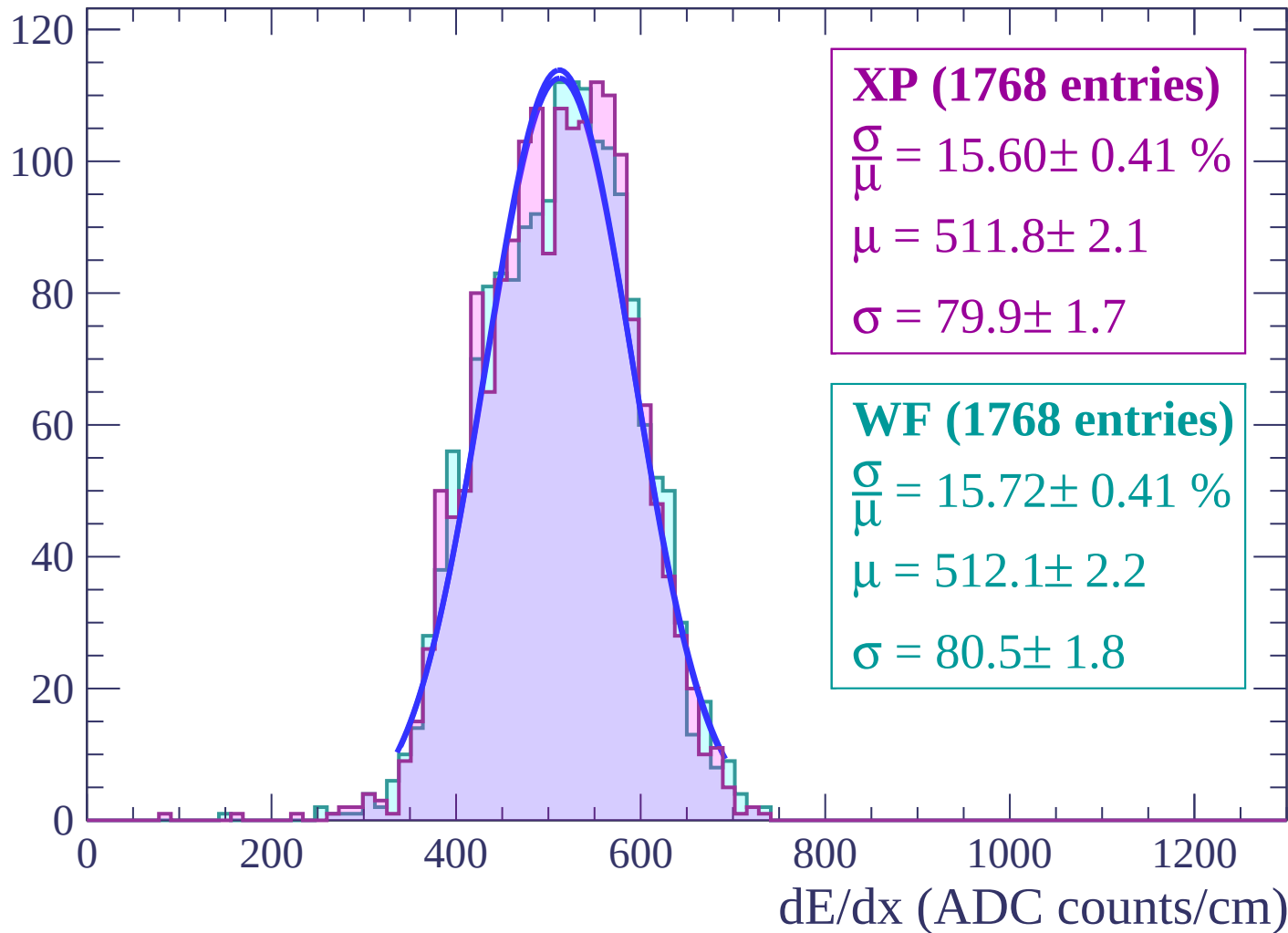
Energy loss | $-44 < \theta < -42$

Count



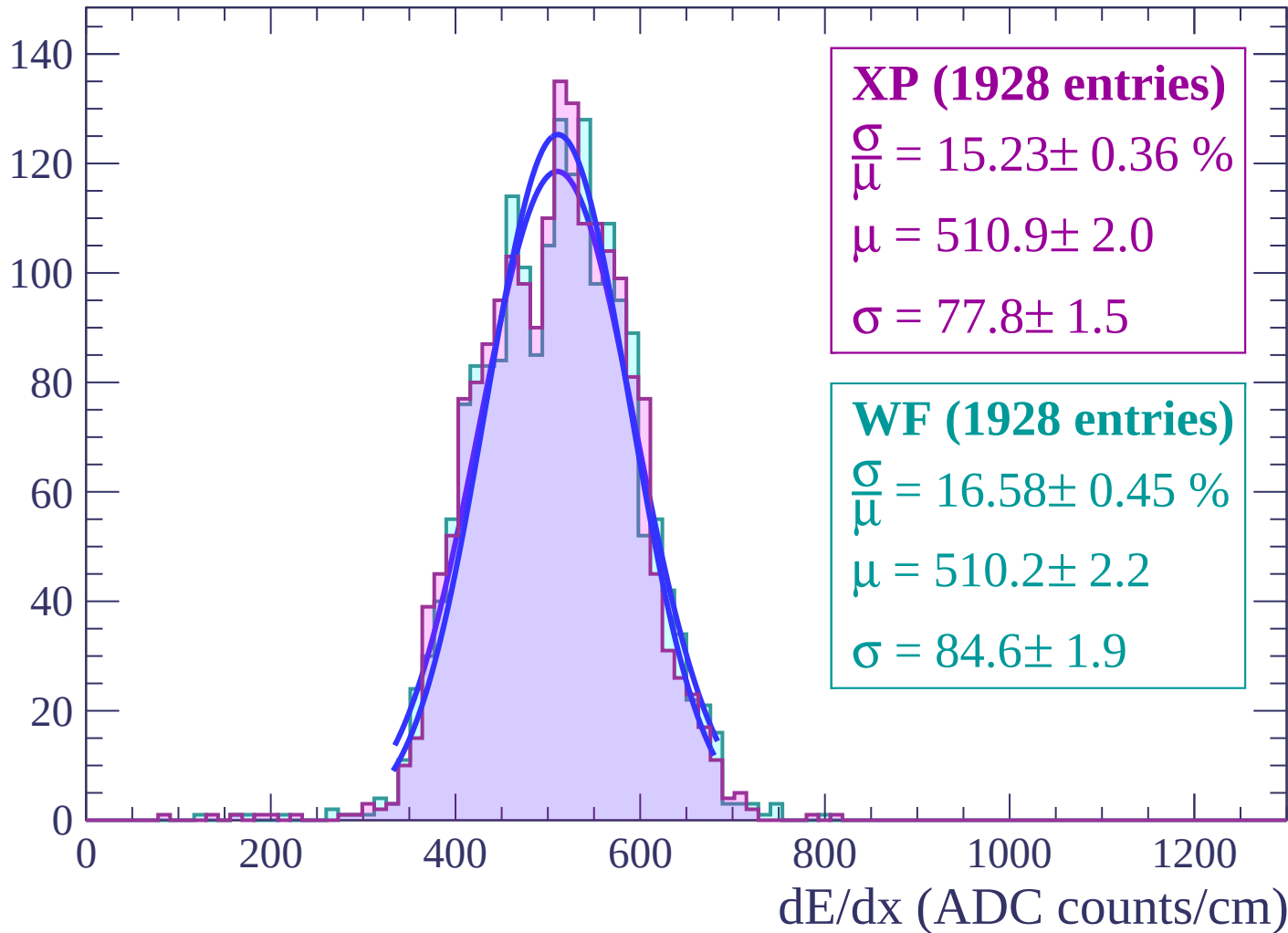
Energy loss | $-42 < \theta < -40$

Count



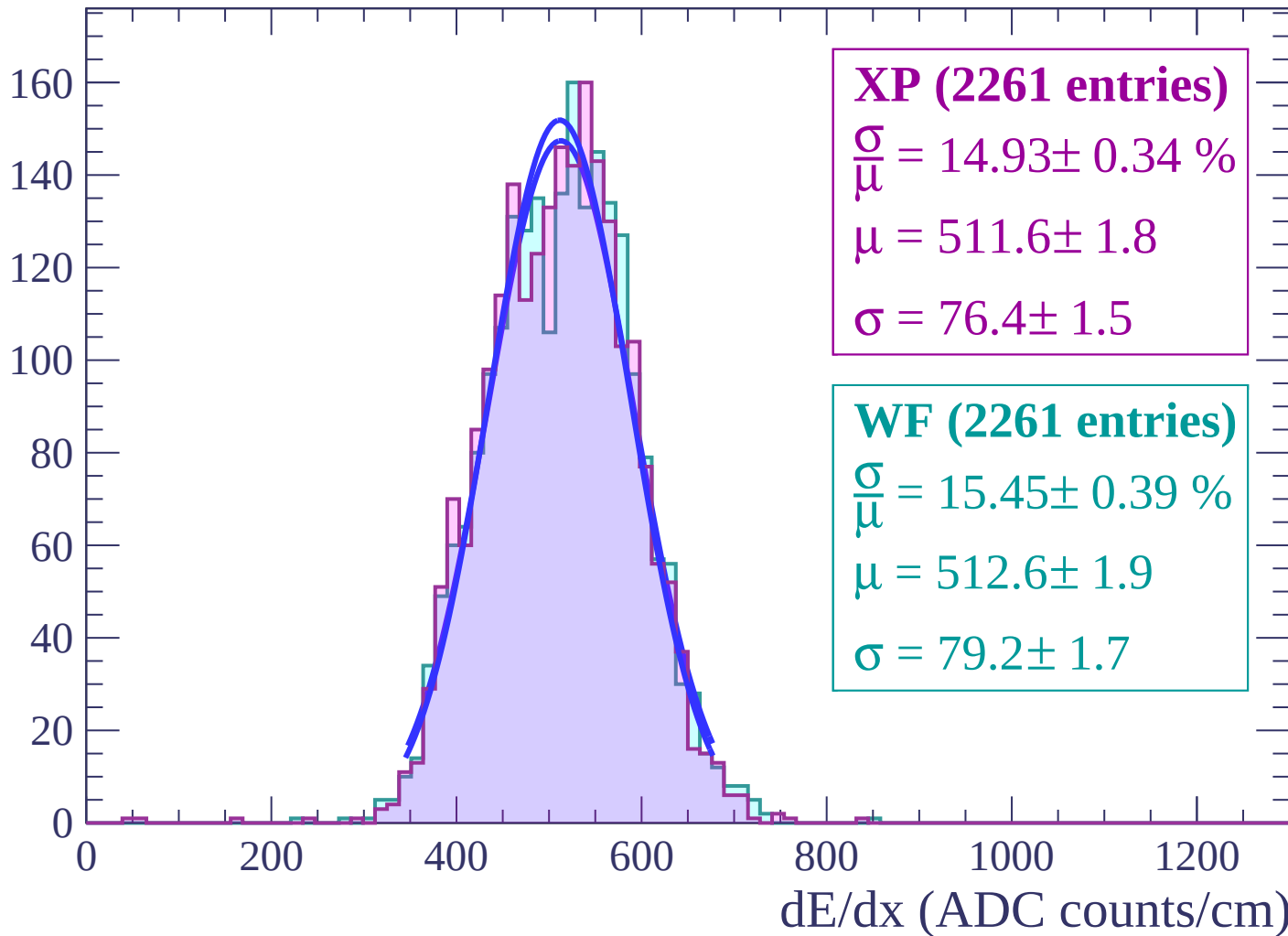
Energy loss | $-40 < \theta < -38$

Count

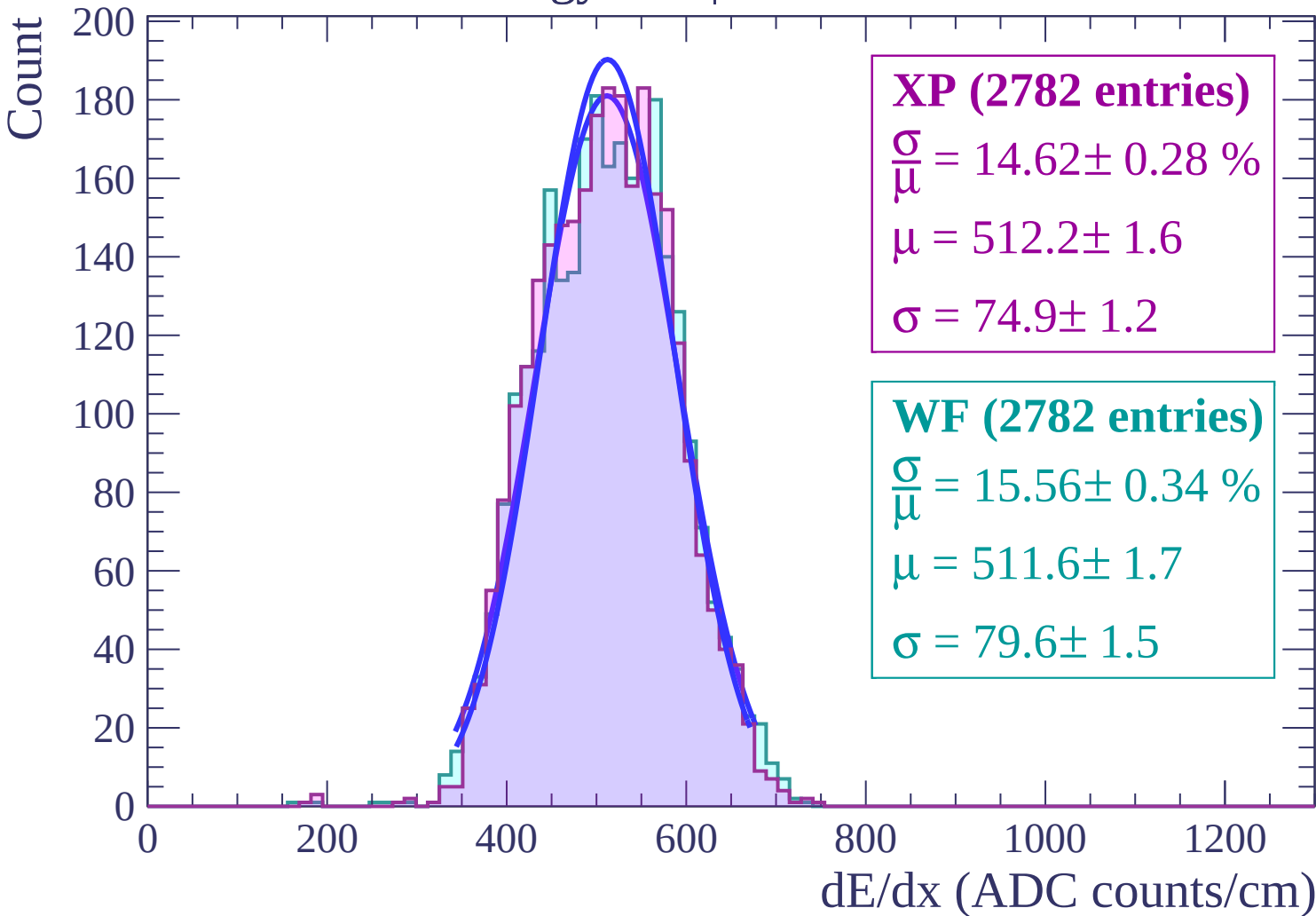


Energy loss | $-38 < \theta < -36$

Count

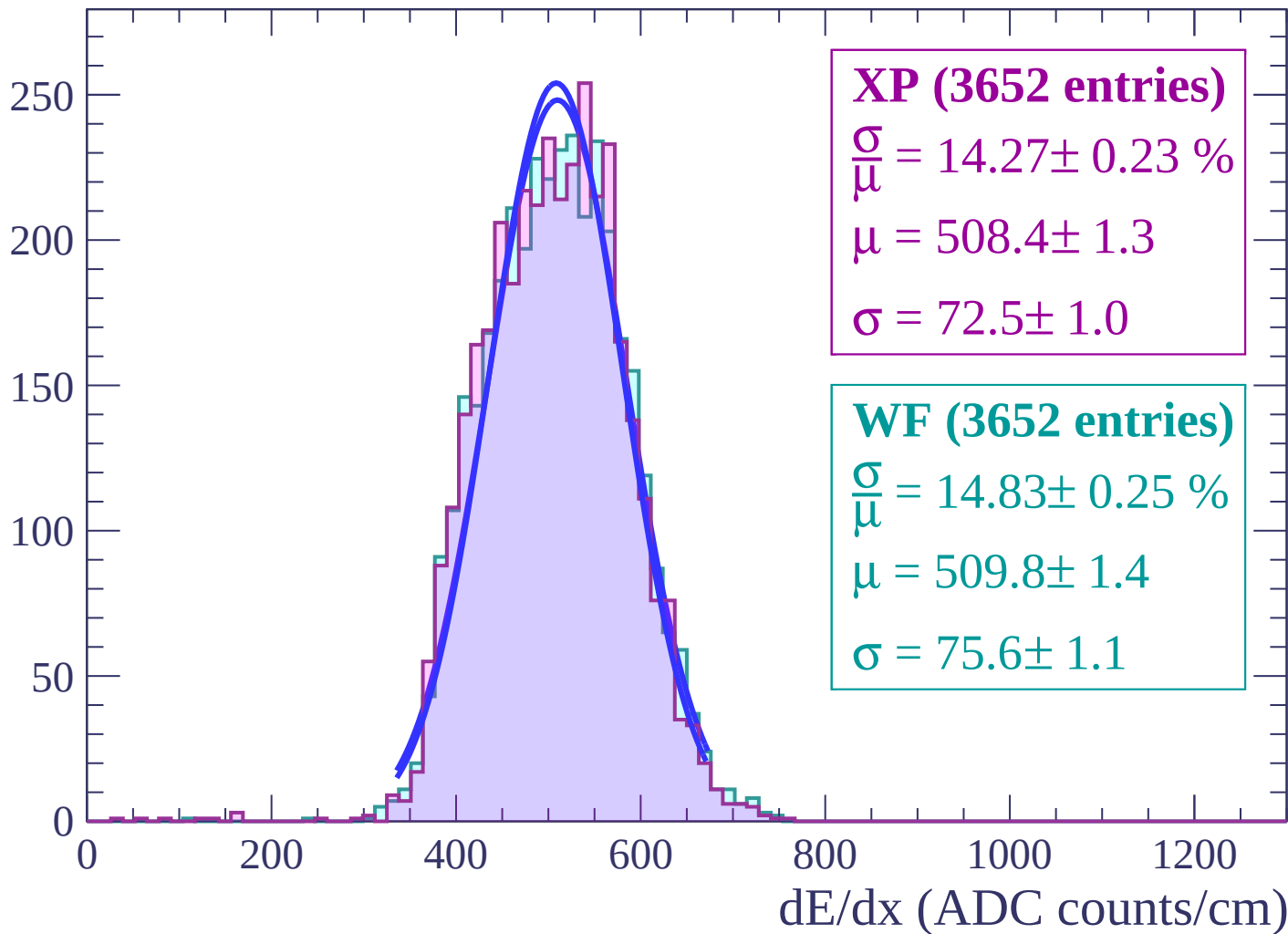


Energy loss | $-36 < \theta < -34$



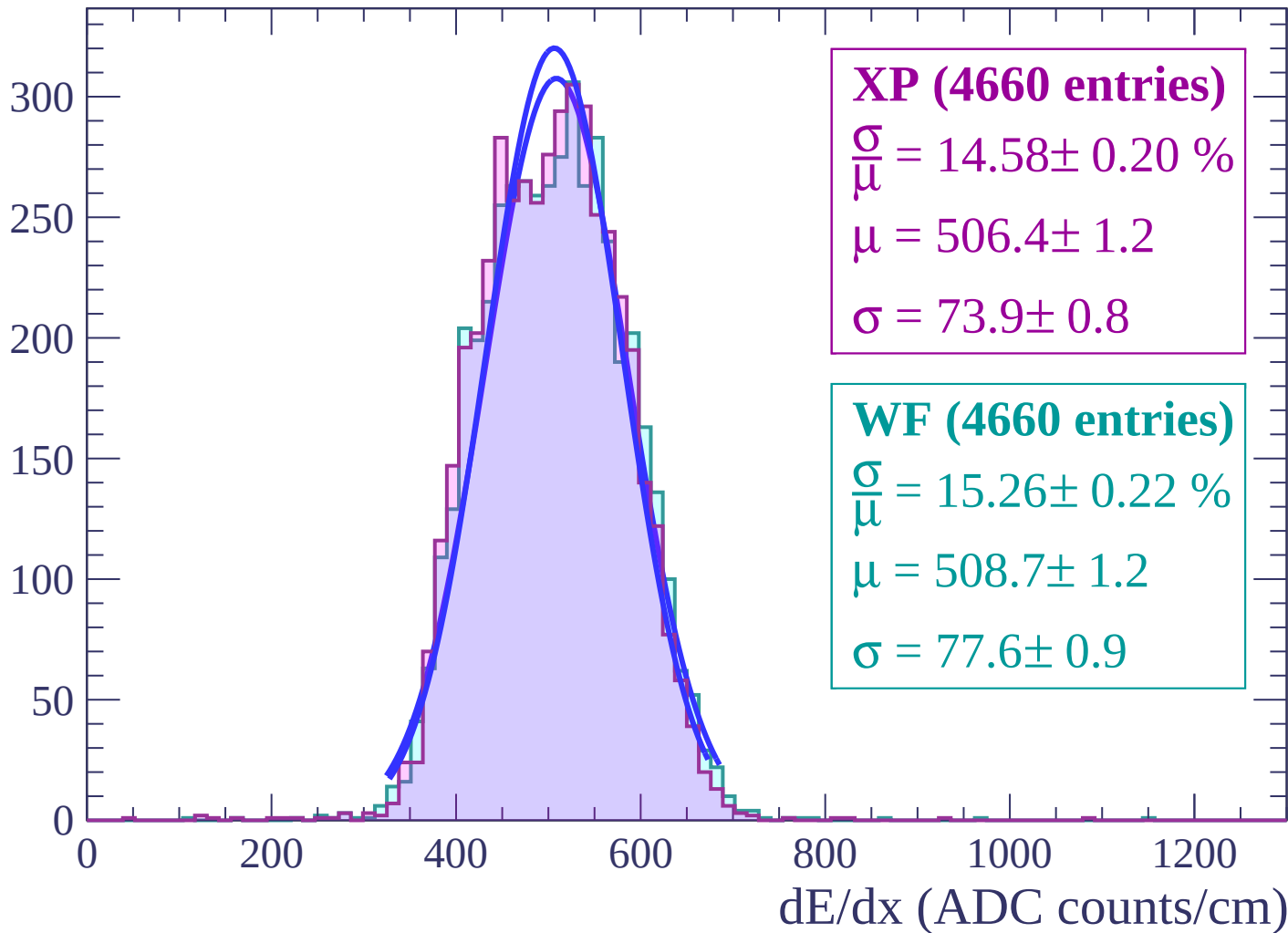
Energy loss | $-34 < \theta < -32$

Count



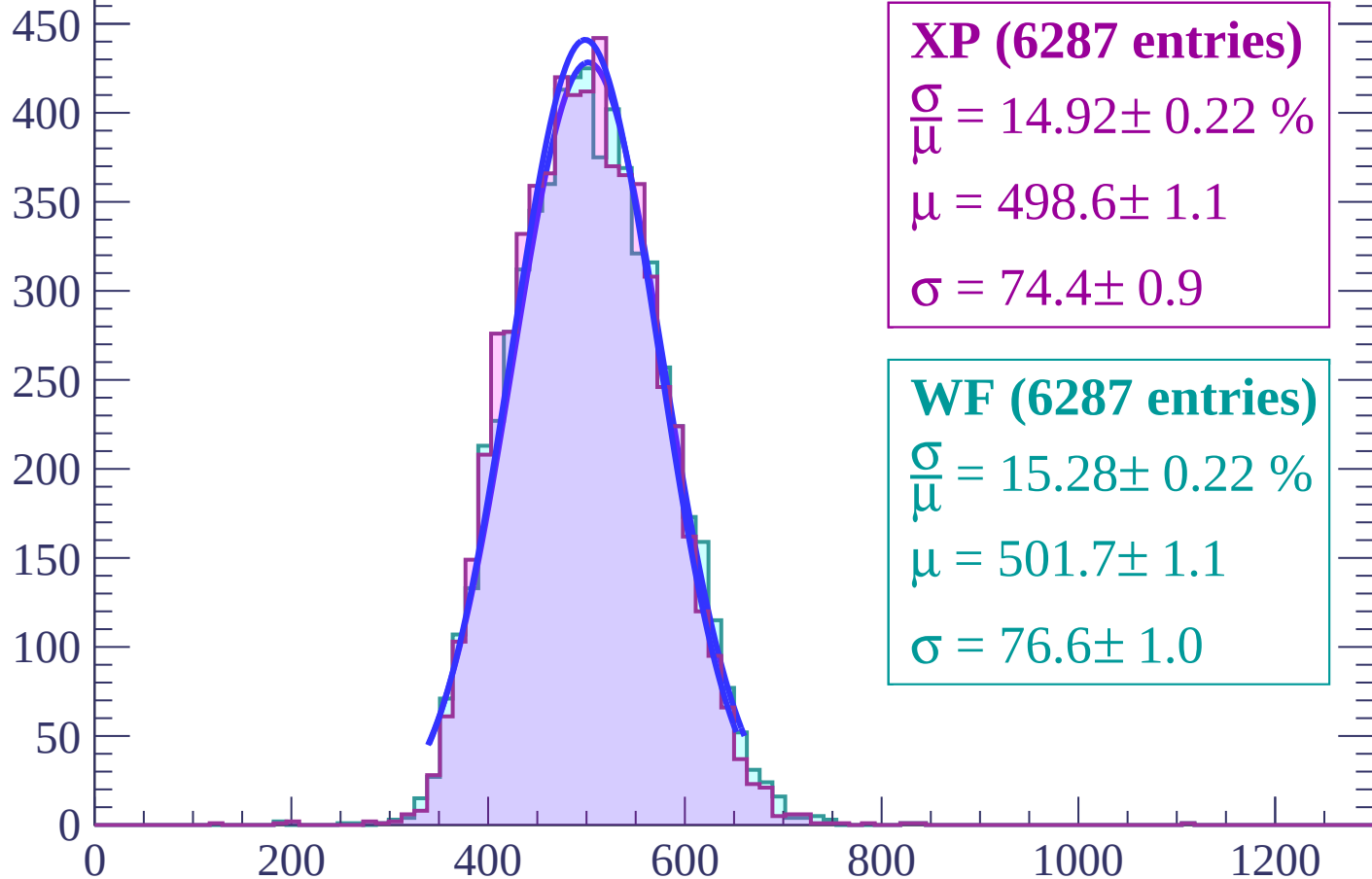
Energy loss | $-32 < \theta < -30$

Count



Energy loss | $-30 < \theta < -28$

Count



XP (6287 entries)

$$\frac{\sigma}{\mu} = 14.92 \pm 0.22 \%$$

$$\mu = 498.6 \pm 1.1$$

$$\sigma = 74.4 \pm 0.9$$

WF (6287 entries)

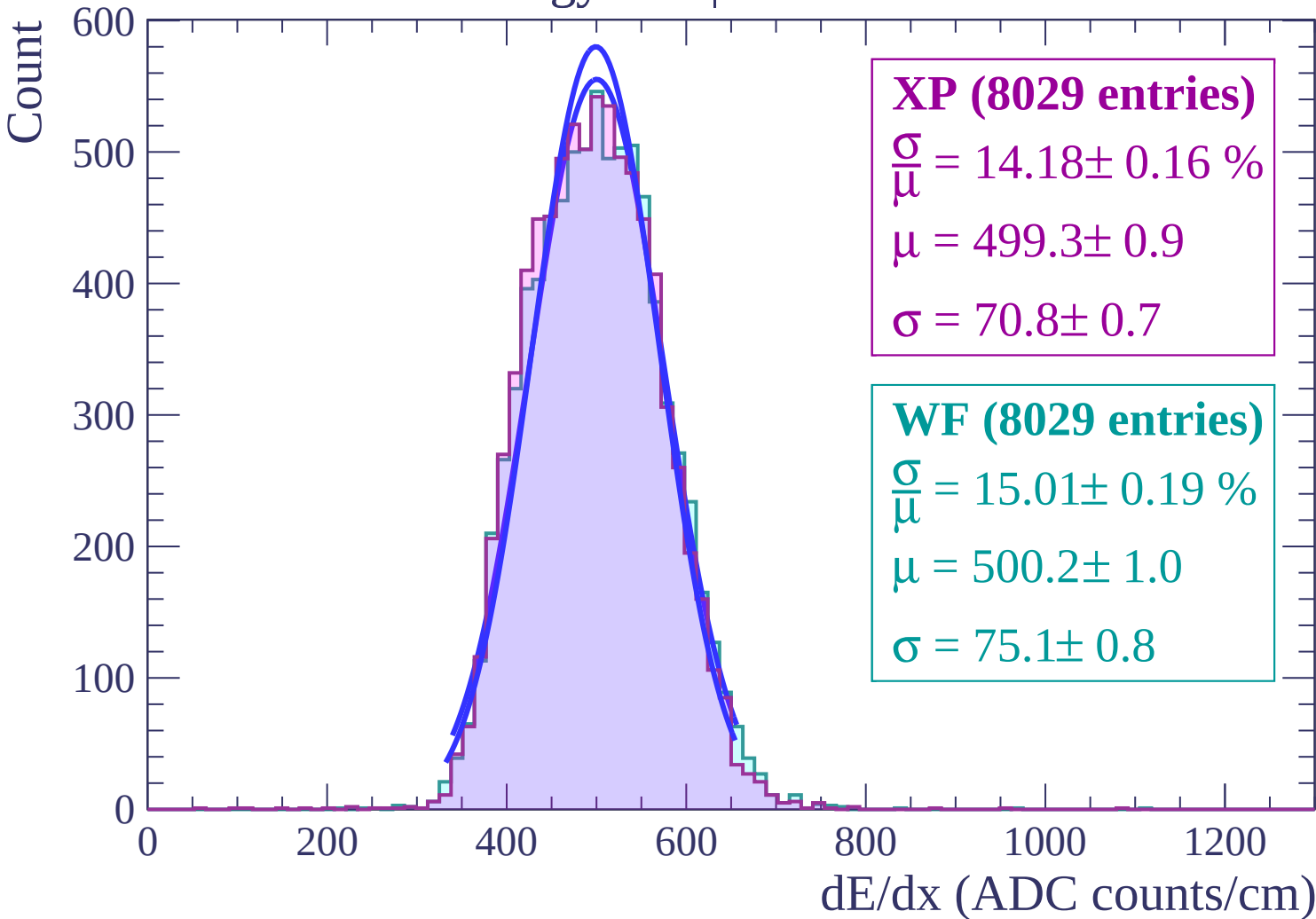
$$\frac{\sigma}{\mu} = 15.28 \pm 0.22 \%$$

$$\mu = 501.7 \pm 1.1$$

$$\sigma = 76.6 \pm 1.0$$

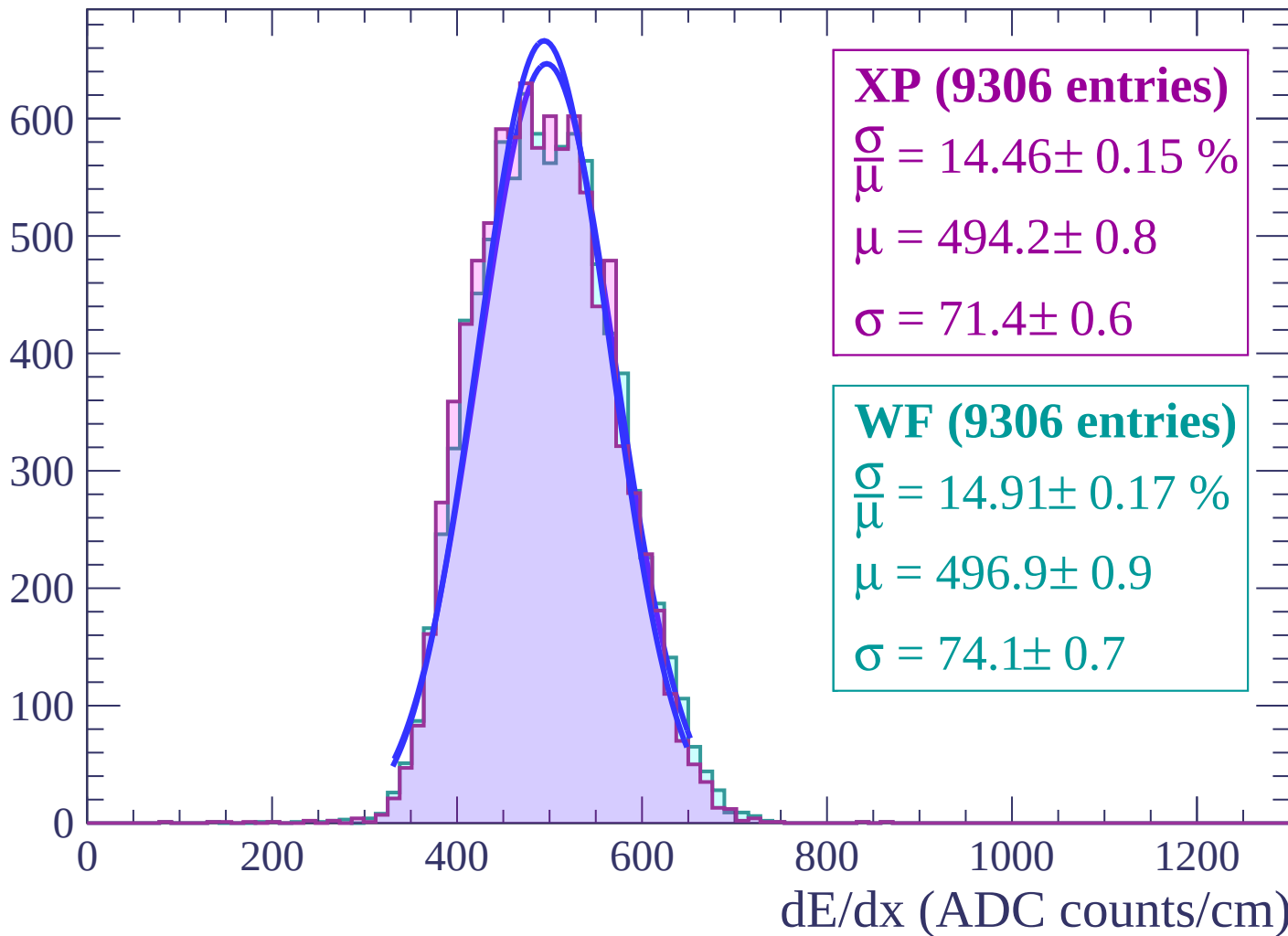
dE/dx (ADC counts/cm)

Energy loss | $-28 < \theta < -26$



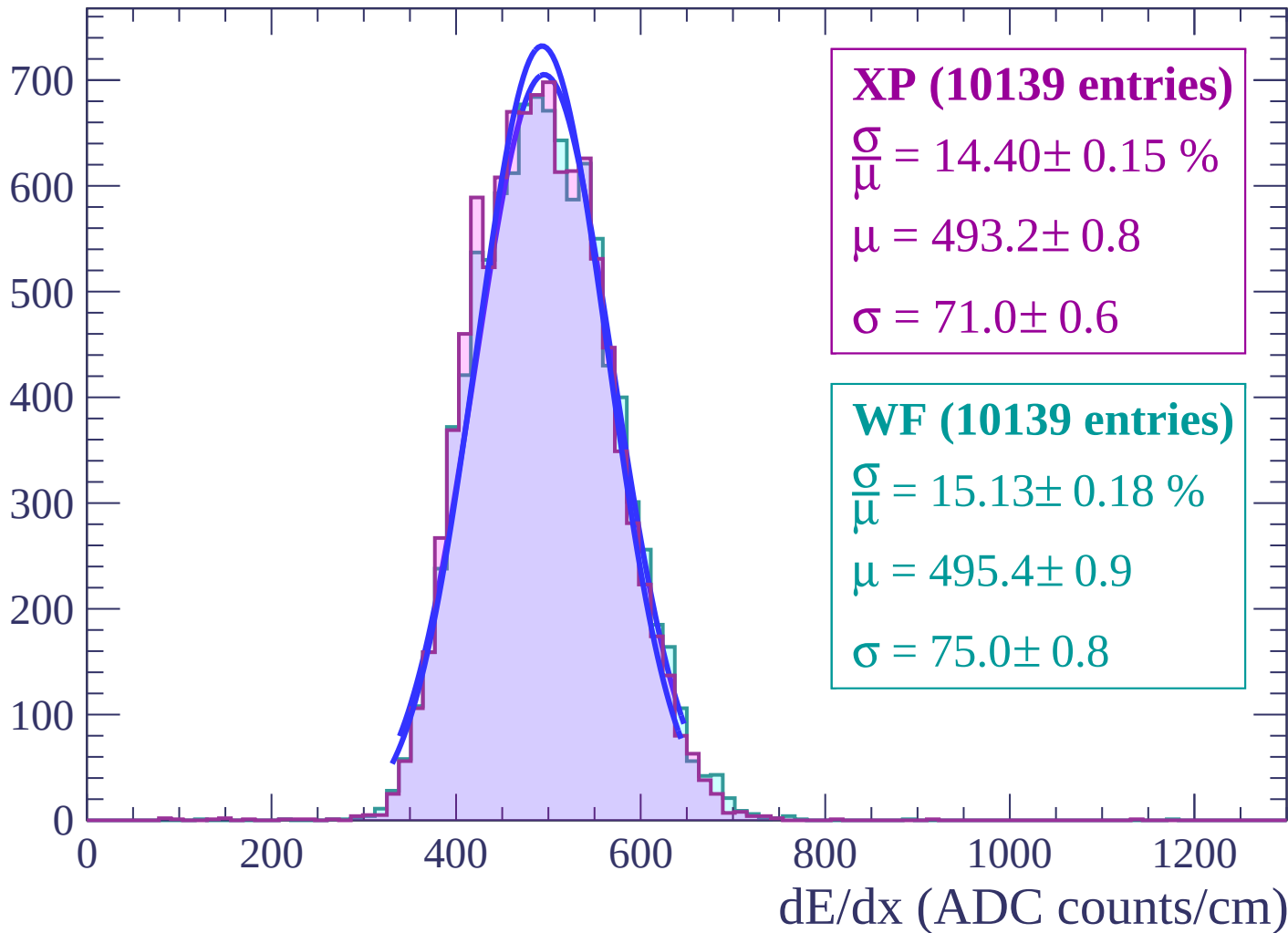
Energy loss | $-26 < \theta < -24$

Count



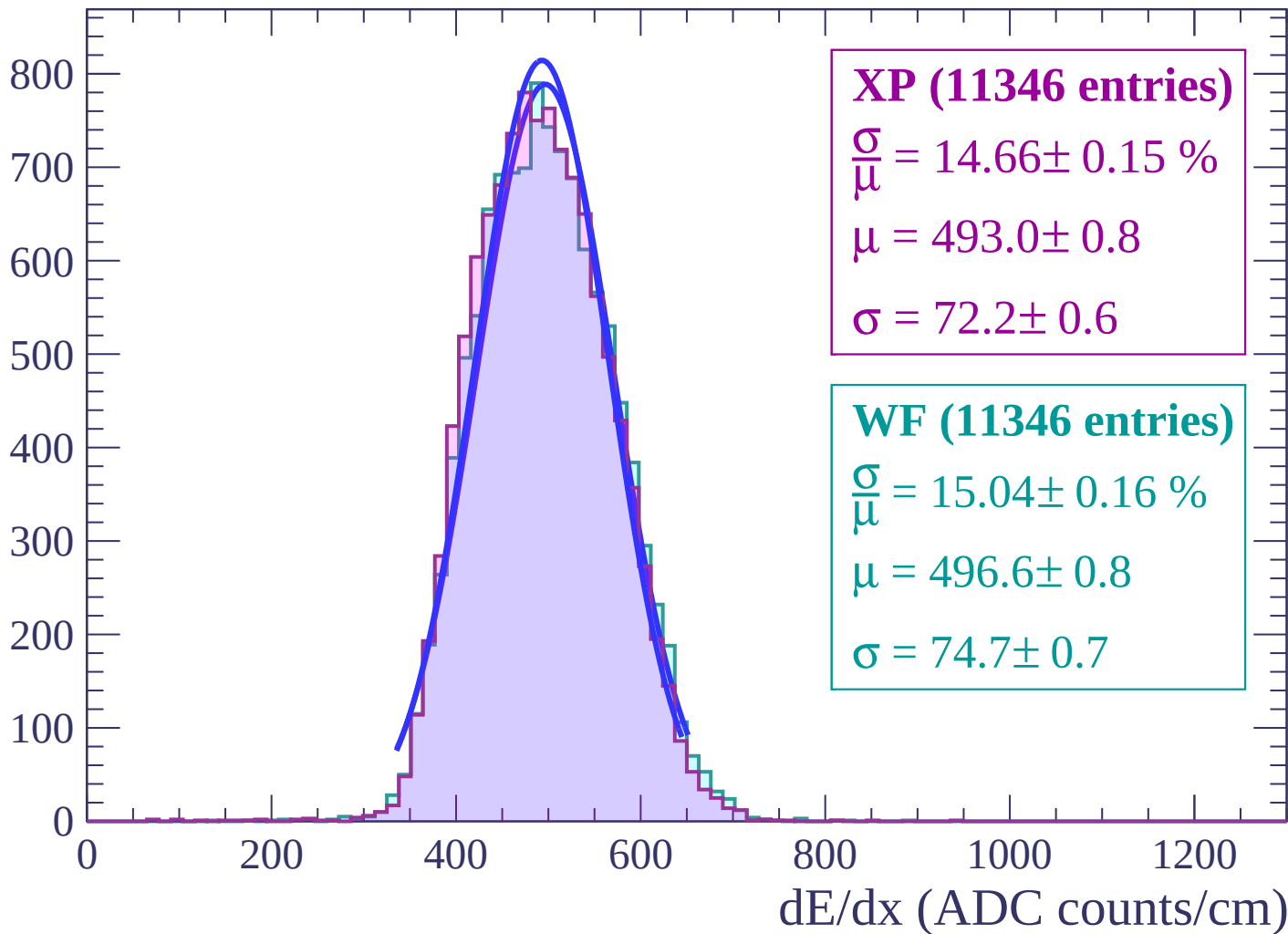
Energy loss | $-24 < \theta < -22$

Count



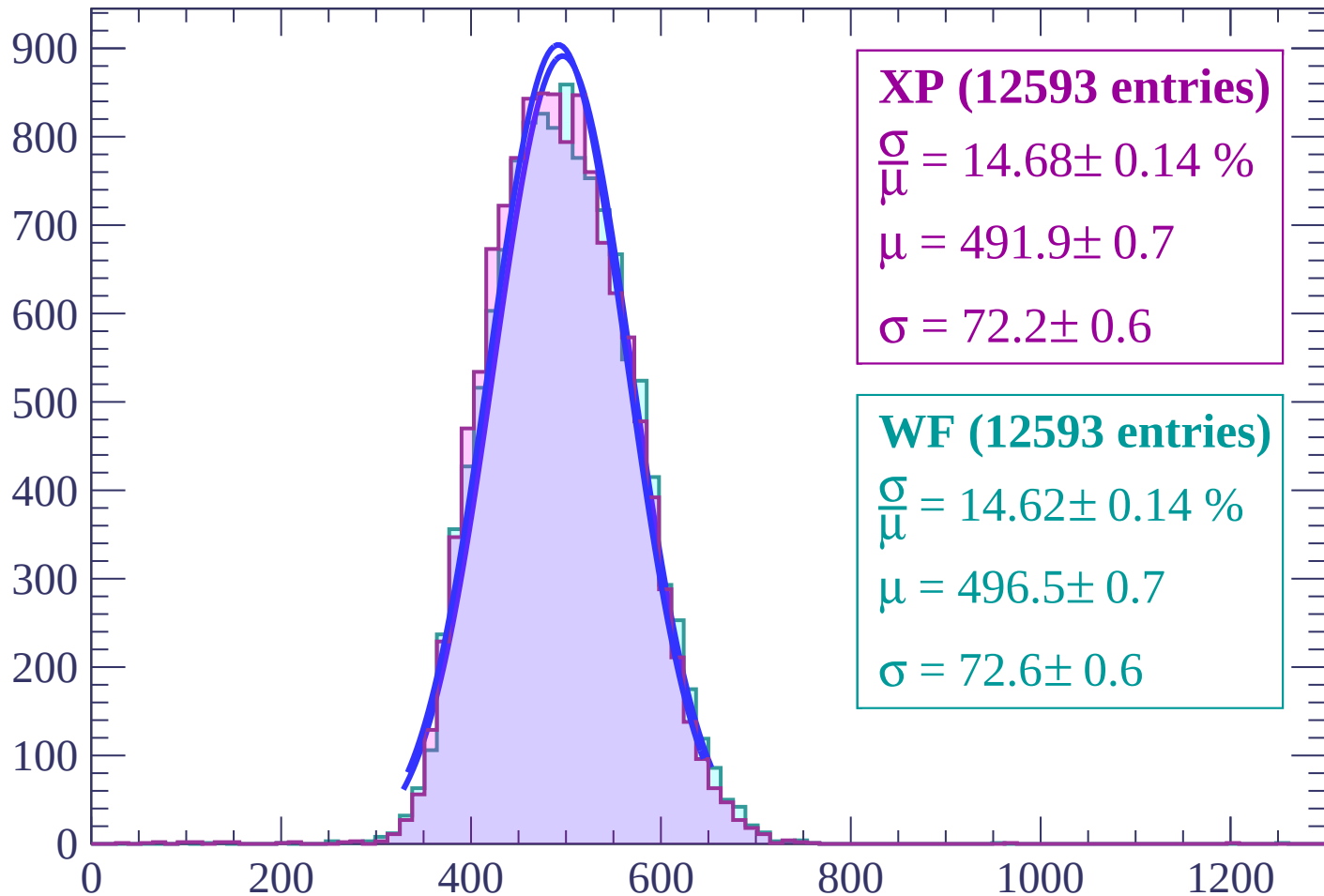
Energy loss | $-22 < \theta < -20$

Count



Energy loss | $-20 < \theta < -18$

Count



XP (12593 entries)

$$\frac{\sigma}{\mu} = 14.68 \pm 0.14 \%$$

$$\mu = 491.9 \pm 0.7$$

$$\sigma = 72.2 \pm 0.6$$

WF (12593 entries)

$$\frac{\sigma}{\mu} = 14.62 \pm 0.14 \%$$

$$\mu = 496.5 \pm 0.7$$

$$\sigma = 72.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-18 < \theta < -16$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (13814 entries)

$\frac{\sigma}{\mu} = 14.62 \pm 0.14 \%$

$\mu = 491.5 \pm 0.7$

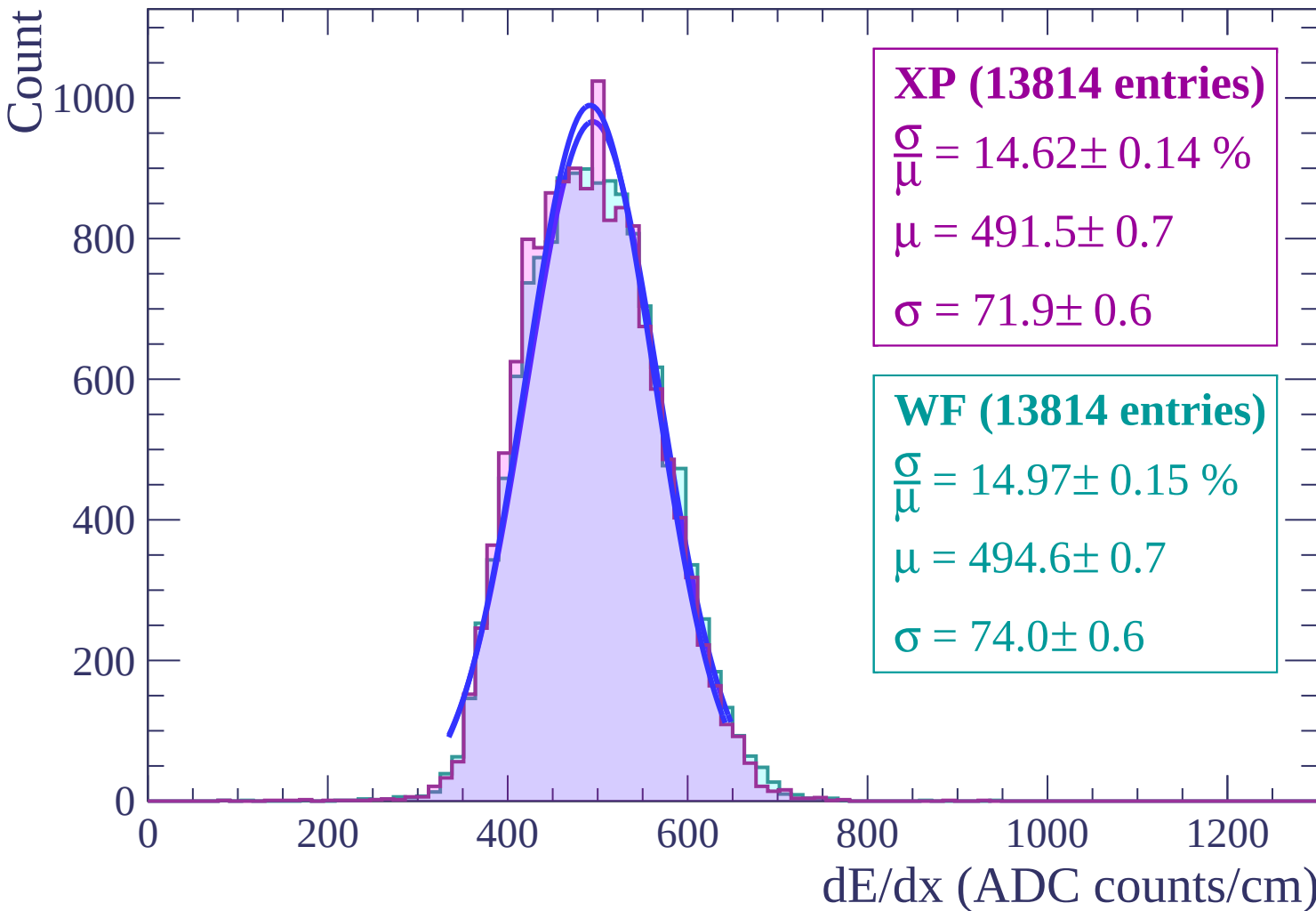
$\sigma = 71.9 \pm 0.6$

WF (13814 entries)

$\frac{\sigma}{\mu} = 14.97 \pm 0.15 \%$

$\mu = 494.6 \pm 0.7$

$\sigma = 74.0 \pm 0.6$



Energy loss | $-16 < \theta < -14$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (14900 entries)

$\frac{\sigma}{\mu} = 14.13 \pm 0.11 \%$

$\mu = 493.1 \pm 0.6$

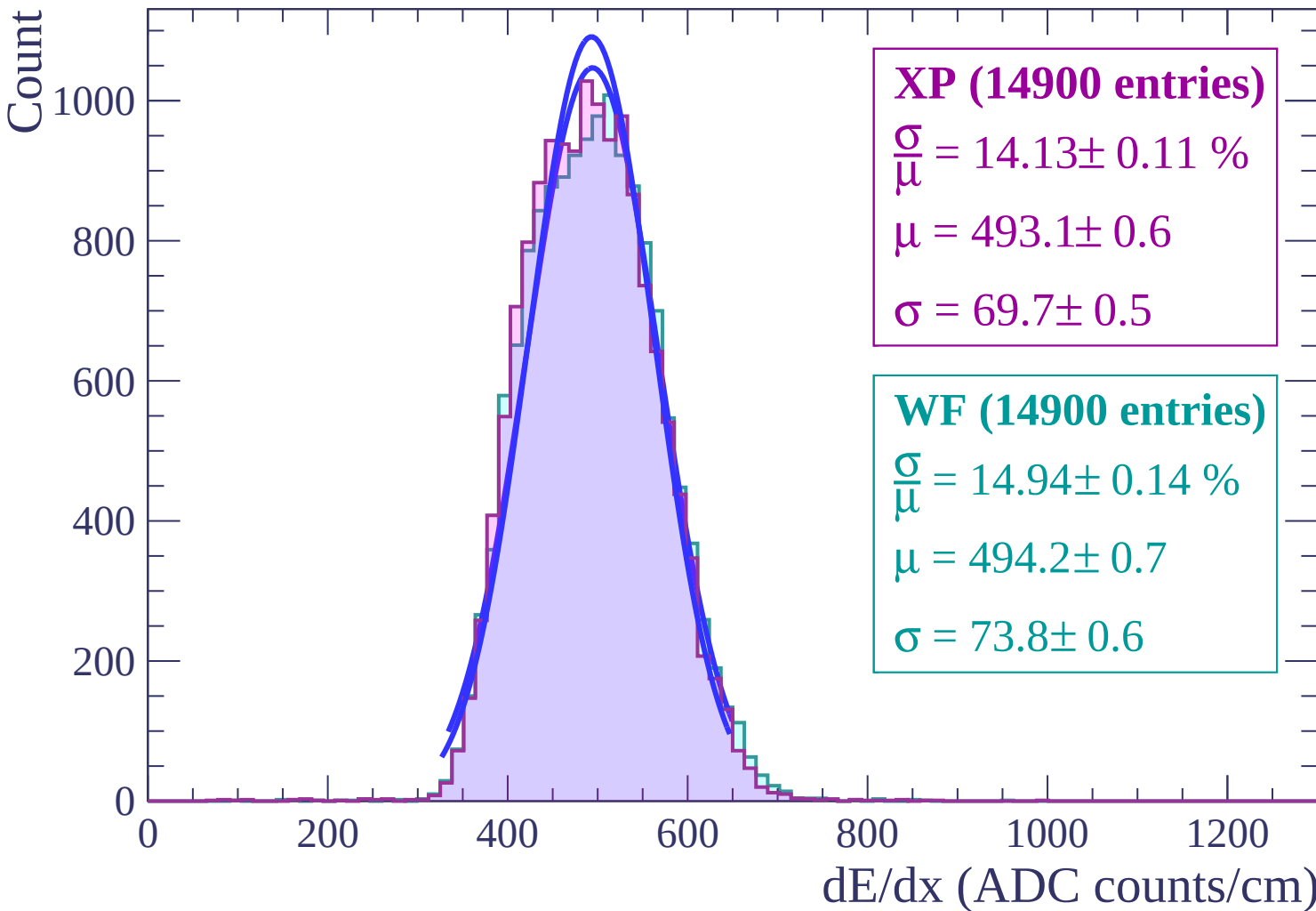
$\sigma = 69.7 \pm 0.5$

WF (14900 entries)

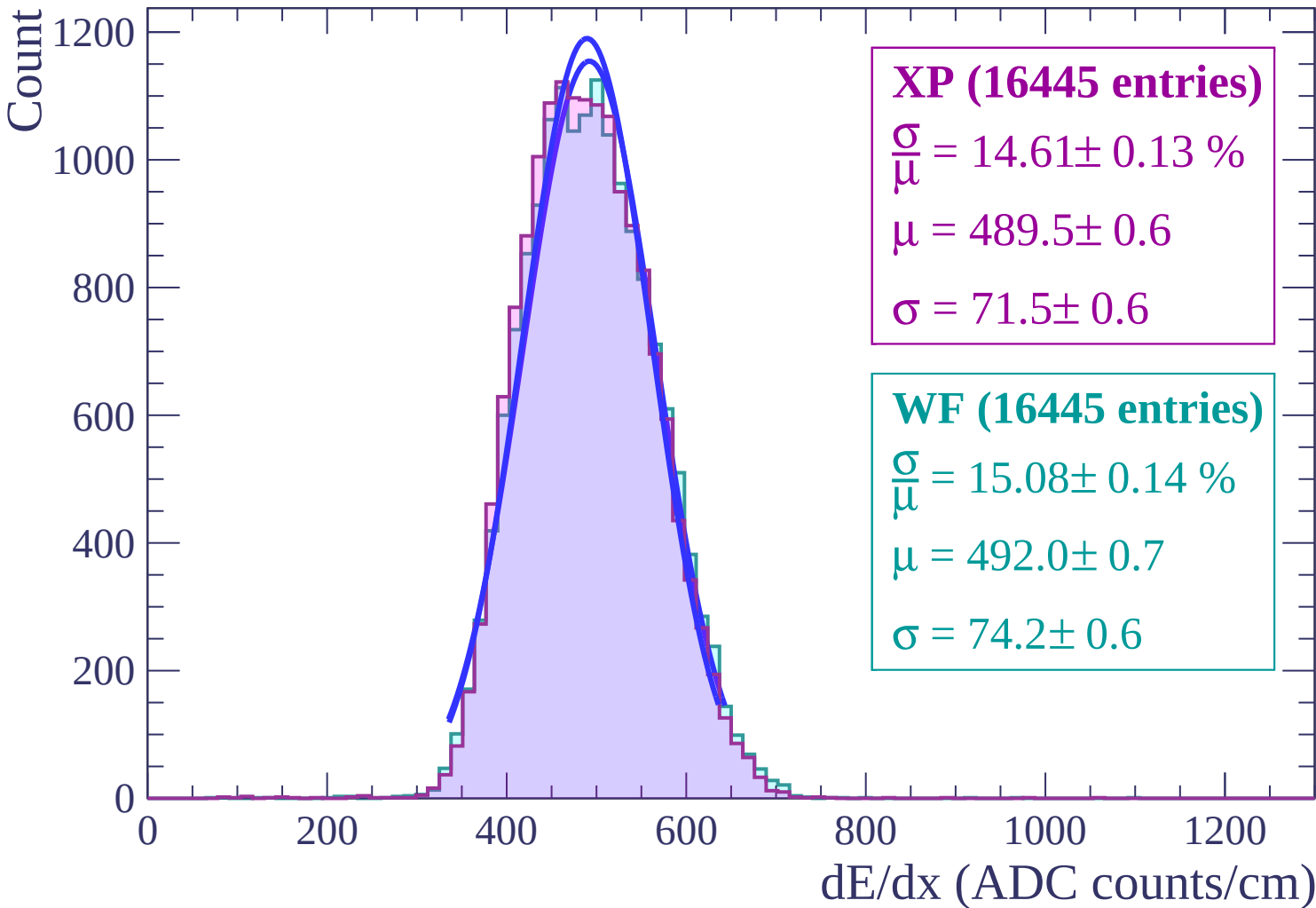
$\frac{\sigma}{\mu} = 14.94 \pm 0.14 \%$

$\mu = 494.2 \pm 0.7$

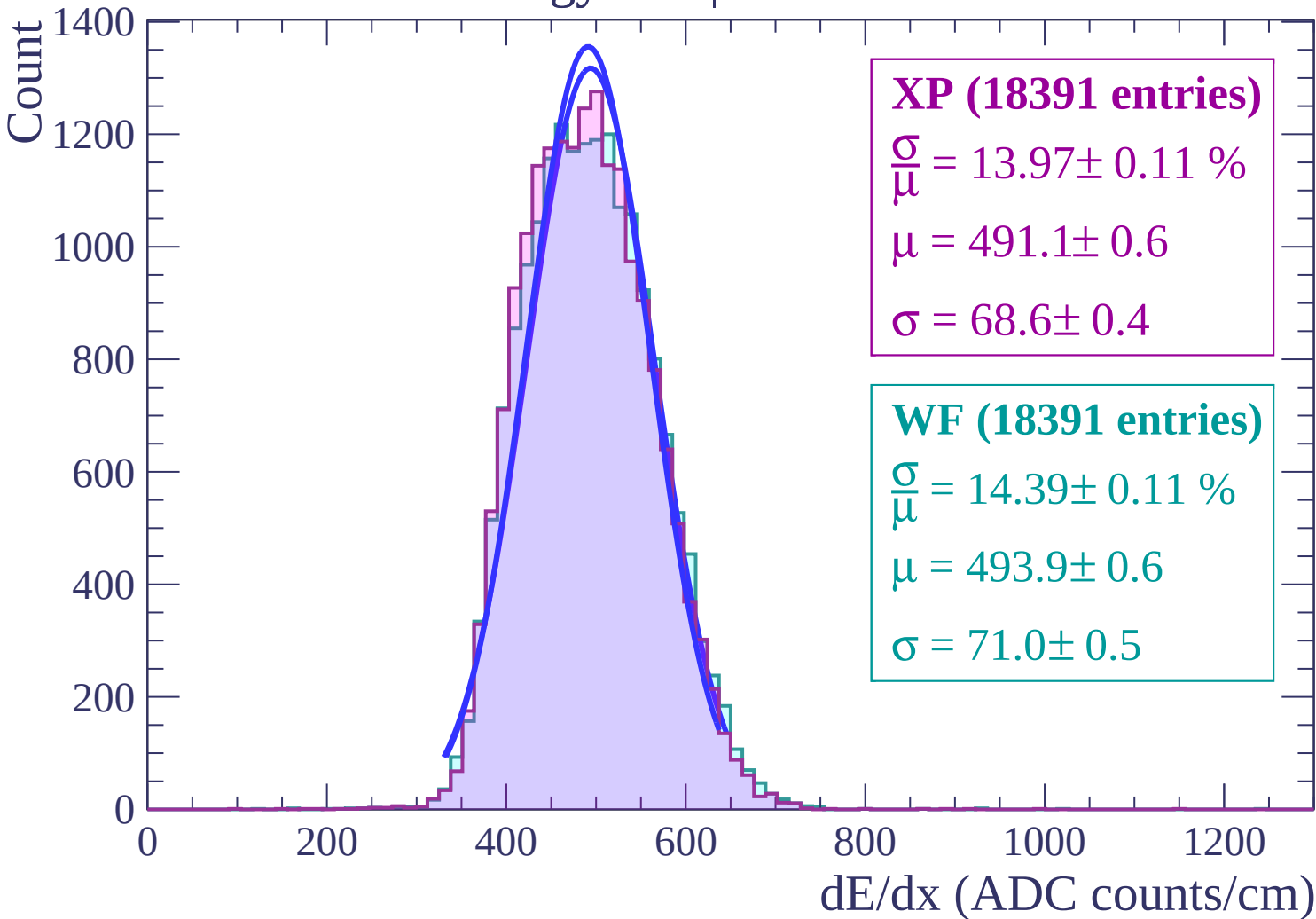
$\sigma = 73.8 \pm 0.6$



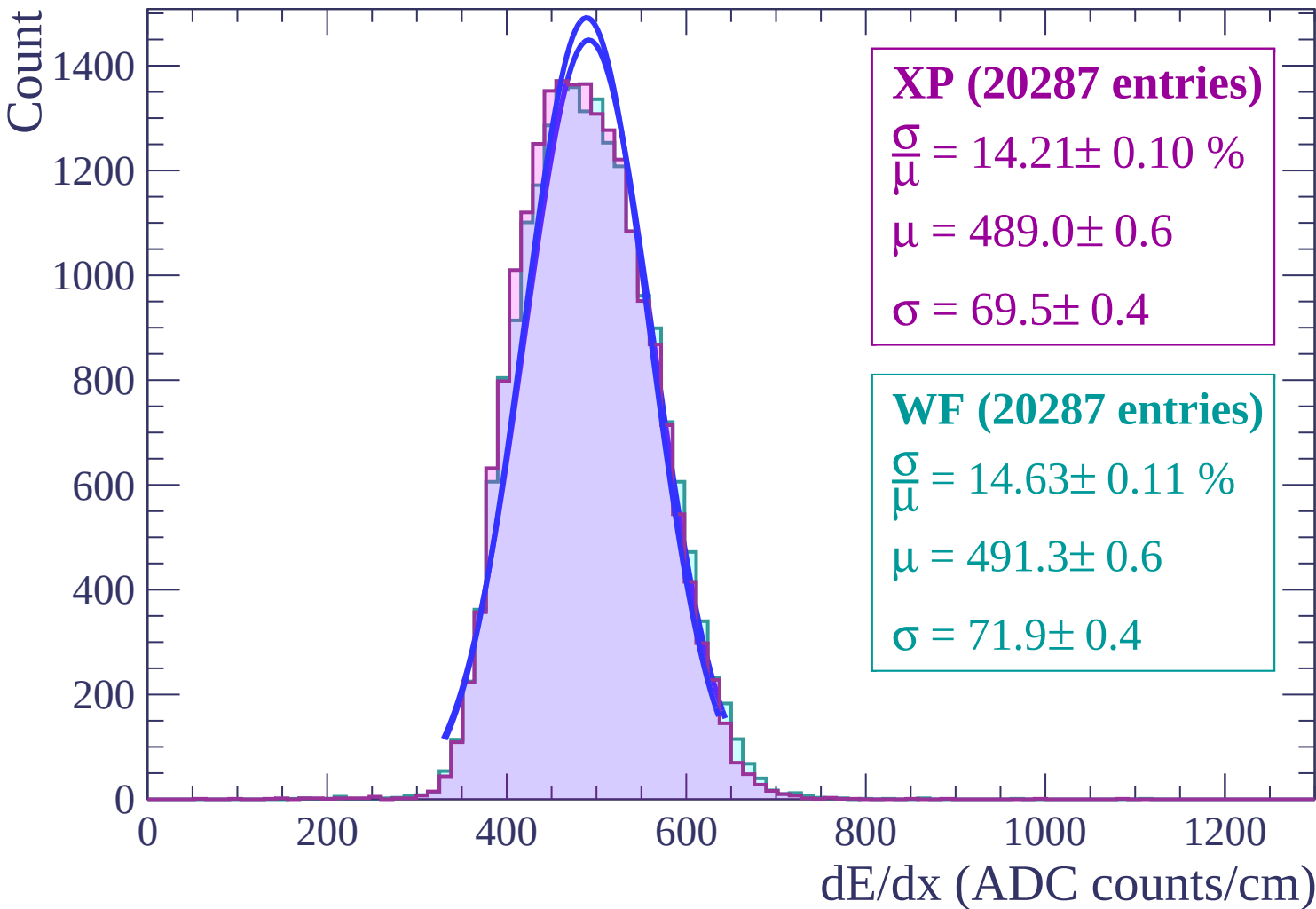
Energy loss | $-14 < \theta < -12$



Energy loss | $-12 < \theta < -10$

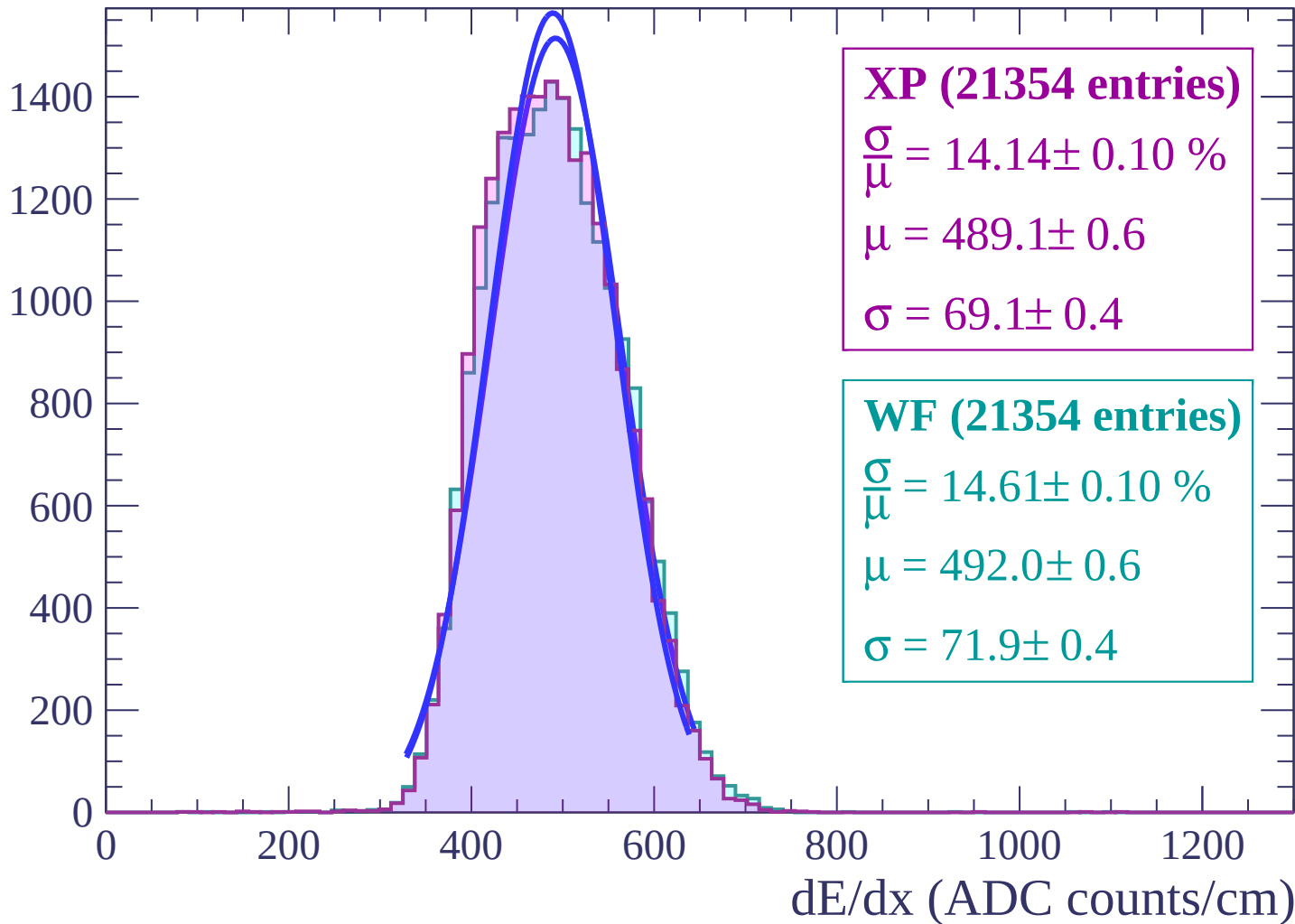


Energy loss | $-10 < \theta < -8$

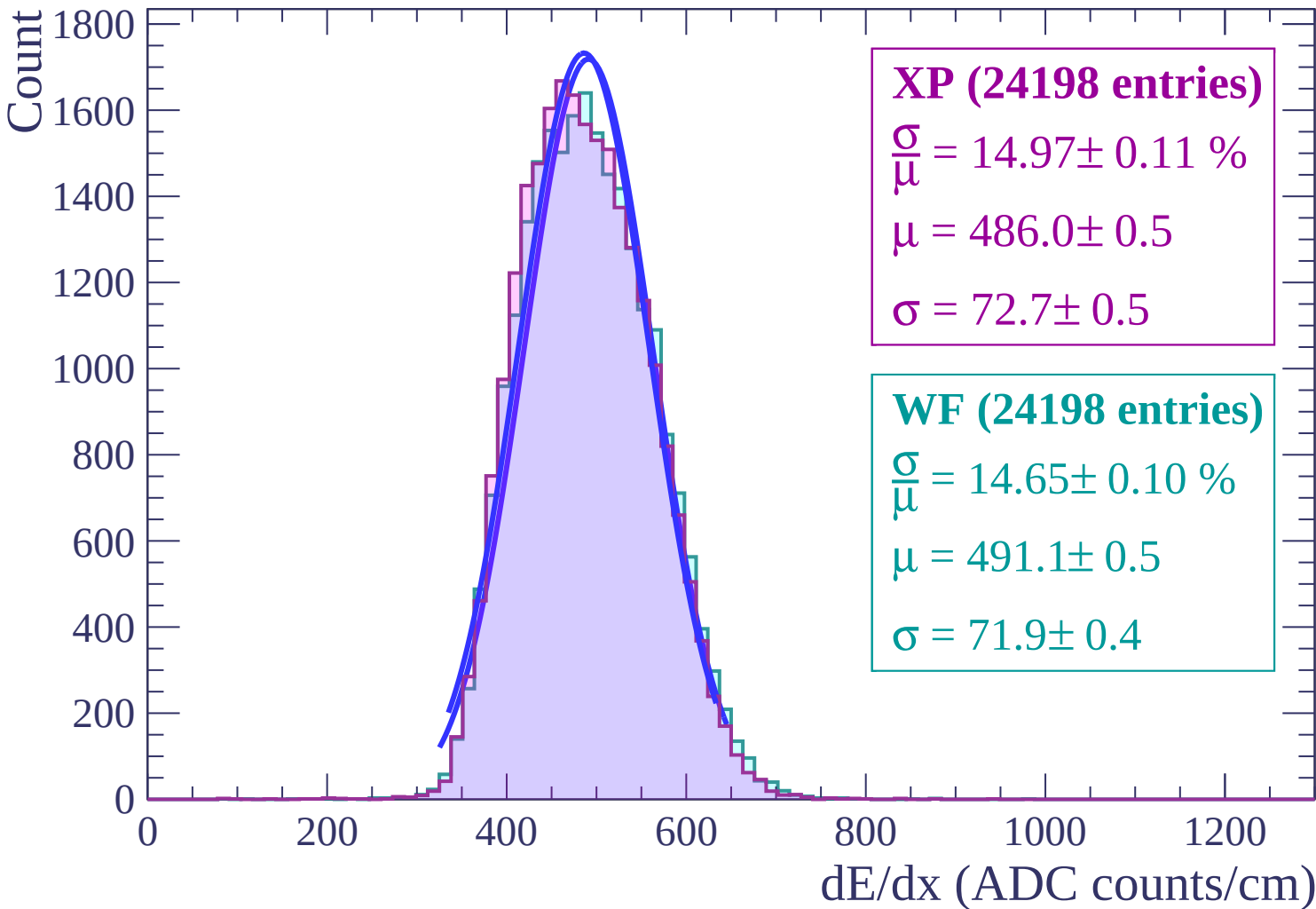


Energy loss | $-8 < \theta < -6$

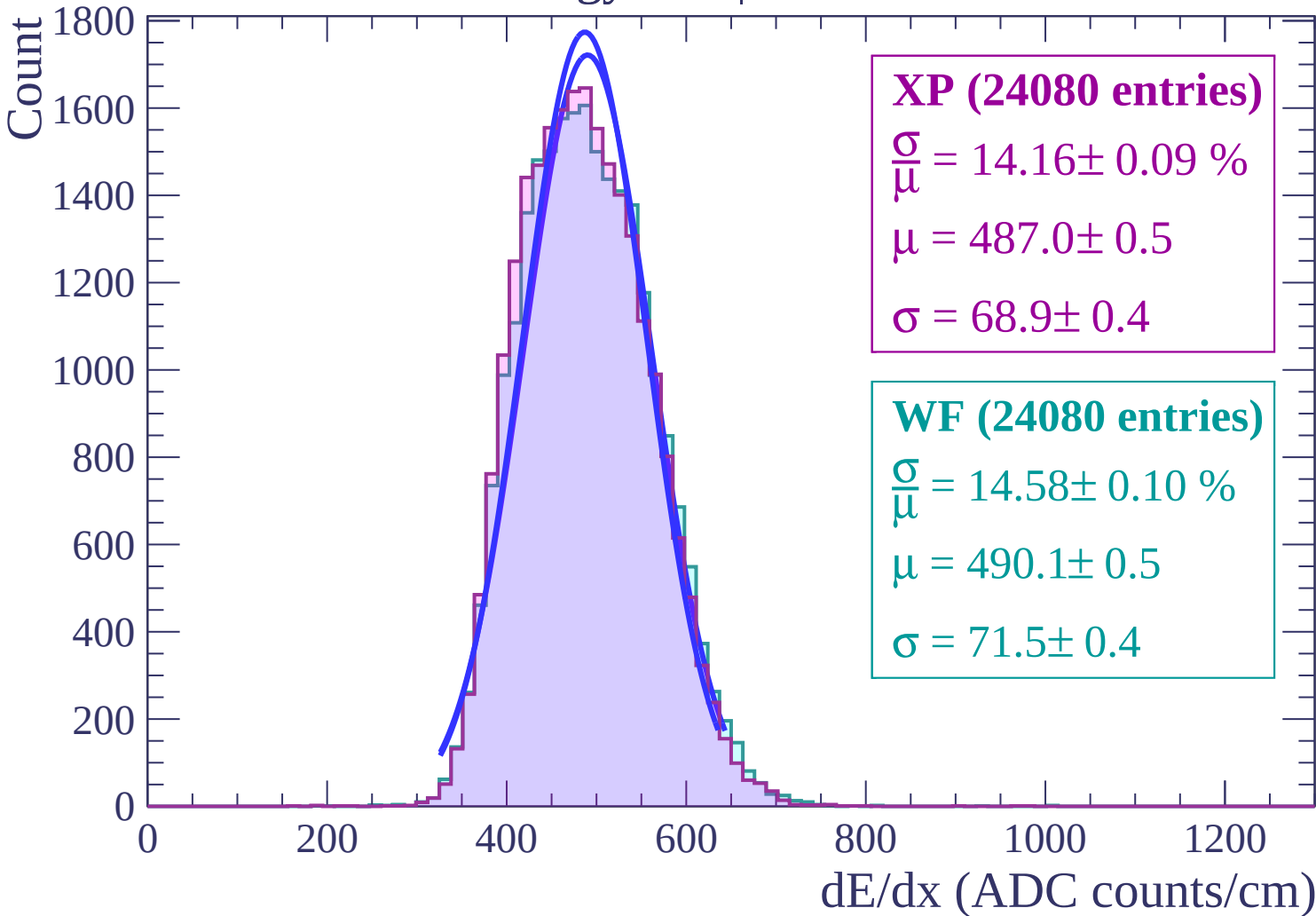
Count



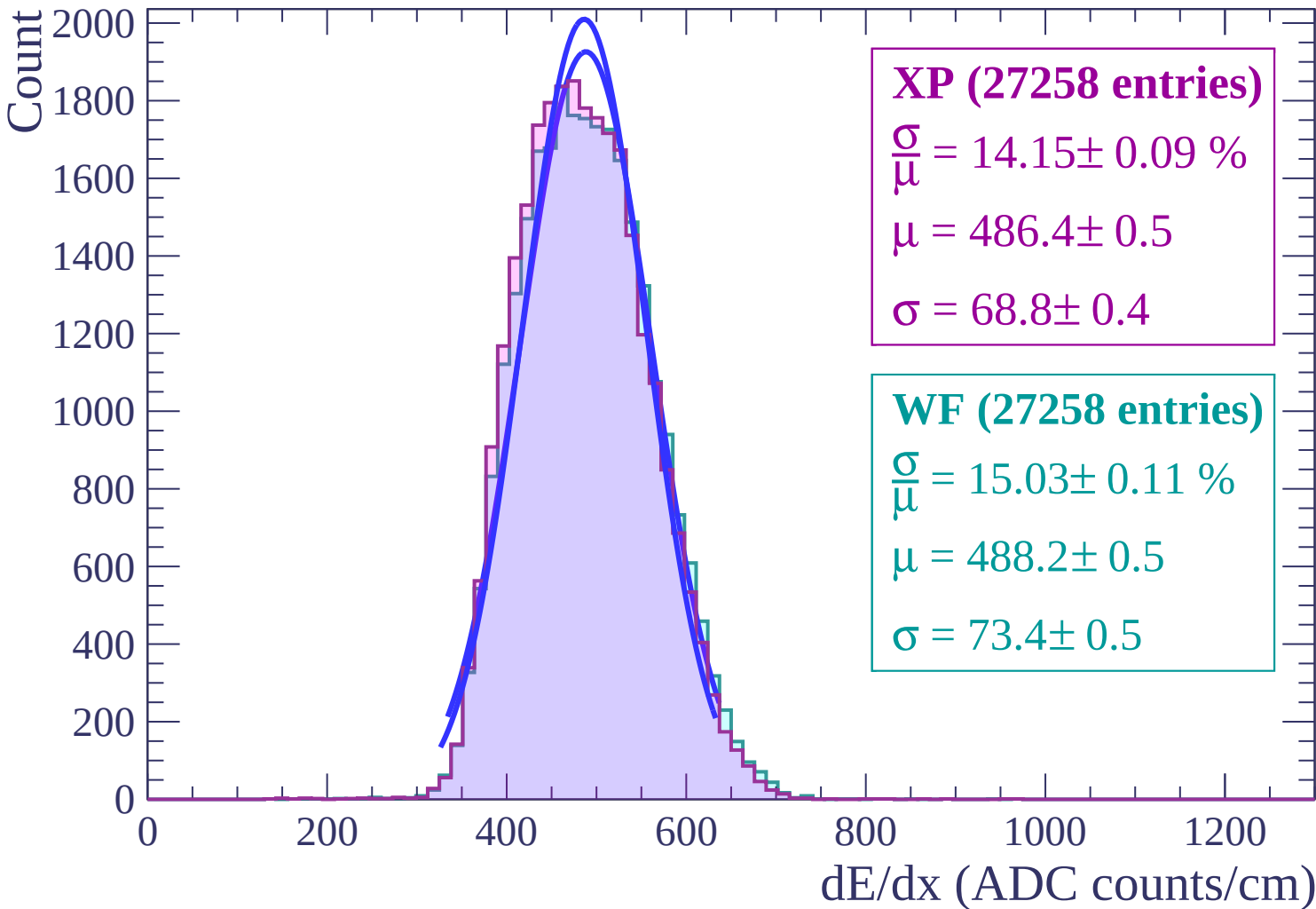
Energy loss | $-6 < \theta < -4$



Energy loss | $-4 < \theta < -2$



Energy loss $-2 < \theta < 0$



Energy loss | $0 < \theta < 2$

Count

3000

2500

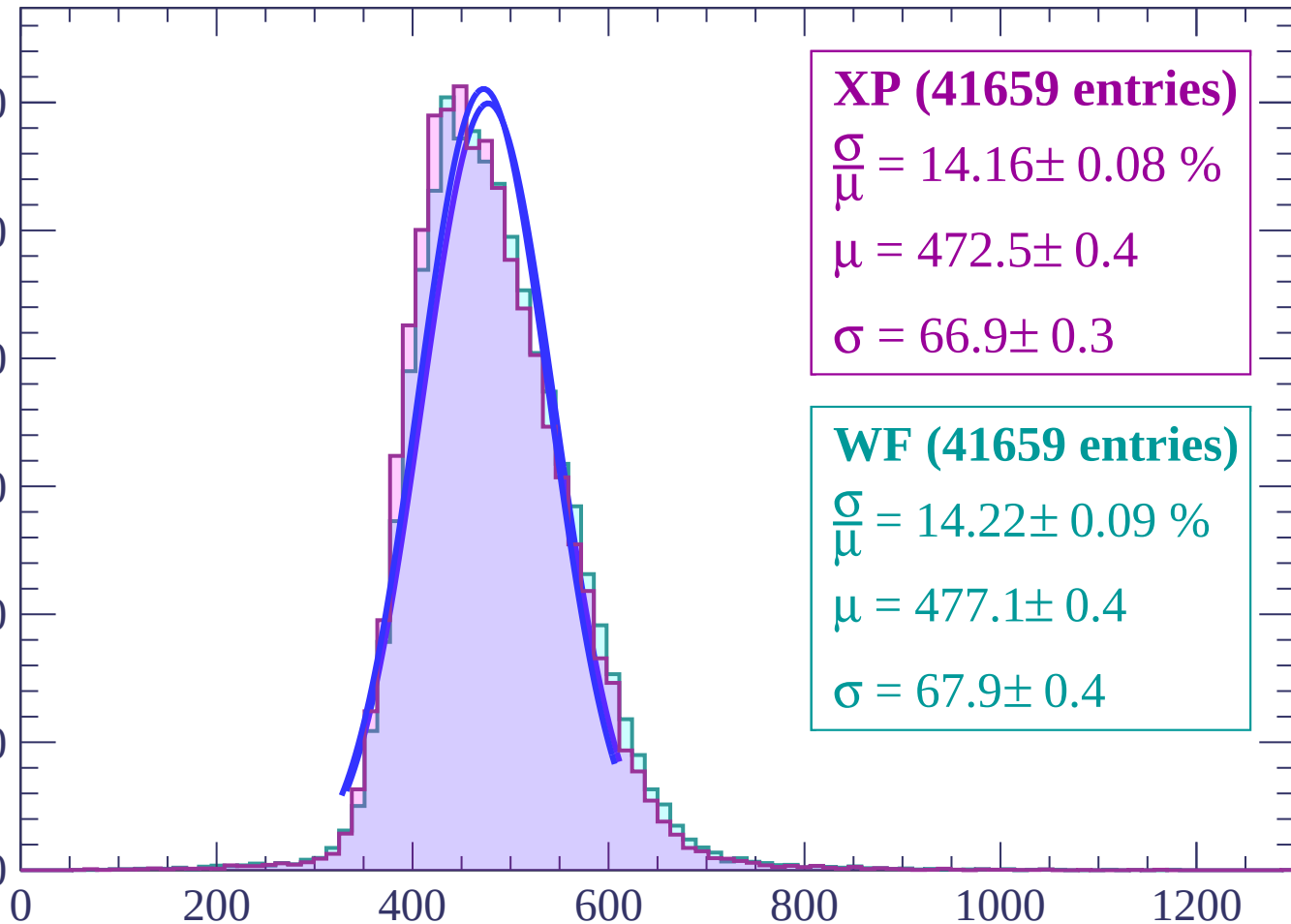
2000

1500

1000

500

0



XP (41659 entries)

$\frac{\sigma}{\mu} = 14.16 \pm 0.08 \%$

$\mu = 472.5 \pm 0.4$

$\sigma = 66.9 \pm 0.3$

WF (41659 entries)

$\frac{\sigma}{\mu} = 14.22 \pm 0.09 \%$

$\mu = 477.1 \pm 0.4$

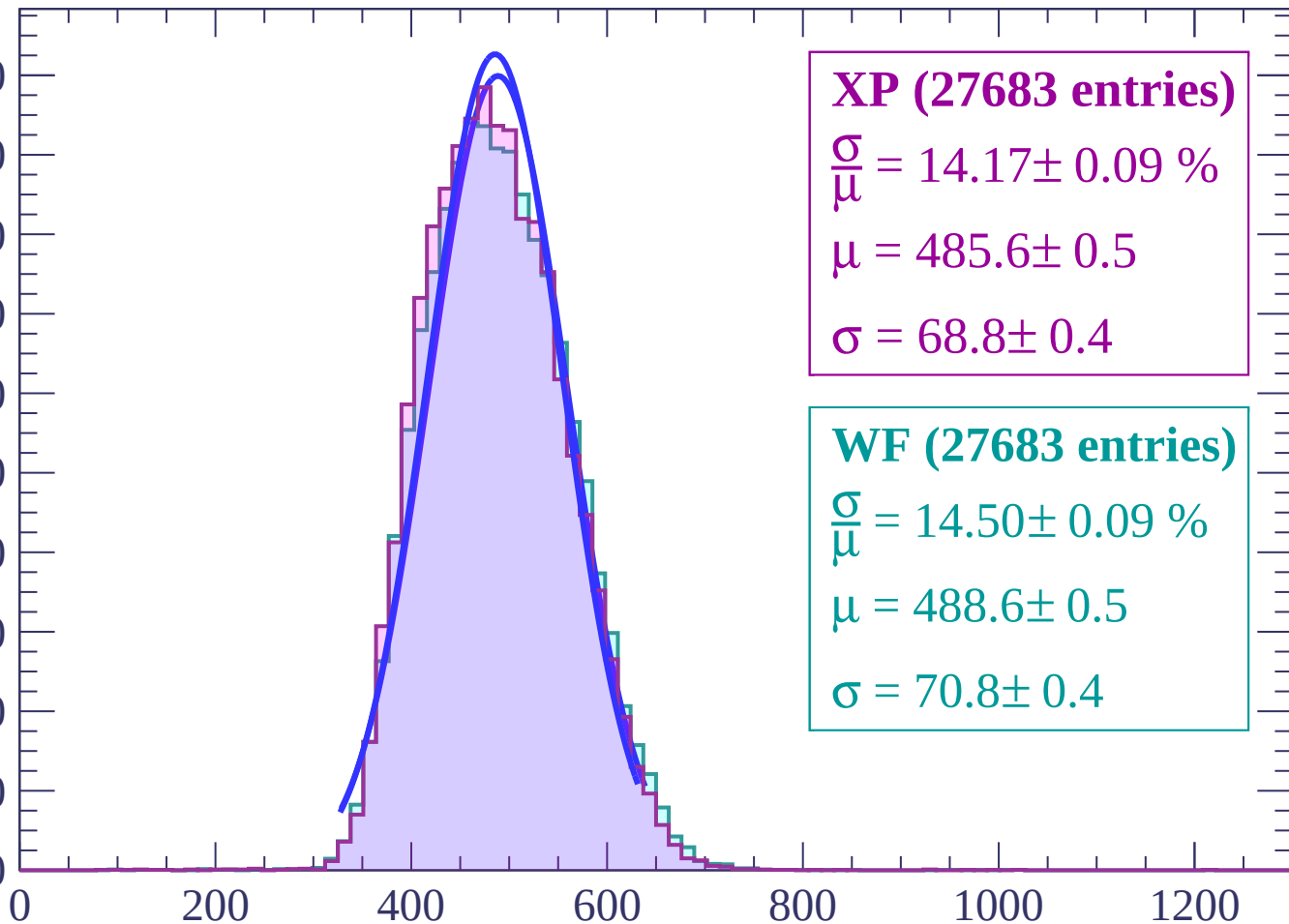
$\sigma = 67.9 \pm 0.4$

dE/dx (ADC counts/cm)

Energy loss | $2 < \theta < 4$

Count

2000
1800
1600
1400
1200
1000
800
600
400
200
0



XP (27683 entries)

$\frac{\sigma}{\mu} = 14.17 \pm 0.09 \%$

$\mu = 485.6 \pm 0.5$

$\sigma = 68.8 \pm 0.4$

WF (27683 entries)

$\frac{\sigma}{\mu} = 14.50 \pm 0.09 \%$

$\mu = 488.6 \pm 0.5$

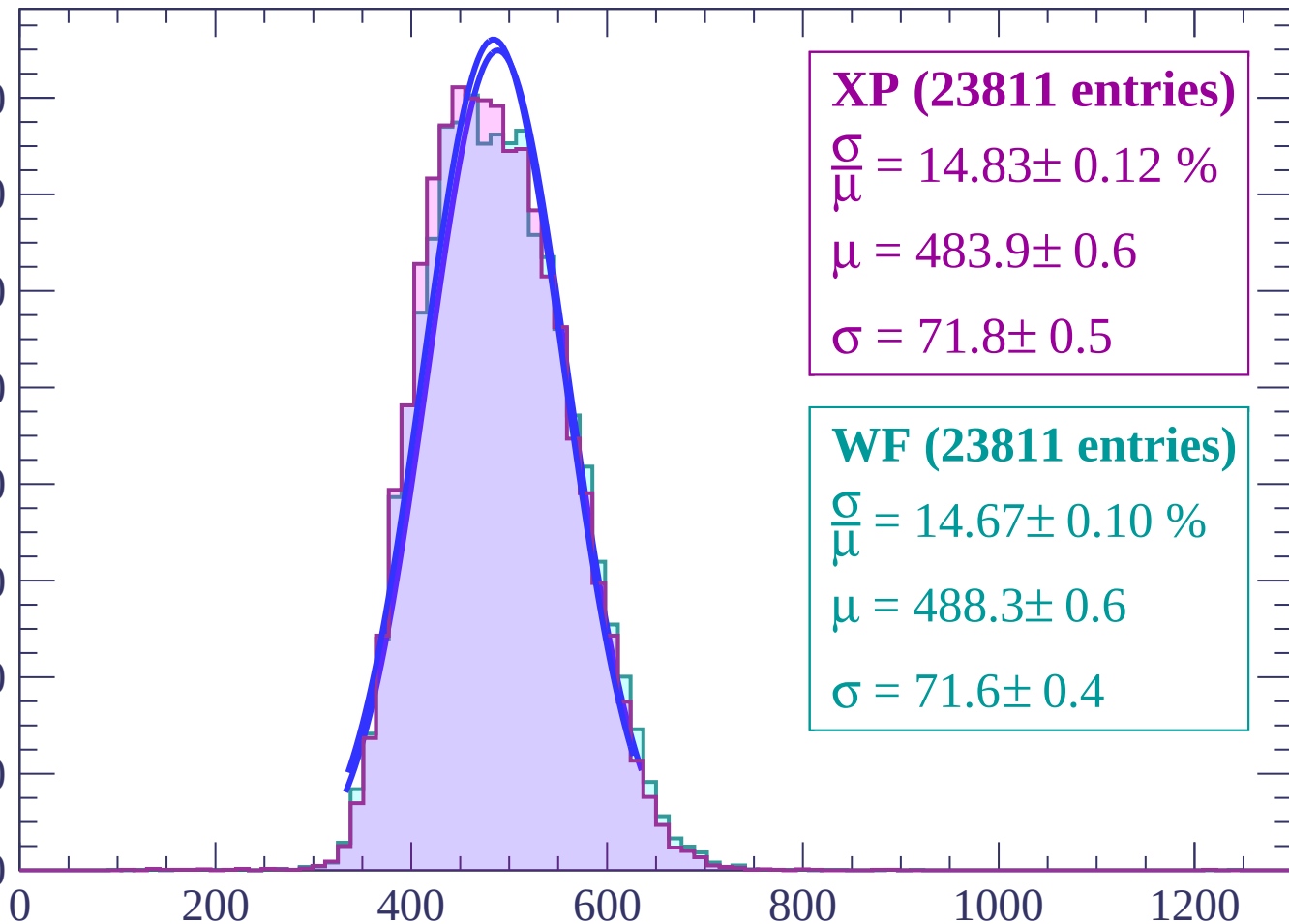
$\sigma = 70.8 \pm 0.4$

dE/dx (ADC counts/cm)

Energy loss | $4 < \theta < 6$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (23811 entries)

$\frac{\sigma}{\mu} = 14.83 \pm 0.12 \%$

$\mu = 483.9 \pm 0.6$

$\sigma = 71.8 \pm 0.5$

WF (23811 entries)

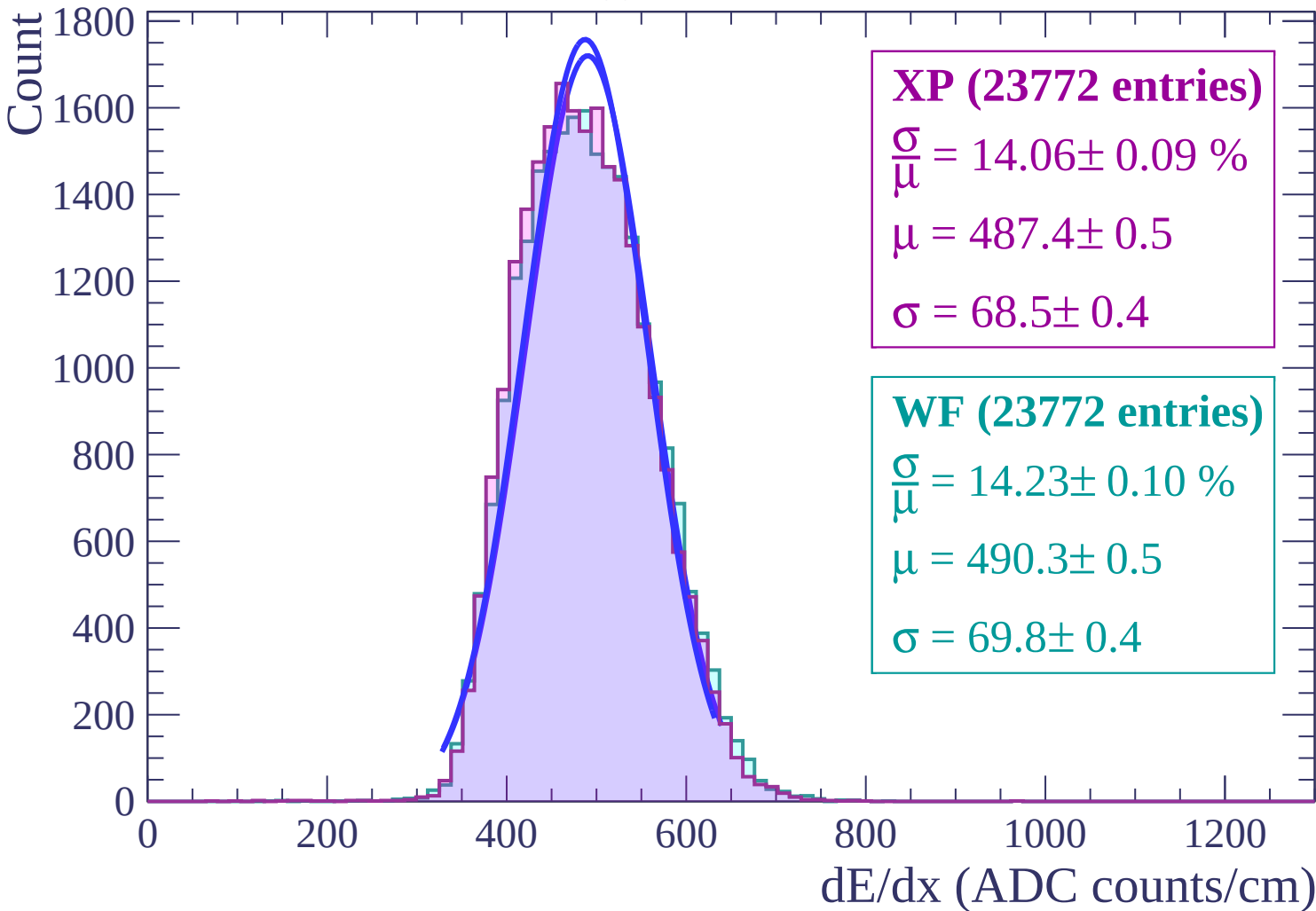
$\frac{\sigma}{\mu} = 14.67 \pm 0.10 \%$

$\mu = 488.3 \pm 0.6$

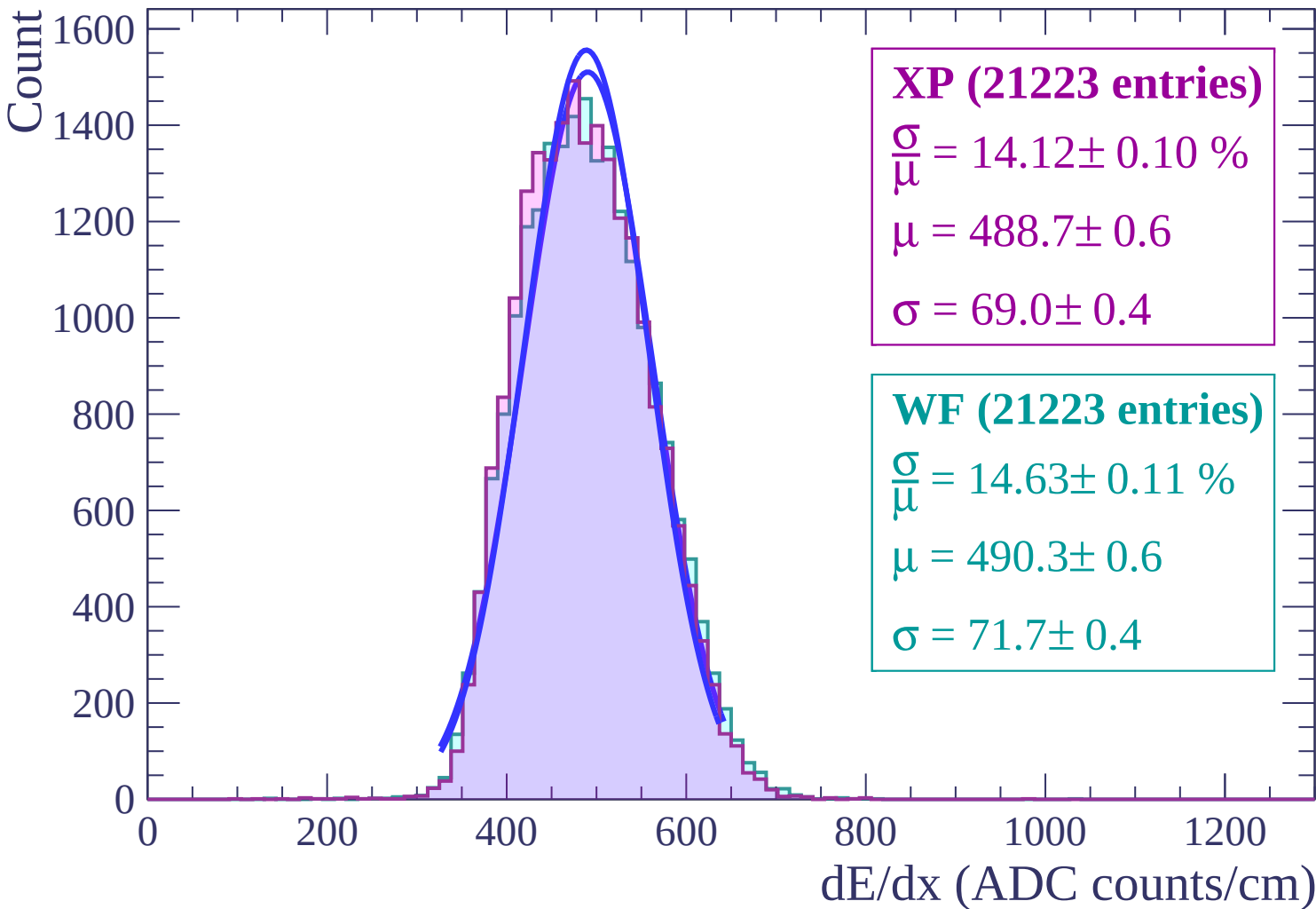
$\sigma = 71.6 \pm 0.4$

dE/dx (ADC counts/cm)

Energy loss | $6 < \theta < 8$

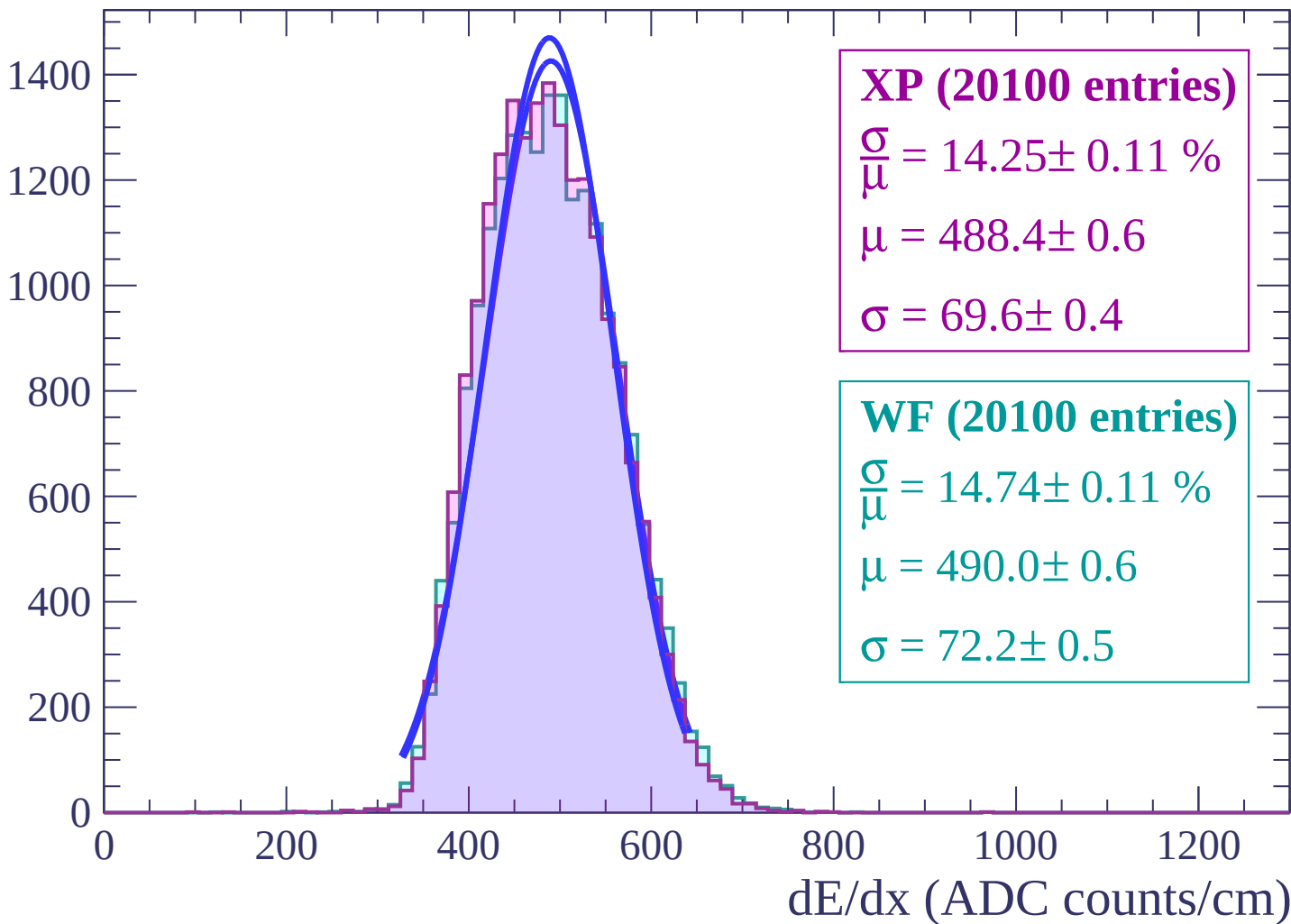


Energy loss | $8 < \theta < 10$



Energy loss | $10 < \theta < 12$

Count



Energy loss | $12 < \theta < 14$

Count

1200

1000

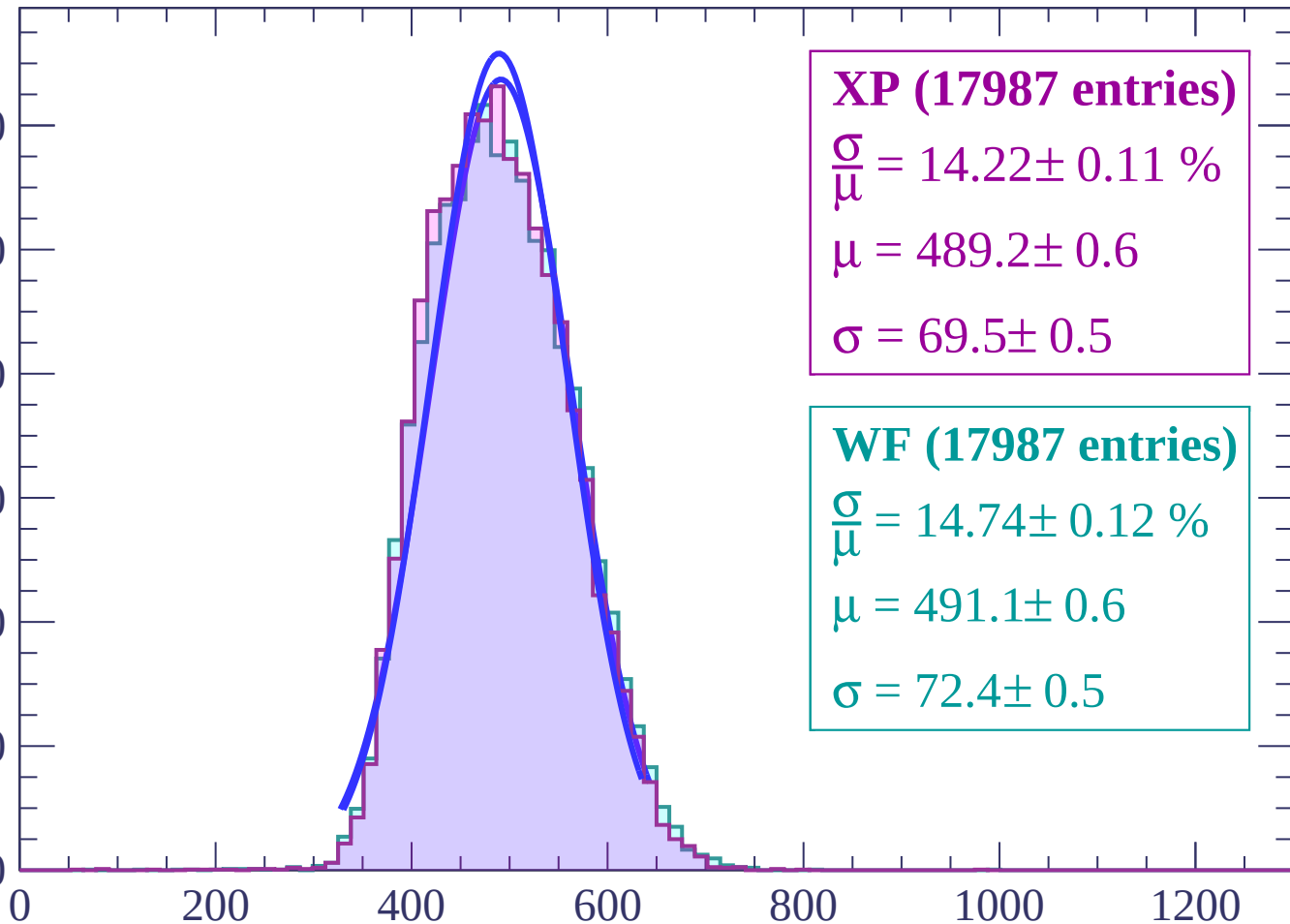
800

600

400

200

0



XP (17987 entries)

$\frac{\sigma}{\mu} = 14.22 \pm 0.11 \%$

$\mu = 489.2 \pm 0.6$

$\sigma = 69.5 \pm 0.5$

WF (17987 entries)

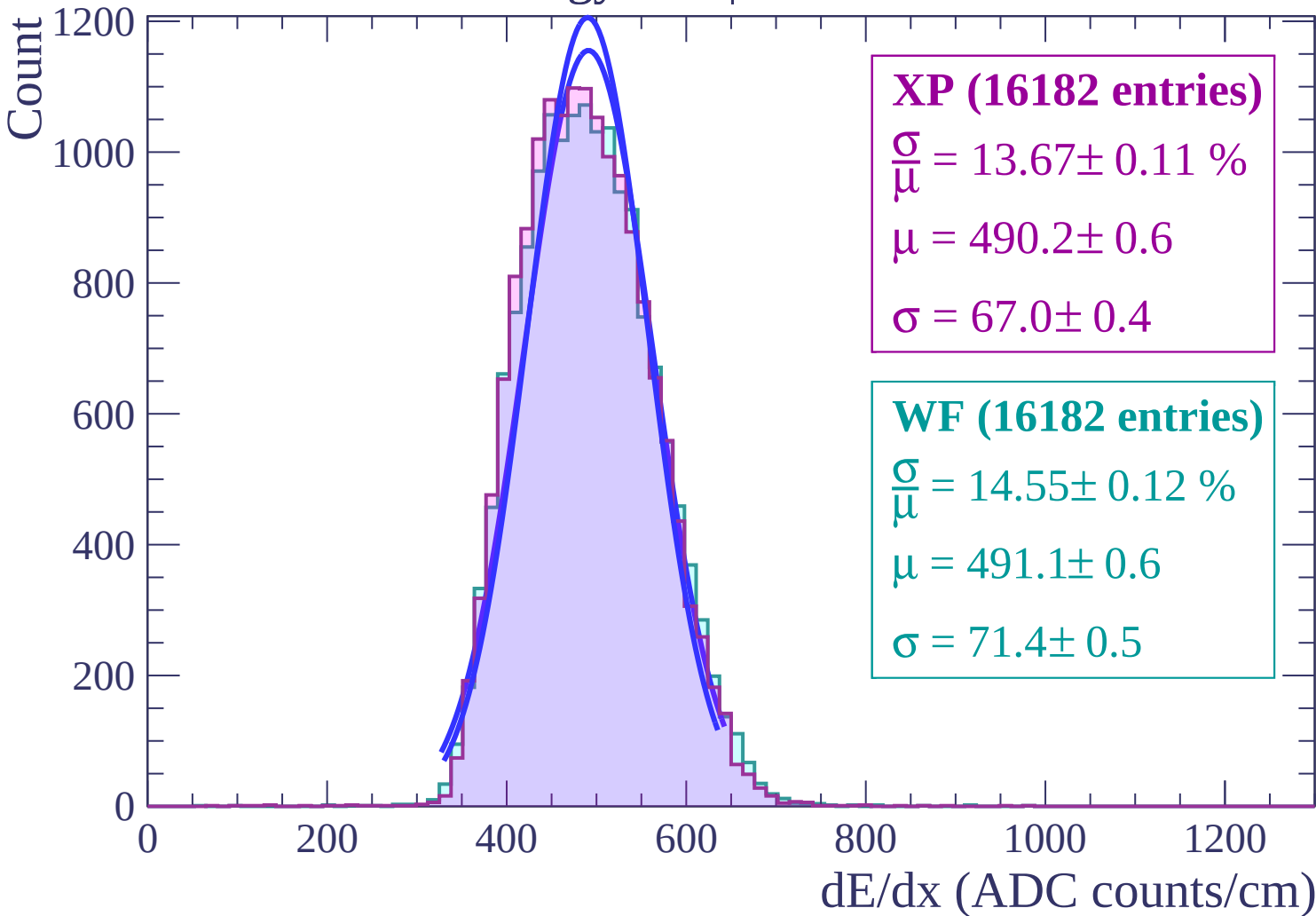
$\frac{\sigma}{\mu} = 14.74 \pm 0.12 \%$

$\mu = 491.1 \pm 0.6$

$\sigma = 72.4 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $14 < \theta < 16$



Energy loss | $16 < \theta < 18$

Count

1000

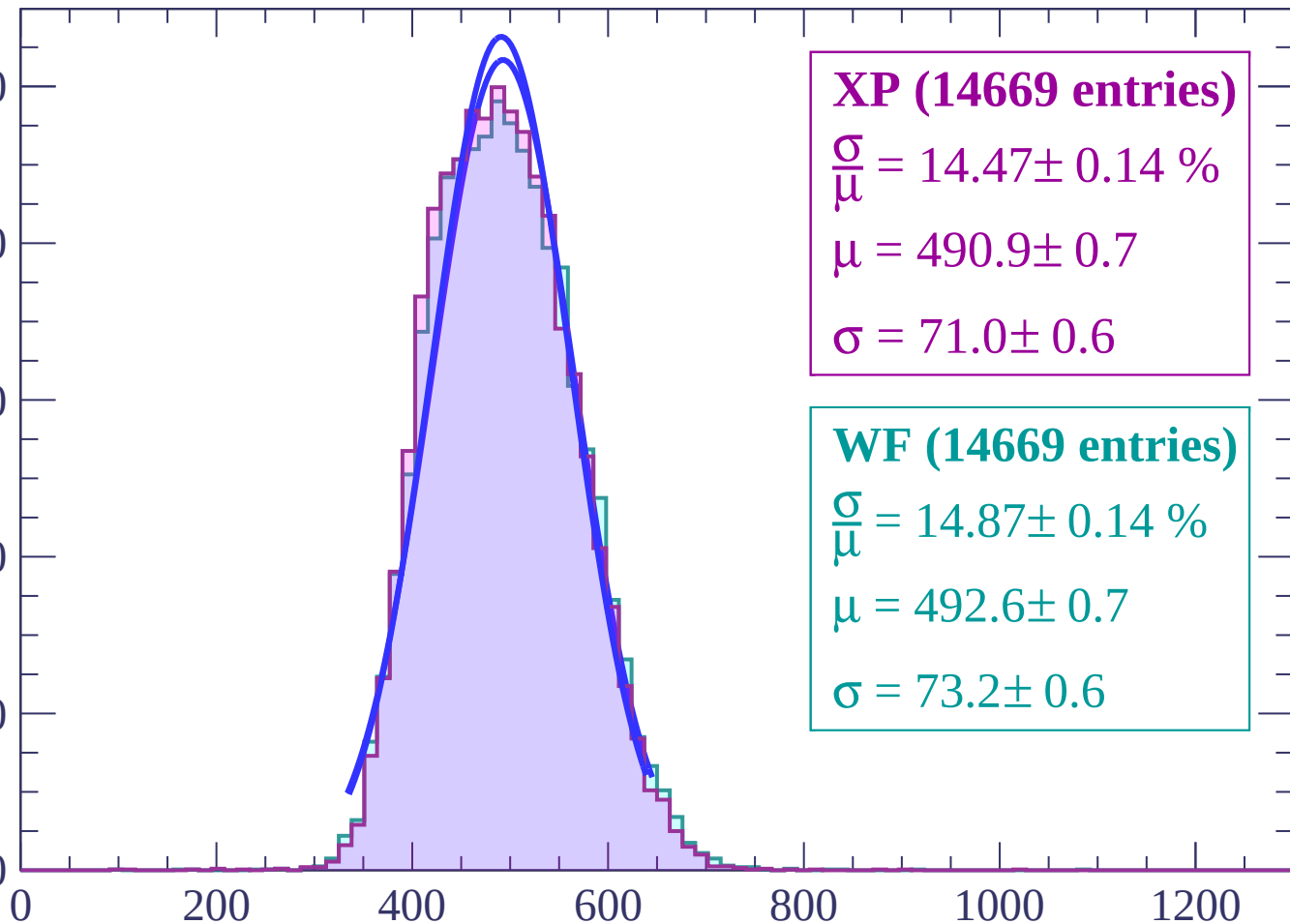
800

600

400

200

0



XP (14669 entries)

$\frac{\sigma}{\mu} = 14.47 \pm 0.14 \%$

$\mu = 490.9 \pm 0.7$

$\sigma = 71.0 \pm 0.6$

WF (14669 entries)

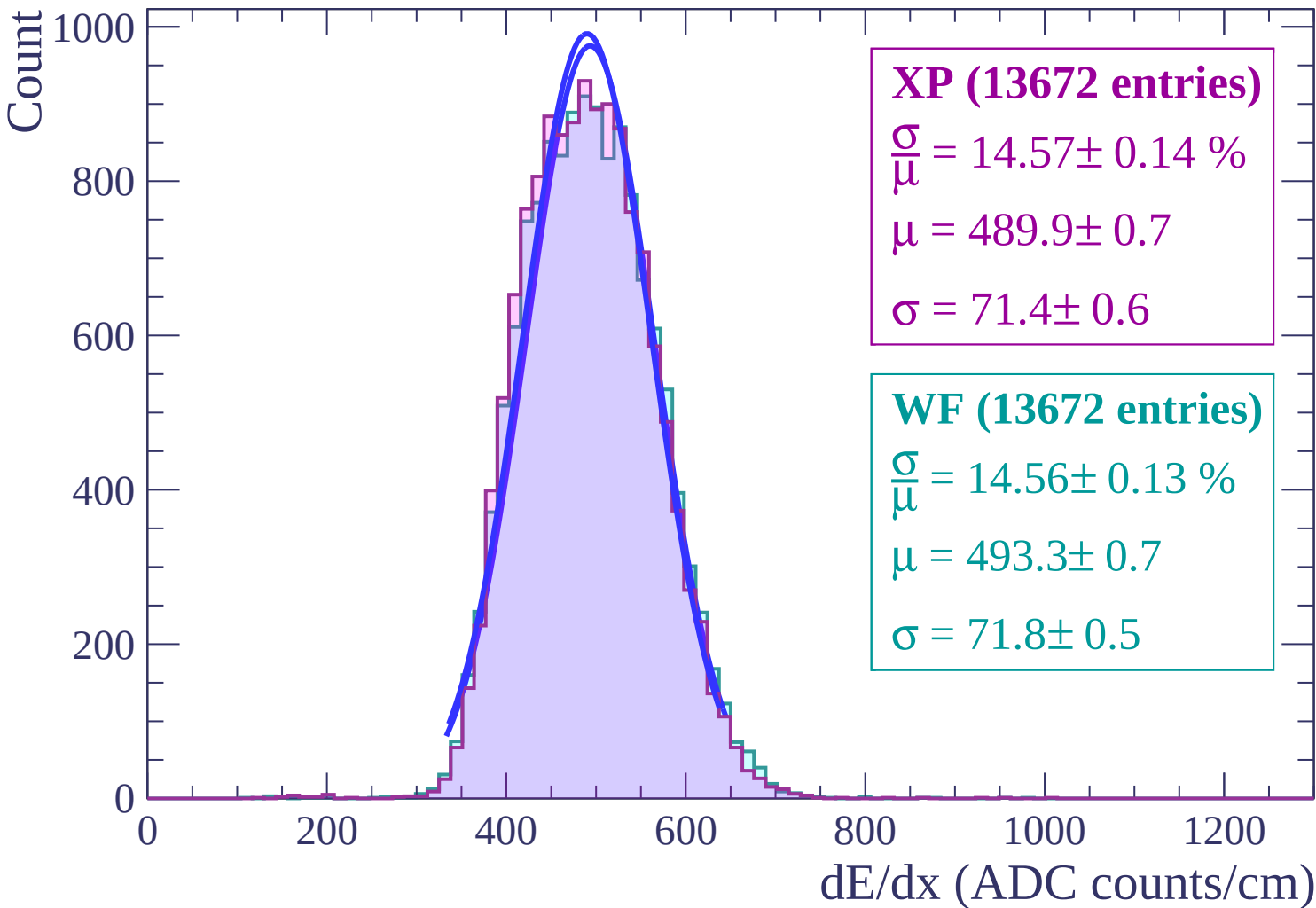
$\frac{\sigma}{\mu} = 14.87 \pm 0.14 \%$

$\mu = 492.6 \pm 0.7$

$\sigma = 73.2 \pm 0.6$

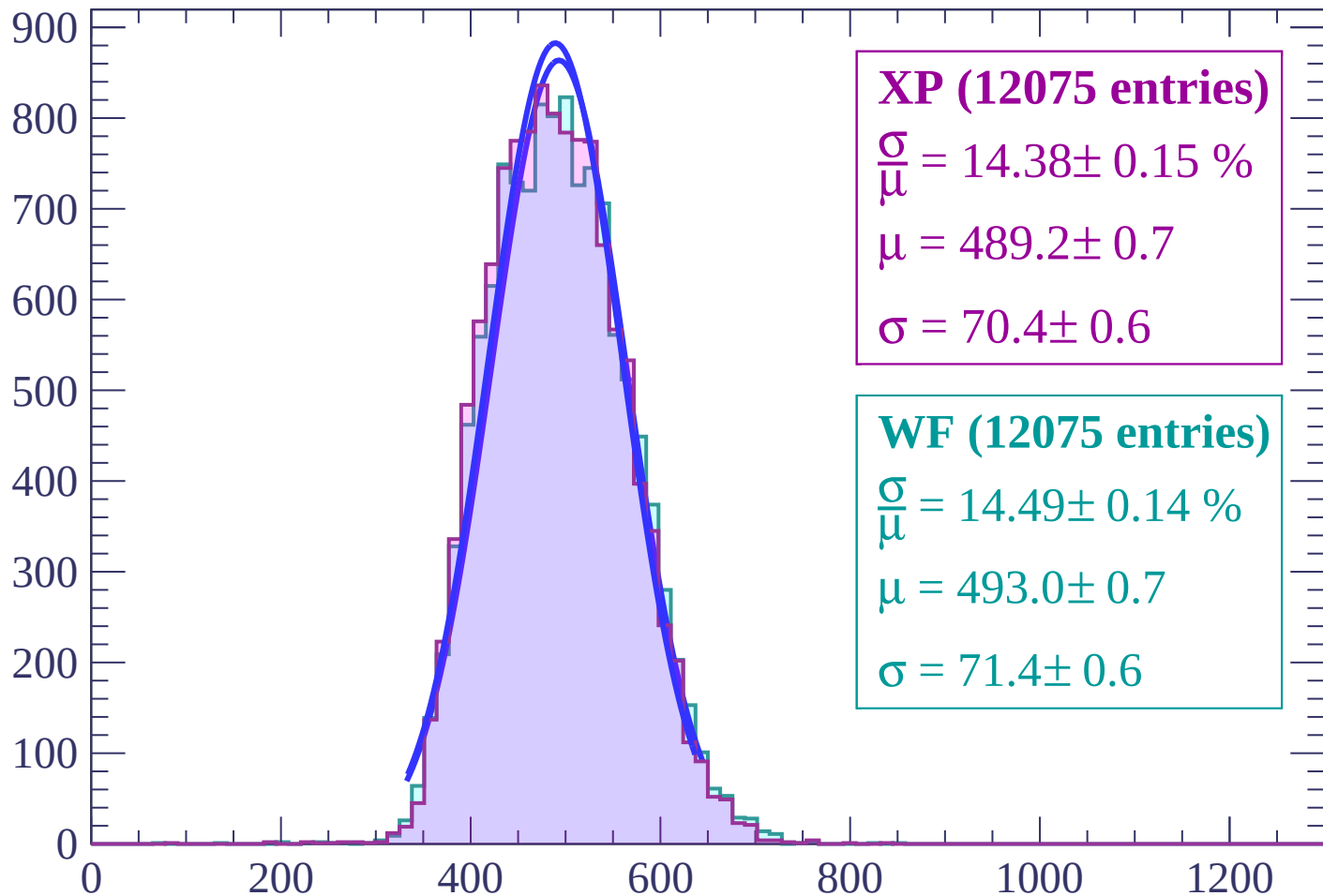
dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 20$



Energy loss | $20 < \theta < 22$

Count



XP (12075 entries)

$$\frac{\sigma}{\mu} = 14.38 \pm 0.15 \%$$

$$\mu = 489.2 \pm 0.7$$

$$\sigma = 70.4 \pm 0.6$$

WF (12075 entries)

$$\frac{\sigma}{\mu} = 14.49 \pm 0.14 \%$$

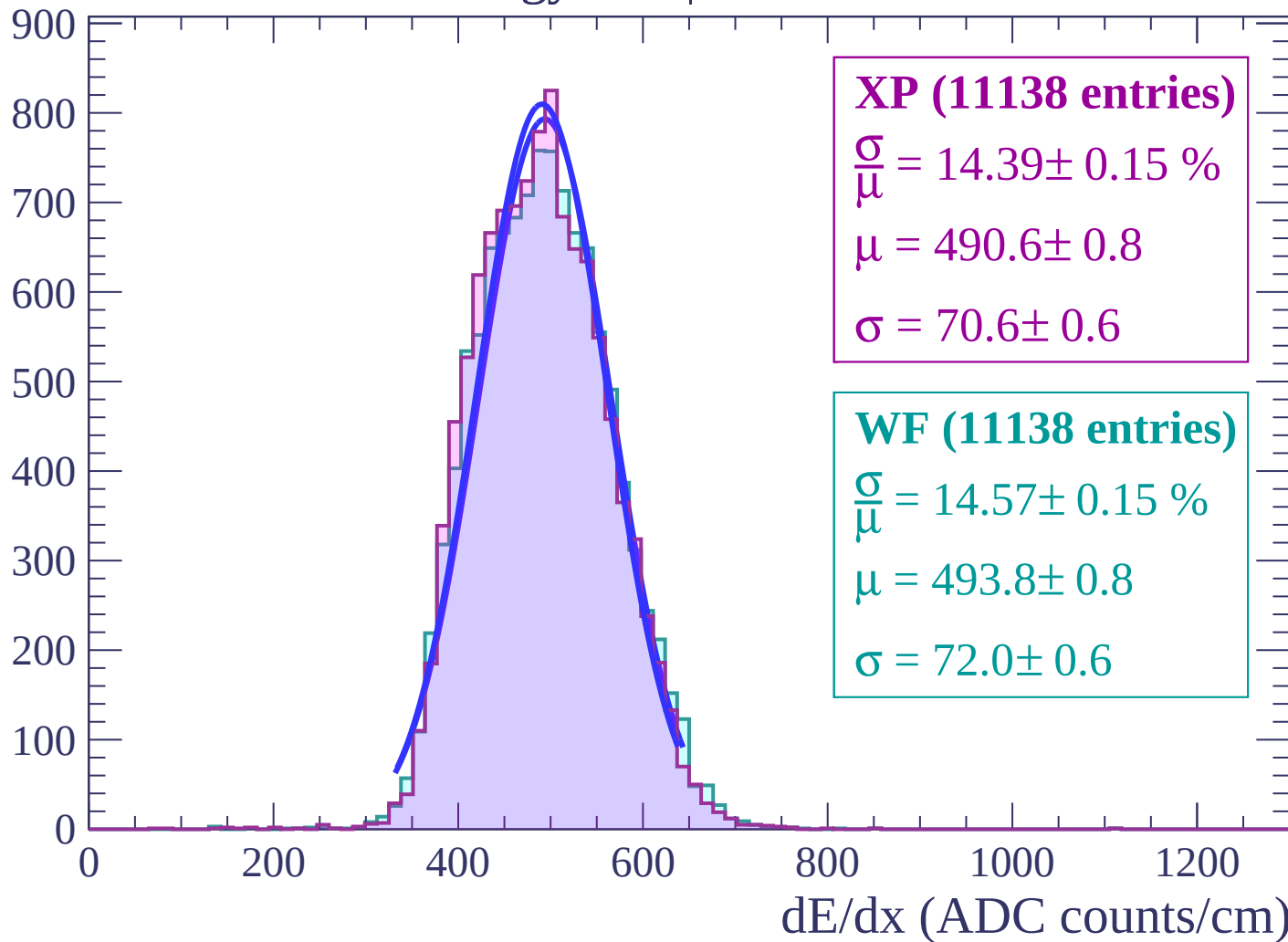
$$\mu = 493.0 \pm 0.7$$

$$\sigma = 71.4 \pm 0.6$$

dE/dx (ADC counts/cm)

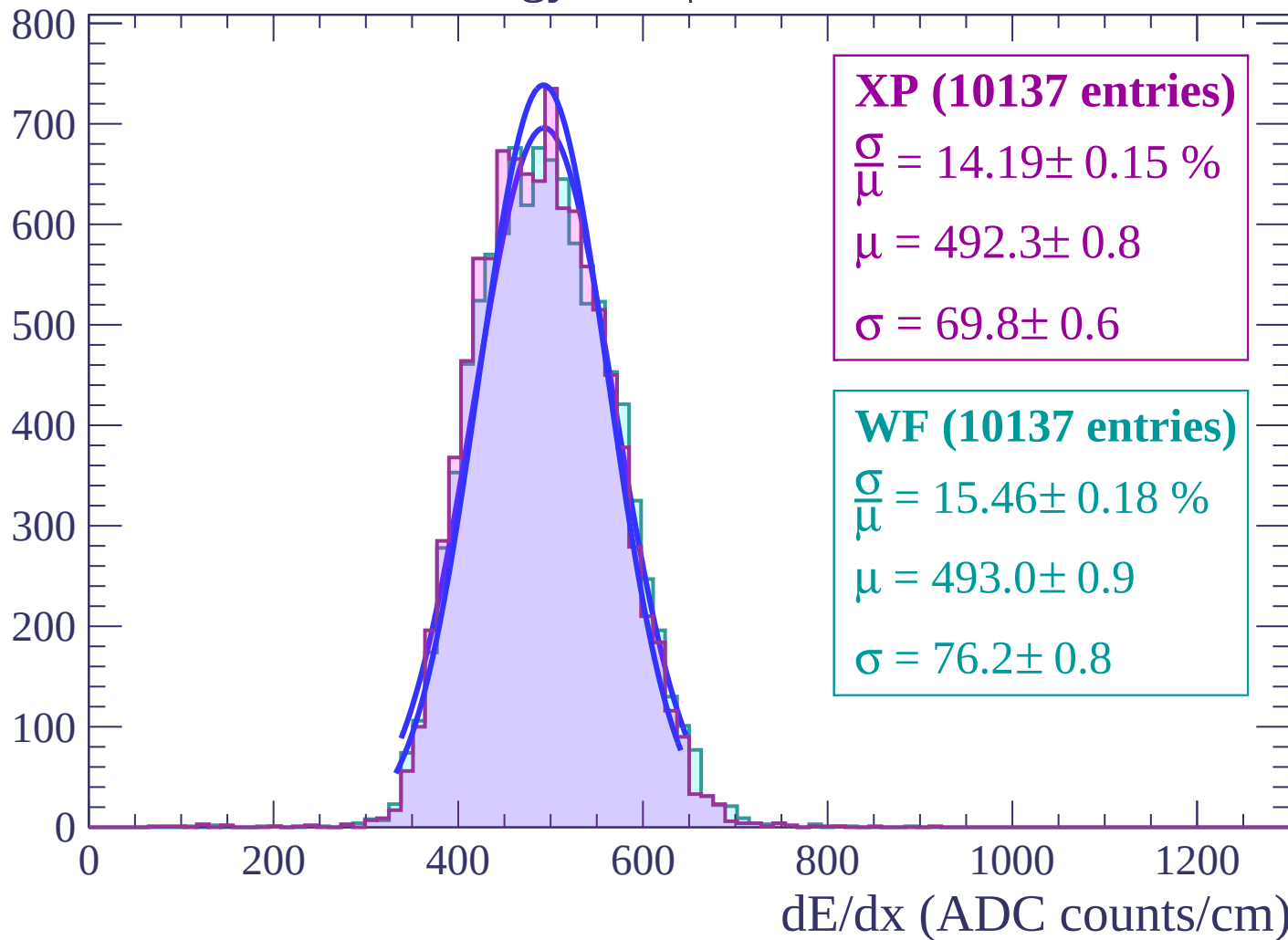
Energy loss | $22 < \theta < 24$

Count



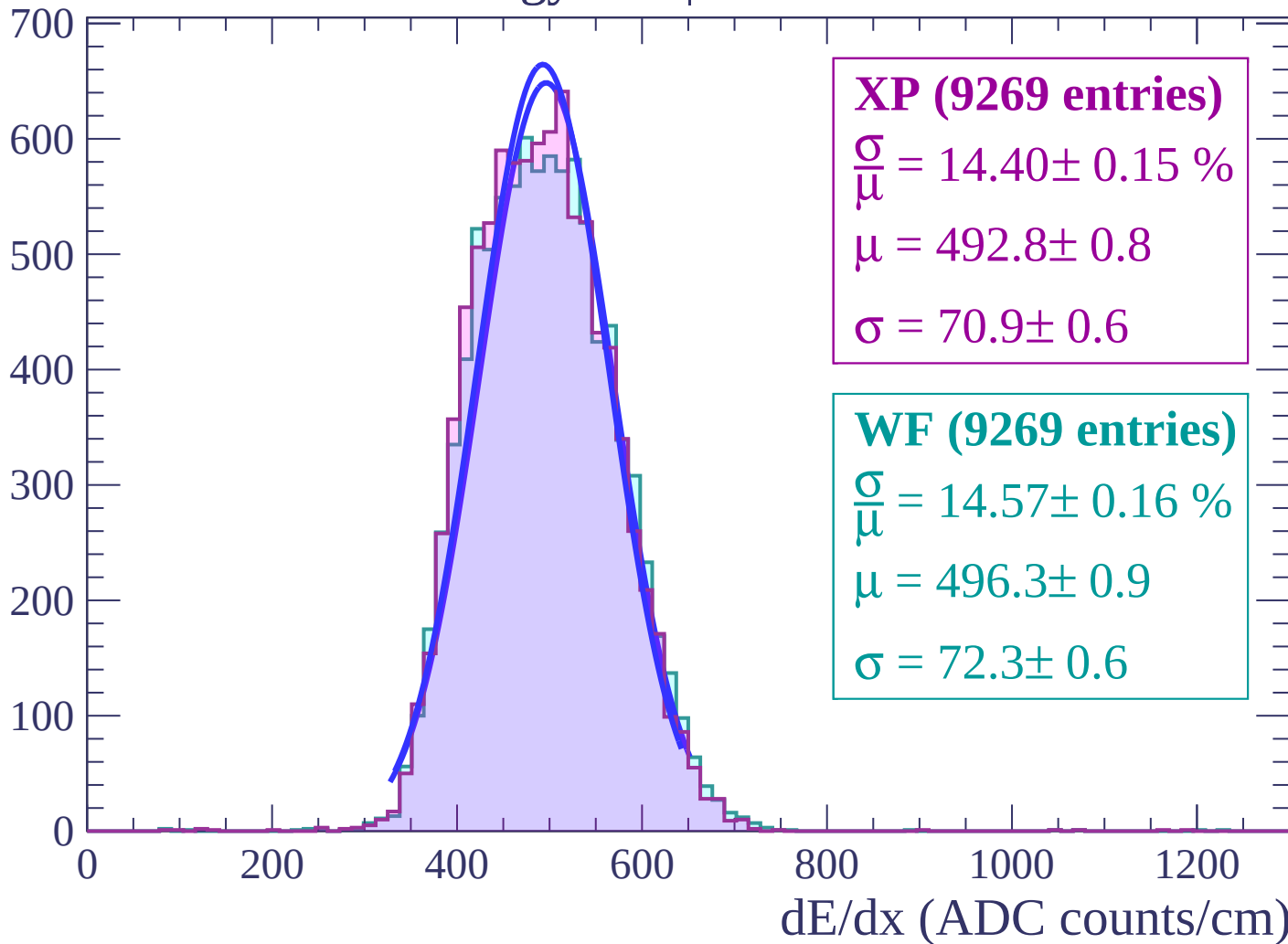
Energy loss | $24 < \theta < 26$

Count



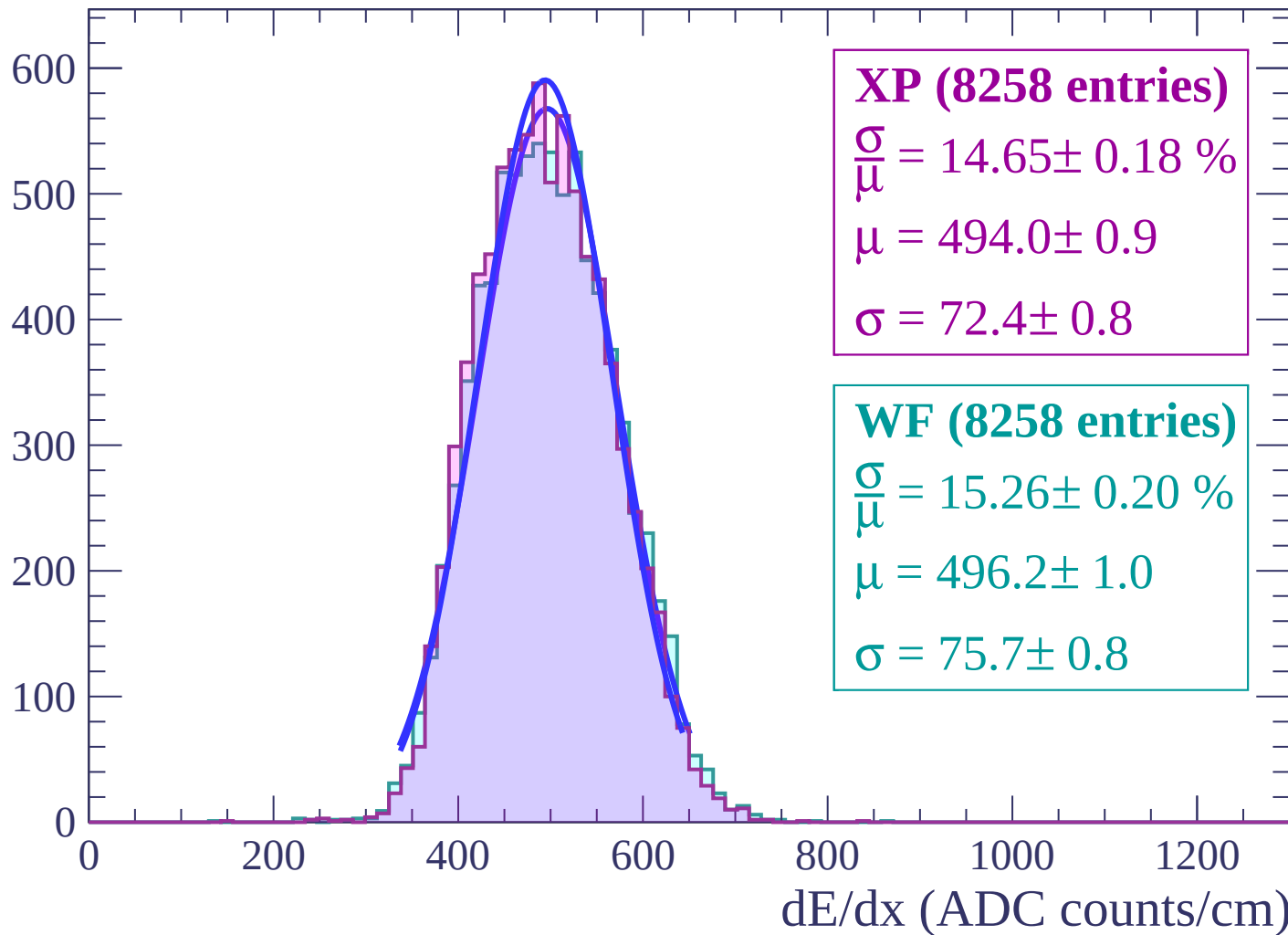
Energy loss | $26 < \theta < 28$

Count



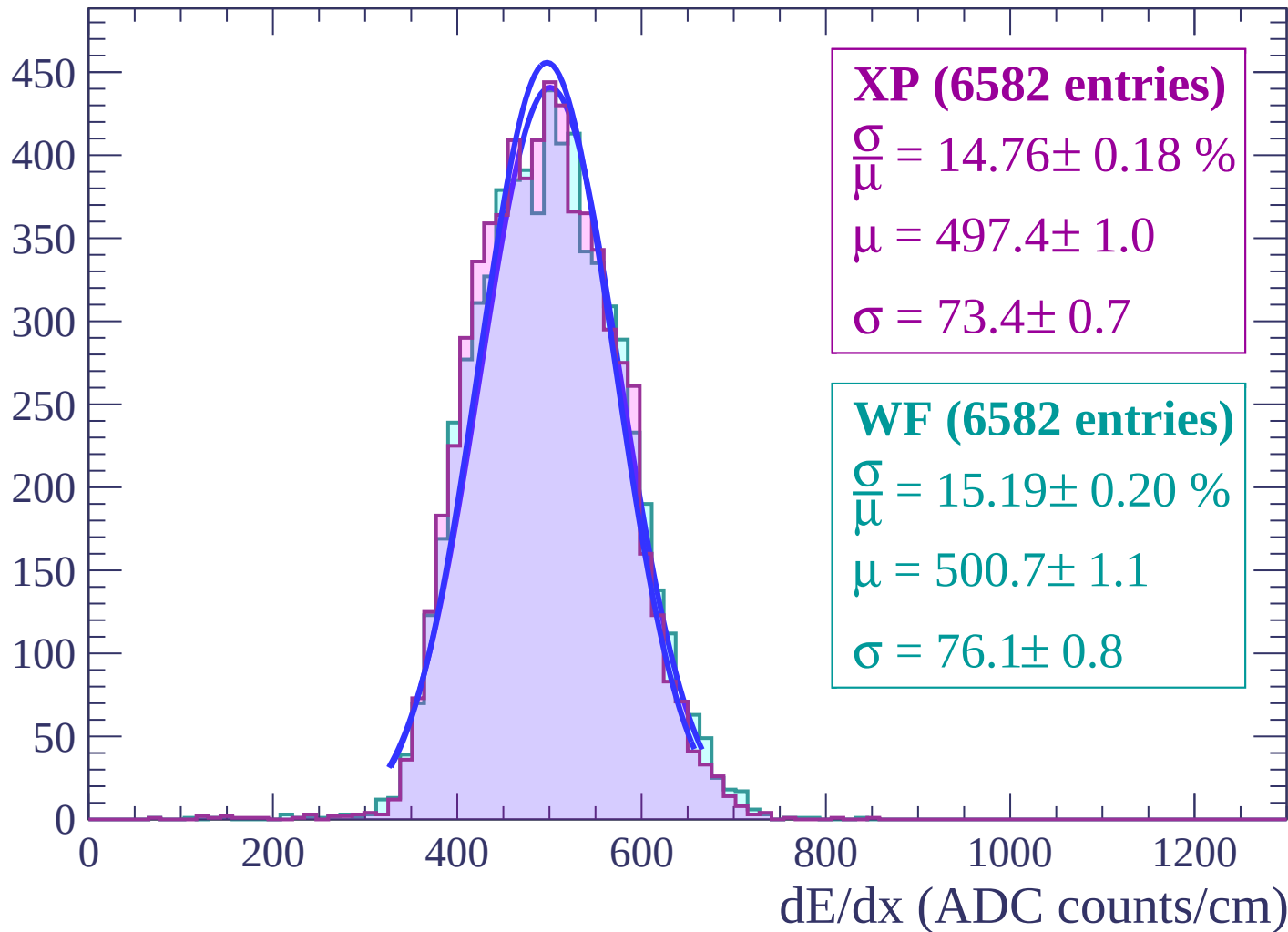
Energy loss | $28 < \theta < 30$

Count



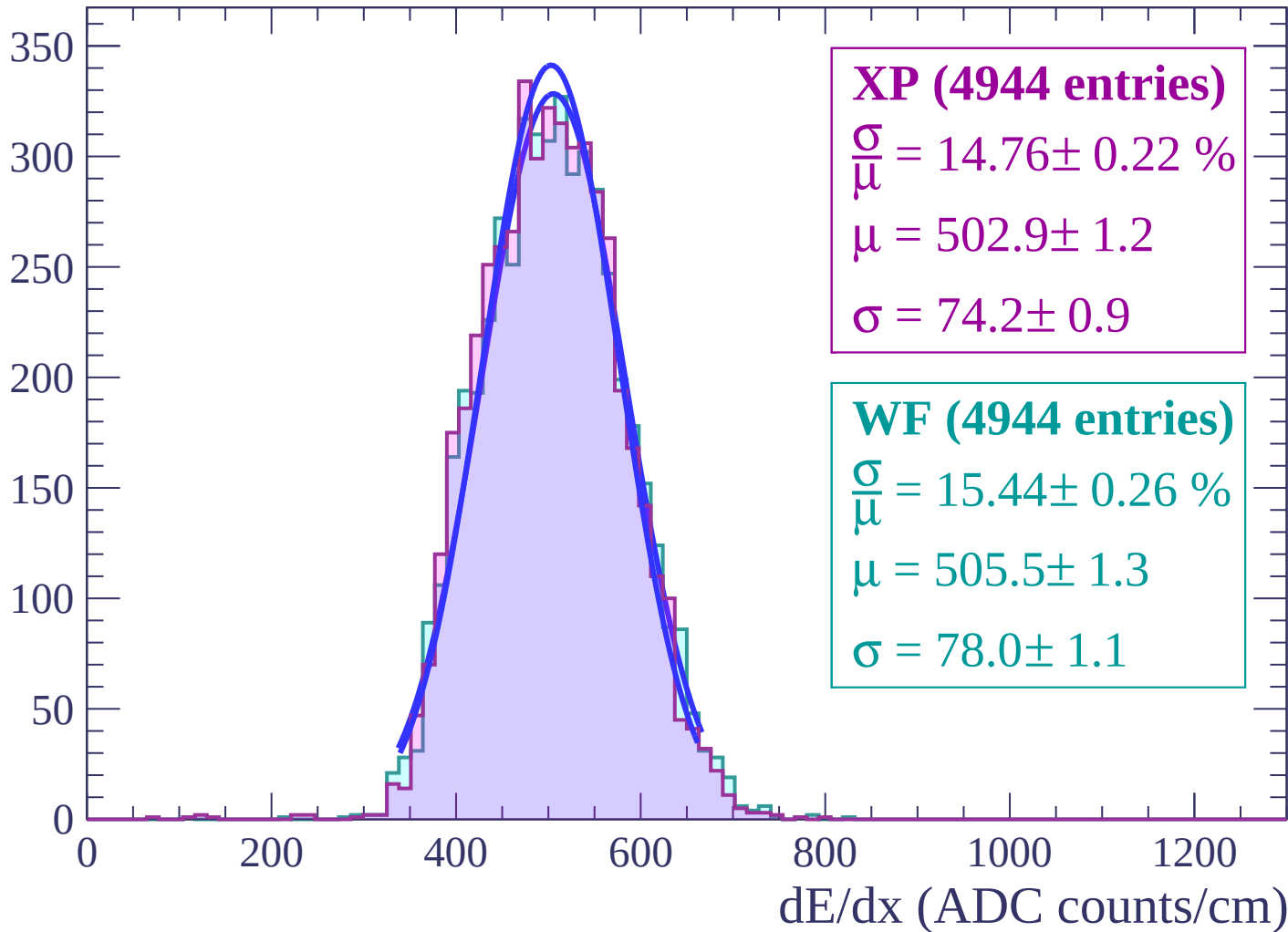
Energy loss | $30 < \theta < 32$

Count



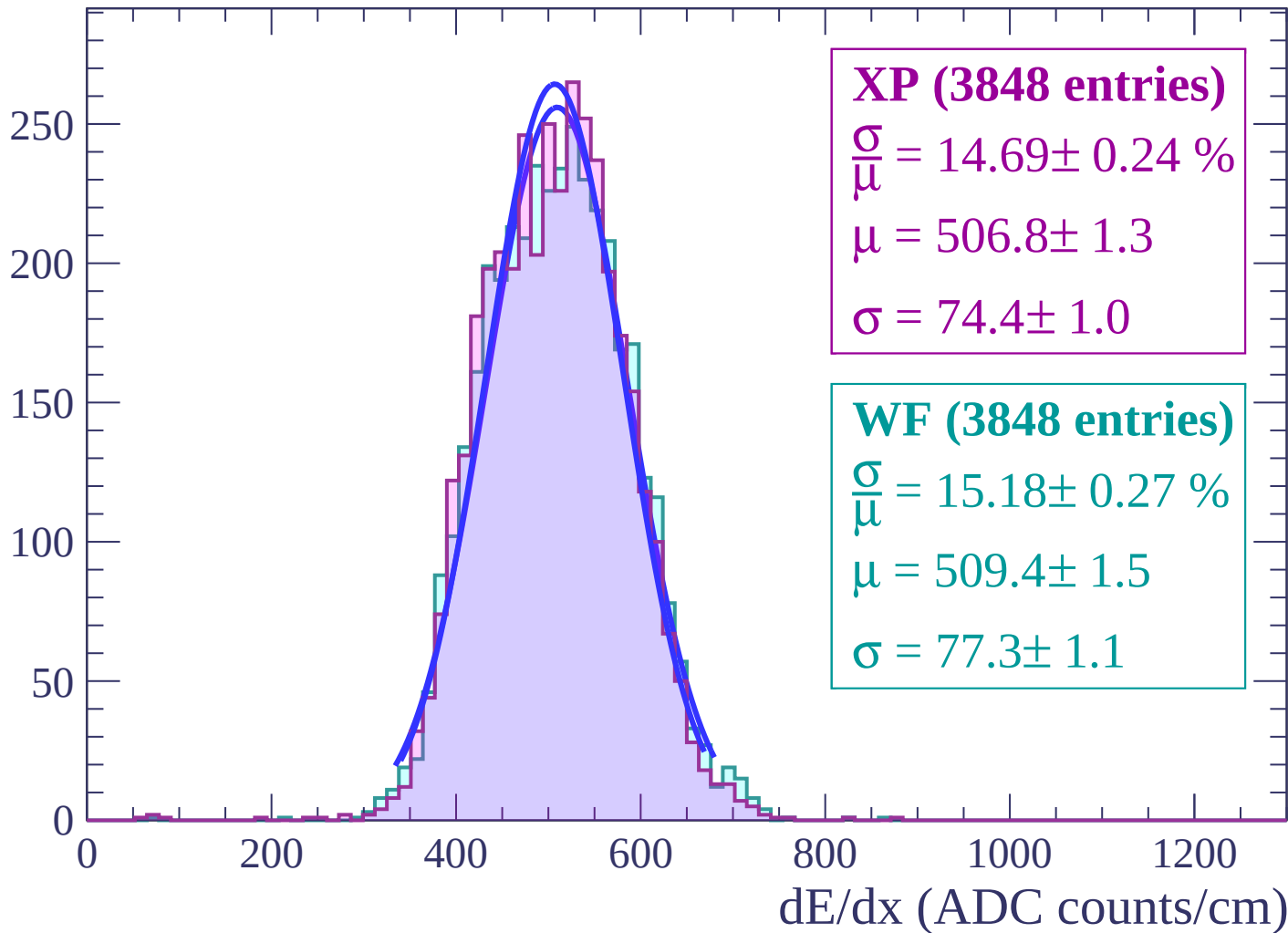
Energy loss | $32 < \theta < 34$

Count



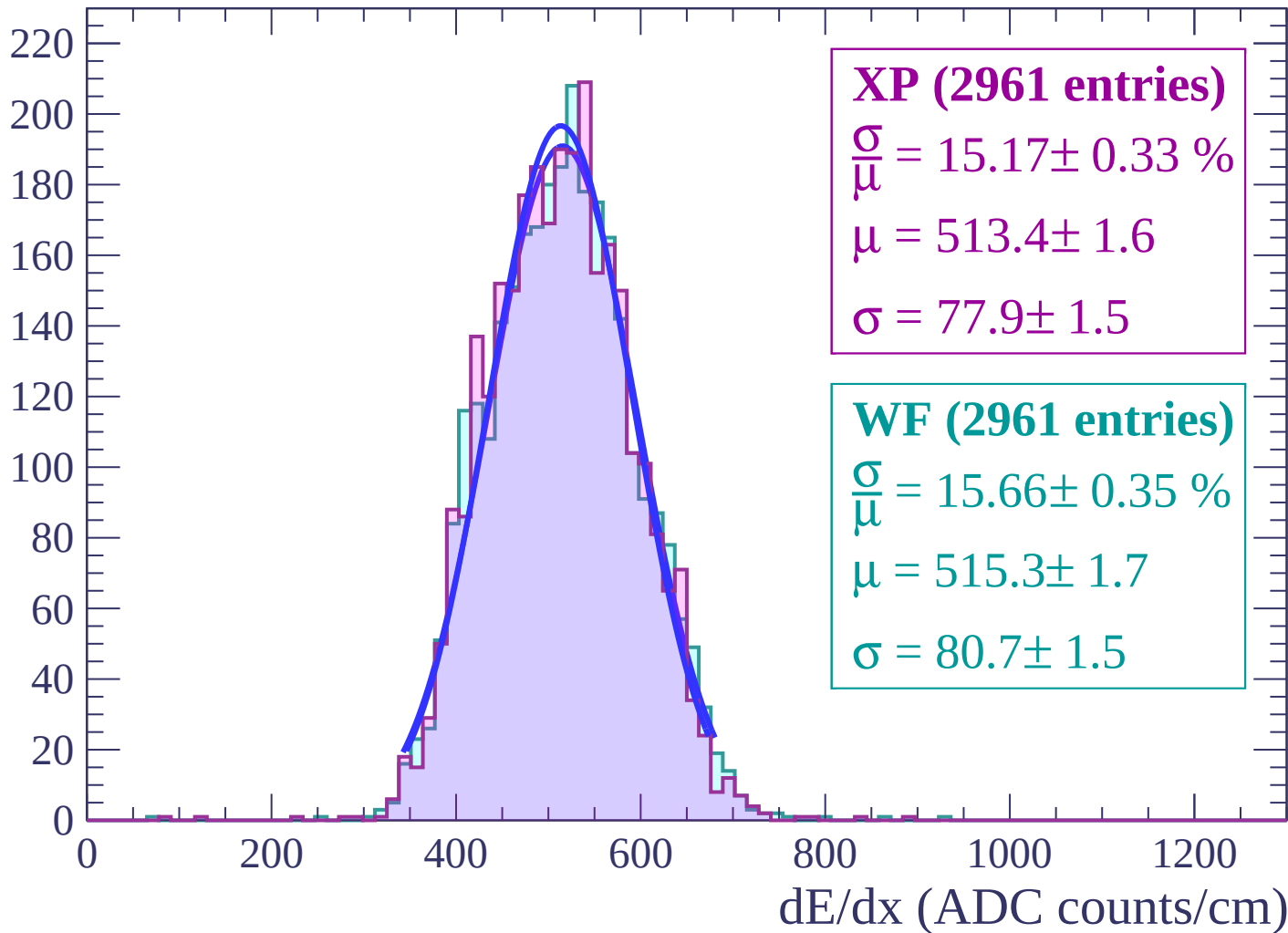
Energy loss | $34 < \theta < 36$

Count



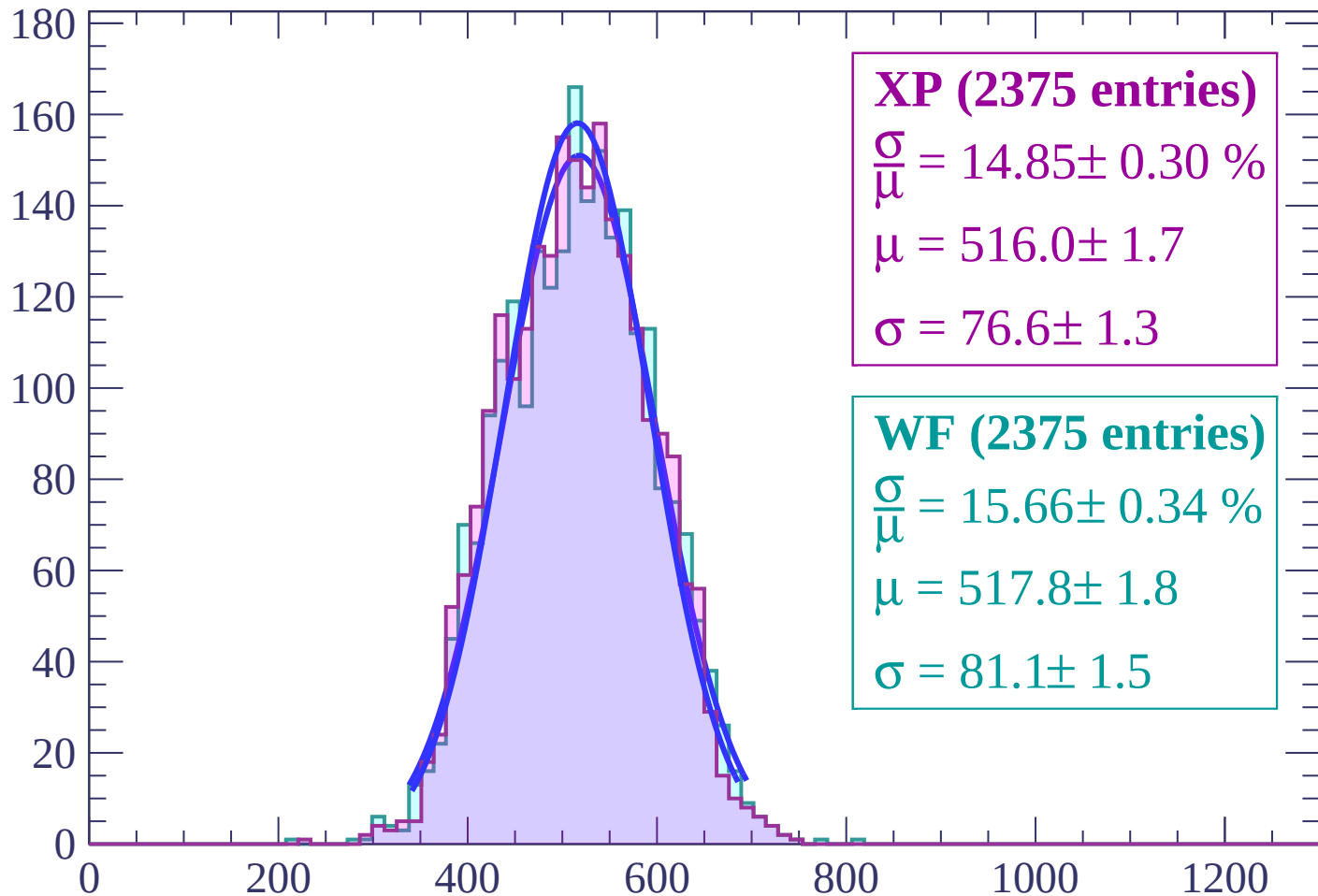
Energy loss | $36 < \theta < 38$

Count



Energy loss | $38 < \theta < 40$

Count



XP (2375 entries)

$\frac{\sigma}{\mu} = 14.85 \pm 0.30 \%$

$\mu = 516.0 \pm 1.7$

$\sigma = 76.6 \pm 1.3$

WF (2375 entries)

$\frac{\sigma}{\mu} = 15.66 \pm 0.34 \%$

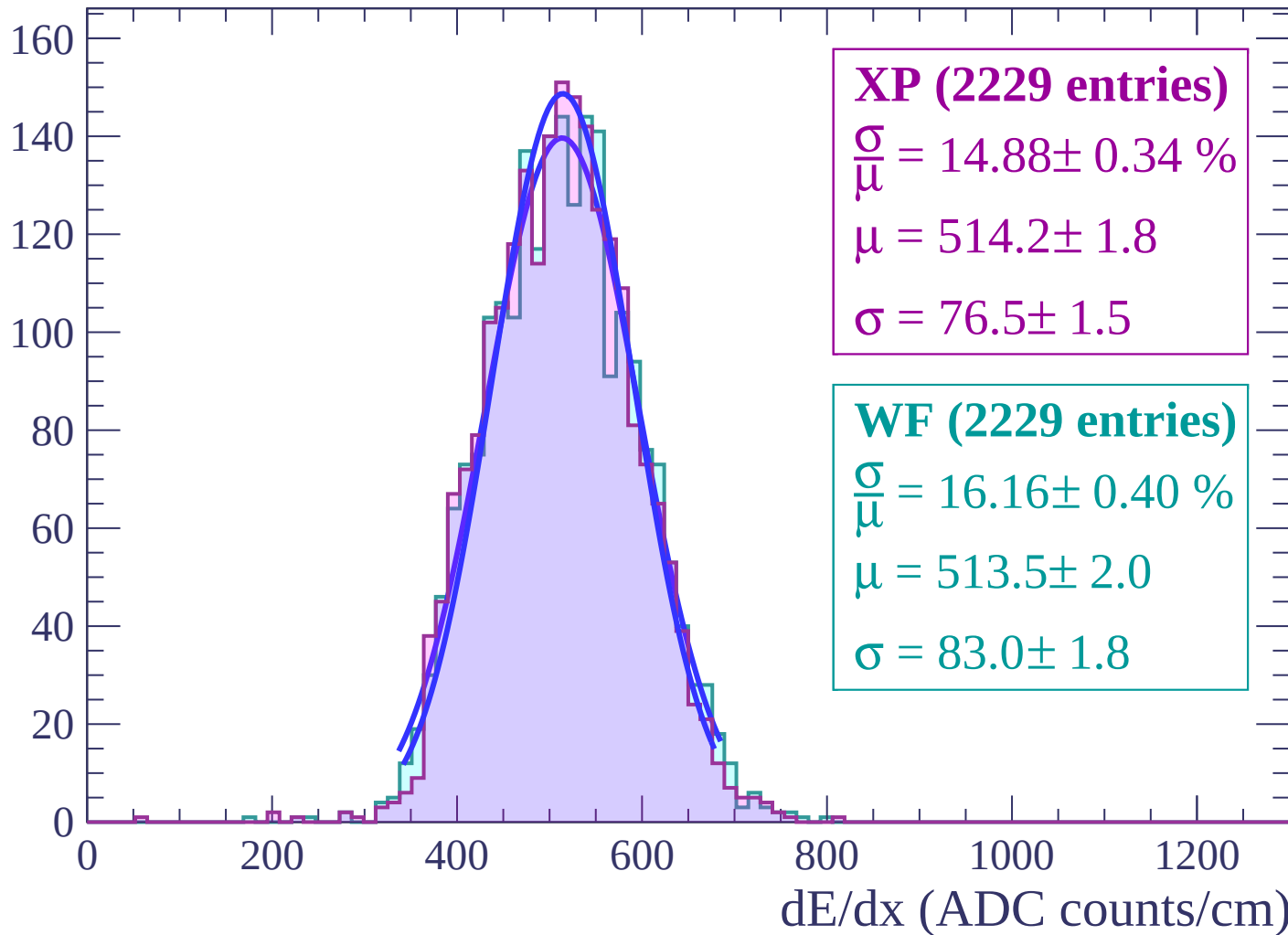
$\mu = 517.8 \pm 1.8$

$\sigma = 81.1 \pm 1.5$

dE/dx (ADC counts/cm)

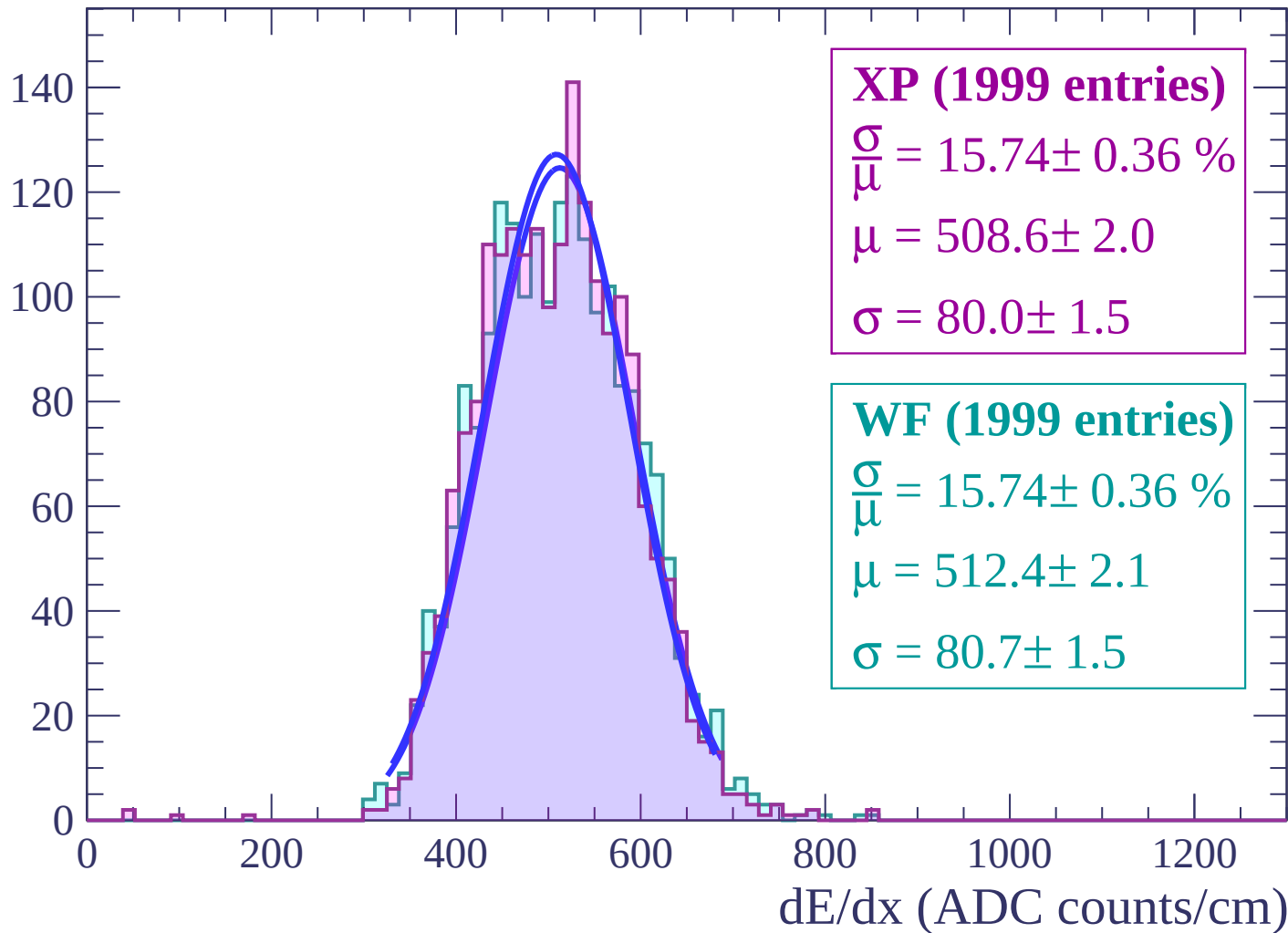
Energy loss | $40 < \theta < 42$

Count



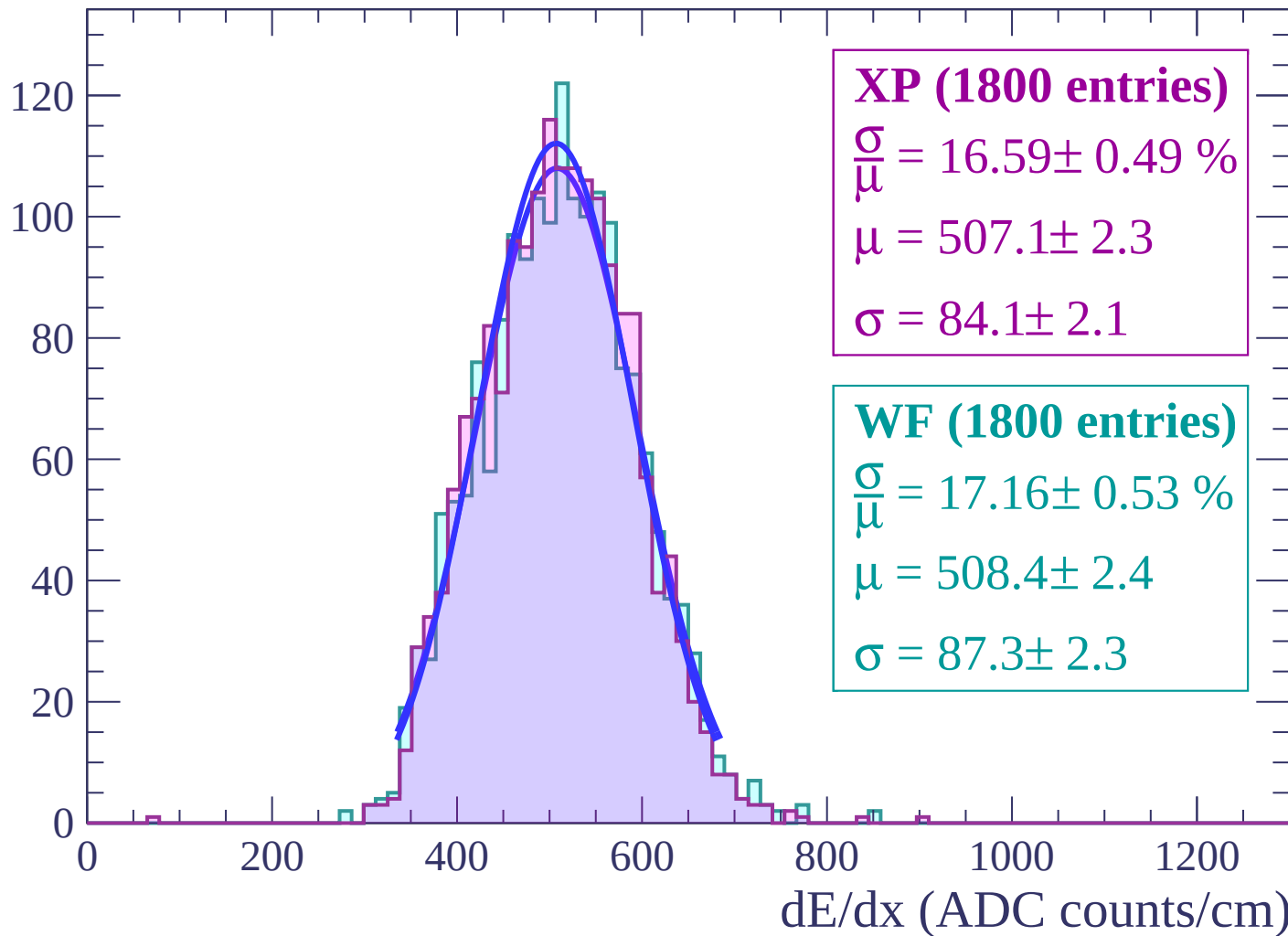
Energy loss | $42 < \theta < 44$

Count



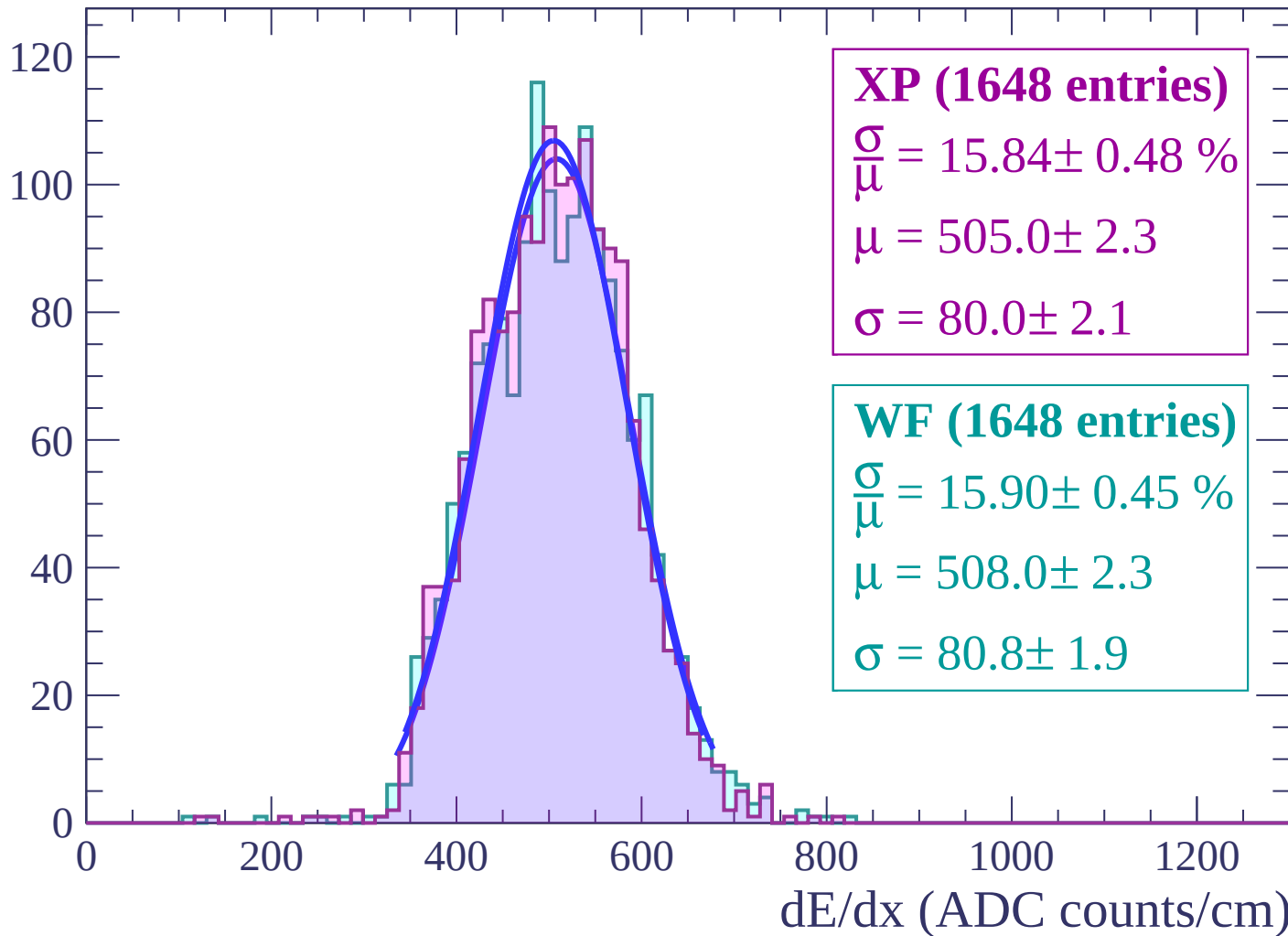
Energy loss | $44 < \theta < 46$

Count



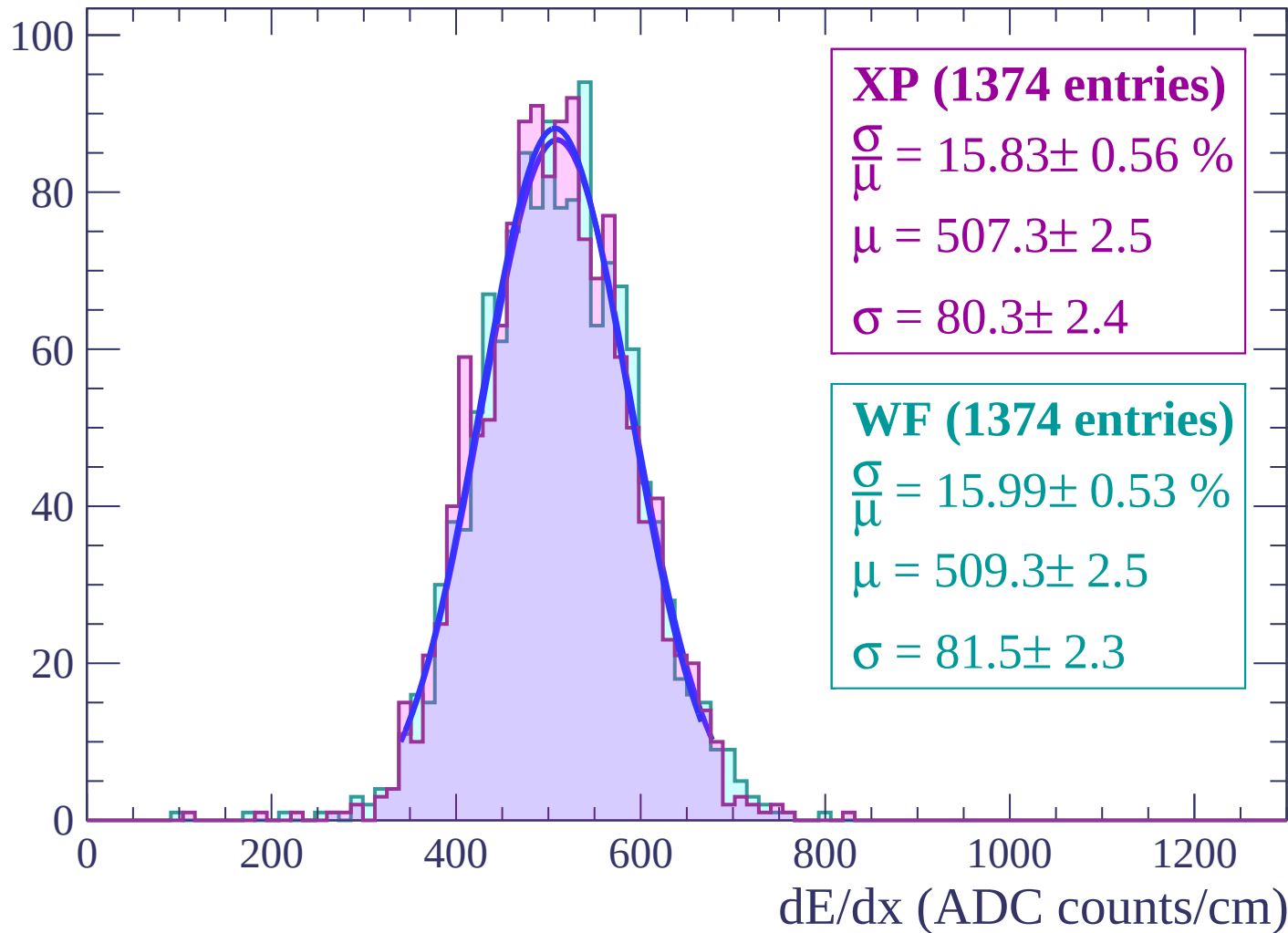
Energy loss | $46 < \theta < 48$

Count



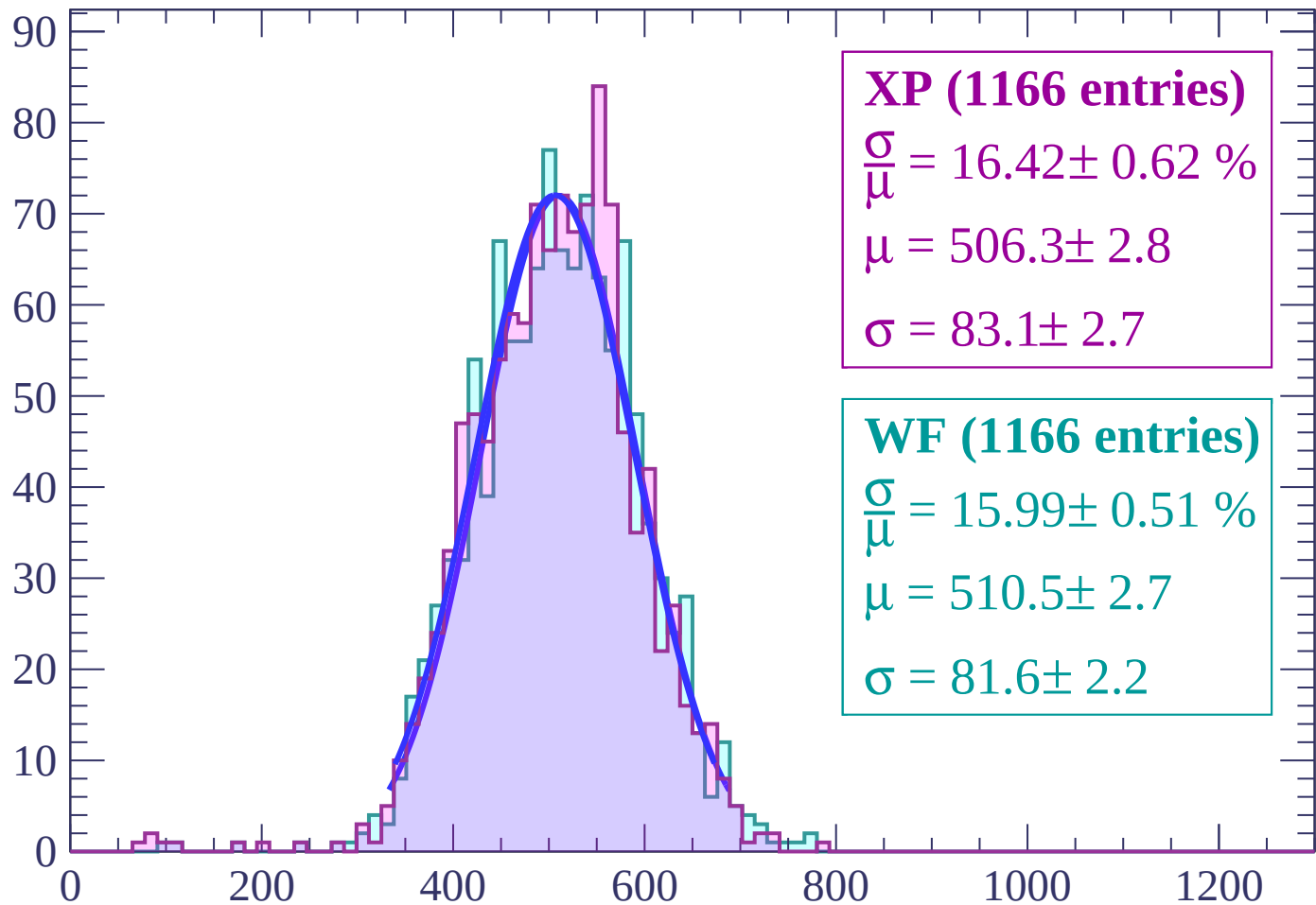
Energy loss | $48 < \theta < 50$

Count



Energy loss | $50 < \theta < 52$

Count



XP (1166 entries)

$$\frac{\sigma}{\mu} = 16.42 \pm 0.62 \%$$

$$\mu = 506.3 \pm 2.8$$

$$\sigma = 83.1 \pm 2.7$$

WF (1166 entries)

$$\frac{\sigma}{\mu} = 15.99 \pm 0.51 \%$$

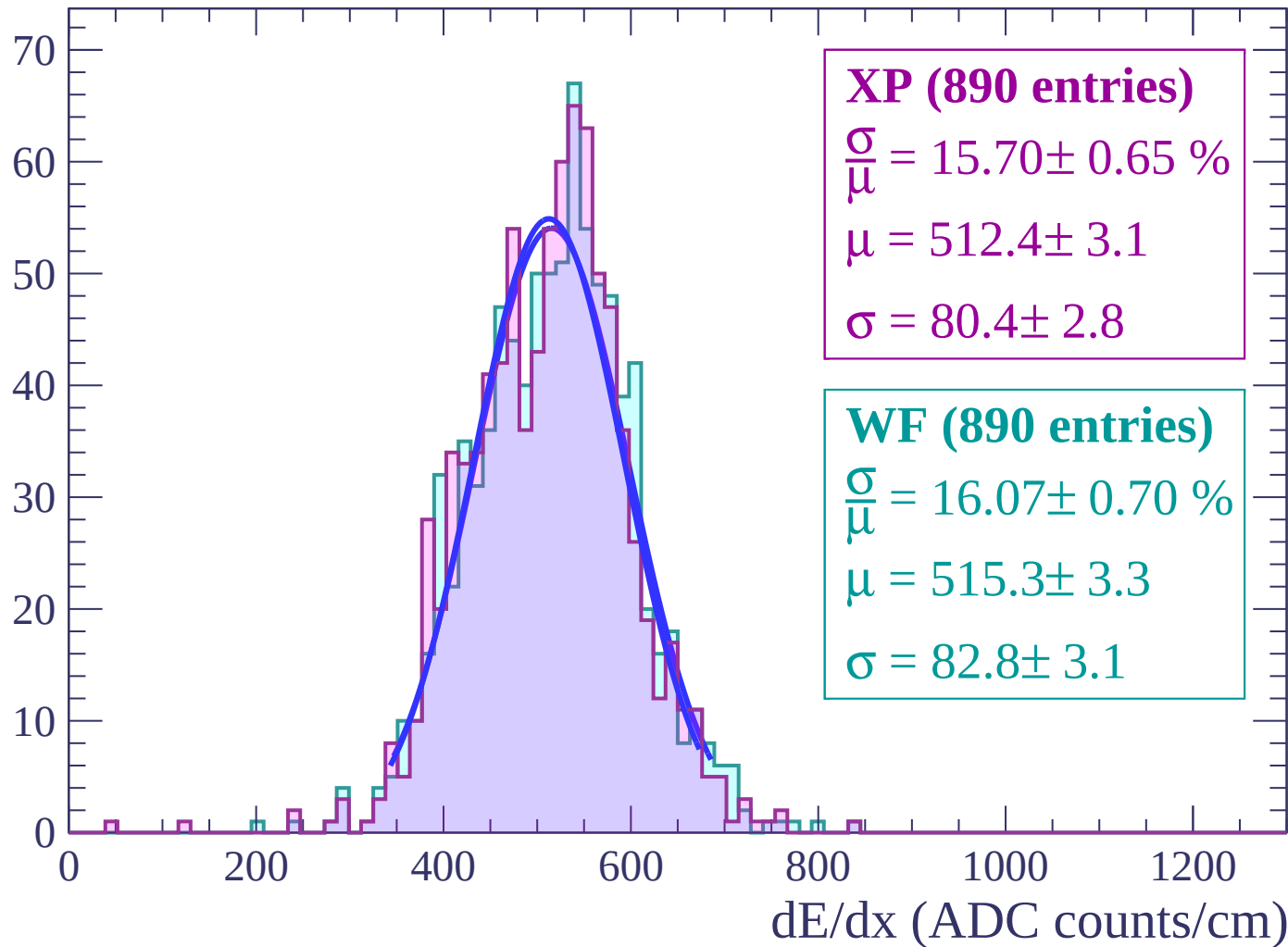
$$\mu = 510.5 \pm 2.7$$

$$\sigma = 81.6 \pm 2.2$$

dE/dx (ADC counts/cm)

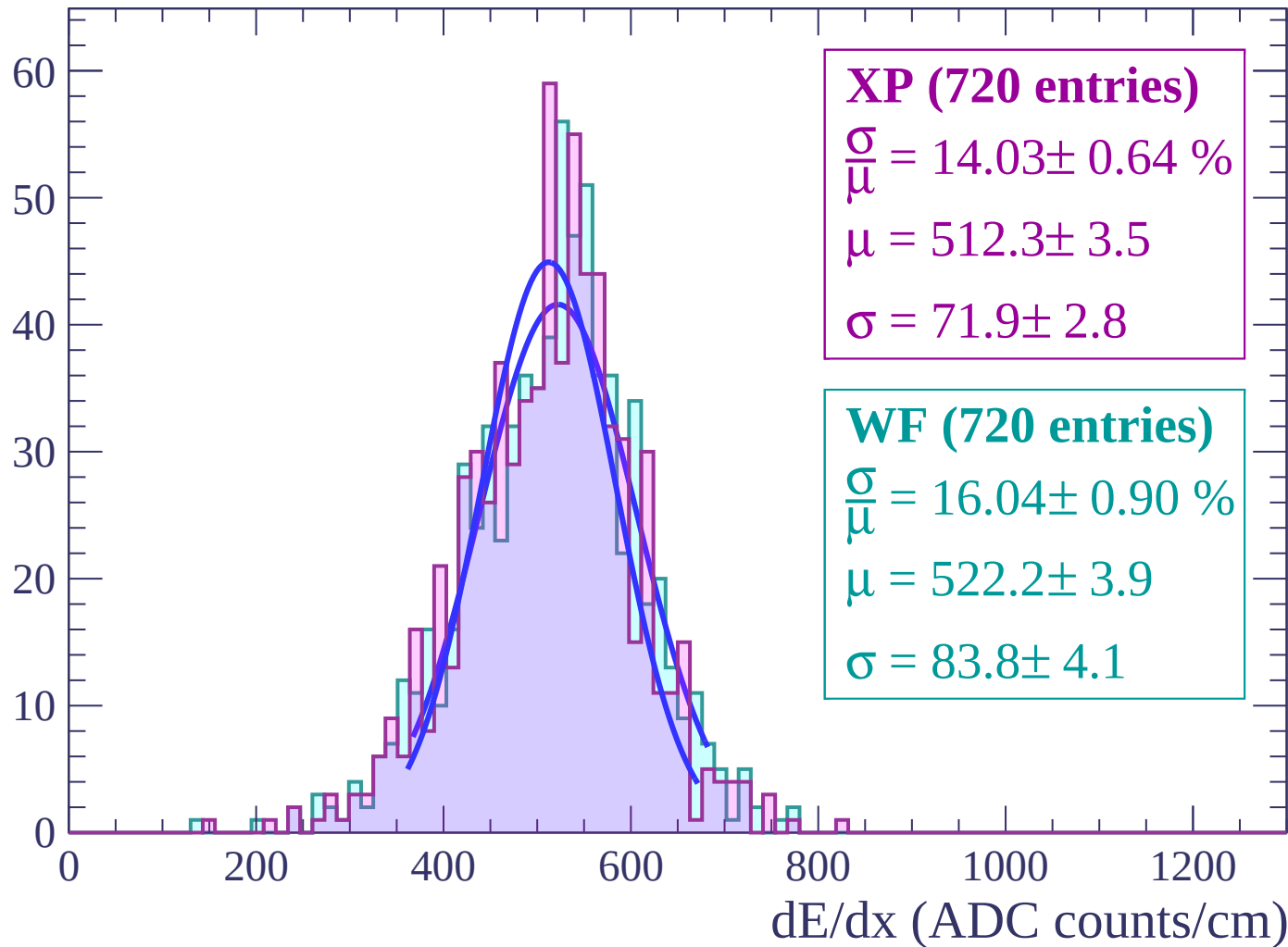
Energy loss | $52 < \theta < 54$

Count



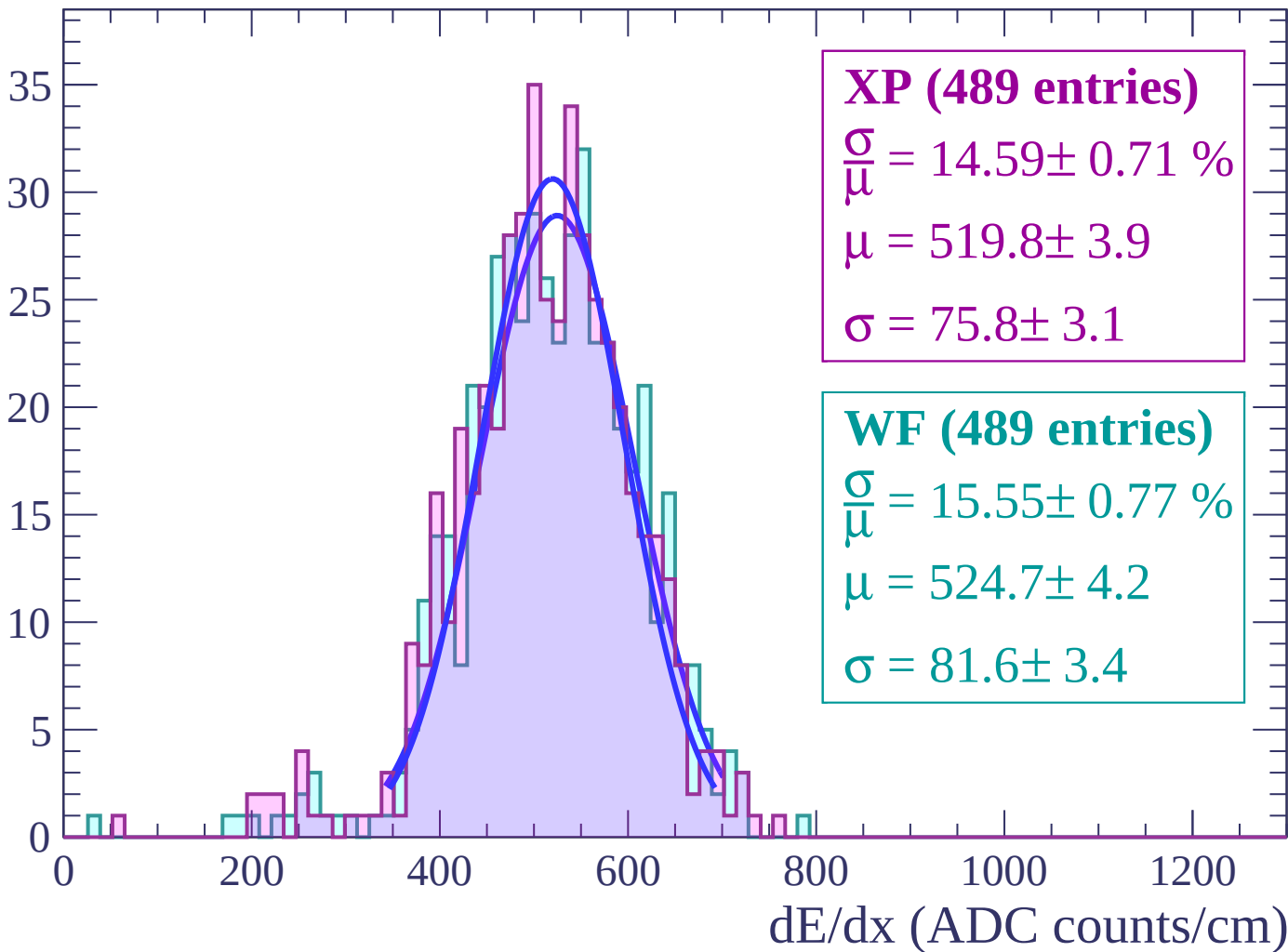
Energy loss | $54 < \theta < 56$

Count



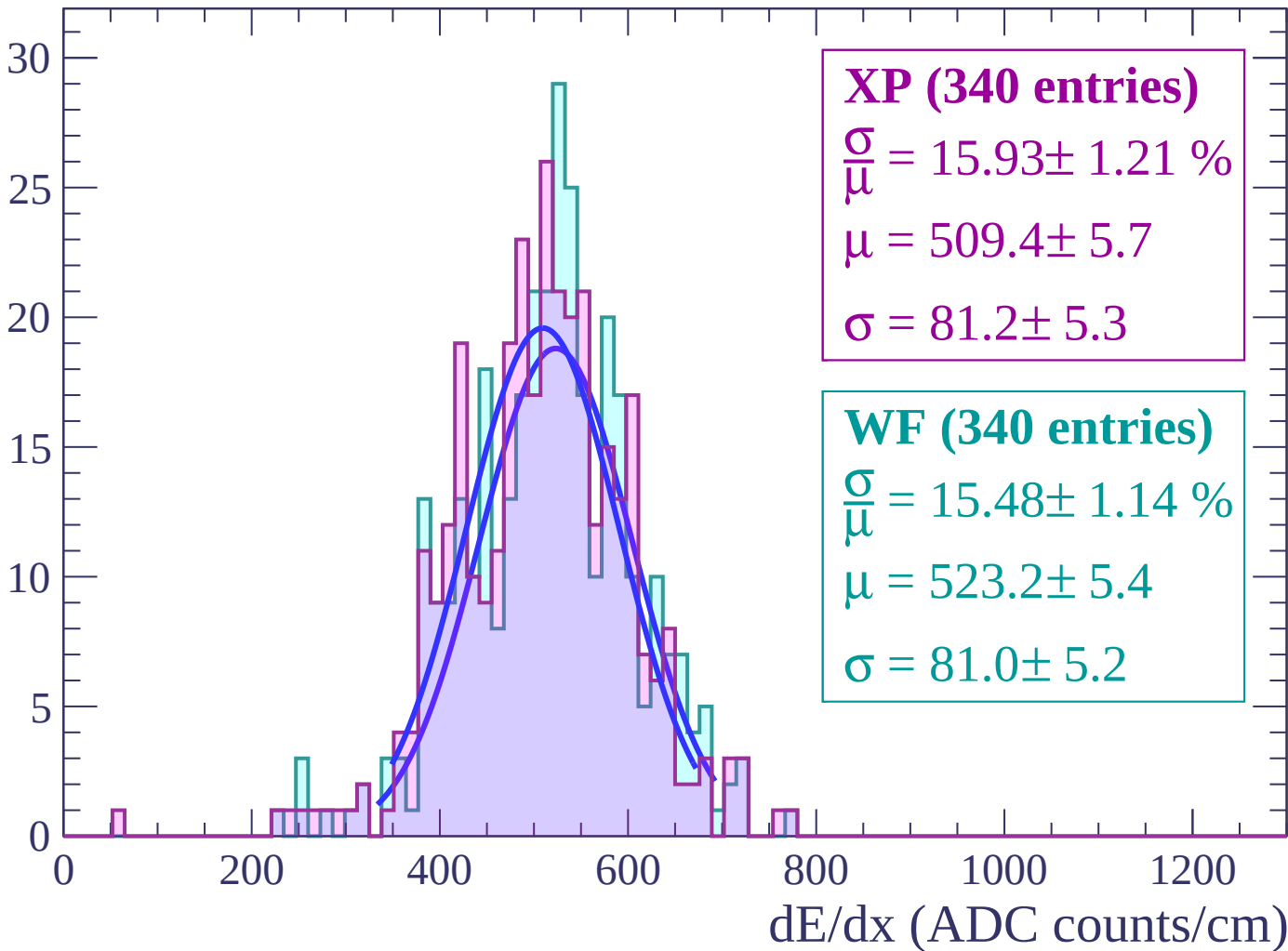
Energy loss | $56 < \theta < 58$

Count



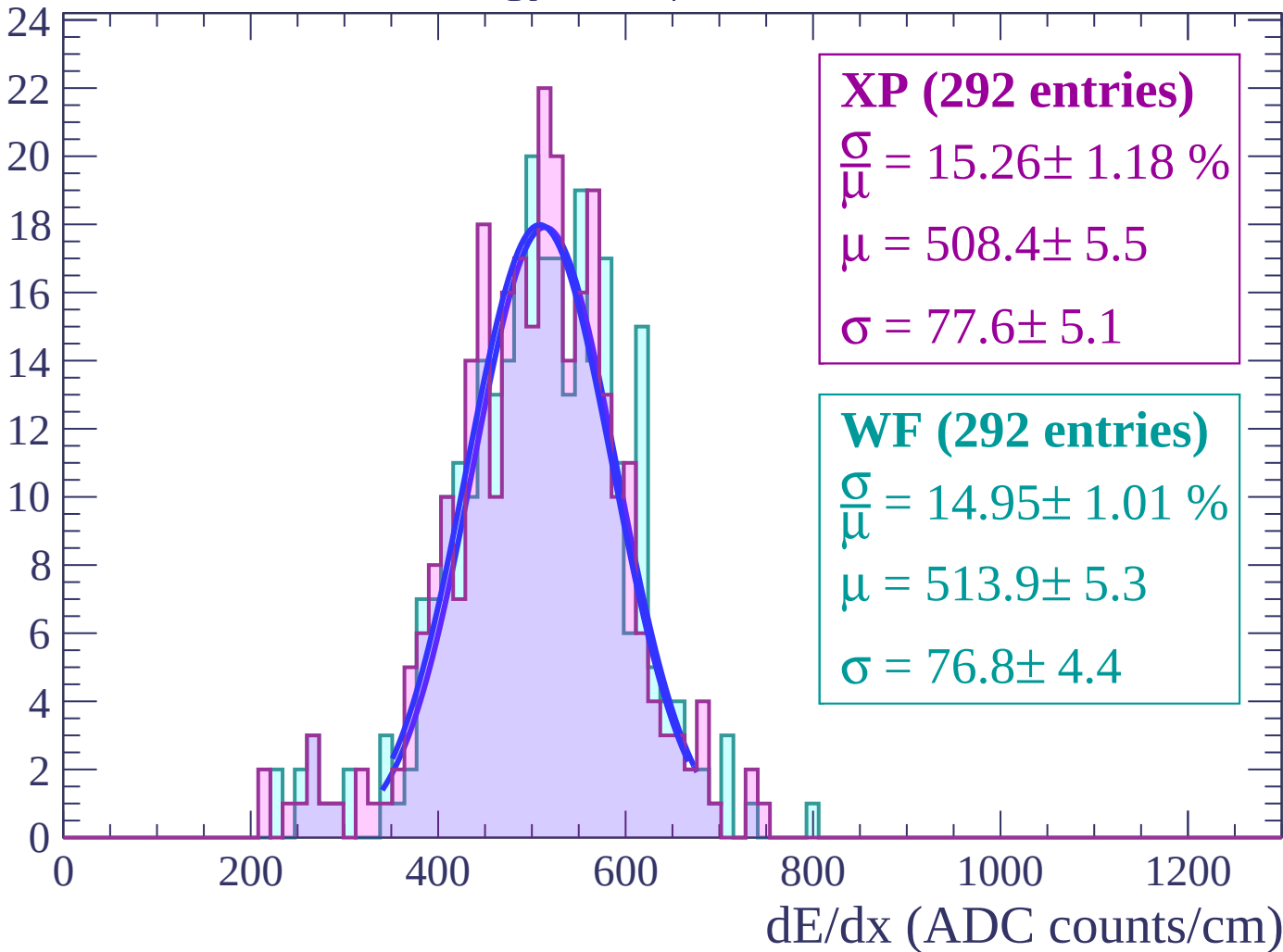
Energy loss | $58 < \theta < 60$

Count



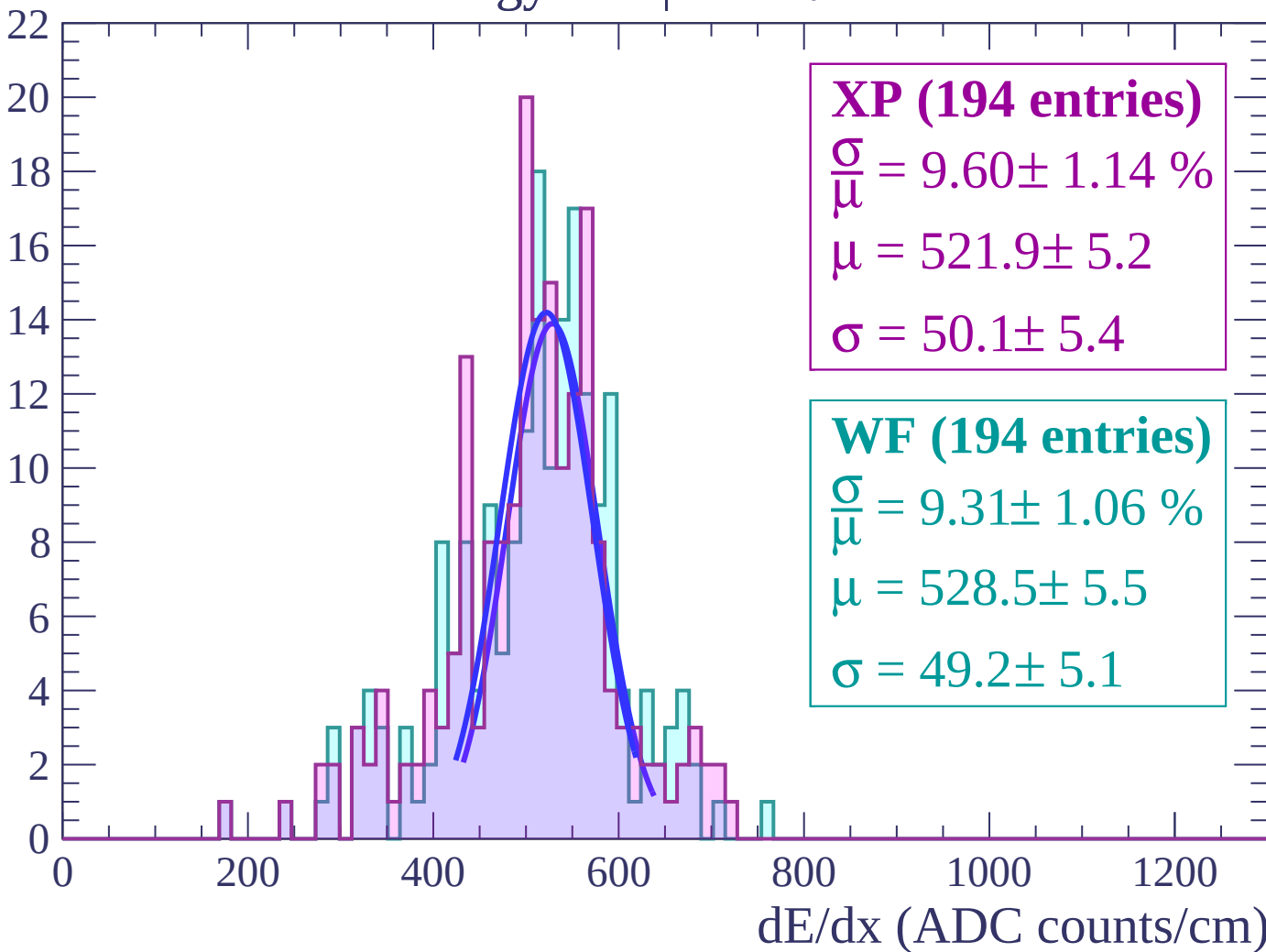
Energy loss | $60 < \theta < 62$

Count



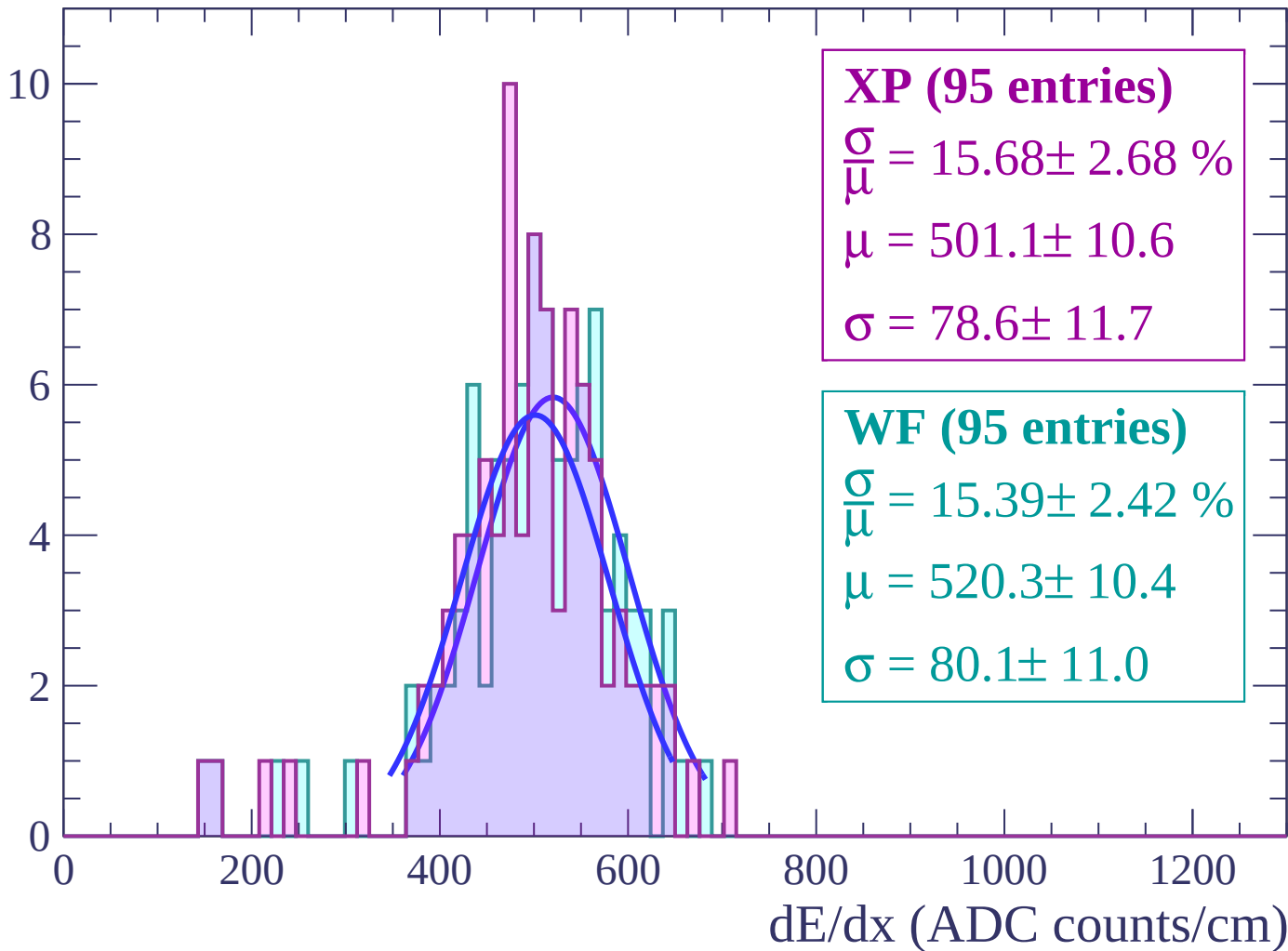
Energy loss | $62 < \theta < 64$

Count



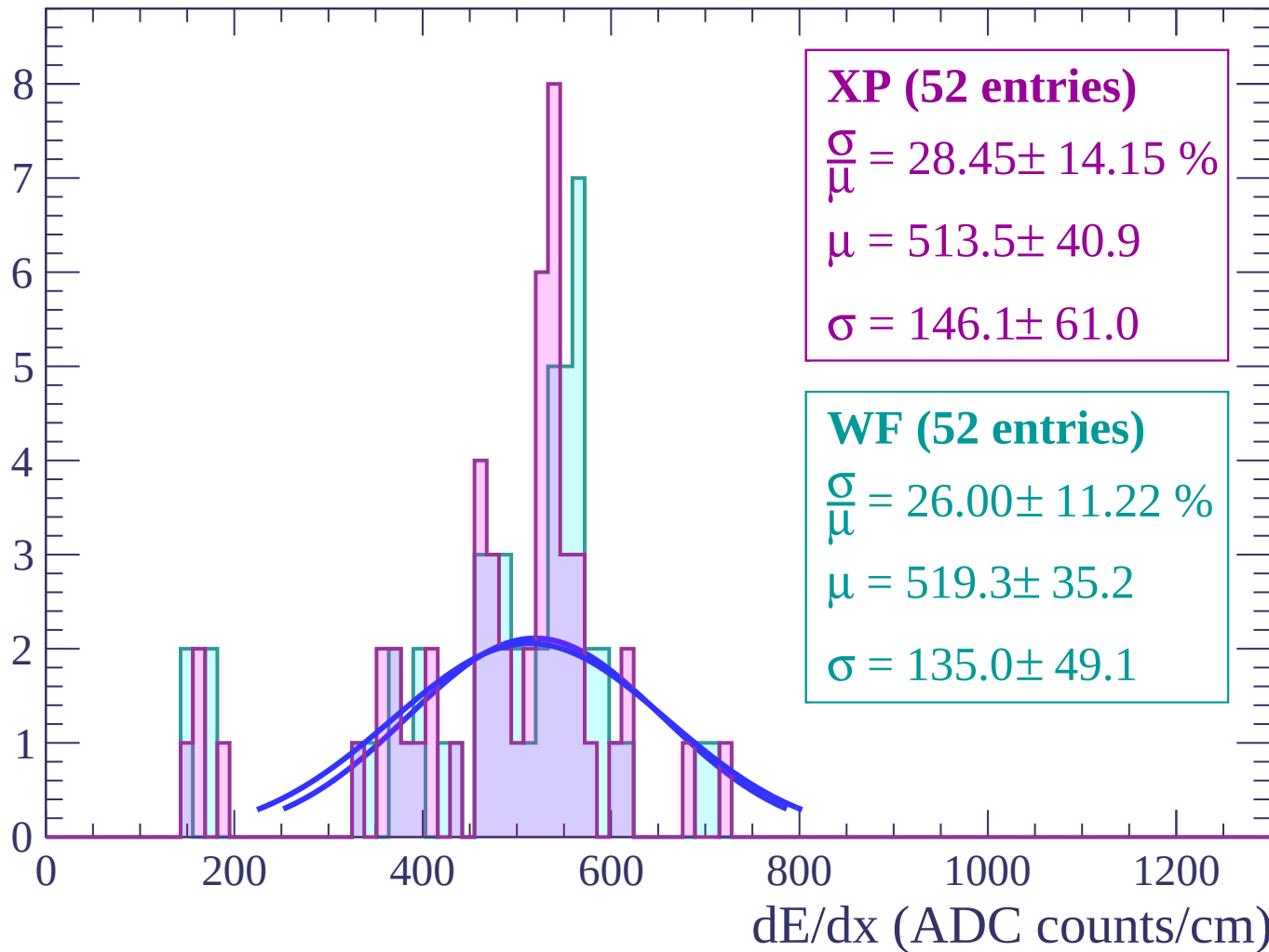
Energy loss | $64 < \theta < 66$

Count



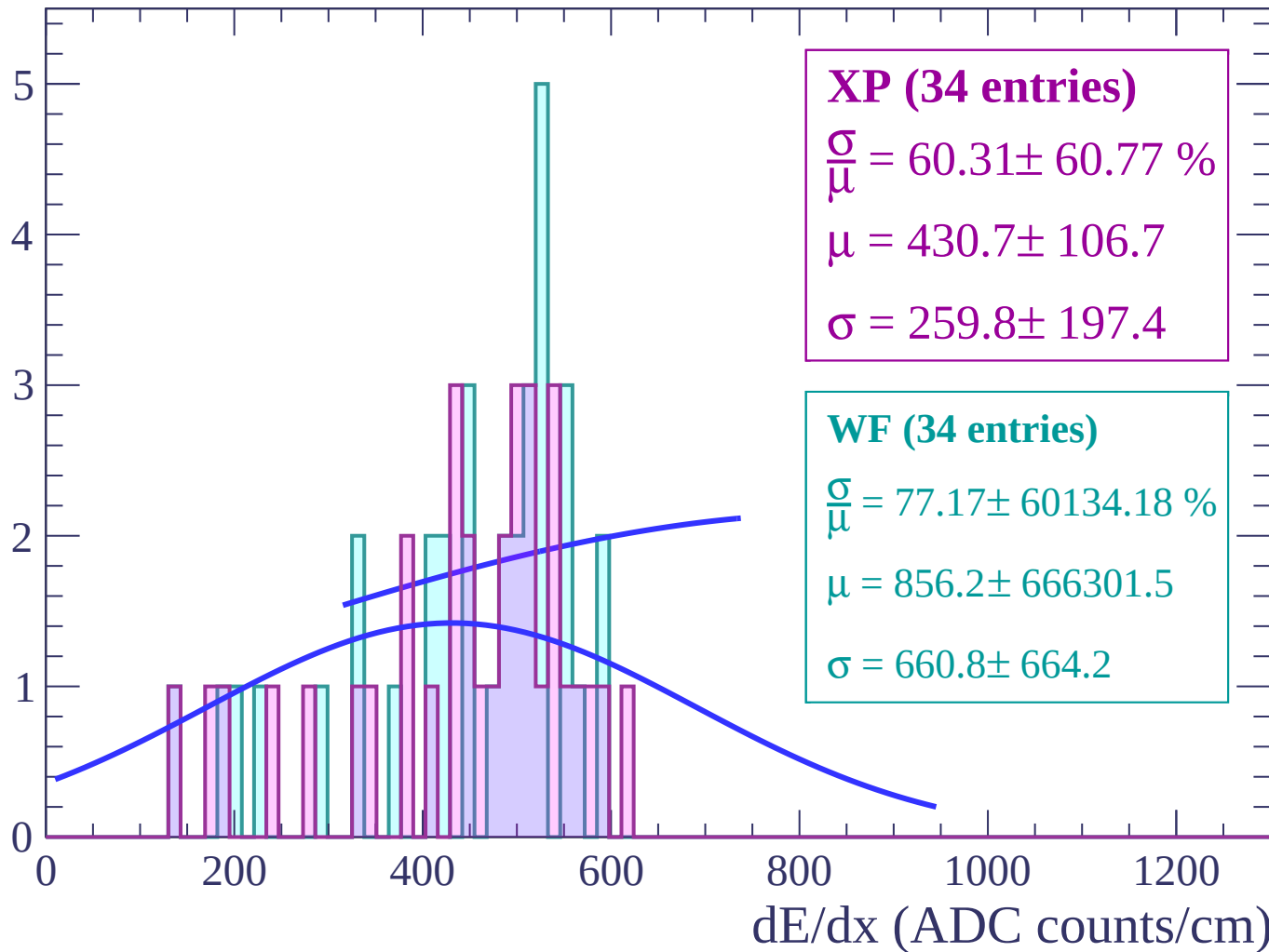
Energy loss | $66 < \theta < 68$

Count



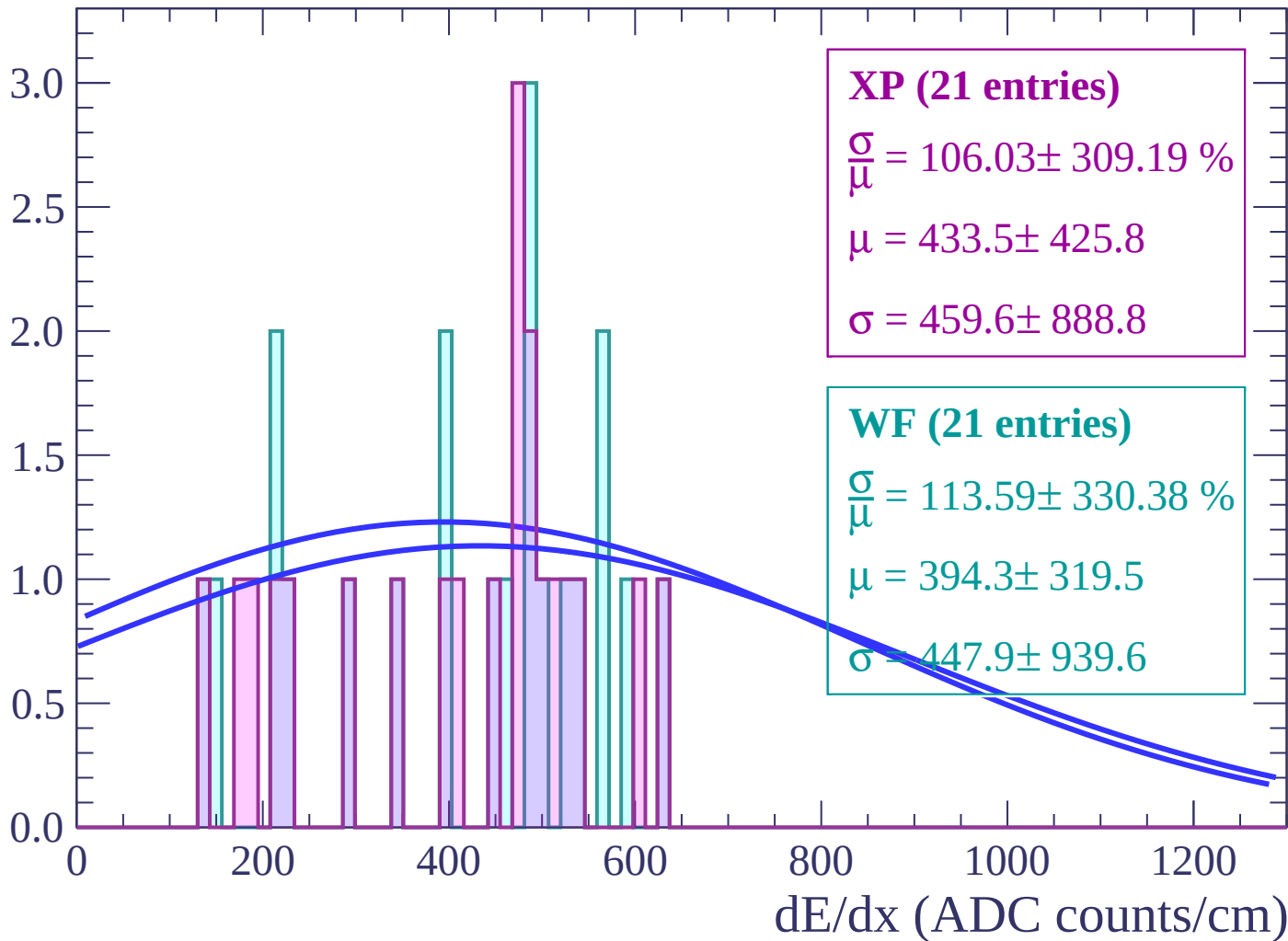
Energy loss | $68 < \theta < 70$

Count



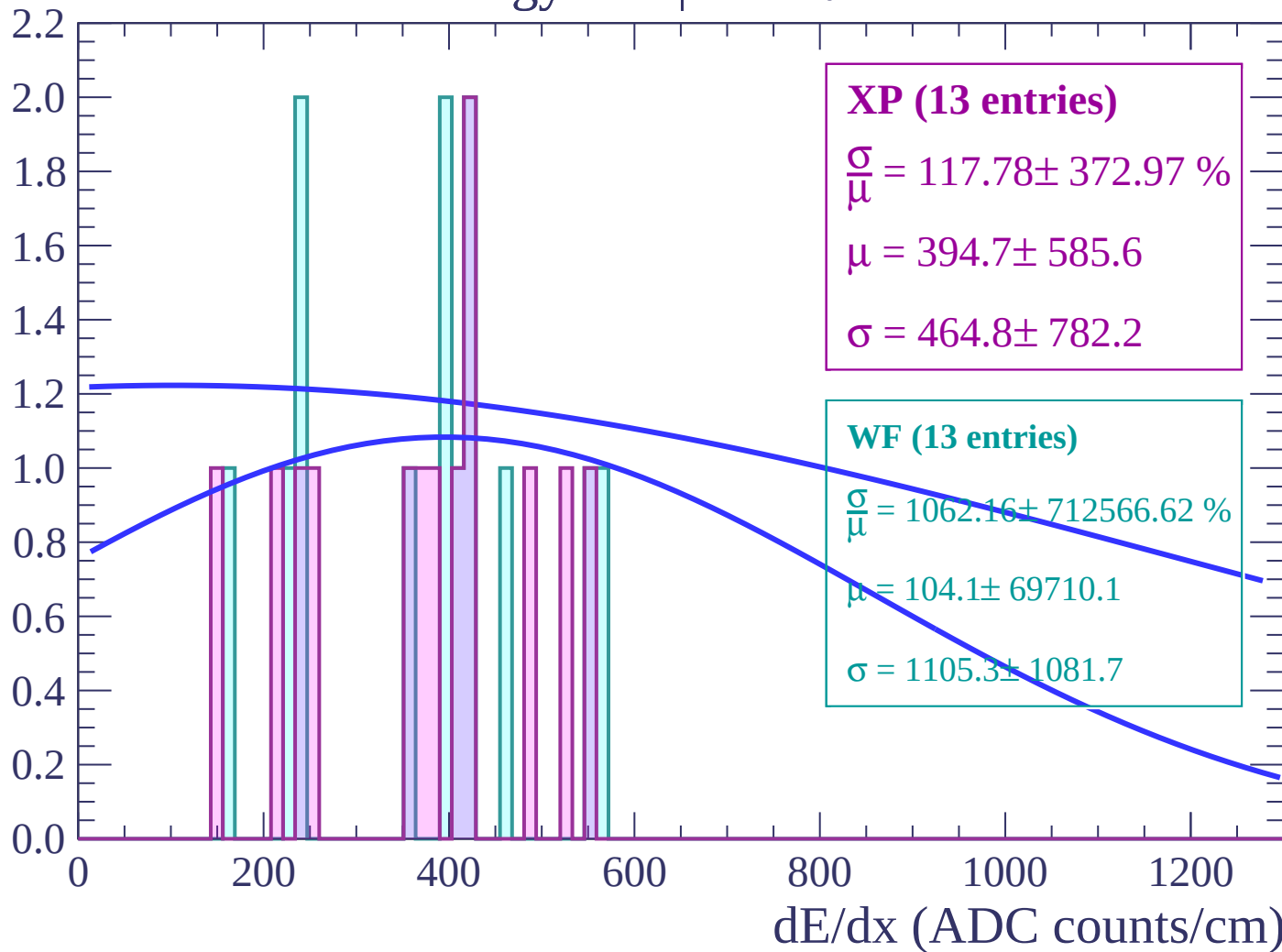
Energy loss | $70 < \theta < 72$

Count



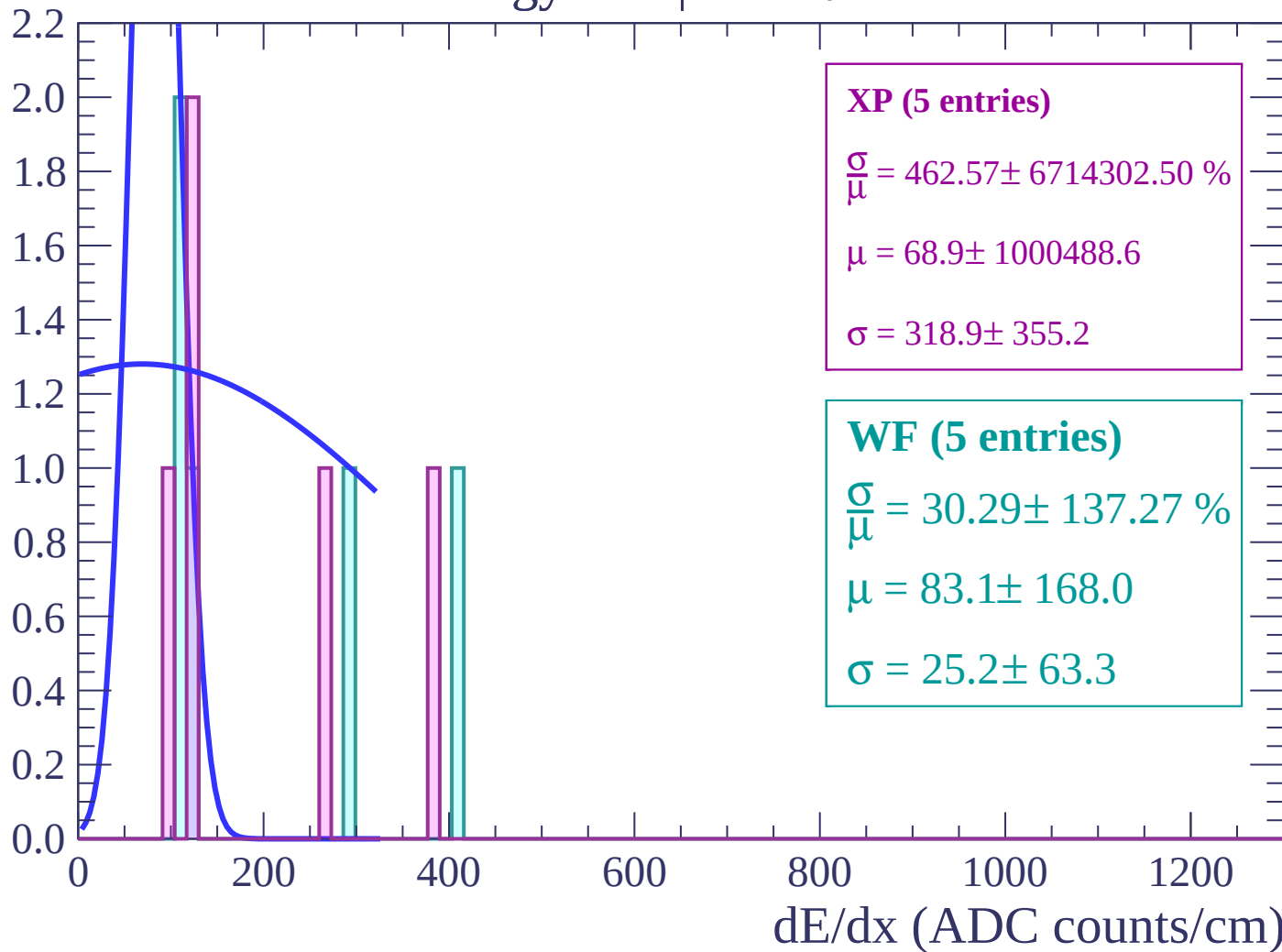
Energy loss | $72 < \theta < 74$

Count



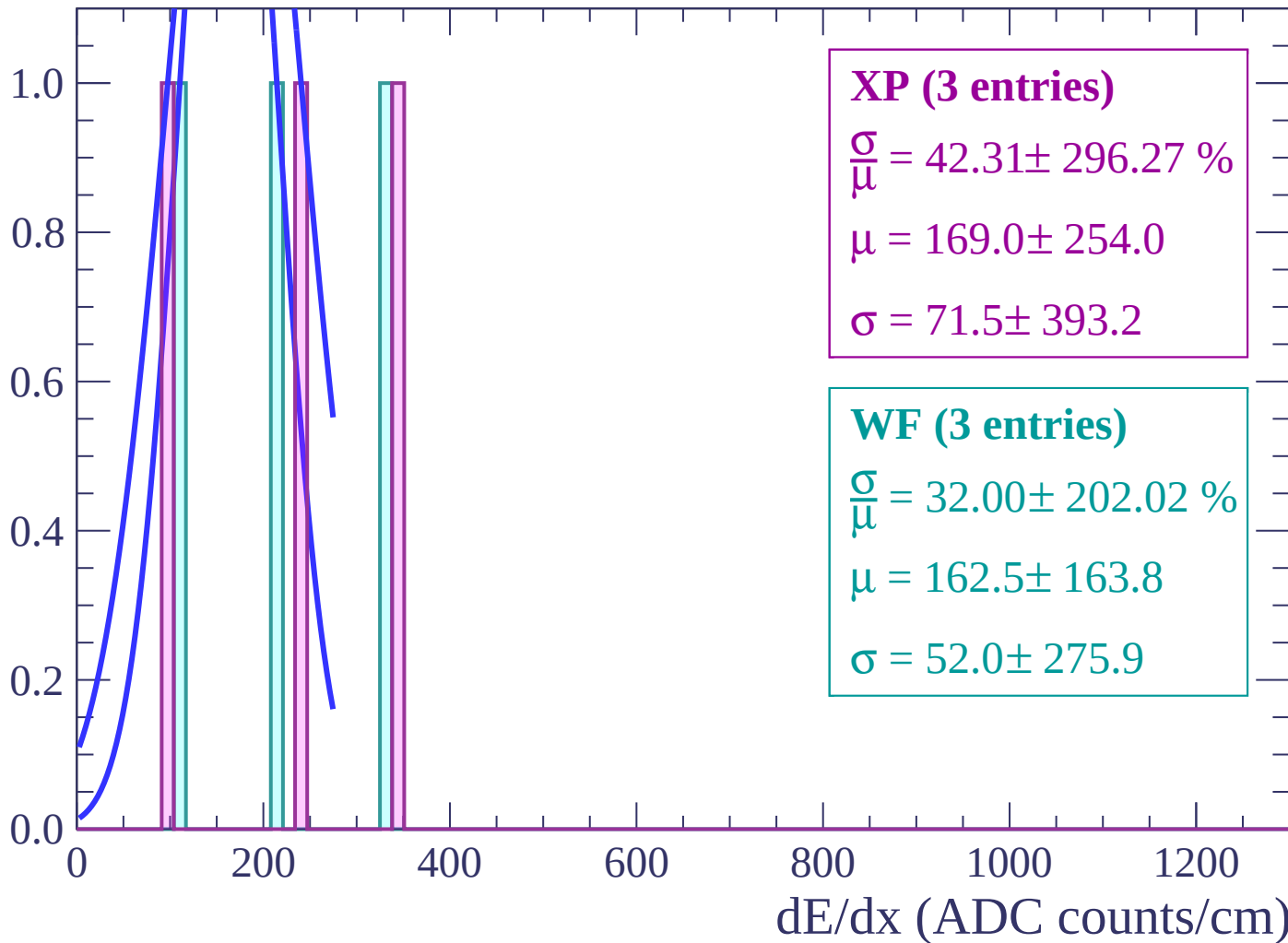
Energy loss | $74 < \theta < 76$

Count



Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

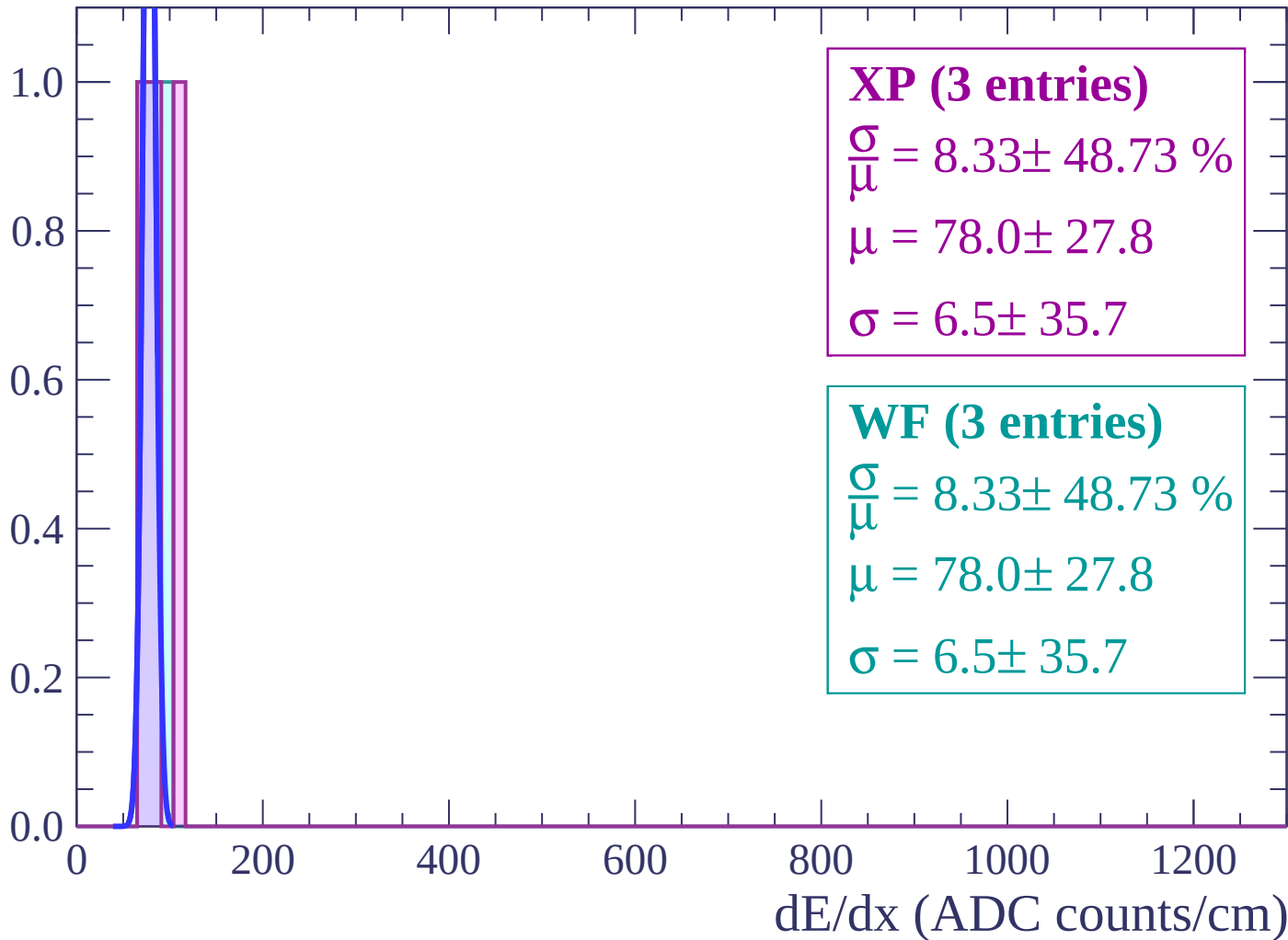
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $80 < \theta < 82$

Count



Energy loss | $82 < \theta < 84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $84 < \theta < 86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $86 < \theta < 88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $88 < \theta < 90$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $90 < \theta < 92$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

